Where Nehari's Theorem Survives

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Preface

This book organizes a body of research notes around one question: *where does Nehari's theorem survive in infinitely many variables, and why?* In one variable every bounded Hankel form has a bounded symbol. On $\mathbb{T}^\infty$ this fails, and the multiplicative Hilbert matrix sat for years exactly on the fault line. The chapters record what is now known on either side of it.

The book assumes the standard theory: Hardy spaces of Dirichlet series and the Bohr lift, Helson's inequality, the Cole–Gamelin estimate, weighted Bergman spaces of Dirichlet series in the sense of Bayart, Brevig, Haimi, Ortega-Cerdà and Perfekt, and Perfekt's recent Fejér–Riesz inequality. Known results are quoted, not reproved, and attributed where we know their source. For the remaining results we have not completed a literature search, so "new" here means only that we did not find them; it is not a priority claim. Failed approaches appear only as brief remarks, where they explain why a result has the shape it has.

How the proofs were checked. The results and most proofs were developed with the assistance of large language models and checked line by line in separate model runs. Results that rest on a computation say so in their statement or proof; rigorous computations use interval or exact rational arithmetic, and uncertified computations are marked as numerical. A *probabilistic certificate* is a computation whose conclusion holds except on an event of explicitly bounded probability, under stated assumptions on the random-number generator and on floating-point arithmetic. It is not a proof. None of these is formal verification, and none replaces reading by a specialist: *no proof in this book has been independently verified.* The proofs most in need of such a reading are the envelope argument of section 8.1, the Schatten characterization Theorem 5.31, and the Hellinger–Bessel chain of chapters 1 and 2 on which most of the book depends.

The parts are arranged as an argument. Part I proves the stability theorem for point evaluations and the Hellinger–Bessel inequality, and applies them to Carleson measures. Part II uses them, with sign averages, to prove Nehari-type theorems for structured Helson matrices. Part III shows where Nehari's theorem fails, in Schatten classes, and why the known constructions stop where they do. Part IV returns to the borderline object, the multiplicative Hilbert matrix, and to the constants and coefficient inequalities that remain open there.

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Overview

The problem

A Helson matrix $M(\alpha) = (\alpha(mn))_{m,n\geq 1}$ is a Hankel form in infinitely many variables: under the Bohr lift it becomes $(F, G) \mapsto \int FG\,\overline{g}\,\mathrm{d}m$ on $H^2(\mathbb{T}^\infty)$, for any g whose analytic coefficients are α. On the circle Nehari's theorem says that every bounded Hankel form has such a g in L^∞, of the same norm. On $\mathbb{T}^\infty$ this fails: the best constant on $\mathbb{T}^d$ grows at least exponentially in d [69], and there are bounded Helson matrices with no bounded symbol at all. The one-variable proof goes through inner–outer factorization, $H^1 = H^2 \cdot H^2$, and that is exactly what is lost.

This book asks which Helson matrices still have bounded symbols, and why. Duality reduces the question to a Carleson embedding: a positive matrix $M(\alpha)$ with $\alpha(n) = \int z^{\kappa(n)}\,\mathrm{d}\mu$ has a bounded symbol as soon as $\int |F|\,\mathrm{d}\mu \lesssim \|F\|_{H^1}$, whereas boundedness of the matrix is the same estimate in H^2. So the question becomes: when does an H^2 Carleson measure on $\mathbb{D}^\infty$ embed H^1? In one variable the answer is always, by factorization; here we need a replacement.

The starting point is Perfekt's theorem [70]: the multiplicative Hilbert matrix $(1/(\sqrt{mn}\log mn))$, which is the moment matrix of Lebesgue measure on the curve $\gamma(s) = (p^{-s})_p$, has a bounded symbol. His proof localizes point evaluations with Möbius maps and Hoeffding's inequality, and works for H^1.

Separation: the Hellinger–Bessel inequality

Part I replaces factorization by a one-line inequality and a Gram estimate. For a point $r \in \mathbb{D}^\infty \cap \ell^2$ with reproducing kernel K_r, every $F \in H^p$ and every $p > 0$ satisfy

$$|F(r)|^{p/2} \leq \int |F|^{p/2} |K_r|\,\mathrm{d}m$$

(Theorem 1.7). Equivalently, a function that is efficient at r must have $|F|^p$ close, in Hellinger distance, to the harmonic measure $\omega_r = |K_r|^2\,\mathrm{d}m/\|K_r\|^2$. Since $|F|^{p/2}$ lies in L^2, the evaluations at many points can then be added up by Bessel's inequality for the vectors $|K_r|$. This gives the *Hellinger–Bessel inequality* (Theorem 2.1): for every positive measure μ and every $p > 0$,

$$\int |F|^p\,\mathrm{d}\mu \leq \|\mathcal{G}_\mu\|\,\|F\|_p^p, \qquad \mathcal{G}(r,s) = \|K_r\|\|K_s\|\,H(\omega_r, \omega_s).$$

The condition does not see p, and it holds in the Hardy, Bergman and Gaussian geometries alike. Everything quantitative then rests on computing Hellinger affinities. In each coordinate they are hypergeometric, and across coordinates they multiply,

so $H(\omega_r, \omega_s) \le e^{-E(r,s)/4}$ with $E = \sum_p \rho_p^2$ the sum of squared pseudo-hyperbolic distances. Along the prime curve E is, up to a bounded error, the hyperbolic distance in the half-plane (Theorem 1.6).

The consequences in Part I:

- the Fejér–Riesz inequality $\int_0^1 |f(1/2+\sigma)|^p \, \mathrm{d}\sigma \le (5+2\sqrt{2})\|f\|^p_{\mathscr{H}^p}$ for every $p > 0$ (Theorem 2.5), and, keeping the phases of one prime, the constant $\Gamma(1/4)^2/\sqrt{2\pi} \approx 5.244$ on $(0, 2)$, which is exactly the limit of the method there (Theorem 2.9);
- Carleson estimates for measures along curves in all three geometries and on prime sets of any density, with constant below 21 for the prime curve (Theorem 2.11);
- a complete description of Carleson measures on families separated in energy: for these, H^p Carleson for one p is the same as for all p, and the same as the one-point condition (Proposition 2.14 and Corollary 2.13);
- on vertical segments of the critical line, the local embedding constant at distance ε is at most $\exp\bigl(C\sqrt{\log(1/\varepsilon)\log\log(1/\varepsilon)}\bigr)$ for every p, smaller than any power of $1/\varepsilon$ (Theorem 3.4), and at least $\sqrt{\log(1/\varepsilon)}/\log\log(1/\varepsilon)$ for suitable $p < 2$ (Proposition 3.5); for $p = 2$ the constant C_0 of $\int_0^1 |f(1/2+it)|^2 \, \mathrm{d}t \le C_0 \|f\|^2_{\mathscr{H}^2}$ satisfies $2\pi < C_0 < 2\pi + 10^{-3}$ and is attained (Proposition 3.8).

Two structural facts organize Part I. The extremals of point evaluation in infinitely many variables are exactly $cK_r^{2/p}$ (Proposition 1.16), and in every geometry the harmonic-measure overlap is squeezed between the reproducing-kernel overlap and its square root, $|\langle k_r, k_s\rangle| \le H(\omega_r, \omega_s) \le |\langle k_r, k_s\rangle|^{1/2}$ (Proposition 1.4), with equality on the right for Gauss. The method has limits. The Gram condition is not necessary: angular averages in one coordinate are Carleson for every p while $\mathcal{G}_\mu$ is unbounded (chapter 2). And near the critical line its constant is exactly $\Gamma(1/4)^2/\sqrt{2\pi} \approx 5.24$, against π for the corresponding H^2 estimate.

Symmetry: sign averages

The second mechanism is independent. For $g \in H^p(\mathbb{T})$ and $|x| < 1$,

$$\frac{|g(x)|^p + |g(-x)|^p}{2} \le \frac{\|g\|_p^p}{1-|x|^4},$$

sharply, and this tensorizes (Proposition 4.6). Averaging signs in a set of coordinates therefore costs $\prod(1-|x_p|^4)^{-1}$, which is finite for $x \in \ell^4$, whereas a single evaluation needs $x \in \ell^2$. Part II combines the two mechanisms. Keep the primes of a set R of density θ on the curve and average the signs of the rest. The resulting family of positive Helson matrices interpolates between the multiplicative Hilbert matrix (R all primes) and its full sign average (R empty). For every member, boundedness, the H^1 Carleson condition and the existence of a bounded symbol are equivalent to $\nu((1/2, 1/2+t]) = O(t^\theta)$, with a constant depending only on R (Theorem 4.2). On the curve, the Nehari constant is below 43; for the full sign average it is at most $\zeta(2)$. Positivity alone is not enough: a countable superposition of measures

on sign-rotated copies of the curve gives a bounded non-negative Helson matrix with no bounded symbol (Theorem 4.21), which answers a question of Perfekt and Pushnitski in the negative.

The same family has a complete compactness theory. There is a Hartman-type theorem: compactness is the little-o condition, and the symbol then lies in $C(\mathbb{T}^\infty) + N$ (Theorem 4.10). There is an essential Nehari theorem: the essential norm is comparable to the essential size of the symbol, and equals $\pi w(1/2)$ for measures with a density (Theorem 4.11). Finally, for the curve matrix compressed to the integers generated by R, the whole distribution of the spectrum is read off from the dyadic profile $a = (e^{j\theta}\nu(I_j))_j$: for every $0 < p < \infty$ and $0 < q \le \infty$, it lies in $S_{p,q}$ if and only if $a \in \ell^{p,q}$ (Theorem 5.38). For R all primes this is the curve matrix itself. In particular, positive curve matrices occur in every Schatten class and all have bounded symbols. Off the curve the spectrum collapses: root-exponential decay for compressions to sparse prime sets, and $1/(n \log n)$ for the full sign average. The constructions and compactness arguments carry over to Bergman and Fock spaces, where Kulikov's contractive embedding, and its Gaussian limit, replace Carleman's inequality (chapter 6).

Failure: the threshold 4

Part III turns to Schatten-class forms with no bounded symbol. Helson's theorem rules them out in S_2. Brevig and Perfekt found them in S_p for $p > 5.74$, and Bampouras was the first to show that they exist for every $p > 4$ [4]. We found the same bound independently, with the same constant-plus-product family; chapter 7 gives both proofs. His proof integrates over scales and is shorter; ours fixes the scale by large deviations in the Gaussian model and is Legendre dual to his.

The point of Part III is the threshold itself. *These methods stop at* 4*; whether the true threshold is* 4 *is open.* For the constant-plus-product family, no test function and no choice of coefficients beats S_4, for any n (Theorem 8.1). For every polynomial on every torus $\|\varphi\|_{L^4} \le \|H_\varphi\|_{S_4}$, with equality in the Gaussian model, so scale integration can never go below 4 (Corollary 7.7). Affine symbols, even quadratic symbols of rank at most two, and zero-free symbols are stuck in the same way. Every family of bounded degree in a bounded number of variables is excluded on an interval above 2 (Theorem 8.14). For the Gaussian method everything comes down to one concrete question: is $|\int f\overline{\varphi}\,\mathrm{d}\mu| \le \|f\|_{L^1(\mu)}\|\varphi\|_{L^4(\mu)}$ for Gaussian polynomials (Problem 7)? A counterexample would go below 4; a proof would show that Gaussian blocks cannot, but would not decide p_*, which is 4 exactly when forms in S_4 have bounded symbols (Proposition 7.13). Bampouras's Paley–Wiener threshold, also 4, appears to have a different source: square-root cancellation in L^1 combined with parabolic scaling, not a Gaussian moment (section 8.5).

The borderline: is the constant π?

Part IV returns to the multiplicative Hilbert matrix. In one variable four inequalities share the constant π: Hilbert's, Fejér and Riesz's, Hardy's, and the bound for $\sum a_k/(k+1)$ on H^1, all because every H^1 function is a product of two H^2 functions (Proposition 9.1). On $\mathbb{T}^\infty$ the first survives with the same constant and factorization

fails, and chapters 9 to 11 measure what that costs. The norm C of the functional $\sum a_n/(\sqrt{n}\log n)$ on $\mathscr{H}^1_0$ is finite by Perfekt's theorem and satisfies $\pi \le C \le 5.515$, by a computer-assisted bound (Proposition 9.14); the functional has an explicit symbol of norm at most $10^5 + 7$ (Theorem 9.18); and, by the same computation, no counterexample to $C = \pi$ uses only primes below 10^{216}; there even the sum with absolute values on the coefficients is below π. Near-extremal vectors for the Hilbert matrix vary slowly under multiplication, so no fixed finite-prime factor can raise an extremal square above π (Theorem 9.3 and Corollary 9.4); the problem is equivalent to a Hankel completion on the ordered rationals (Theorem 9.5); and the six per cent excess over the H^2 benchmark seen in two to four primes is an Ortega-Cerdà–Seip effect of linear symbols, which tends to $\pi/(2\sqrt{2})$ for two large primes and says nothing about π (Proposition 9.10).

Chapter 10 turns to the relatives of ℓ. The embedding of Part I settles a conjecture of Bondarenko, Brevig, Saksman and Seip, who had treated $\beta \neq 1/p$ for $p \le 2$, and more: for every $0 < p < \infty$, the fractional primitive $\sum a_n/(\sqrt{n}(\log n)^\beta)$ is bounded on $\mathscr{H}^p$ if and only if $\beta > 1/p$, or $\beta = 1/p$ and $p \le 1$, exactly as on the disc; on a prime set of density θ the threshold is θ/p (Corollary 10.13 and Theorem 10.12). The companion question with absolute values on the coefficients is proved on every piece of $\mathscr{H}^1$ that a degree constraint visible to the Euler product cuts out (Theorem 10.2). On a prime set of density θ the absolute sum is bounded for every logarithmic exponent $\beta > \theta$, and the signed sum at $\beta = \theta$ (Corollary 10.10), so only the endpoint is open.

Is $C = \pi$? Very probably not. For quadratic combinations of Euler kernels at many scales near the critical line, the H^1 norm has an exact Gaussian limit (Theorem 11.3), so C is at least a constant of a log-correlated Gaussian field, and we conjecture equality (Conjecture 1). A probabilistic certificate puts that constant at 3.1450 or more, with failure probability at most $2 \cdot 10^{-30}$ (remark 11.10); a deterministic proof is open. The same Gaussian witnesses show that this excess would survive removing every continuous symbol (Corollary 11.7), and if ℓ does not attain its norm, then C equals its distance to the functionals of finitely many coefficients (Corollary 11.14).

Reading guide

Chapters 1 and 2 are the foundation. Part II depends on them throughout; chapter 3 is not needed later except for Theorem 3.1, used in chapter 10. Part III is independent of Parts I and II. In Part IV, chapter 9 uses chapter 2; chapter 10 needs chapter 2, Theorem 3.1 and Helson's inequality; chapter 11 uses chapter 9 and the essential norms of Part II. Each chapter ends with what remains open, and chapter B collects the open problems.

Part 1

Replacing factorization: point evaluations

CHAPTER 1

Harmonic measure, energy and stability

How much of the torus does a point evaluation use, and when can two point evaluations share it? Section 1.2 answers this for one point: the mass of an extremal function is exactly the harmonic measure of the point. Section 1.3 answers it for two points: harmonic measures overlap by an amount exponentially small in an energy, with an exact formula in one variable and a quantitative form of Kakutani's dichotomy. Section 1.5 identifies that energy, along the Dirichlet half-plane, with hyperbolic distance. Section 1.6 proves the converse: a function that is nearly extremal at a point distributes its mass like that point's harmonic measure, in every H^p. Chapter 2 turns this into the Hellinger–Bessel inequality.

Three geometries run in parallel: the Hardy space of the torus, Bergman spaces of the polydisc, and the Fock space. The Gaussian case is the flat model, where every overlap is an exact exponential. Statements are made on finitely many coordinates, with constants independent of their number, and pass to $\mathbb{T}^\infty$ by approximation.

What is new, to our knowledge, is this point of view: one change-of-variables rule for three geometries reduces every statement about one or two point evaluations to a computation with harmonic measures. In this form the stability theorem, its two-point and many-point consequences, and the two-sided identification of energy with hyperbolic distance are new; the real-axis case of the latter, a lower bound for the energy along the prime curve, is in Perfekt's paper [70]. The basic inequality $|F(r)|^{p/2} \le \int |F|^{p/2} |K_r| \, \mathrm{d}\lambda$ is elementary: it is the mean-value argument behind the Cole–Gamelin estimate, and because $|K_r|$ is a product over coordinates it tensorizes, so it holds in infinitely many variables with no loss. What is new is reading it as a lower bound for a Hellinger affinity, which turns questions about many point evaluations into Hilbert-space geometry (chapter 2).

1.1. Three geometries

Each geometry consists of a product probability measure λ, the closure X^p of the analytic polynomials in $L^p(\lambda)$, a reproducing kernel K_r of X^2, and a coordinatewise family of maps T_r with $T_r(0) = r$.

	λ per coordinate	$K_r(z)$ per coordinate	$T_r(w)$ per coordinate
Hardy	arc length $\mathrm{d}m$ on $\mathbb{T}$	$(1-\overline{r}z)^{-1}$	$(r+w)/(1+\overline{r}w)$
Bergman	$m_\beta = (\beta-1)(1-\lvert z\rvert^2)^{\beta-2}\,\mathrm{d}A$	$(1-\overline{r}z)^{-\beta}$	$(r+w)/(1+\overline{r}w)$
Gauss	$\gamma = \pi^{-1}e^{-\lvert z\rvert^2}\,\mathrm{d}A$ on $\mathbb{C}$	$e^{\overline{r}z}$	$r+w$

Here $\beta > 1$, and $\mathrm{d}A$ is area measure on $\mathbb{D}$, normalized to have mass one in the Bergman row. Write $k_r = K_r/\|K_r\|_2$ for the normalized kernel. Then

$$\|K_r\|_2^2 = \prod_p (1-|r_p|^2)^{-1}, \quad \prod_p (1-|r_p|^2)^{-\beta}, \quad \exp\Big(\sum_p |r_p|^2\Big)$$

in the three cases. All three have the same change-of-variables rule.

LEMMA 1.1. *In each geometry, for every r,*

$$\lambda \circ T_r = \|K_r\|_2^2 \, |K_r \circ T_r|^{-2} \, \lambda,$$

and $K_r \circ T_r$ is holomorphic and zero-free on a neighbourhood of the closed domain.

PROOF. In one coordinate:

- *Hardy.* $K_r(T_r w) = (1+\overline{r}w)/(1-|r|^2)$, and the Jacobian of T_r on $\mathbb{T}$ is $(1-|r|^2)/|1+\overline{r}w|^2$.
- *Bergman.* The density of $m_\beta \circ T_r$ is $(1-|r|^2)^\beta |1+\overline{r}w|^{-2\beta}$, and $K_r(T_r w) = \big((1+\overline{r}w)/(1-|r|^2)\big)^\beta$.
- *Gauss.* $\gamma(r+dw) = e^{-|r|^2 - 2\Re(\overline{r}w)}\gamma(dw)$, and $K_r(r+w) = e^{|r|^2+\overline{r}w}$.

In each case the identity is checked by multiplying out. Products over coordinates preserve it. □

1.2. One point

A *representative* of evaluation at r is a function ψ with $\int F\overline{\psi}\,\mathrm{d}\lambda = F(r)$ for every analytic polynomial F. Every representative has $\|\psi\|_\infty$ at least the norm of evaluation on X^1. Two representatives sit at opposite extremes.

- *The positive representative* $|k_r|^2 = |K_r|^2/\|K_r\|_2^2$. In the Hardy case this is the Poisson kernel P_r. It has L^1 norm one, but it is unbounded, with $\sup P_r = \prod_p (1+|r_p|)/(1-|r_p|)$, which is infinite unless $\sum|r_p| < \infty$.
- *The flat representative* $\Phi_r = \|K_r\|_2^2 \, K_r/\overline{K_r}$, of constant modulus $\|K_r\|_2^2$, introduced in the Hardy case by Perfekt [70]. The next lemma proves the representing property, together with its localized version.

In the Hardy case the norm of evaluation is due to Cole and Gamelin [30], who identified $\mathbb{D}^\infty \cap \ell^2$ as the set of points where evaluation on $H^1(\mathbb{T}^\infty)$ is bounded.

LEMMA 1.2 (Localization). *Let D have positive measure and be invariant under the common rotations $w \mapsto e^{i\vartheta}w$. Then*

$$\psi_{r,D} = \frac{\|K_r\|_2^2}{\lambda(D)} \, \frac{K_r}{\overline{K_r}} \, \mathbf{1}_{T_r(D)}$$

represents evaluation at r, and $|\psi_{r,D}| = \|K_r\|_2^2/\lambda(D)$ on its support $T_r(D)$.

PROOF. Substitute $z = T_r w$ and use Lemma 1.1. Since $\overline{K_r}/K_r \cdot |K_r|^{-2} = K_r^{-2}$,

$$\int F\,\overline{\psi_{r,D}}\,\mathrm{d}\lambda = \frac{\|K_r\|_2^4}{\lambda(D)} \int_D G\,\mathrm{d}\lambda, \qquad G = \frac{F\circ T_r}{(K_r \circ T_r)^2}.$$

The function G is holomorphic near the closed domain (in the Gauss case it is entire, with growth at most $e^{c|w|}$). Since D and λ are invariant under common rotations, averaging $G(e^{i\vartheta}w)$ over ϑ shows $\int_D G\,\mathrm{d}\lambda = G(0)\lambda(D) = F(r)\lambda(D)/\|K_r\|_2^4$. □

PROPOSITION 1.3. *In each geometry, evaluation at r has norm exactly $\|K_r\|_2^2$ on X^1. The flat representative is extremal:*

$$\min\{\|\psi\|_\infty : \psi \text{ represents evaluation at } r\} = \|K_r\|_2^2 = \|\Phi_r\|_\infty.$$

The function $F_r = K_r^2/\|K_r\|_2^2$ attains the norm, with $F_r(r) = \|K_r\|_2^2$, $\|F_r\|_1 = 1$, and

$$|F_r|\,\mathrm{d}\lambda = \omega_r \coloneqq |k_r|^2\,\mathrm{d}\lambda = (T_r)_*\lambda.$$

PROOF. The flat representative gives $|F(r)| \le \|K_r\|_2^2\|F\|_1$. For F_r we have $\|F_r\|_1 = \|K_r\|_2^2/\|K_r\|_2^2 = 1$ and $F_r(r) = \|K_r\|_2^4/\|K_r\|_2^2$. So the norm is $\|K_r\|_2^2$, and no representative has smaller sup norm. The identity $|k_r|^2\lambda = (T_r)_*\lambda$ is Lemma 1.1, read backwards. □

We call ω_r the *harmonic measure* of r. In the Hardy case it is the usual product of Poisson measures, and in the Gaussian case it is the standard Gaussian translated to r. Proposition 1.3 says that the mass of the extremal function at r is distributed exactly like ω_r. So the question of where near-extremal functions live becomes a question about harmonic measures.

1.3. Two points

Let $\rho(a,b) = |a-b|/|1-\overline{a}b|$ be the pseudo-hyperbolic distance on $\mathbb{D}$, and define the *energy* between two points:

$$E(r,s) = \sum_p \rho(r_p,s_p)^2 \quad \text{(Hardy, Bergman)}, \qquad E(r,s) = \sum_p |r_p - s_p|^2 \quad \text{(Gauss)}.$$

Two numbers measure how much two points share. In X^2 it is the overlap $|\langle k_r, k_s\rangle|$ of the normalized kernels. In X^1 it is the Hellinger affinity

$$H(r,s) = \int \sqrt{\mathrm{d}\omega_r\,\mathrm{d}\omega_s} = \int |k_r||k_s|\,\mathrm{d}\lambda$$

of the harmonic measures.

PROPOSITION 1.4. *The two overlaps compare as follows.*

(1) Exact X^2 overlaps.

$$|\langle k_r, k_s\rangle|^2 = \prod_p (1-\rho_p^2), \quad \prod_p (1-\rho_p^2)^\beta, \quad e^{-E(r,s)}$$

in the three cases, where $\rho_p = \rho(r_p, s_p)$.

(2) Comparison. *In every geometry, $H(r,s) \ge |\langle k_r, k_s\rangle|$.*

(3) Exact X^1 overlaps. *In the Gauss case $H(r,s) = e^{-E(r,s)/4}$. In the Hardy and Bergman cases $H(r,s) = \prod_p h_\beta(\rho_p)$, with $\beta = 1$ for Hardy and*

$$h_\beta(\rho) = (1-\rho^2)^{\beta/2}\,{}_2F_1\Big(\frac{\beta}{2}, \frac{\beta}{2}; \beta; \rho^2\Big);$$

for $\beta = 1$ we write $h = h_1 = (2/\pi)(1-\rho^2)^{1/2}\mathbf{K}(\rho)$, with $\mathbf{K}$ the complete elliptic integral of the first kind.

(4) Two-sided comparison. *In every geometry,*

$$|\langle k_r, k_s\rangle| \le H(r,s) \le |\langle k_r, k_s\rangle|^{1/2},$$

with equality on the right in the Gauss case. Consequently $H(r,s) \le e^{-E(r,s)/4}$ *in the Hardy and Gauss cases, and* $H(r,s) \le e^{-\beta E(r,s)/4}$ *in the Bergman case; in the Hardy case* $h(\rho) \le (1-\rho^2)^{1/4} \le 1 - \rho^2/4$.

PROOF. (1) In one coordinate, $|\langle k_a, k_b\rangle|^2 = (1-|a|^2)(1-|b|^2)/|1-\overline{a}b|^2 = 1 - \rho(a,b)^2$, and the Bergman kernel is the βth power of the Hardy kernel. In the Gauss case, $|\langle k_r, k_s\rangle|^2 = e^{2\Re\langle s,r\rangle - |r|^2 - |s|^2} = e^{-|r-s|^2}$.

(2) $\int |k_r||k_s|\,\mathrm{d}\lambda \ge |\int k_r \overline{k_s}\,\mathrm{d}\lambda|$.

(3) In the Gauss case ω_r is the complex Gaussian centered at r with identity covariance, and the Hellinger affinity of two such measures is $e^{-|r-s|^2/4}$. In the other two cases, Möbius invariance of the pseudo-hyperbolic distance and of the construction $\omega_a = (T_a)_* m$ reduce one coordinate to $a = 0$, $b = c$ with $|c| = \rho$, and $h_\beta(\rho) = (1-\rho^2)^{\beta/2}\|(1-\overline{c}z)^{-\beta/2}\|^2_{A^2_\beta}$, which is the stated hypergeometric expression.

(4) The left inequality is (2). On the right both sides are products over coordinates. In the Gauss case $H = e^{-E/4} = |\langle k_r, k_s\rangle|^{1/2}$. In the other two cases, with $a = \beta/2$ and $x = \rho^2$, the claim is ${}_2F_1(a,a;2a;x) \le (1-x)^{-a/2}$, which holds coefficient by coefficient: both series start at 1, and the ratios of consecutive coefficients are $(a+k)^2/((2a+k)(k+1))$ and $(a/2+k)/(k+1)$, where $(a+k)^2 \le (2a+k)(a/2+k)$ reduces to $0 \le ak/2$. Finally $(1-x)^{a/2} \le e^{-ax/2}$, and $(1-x)^{1/4} \le 1 - x/4$ by Bernoulli's inequality. □

So in every geometry the overlap of harmonic measures decays exponentially in the energy, and it is never smaller than the X^2 overlap. The Gauss case is the exact model. In the Hardy and Bergman cases the upper rates are $1/4$ and $\beta/4$, respectively; both are optimal at small energy, and near $\rho = 1$ the affinities are far smaller. For constants near the boundary the relevant Hardy bound is $h \le (1-\rho^2)^{1/4}$, from part (4).

COROLLARY 1.5 (Kakutani's dichotomy, quantitatively). *In the Hardy geometry on infinitely many coordinates,* ω_r *and* ω_s *are mutually singular if and only if* $E(r,s) = \infty$. *In particular* ω_r *is absolutely continuous with respect to Haar measure exactly when* $r \in \ell^2$, *and any two points of* $\mathbb{D}^\infty \cap \ell^2$ *have equivalent harmonic measures, with*

$$2\big(1 - H(r,s)\big) \le \|\omega_r - \omega_s\| \le 2\sqrt{1 - H(r,s)^2}.$$

PROOF. By Kakutani's theorem [51], two product measures with equivalent factors are equivalent or mutually singular according as the product of the factor affinities is positive or zero. By Proposition 1.4 that product is positive exactly when $\sum \rho_p^2 < \infty$. Take $s = 0$ for the second statement. The total variation bounds are the standard Hellinger inequalities. □

This explains two things. First, it explains the domain $\mathbb{D}^\infty \cap \ell^2$ of Cole and Gamelin: it is exactly where harmonic measure is absolutely continuous with respect to Haar measure. Second, it shows that between two points of $\mathbb{D}^\infty \cap \ell^2$ mutual

singularity is only ever approximate. Any argument that separates point evaluations must therefore be quantitative in the energy. The rest of the chapter makes it so.

1.4. Localization

The flat representative spreads its mass over the whole space. The key observation, due to Perfekt [70] in the Hardy case, is the localization identity of Lemma 1.2: the representative can be cut down to the image under T_r of any rotation-invariant set, at the price of the measure of that set.

The cost of localization is the factor $1/\lambda(D)$. As long as D has measure at least $1/2$, the localized representative is at most twice the flat one, and it lives on $T_r(D)$. That set carries at least half of the harmonic measure $\omega_r = (T_r)_*\lambda$.

1.5. The hyperbolic plane on the prime curve

Lift the right half-plane $\{\Re s > 1/2\}$ to the torus by $s \mapsto \gamma(s) = (p^{-s})_p$. Write d for the hyperbolic distance in the half-plane $\{\Re w > 0\}$, $w = s - 1/2$, with metric $|\mathrm{d}w|/\Re w$.

THEOREM 1.6. *Uniformly for $0 < \Re w, \Re w' \le 1$ and $|\Im w - \Im w'| \le 1$,*

$$E\big(\gamma(\tfrac12 + w), \gamma(\tfrac12 + w')\big) = 2\log\cosh\frac{d(w,w')}{2} + O(1).$$

PROOF. Put $a_p = p^{-1/2-w}$, $b_p = p^{-1/2-w'}$. Then $\rho_p^2 = |a_p - b_p|^2/|1 - \overline{b_p}a_p|^2$, and $|a_p - b_p|^2(1 + |a_pb_p|)^{-2} \le \rho_p^2 \le |a_p - b_p|^2(1 - |a_pb_p|)^{-2}$. The difference contributes $O(\sum_p p^{-2}) = O(1)$. Expanding,

$$\sum_p |a_p - b_p|^2 = \mathcal{P}(1 + 2\Re w) + \mathcal{P}(1 + 2\Re w') - 2\Re\,\mathcal{P}(1 + w + \overline{w'}),$$

where $\mathcal{P}(1+z) = \sum_p p^{-1-z} = \log\zeta(1+z) + O(1) = \log(1/z) + O(1)$ uniformly for $0 < \Re z \le 2$ and $|\Im z| \le 1$, since ζ has only the simple pole at 1 there. The sum is therefore $\log\big(|w + \overline{w'}|^2/(4\Re w\Re w')\big) + O(1)$. For the right half-plane, $\cosh^2(d/2) = |w + \overline{w'}|^2/(4\Re w\Re w')$. □

So near the critical line, the geometry that governs harmonic measures on $\mathbb{T}^\infty$ is the hyperbolic plane, and $H(\omega,\omega') \lesssim \cosh(d/2)^{-1/2} \approx e^{-d/4}$. Hyperbolic area grows like e^d, while the number of points within distance d along a curve going to the boundary grows only like d. That difference explains why curves carry H^p-Carleson measures for every p (chapter 2), while vertical segments, which are horocycles with $e^{d/2}$ growth, fail at $p = 1$ (chapter 3).

1.6. Stability of point evaluation

For $F \in X^p$ define the efficiency at r, $\mathfrak{e}_r^{(p)}(F) = |F(r)|^p/(\|K_r\|_2^2\|F\|_p^p) \in [0,1]$.

THEOREM 1.7 (Stability). *In each geometry, for $0 < p < \infty$, nonzero $F \in X^p$ and every point r,*

$$H\Big(\frac{|F|^p\lambda}{\|F\|_p^p}, \omega_r\Big) \ge \mathfrak{e}_r^{(p)}(F)^{1/2}, \qquad \textit{equivalently} \qquad \int |F|^{p/2}|K_r|\,\mathrm{d}\lambda \ge |F(r)|^{p/2}.$$

PROOF. We first take F to be a cylinder polynomial and r to have finite support, so only finitely many coordinates are involved.

By Lemma 1.1, $K_r \circ T_r$ is holomorphic and zero-free near the closed domain, so $G = (F \circ T_r)\left(\|K_r\|_2^2/(K_r \circ T_r)^2\right)^{1/p}$ is holomorphic there, and $|G|^{p/2} = |F \circ T_r|^{p/2}\|K_r\|_2 |K_r \circ T_r|^{-1}$. Substituting $z = T_r w$ and using the cocycle $\lambda \circ T_r = \|K_r\|_2^2 |K_r \circ T_r|^{-2}\lambda$,

$$\int |F|^{p/2}|K_r|\,\mathrm{d}\lambda = \int |F \circ T_r|^{p/2}\,\frac{\|K_r\|_2^2}{|K_r \circ T_r|}\,\mathrm{d}\lambda = \|K_r\|_2 \int |G|^{p/2}\,\mathrm{d}\lambda.$$

Since $\log|G|$ is plurisubharmonic, $|G|^{p/2}$ is subharmonic on complex lines through 0, and λ is invariant under common rotations. So $\int |G|^{p/2}\,\mathrm{d}\lambda \geq |G(0)|^{p/2} = |F(r)|^{p/2}\|K_r\|_2^{-1}$, using $K_r(r) = \|K_r\|_2^2$. Hence $\int |F|^{p/2}|K_r|\,\mathrm{d}\lambda \geq |F(r)|^{p/2}$.

For a general allowed r, its coordinate truncations $r^{(N)}$ satisfy $K_{r^{(N)}} \to K_r$ in L^2, while the polynomial values converge to $F(r)$. Cauchy–Schwarz passes the integral inequality to r. For $F \in X^p$, take cylinder polynomials $P_n \to F$ in L^p. Then $|P_n|^{p/2} \to |F|^{p/2}$ in L^2, so the weighted integrals converge again by Cauchy–Schwarz. The polynomial inequality also gives $|P(r)|^p \leq K_r(r)\|P\|_p^p$, so evaluation extends continuously and $P_n(r) \to F(r)$. This argument includes $p < 1$. The affinity form follows on dividing by $\|K_r\|_2 \|F\|_p^{p/2}$. □

Taking $H \leq 1$ recovers the point estimate $|F(r)|^p \leq \|K_r\|_2^2\|F\|_p^p$. The theorem is that estimate with a defect term. For Banach-valued F the same proof applies with $\log\|F\|_X$ in place of $\log|F|$. In one variable it is a stability form of the factorization $F = Ih^{2/p}$. In several variables F need not factor, but the inequality is linear in $|F|^{p/2}$ and the weight $|K_r|$ is a product, so it tensorizes; the square root $|F|^{p/2}$ plays the role of the H^2 vector h.

COROLLARY 1.8 (Two-point necessity). *In the Hardy and Gauss geometries, if* $\mathfrak{e}_r^{(p)}(F), \mathfrak{e}_s^{(p)}(F) \geq 1-\eta$ *with* $\eta < 1/2$, *then* $E(r,s) \leq 4\log\big(1/(1-2\eta)\big) = 8\eta + O(\eta^2)$. *In the Bergman geometry the same holds with* E *replaced by* βE.

PROOF. Put $u = |F|^{p/2}/\|F\|_p^{p/2}$, $v_r = |k_r|$ and $v_s = |k_s|$. These are unit vectors in $L^2(\lambda)$, and Theorem 1.7 gives $\langle u, v_r\rangle, \langle u, v_s\rangle \geq \sqrt{1-\eta}$. By Cauchy–Schwarz,

$$4(1-\eta) \leq \langle u, v_r + v_s\rangle^2 \leq \|v_r + v_s\|_2^2 = 2 + 2H(\omega_r, \omega_s).$$

Hence $H(\omega_r, \omega_s) \geq 1 - 2\eta$. Combine this with Proposition 1.4. □

The preceding argument loses a factor 2 in the leading term; Theorem 1.9 removes that loss. The function $K_t^{2/p}$ at the coordinatewise hyperbolic midpoint t of r and s (the Euclidean midpoint in Gaussian geometry) has efficiency $1 - \kappa E/4 + O(E^2)$ at both points, where $\kappa = \beta$ in Bergman geometry and $\kappa = 1$ otherwise. Along the prime curve, a Dirichlet series is $(1-\eta)$-efficient only on a set of scales of logarithmic diameter at most $7 + 4\log(1/(1-2\eta))$. So separation in energy is not only sufficient for adding up evaluations (chapter 2) but necessary.

The sharp leading two-point constant

The leading coefficient 8 can be replaced by 4, uniformly in the number of variables. Write $e_z(F) = \mathfrak{e}_z^{(p)}(F)$ and $c(r,s) = |K_s(r)|/\sqrt{K_r(r)K_s(s)}$, and set $\kappa = \beta$ in the Bergman geometry and $\kappa = 1$ otherwise.

THEOREM 1.9. *Fix $p > 0$, put $k = \lceil p/2 \rceil$, and $a = 2k/p$. In all three geometries, in any finite number of variables or countably many variables, if $e_r(F), e_s(F) \geq 1-\eta$ and $0 \leq \eta < 1 - 2^{-1/a}$, then*

$$c(r,s) \geq \big(2(1-\eta)^a - 1\big)^{1/a}, \qquad \kappa E(r,s) \leq B_a(\eta) \coloneqq -\frac{2}{a}\log\big(2(1-\eta)^a - 1\big). \tag{1.1}$$

In particular, uniformly in the dimension, points, and function,

$$\kappa E(r,s) \leq 4\eta + 2(a+1)\eta^2 + O_p(\eta^3).$$

The leading coefficient 4 is optimal for each fixed p and geometry. For even integer p, $a = 1$ gives the exact improved logarithmic upper bound.

To retain the analytic phases, we take an integer power of F and estimate it in a Hilbert space. The following change of measure makes that power contractive without requiring a fractional power of F across its zeros.

LEMMA 1.10. *For $0 < p \leq q < \infty$, there is a contractive inclusion from X^p to the L^q analytic space with coordinate measure $m_{q\beta/p}$ in the Bergman case, $m_{q/p}$ in the Hardy case when $q > p$, and $\gamma_{q/p} = (q/p)\pi^{-1}e^{-(q/p)|z|^2}\,\mathrm{d}A_{\mathbb{C}}$ in the Gaussian case. When $q = p$, keep the original measure.*

PROOF. The one-variable Bergman inclusion is Kulikov [55, Corollary 1.3]. For Hardy polynomials it follows by letting the source parameter β decrease to 1 in the same inequality; the radial probability measures m_β tend to boundary Haar measure. The target parameter tends to $q/p > 1$. The Gaussian inclusion is Janson [49, Theorem 11], rescaled from his variance normalization: $\|f(\sqrt{p/q}\,\cdot)\|_{L^q(\gamma)} \leq \|f\|_{L^p(\gamma)}$.

Here is the tensorization, including $p < 1$. If a one-variable analytic map T has the indicated p-to-q contraction, applying it in variables x, y and using Minkowski with exponent $q/p \geq 1$ gives

$$\|T_xT_yf\|_{L^q_y(L^q_x)} \leq \|T_yf\|_{L^q_y(L^p_x)} \leq \|T_yf\|_{L^p_x(L^q_y)} \leq \|f\|_{L^p_x(L^p_y)}.$$

The source and target measures in each variable are understood separately. For the inclusions T is the identity on analytic functions; in the Gaussian description it may instead be dilation. Induction proves the finite-product claim. The same bound holds for cylinder polynomials in a countable product, since unused coordinates have total mass one. Polynomial density extends both claims to the analytic closures, also when $p < 1$. □

PROOF OF THEOREM 1.9. Normalize $\|F\|_p = 1$. Apply Lemma 1.10 with $q = 2k$, and set $G = F^k$. Then G is in the target Hilbert space Y, and $\|G\|_Y^2 \leq 1$. Indeed polynomial approximations to F converge in the target L^{2k}, so their kth powers converge in L^2. The target reproducing kernel K^Y is

$$K^Y_z(w) = \prod_j (1 - \overline{z}_j w_j)^{-a\kappa} \quad \text{or} \quad e^{a\langle w,z\rangle}.$$

Consequently $K_z^Y(z) = K_z(z)^a$, and its normalized overlap is $c(r,s)^a$. The two Hilbert efficiencies of G are at least $(1-\eta)^a$.

For any two unit vectors u, v with overlap $b = |\langle u, v\rangle|$, the sum of their rank-one projections has norm $1+b$: its two possibly nonzero eigenvalues are $1+b$ and $1-b$. Thus for a unit vector h,

$$\min\{|\langle h,u\rangle|^2, |\langle h,v\rangle|^2\} \le \frac{1+b}{2}.$$

Apply this with the target evaluation vectors and $h = G/\|G\|_Y$. This proves $c(r,s)^a \ge 2(1-\eta)^a - 1$. Direct kernel evaluation gives

$$-2\log c(r,s) = \kappa \sum_j -\log(1-\rho_j^2) \ge \kappa E(r,s)$$

for Hardy and Bergman; in the Gaussian case $-2\log c = E$. This proves (1.1); its Taylor expansion gives the asserted remainder.

For sharpness take $F = 1$ in one variable at $r = -t, s = t$ with $t > 0$ small. In the Gaussian case $\eta = 1-e^{-t^2}$ and $E = 4t^2$. In Hardy and Bergman, $\eta = 1-(1-t^2)^\kappa$ and $E = 4t^2/(1+t^2)^2$. Hence $\kappa E/\eta \to 4$ in all cases, for every p. □

THEOREM 1.11. *In one-variable Gaussian X^1, the stronger assertion $E \le -2\log(1-2\eta)$ is false, even for arbitrarily small positive E.*

PROOF. Fix $r > 0$, put $b = \pi/(2r)$ and $H(z) = 2\cosh(rz)$, and define

$$F_t(z) = H(z)^2 + it\frac{H(z)}{z-ib}, \qquad Q_t = \frac{|F_t(r)|}{\|F_t\|_{L^1(\gamma)}}.$$

The quotient is entire because $H(ib) = 0$. Put $L = \cosh(r^2)$, $N_t = |F_t(r)|$ and $D_t = \|F_t\|_1$. Reproduction gives $N_0 = 4L^2$, $D_0 = 4L$, and direct differentiation gives $N_0' = -2Lb/(r^2+b^2)$. Dominated convergence gives $D_0' = \Re\int i\overline{H}(z)/(z-ib)\,\mathrm{d}\gamma(z)$. Expand $(z-ib)^{-1}$ on circles of radius below and above b. Angular integration leaves only the circles below b, so

$$D_0' = -\frac{2}{b}\int_0^{b^2} e^{-u}\cos(ru/b)\,\mathrm{d}u = -\frac{2(b+re^{-b^2})}{r^2+b^2}.$$

The last equality uses $rb = \pi/2$. Consequently

$$Q_0 = L, \qquad Q_0' = \frac{N_0' - LD_0'}{4L} = \frac{re^{-b^2}}{2(r^2+b^2)} > 0.$$

The values at r and $-r$ are complex conjugates, so their common efficiency is $e_t = e^{-r^2}Q_t$. Hence, for sufficiently small $t > 0$, $1/2 < e_t < 1$ and $2e_t - 1 > e^{-2r^2}$. With $\eta_t = 1-e_t$, this contradicts $E(-r,r) = 4r^2 \le -2\log(1-2\eta_t)$. Since $r > 0$ is arbitrary, the failure occurs at arbitrarily small energy. Finally, $H(z)/(z-ib) = \int_0^1 H'(ib + u(z-ib))\,\mathrm{d}u$ shows that $|F_t(z)| \le Ce^{2r|z|}$, a bound we use below. Exponential Taylor series and dominated convergence also give $F_t \in X^1$. □

THEOREM 1.12. *Let $\beta \ge 1$, where $\beta = 1$ denotes the Hardy geometry (coordinate measure m_1 is Haar measure on $\mathbb{T}$). For $p = 1$ the assertion $\beta E(r,s) \le -2\log(1-2\eta)$*

is false in N variables for all sufficiently large N, hence also in countably many variables, with βE arbitrarily small.

PROOF. Fix $r > 0$ and $t > 0$ as in the proof of Theorem 1.11, so that the Gaussian efficiency e_t of $F = F_t$ at $\pm r$ satisfies $1/2 < e_t < 1$ and $2e_t - 1 > e^{-2r^2}$. The preceding proof gives $|F(w)| \le Ce^{2r|w|}$.

For $N\beta > r^2$ put $x_N = r/\sqrt{N\beta}$ and, on $\mathbb{D}^N$,

$$G_N(z) = F(W_N(z)), \qquad W_N(z) = \sqrt{\beta/N}\sum_{j=1}^{N} z_j,$$

with points $r_\pm = (\pm x_N, \dots, \pm x_N)$. Then G_N is entire on $\mathbb{C}^N$, so its Taylor polynomials converge uniformly on the closed polydisc and $G_N \in X^1$. Moreover $G_N(r_\pm) = F(\pm r)$, which are complex conjugates of equal modulus, and $K_{r_\pm}(r_\pm) = (1 - r^2/(N\beta))^{-N\beta} \to e^{r^2}$. Each coordinate pseudohyperbolic distance is $2x_N/(1 + x_N^2)$, so

$$\beta E(r_+, r_-) = \frac{4r^2}{(1 + x_N^2)^2} \to 4r^2.$$

Under $m_\beta^{\otimes N}$ the coordinates z_j are independent, bounded by 1, rotation invariant, and $\int |z|^2 \, \mathrm{d}m_\beta = 1/\beta$ (the coefficient of $|z|^2$ in $(1 - |z|^2)^{-\beta}$ is β). Thus $\Re z_j$, $\Im z_j$ have mean 0, variance $1/(2\beta)$ and covariance 0. By the multivariate central limit theorem [14, §29], W_N converges in distribution to the standard complex Gaussian γ. For uniform integrability, let X denote either $\Re z_j$ or $\Im z_j$. Rotation invariance makes X symmetric, and $|X| \le 1$. Hence

$$\int e^{\lambda X} = \int \cosh(\lambda X) \le \cosh \lambda \le e^{\lambda^2/2}, \qquad \lambda \in \mathbb{R}.$$

The last inequality follows termwise from $(2k)! \ge 2^k k!$. Independence therefore gives $\int e^{\pm 8r\Re W_N}, \int e^{\pm 8r\Im W_N} \le e^{32r^2\beta}$ for every N. Since $e^{4r|w|} \le (e^{8r|\Re w|} + e^{8r|\Im w|})/2$, the integrals of $|G_N|^2 \le C^2 e^{4r|W_N|}$ are bounded uniformly in N. Hence $|F(W_N)|$ is uniformly integrable, and [14, §25] gives

$$\|G_N\|_{L^1(m_\beta^{\otimes N})} = \int |F(W_N)| \to \|F\|_{L^1(\gamma)}.$$

Therefore the common efficiency $e^{(N)}$ of G_N at $r_\pm$ tends to e_t. With $\eta_N = 1 - e^{(N)}$ we get $\eta_N \to 1 - e_t \in (0, 1/2)$ and

$$1 - 2\eta_N = 2e^{(N)} - 1 \to 2e_t - 1 > e^{-2r^2} = \lim_N e^{-\beta E(r_+, r_-)/2}.$$

So for large N, $0 < \eta_N < 1/2$ and $\beta E(r_+, r_-) > -2\log(1 - 2\eta_N)$. Padding by zero coordinates gives the same in countably many variables, and $\beta E \to 4r^2$ with r arbitrary. □

REMARK 1.13. In one-variable Hardy the halved bound holds for every p: factor $F = Ih^{2/p}$ with I inner and $h \in H^2$ outer. Then $\|h\|_2^2 = \|F\|_p^p$ and $|h(z)|^2 \ge |F(z)|^p$, so the two-projection estimate applies without changing the kernel. Thus Theorem 1.12 genuinely needs many variables in the Hardy case.

Many points

For more than two points the same Hilbert-space picture does better than applying the two-point estimate separately.

COROLLARY 1.14 (Many points). *In each geometry let $F \in X^p$ be nonzero, let $r_1, \dots, r_N$ be points, $\mathfrak{e}_i = \mathfrak{e}_{r_i}^{(p)}(F)$ and $H_{ij} = H(\omega_{r_i}, \omega_{r_j})$. Then for all $a_1, \dots, a_N \geq 0$,*

$$\Big(\sum_i a_i \sqrt{\mathfrak{e}_i}\Big)^2 \leq \sum_{i,j} a_i a_j H_{ij}.$$

In particular, if $N \geq 2$ and every $\mathfrak{e}_i \geq 1 - \eta$, the average of H_{ij} over the pairs $i < j$ is at least $1 - N\eta/(N-1)$.

PROOF. The square roots v of $|F|^p\lambda/\|F\|_p^p$ and w_i of ω_{r_i} are unit vectors in $L^2(\lambda)$ with $\langle v, w_i\rangle \geq \sqrt{\mathfrak{e}_i}$ by Theorem 1.7. So $\sum_i a_i\sqrt{\mathfrak{e}_i} \leq \langle v, \sum_i a_i w_i\rangle \leq \|\sum_i a_i w_i\|_2$. For the second statement take equal weights. □

The two-point estimate gives only $1 - 2\eta$ for each pair. For $N > 2$ the average bound is better. If $0 \leq \eta < (N-1)/N$, combining it with $H \leq e^{-\kappa E/4}$ forces some pair to satisfy $\kappa E \leq -4\log(1 - N\eta/(N-1))$.

Equality and near-extremizers

COROLLARY 1.15. *In each geometry, let $0 < p < \infty$ and $F \in X^p$ with $F(r) \neq 0$.*

(1) In finitely many variables, $\mathfrak{e}_r^{(p)}(F) = 1$ if and only if $F = cK_r^{2/p}$ for a constant c.

(2) With total variation $\|\cdot\|_{\mathrm{TV}}$ the total mass of the difference,

$$\Big\| \frac{|F|^p\lambda}{\|F\|_p^p} - \omega_r \Big\|_{\mathrm{TV}} \leq 2\sqrt{1 - \mathfrak{e}_r^{(p)}(F)}.$$

PROOF. (1) We first work in finitely many variables. For the transformed function G in the stability proof, the cocycle gives $\|G\|_p^p = \|F\|_p^p$ and $|G(0)|^p = |F(r)|^p/K_r(r)$. Equality in evaluation therefore means $\|G\|_p^p = |G(0)|^p$. Since $G(0) \neq 0$, a local analytic branch of $G^{p/2}$ exists near zero. If G is nonconstant, Parseval on a sufficiently small torus gives

$$\int_{\mathbb{T}^d} |G(\rho\zeta)|^p \, \mathrm{d}m(\zeta) > |G(0)|^p.$$

The torus means of $|G|^p$ increase in each radius by subharmonicity. In Hardy the boundary mean is therefore strictly larger than $|G(0)|^p$. In Bergman and Gaussian, the product radial measure assigns positive mass to radii all at least ρ; all torus means are at least $|G(0)|^p$, so radial integration gives the same strict inequality. This contradicts equality. Hence G is constant and $F = cK_r^{2/p}$. The converse follows from the kernel norm.

(2) For probability measures, $\|\mu - \nu\|_{\mathrm{TV}} \leq 2\sqrt{1 - H(\mu,\nu)^2}$. Apply this with $\mu = |F|^p\lambda/\|F\|_p^p$ and $\nu = \omega_r$, using $H(\mu,\nu) \geq (\mathfrak{e}_r^{(p)}(F))^{1/2}$ from Theorem 1.7. □

So a $(1-\eta)$-efficient function at r distributes its mass like ω_r up to $2\sqrt{\eta}$ in total variation. The converse fails. In the Hardy geometry, let I be a Möbius map in one coordinate with $|I(r)| = (1-\eta)^{1/p}$ and put $F = IK_r^{2/p}$. Then $|F|^p\,\mathrm{d}m/\|F\|_p^p = \omega_r$ exactly, while $\mathfrak{e}_r^{(p)}(F) = 1-\eta$. Modulus information cannot distinguish near-extremizers from inner perturbations of the extremizer; this is the phase blindness that limits the method in chapter 9.

The proofs above use T_r in every coordinate, which is harmless when r has finitely many nonzero coordinates. The equality statement holds on the whole domain of bounded point evaluations, in each geometry.

PROPOSITION 1.16 (Equality in infinitely many variables). *In each geometry, let $0 < p < \infty$, let $r \in \ell^2$, with $r \in \mathbb{D}^\infty$ in the Hardy and Bergman cases, and let $F \in X^p$ be nonzero. Then $\mathfrak{e}_r^{(p)}(F) = 1$ if and only if $F = cK_r^{2/p}$ for a constant c.*

PROOF. Equality on the whole product forces equality on every finite block. We first justify the partial evaluations used to make this precise. For any block and any allowed point a in that block, the point-evaluation estimate on polynomial slices gives

$$\|F(a,\cdot)\|_p^p \le K_a(a)\|F\|_p^p.$$

Polynomial density extends this map to X^p, also when $p < 1$. For an infinite block the finite kernel diagonals are bounded by $K_a(a)$, so the same argument applies. Almost every finite-block slice belongs to its analytic L^p closure: choose polynomial approximations with summable L^p errors to the power p, and apply Tonelli to the slice errors. The slice evaluations agree with the partial evaluation just defined; evaluating two disjoint blocks successively agrees with full evaluation, again by density.

Suppose that $\mathfrak{e}_r^{(p)}(F) = 1$. Split the coordinates into $H = \{1,\dots,N\}$ and its complement T, write $r = (r_H, r_T)$, and put $A_N = K_{r_H}(r_H)$ and $B_N = K_{r_T}(r_T)$. Then

$$|F(r)|^p \le B_N \int |F(r_H,w)|^p\,\mathrm{d}\lambda_T(w) \le B_N A_N \|F\|_p^p.$$

Since $A_N B_N = K_r(r)$, both inequalities are equalities. The nonnegative deficit in the finite-block evaluation bound therefore vanishes for almost every w. The finite-dimensional case of Corollary 1.15, including the zero slice, gives

$$F(z_H,w) = K_{r_H}(z_H)^{2/p} h_N(w), \qquad h_N(w) = F(0_H,w),$$

where the kernel power has value one at zero. Let $Q_N F$ be F with all coordinates after N evaluated at zero. These maps are contractions, and $Q_N F \to F$ in X^p by cylinder-polynomial density. Evaluating the factorization in the tail at zero yields $Q_N F = F(0)K_{r_H}^{2/p}$. Thus $F(0) = 0$ would force $F = 0$. The finite kernel products converge to K_r at every allowed point, so bounded point evaluation gives $F = F(0)K_r^{2/p}$.

For the converse, we record why the kernel power belongs to X^p. Put $r^{(N)} = (r_1,\dots,r_N,0,\dots)$, let $H_N = K_{r^{(N)}}^{2/p}$, and choose $q > \max(1,p)$. Each coordinate factor has mean one, so (H_N) is a martingale. Taylor expansion and rotation invariance give $\int |K_{r_j}|^{2q/p}\,\mathrm{d}\lambda_j = 1 + O(|r_j|^2)$ as $r_j \to 0$. Since $r \in \ell^2$ and each of

the finitely many initial factors has every finite moment, $\sup_N \|H_N\|_q < \infty$. The L^q martingale convergence theorem gives convergence in L^q, hence in L^p; bounded point evaluations identify the limit as $H = K_r^{2/p}$. In particular $H \in X^p$, and

$$\|H\|_p^p = \lim_N K_{r^{(N)}}(r^{(N)}) = K_r(r), \qquad H(r) = K_r(r)^{2/p}.$$

These identities prove equality. □

REMARK 1.17 (Density of evaluations). Point evaluations at the curve points $\gamma(\sigma)$, $\sigma > 1$, already span a weak-$*$ dense subspace of $(H^1(\mathbb{T}^\infty))^*$, by uniqueness of Dirichlet coefficients. So superpositions of point evaluations, the functionals controlled in chapter 2, are weak-$*$ dense in the dual (Proposition 2.22).

Open

QUESTION 1. What is the sharp many-point version of Corollary 1.14?

The two-point theory also leaves two questions. For which p other than the even integers does the halved bound $\kappa E \le -2\log(1-2\eta)$ hold in every geometry and dimension (Theorems 1.11 and 1.12)? What is the optimal finite-defect function in Theorem 1.9 for general p?

CHAPTER 2

The Hellinger–Bessel inequality

When can point evaluations be added up in H^p, for all p at once? The answer has a single source. By the stability theorem of chapter 1, the square root of $|F|^p$ is at least as aligned with the square root of the harmonic measure ω_r as the efficiency of F at r requires. Bessel's inequality then bounds the sum over many points by the Gram norm of the square roots $\sqrt{\omega_r}$. The results below are consequences of this one inequality; they hold for every $p > 0$, and the constants are explicit and small. To our knowledge the inequality itself, its sharpness on prime curves (Corollary 2.12) and the analysis of what it loses (Proposition 2.18) are developed here. The angular-average and grid examples in that proposition were suggested by an external reviewer. The inequality and its consequences for Fejér–Riesz inequalities and prime curves also appear in the preprint [83].

Throughout, the setting is one of the three geometries of section 1.1, and X^p is the analytic closure in $L^p(\lambda)$. The modulus of the reproducing kernel satisfies $|K_r|^2 = \|K_r\|_2^2 \, \mathrm{d}\omega_r / \mathrm{d}\lambda$.

2.1. The inequality

Why this replaces factorization. In one variable a Carleson estimate in H^p is proved by writing $F = Ih^{2/p}$ with I inner and $h \in H^2$, so that $|h|^2 = |F|^p$, and applying the H^2 estimate to h. On $\mathbb{T}^\infty$ the analytic H^2 function h does not exist. The stability theorem uses instead the vector $|F|^{p/2} \in L^2(\lambda)$, which always exists and is not analytic. Analyticity enters once, in the pointwise inequality $|F(r)|^{p/2} \le \langle |F|^{p/2}, |K_r| \rangle$ of Theorem 1.7; everything after it is Hilbert-space geometry. If F is to be efficient at many points, the one vector $|F|^{p/2}$ must be correlated with many vectors $|K_r|$, and Bessel's inequality forbids this unless the $|K_r|$ are nearly orthogonal.

The price is the passage from K_r to $|K_r|$. The moduli keep positivity and lose the cancellation between kernels: $\langle |K_r|, |K_s| \rangle \ge |\langle K_r, K_s \rangle|$, and by Proposition 1.4 it can be as large as the square root of the kernel overlap. This is why the Gram condition below is not necessary (Proposition 2.18), and why near the critical line it gives the constant $\Gamma(1/4)^2/\sqrt{2\pi} \approx 5.244$ (Theorem 2.9) where the H^2 estimate gives π.

THEOREM 2.1 (Hellinger–Bessel). *Let μ be a positive Borel measure on the set of admissible points, with the ℓ^2 topology. Let $\mathcal{G}_\mu$ be the integral operator on $L^2(\mu)$ with kernel*

$$\mathcal{G}(r, s) = \langle |K_r|, |K_s| \rangle = \|K_r\|_2 \|K_s\|_2 \, H(\omega_r, \omega_s).$$

Then for every $0 < p < \infty$ and $F \in X^p$,

$$\int |F(r)|^p \, \mathrm{d}\mu(r) \le \|\mathcal{G}_\mu\| \, \|F\|_p^p.$$

Moreover, for every positive weight w on $\operatorname{supp}\mu$,

$$\|\mathcal{G}_\mu\| \le \operatorname*{ess\,sup}_{r \in \operatorname{supp}\mu} \frac{1}{w(r)} \int \mathcal{G}(r,s)\, w(s)\, \mathrm{d}\mu(s).$$

With $w(r) = \|K_r\|_2$ the right side becomes $\sup_r \int H(\omega_r,\omega_s)\|K_s\|_2^2 \, \mathrm{d}\mu(s)$.

Proof. Put $u = |F|^{p/2}$, so $\|u\|_2^2 = \|F\|_p^p$. By the stability theorem (Theorem 1.7), in the form $\int |F|^{p/2}|K_r| \, \mathrm{d}\lambda \ge |F(r)|^{p/2}$, we have $|F(r)|^p \le \langle u, |K_r|\rangle^2$. The map $r \mapsto |K_r| \in L^2(\lambda)$ is continuous, because $\big\| |K_r|/\|K_r\|_2 - |K_s|/\|K_s\|_2 \big\|_2^2 = 2 - 2H(\omega_r,\omega_s)$ and $H \ge |\langle k_r, k_s\rangle|$. Let $V : L^2(\mu) \to L^2(\lambda)$, $Vc = \int c(r)|K_r| \, \mathrm{d}\mu(r)$. Then $V^*u(r) = \langle u, |K_r|\rangle$ and $V^*V = \mathcal{G}_\mu$. Hence

$$\int |F(r)|^p \, \mathrm{d}\mu \le \|V^*u\|^2 \le \|V\|^2 \|u\|^2 = \|\mathcal{G}_\mu\| \, \|F\|_p^p.$$

The weighted Schur test needs the row bound only μ-almost everywhere, since the kernel is non-negative. □

In X^2 the corresponding statement is Bessel's inequality for the kernels themselves, with $\langle K_r, K_s\rangle$ in place of $\mathcal{G}(r,s)$. Since $H(\omega_r,\omega_s) \ge |\langle k_r, k_s\rangle|$ (Proposition 1.4), the X^p condition is the stronger one. It is a single, p-independent condition that is sufficient for every p simultaneously.

Corollary 2.2 (Signed and complex measures). *If μ is a complex measure with $S = \operatorname{ess\,sup}_{|\mu|} \int H(\omega_r,\omega_s)\|K_s\|_2^2 \, \mathrm{d}|\mu|(s) < \infty$, then $h \mapsto \int h \, \mathrm{d}\mu$ is bounded on X^1 with norm at most S. In Hardy geometry, the Helson matrix with moments $\alpha(n) = \int r^{\kappa(n)} \, \mathrm{d}\mu(r)$ has a bounded symbol of norm at most S.*

Proof. $|\int h \, \mathrm{d}\mu| \le \int |h| \, \mathrm{d}|\mu|$. Apply Theorem 2.1 with $p = 1$. In Hardy geometry, Hahn–Banach gives the bounded symbol. □

2.2. The Gram kernel along the curve

Let $\gamma(s) = (p^{-s})_p$ and $r(\sigma) = \gamma(1/2 + \sigma)$.

Lemma 2.3. *The one-coordinate Hardy affinity satisfies $h(\rho) \le (1 - \rho^2)^{1/4}$.*

Proof. This is the case $\beta = 1$ of part (4) of Proposition 1.4. □

Lemma 2.4 (Gram kernel). *In the Hardy geometry,*

$$\mathcal{G}(r(\sigma), r(\tau)) \le \zeta(1 + 2\sigma)^{1/4} \zeta(1 + 2\tau)^{1/4} \zeta(1 + \sigma + \tau)^{1/2}.$$

Proof. By Proposition 1.4, $H(\omega_r,\omega_s) \le |\langle k_r, k_s\rangle|^{1/2}$. Hence

$$\mathcal{G}(r,s) \le \big(\|K_r\|_2 \|K_s\|_2 |K_s(r)| \big)^{1/2}.$$

Insert $\|K_{r(\sigma)}\|_2^2 = \zeta(1 + 2\sigma)$ and $K_{r(\tau)}(r(\sigma)) = \zeta(1 + \sigma + \tau)$. □

Near the boundary the kernel is homogeneous of degree -1. With $\zeta(1+x) \approx 1/x$,

$$\mathcal{G} \approx \frac{(\sigma\tau)^{-1/4}}{\sqrt{2}\,(\sigma+\tau)^{1/2}},$$

and in the variable $t = \log(\sigma/\tau)$, after the unitary $f \mapsto \sqrt{\sigma} f$, it becomes the convolution kernel $1/(2\sqrt{\cosh(t/2)})$. Its integral,

$$\int_{\mathbb{R}} \frac{\mathrm{d}t}{2\sqrt{\cosh(t/2)}} = B\big(\tfrac{1}{4},\tfrac{1}{2}\big) = \frac{\Gamma(1/4)^2}{\sqrt{2\pi}} \approx 5.244,$$

is the best constant the inequality can give near the boundary. The same chain with $|\langle K_r, K_s\rangle|$ in place of $\mathcal{G}$, which is the H^2 statement, gives the Carleman kernel with norm exactly π. The factor $5.244/\pi \approx 1.67$ is the price of discarding the phases of F (section 9.4).

2.3. Fejér–Riesz inequalities in every $\mathscr{H}^p$

Perfekt [70] proved the case $p = 1$ of the next theorem, with an unspecified constant (of order 10^6 if traced through the proof), by localizing the flat representative of chapter 1 and controlling the overlaps through separation and concentration estimates. The Hellinger–Bessel inequality replaces that machinery by a single Bessel inequality and gives every p at once. Since $p < 1$ is included and $H^p(\mathbb{T}^\infty)$ admits no factorization, interpolation from $p = 1$ and $p = 2$ is not available. Applying Perfekt's inequality to f^m gives every positive integer $p = m$ with an unspecified constant; for non-integer p these inequalities seem not to have been recorded before.

THEOREM 2.5. *For every $0 < p < \infty$ and $f \in \mathscr{H}^p$,*

$$\int_0^1 |f(1/2+\sigma)|^p \,\mathrm{d}\sigma \le (5+2\sqrt{2})\,\|f\|_{\mathscr{H}^p}^p.$$

PROOF. Apply Theorem 2.1 to Lebesgue measure on $(0,1)$, pushed forward by r. It suffices to bound the operator norm of the kernel $\overline{\mathcal{G}}(\sigma,\tau) = (1+1/(2\sigma))^{1/4}(1+1/(2\tau))^{1/4}(1+1/(\sigma+\tau))^{1/2}$, which dominates $\mathcal{G}$ by Lemma 2.4 and $\zeta(1+x) \le 1+1/x$. Split $(0,1) = I_0 \cup I_1$ with $I_0 = (0,1/10)$.

- *On I_0^2:* $\overline{\mathcal{G}} \le (6/5)2^{-1/2}(\sigma\tau)^{-1/4}(\sigma+\tau)^{-1/2}$. The latter kernel is unitarily a convolution by $1/(2\sqrt{\cosh(t/2)}) \le (\sqrt{2}/2)e^{-|t|/4}$, of norm at most $4\sqrt{2}$. So the block has norm $< 24\sqrt{2}/5 < 7$.
- *Off the diagonal block:* with $q(s) = 1 + (2s)^{-1/2}$ we have $\overline{\mathcal{G}}(s,t)^2 \le q(s)q(t)(1+1/(s+t))$. For $s \in I_0$, $t \in I_1$ use $(s+t)^{-1} \le t^{-1}$. The Hilbert–Schmidt norm squared is at most $\int_{I_0} q \cdot \int_{I_1} q(1+t^{-1}) = (1/10 + 1/\sqrt{5})(9/10 + \log 10 + \sqrt{20} - 1/\sqrt{5}) < 4$.
- *On I_1^2:* use $(s+t)^{-1} \le (2\sqrt{st})^{-1}$. The Hilbert–Schmidt norm squared is at most $A_1^2 + (1/2)C_1^2 < 8$, with $A_1 = \int_{I_1} q < 1.87$ and $C_1 = \int_{I_1} q(s)s^{-1/2}\,\mathrm{d}s < 3$.

The norm of the block operator is at most the top eigenvalue of the symmetric matrix with rows $(7,2)$ and $(2,3)$, which is $5+2\sqrt{2}$. Positivity of the kernels gives $\|\mathcal{G}\| \le \|\overline{\mathcal{G}}\|$. □

For $p = 1$ this is Perfekt's inequality with an explicit constant, and for the multiplicative Hilbert matrix it gives a bounded symbol with an explicit norm.

COROLLARY 2.6 (Perfekt's inequality [70]). *For every $f \in \mathscr{H}^1$,*

$$\Big|a_1 + \sum_{n\ge 2} \frac{a_n}{\sqrt{n}\log n}\Big| \le \Big(5 + 2\sqrt{2} + \sum_{n\ge 2} \frac{1}{n^{3/2}\log n}\Big)\|f\|_{\mathscr{H}^1} < 9\,\|f\|_{\mathscr{H}^1},$$

where the series converges in its natural order. In particular the norm C of this functional on $\mathscr{H}^1_0$ satisfies $\pi \le C < 9$, the lower bound coming from the multiplicative Hilbert matrix on $\mathscr{H}^2$ [22], and by the Hahn–Banach theorem the multiplicative Hilbert matrix has a bounded symbol of norm less than 9. Proposition 9.14 lowers 9 to 5.515.

PROOF. The left side is $|\int_0^1 f(1/2+\sigma)\,\mathrm{d}\sigma + \int_1^\infty (f(1/2+\sigma) - a_1)\,\mathrm{d}\sigma|$. Use Theorem 2.5 with $p = 1$ on the first integral and $|a_n| \le \|f\|_1$ on the second. The series is about 1.113. For natural-order convergence we use Perfekt's Tauberian argument. Put $\varepsilon = 1/\log N$. The divisor estimate $\sum_{e^u \le n < e^{2u}} d(n)/n \le e^2\zeta(1+1/u)^2 \le Cu^2$ for $u \ge 1$ gives, by splitting into geometric intervals in $\log n$,

$$\sum_{n\ge 2} \frac{d(n)}{n\log^2 n} |\mathbf{1}_{n\le N} - n^{-\varepsilon}|^2 \le C.$$

Below $\log N$ use $1 - e^{-\varepsilon \log n} \le \varepsilon \log n$; above it use exponential decay. Helson's inequality gives $\sum_n |a_n|^2/d(n) < \infty$. Cauchy–Schwarz therefore bounds the coefficient tail of the difference between the partial sum and the Abel sum by $C(\sum_{n>M} |a_n|^2/d(n))^{1/2}$, uniformly in N. Each of the finitely many remaining terms tends to zero. The Abel sums converge to the bounded functional by continuity of horizontal translation on $\mathscr{H}^1$, proving the assertion. □

The sharp constant for $p = 2$

For $p = 2$ we can retain the kernel phases. The following bound includes the constant coefficient of the Dirichlet series.

PROPOSITION 2.7. *For every $f \in \mathscr{H}^2$,*

$$\int_0^1 |f(1/2+\sigma)|^2\,\mathrm{d}\sigma \le \pi \|f\|^2_{\mathscr{H}^2}.$$

The constant π is optimal.

PROOF. We first bound the zeta kernel. Convexity of x^{-1-t} gives $\sum_{n\ge 2} n^{-1-t} \le \int_{3/2}^\infty x^{-1-t}\,\mathrm{d}x = (2/3)^t/t$. Since $(1 - e^{-at})/t$ decreases in t, taking $a = \log(3/2)$ gives

$$\zeta(1+t) \le \frac{1}{t} + 1 - \frac{1 - e^{-at}}{t} \le \frac{1}{t} + \frac{13}{18} \qquad (0 < t \le 2).$$

Let $T_N : \ell^2(\{1,\dots,N\}) \to L^2(0,1)$ be given by $T_N b(\sigma) = \sum_{n\le N} b_n n^{-1/2-\sigma}$. The kernel $k_N(\sigma,\tau)$ of $T_N T_N^*$ is bounded above by $(\sigma+\tau)^{-1} + 13/18$. With $w(\sigma) = \sigma^{-1/2}$,

$$\frac{1}{w(\sigma)} \int_0^1 k_N(\sigma,\tau) w(\tau)\,\mathrm{d}\tau \le 2\arctan(1/\sqrt{\sigma}) + \frac{13}{9}\sqrt{\sigma} \le \pi.$$

Indeed, $\arctan x \geq \pi x/4$ for $0 \leq x \leq 1$, and $13/9 < \pi/2$. Schur's test gives $\|T_N\|^2 \leq \pi$, uniformly in N. By density, the trace map extends to $T : \mathscr{H}^2 \to L^2(0,1)$ with $\|T\|^2 \leq \pi$. Continuity of point evaluation identifies this extension with $f \mapsto f(1/2 + \cdot)$.

For sharpness, use the unit vectors $g_L(\sigma) = L^{-1/2}\sigma^{-1/2}\mathbf{1}_{(e^{-2L},e^{-L})}(\sigma)$. Monotone convergence of the positive kernels k_N gives

$$\langle TT^* g_L, g_L\rangle = \iint_{(0,1)^2} \zeta(1+\sigma+\tau) g_L(\sigma) g_L(\tau)\,\mathrm{d}\sigma\,\mathrm{d}\tau.$$

Integral comparison gives $\zeta(1+t) \geq 1/t$. Under $x = -\log\sigma$, the Carleman kernel $(\sigma+\tau)^{-1}$ becomes convolution by $k(x) = 1/(2\cosh(x/2))$, and g_L becomes $L^{-1/2}\mathbf{1}_{(L,2L)}$. Consequently

$$\|T\|^2 \geq \int_{\mathbb{R}} k(x)(1-|x|/L)_+\,\mathrm{d}x \longrightarrow \int_{\mathbb{R}} k(x)\,\mathrm{d}x = \pi.$$

□

One exact prime, and the local constant of the method

The Gram bound loses the phases of all coordinates. In one coordinate they can be kept, by factorizing slices.

THEOREM 2.8. *Write $z = (z_2, w)$ and $a_\sigma = 2^{-1/2-\sigma}$, and let*

$$\mathcal{K}(\sigma,\tau) = \zeta(1+2\sigma)^{1/4}\zeta(1+2\tau)^{1/4}\zeta(1+\sigma+\tau)^{1/2}\Big(\frac{(1-a_\sigma^2)(1-a_\tau^2)}{(1-a_\sigma a_\tau)^2}\Big)^{1/4}.$$

For $0 < \delta \leq \infty$, $0 < p < \infty$ and $F \in H^p(\mathbb{T}^\infty)$, $\int_0^\delta |F(r(\sigma))|^p\,\mathrm{d}\sigma \leq \|\mathcal{K}_\delta\|\,\|F\|_p^p$, where $\mathcal{K}_\delta$ is the integral operator with kernel $\mathcal{K}$ on $L^2(0,\delta)$.

PROOF. For a polynomial F, factor each slice $F(\cdot,w) = I_w G_w^{2/p}$ with I_w inner and $G_w \in H^2(\mathbb{T})$ outer. Normalize $G_w(0) > 0$ unless the slice is identically zero, in which case set $G_w = 0$. This gives a measurable choice of $U(w) = G_w$, so $U \in L^2(\mathbb{T}^{\{p\geq 3\}}; H^2(\mathbb{T}))$ with $\|U\|^2 = \|F\|_p^p$. Stability in the variables w (Theorem 1.7) and $|F(a,w)|^{p/2} \leq |G_w(a)| = \|K_a\|\,|\langle G_w, k_a\rangle|$ give $|F(r(\sigma))|^{p/2} \leq \|K_{r(\sigma)}\|\langle U, v_\sigma\rangle$ with the unit vector $v_\sigma(w) = \theta_\sigma(w) P_{r'(\sigma)}(w)^{1/2} k_{a_\sigma}$, θ_σ chosen measurably so that the slice inner products are nonnegative. Then $|\langle v_\sigma, v_\tau\rangle| \leq |\langle k_{a_\sigma}, k_{a_\tau}\rangle|\,H(\omega_{r'(\sigma)}, \omega_{r'(\tau)})$: the prime 2 contributes $(1-\rho_2^2)^{1/2}$ instead of the affinity $h(\rho_2) \leq (1-\rho_2^2)^{1/4}$. With Lemma 2.3 for the primes other than 2, this gives $\|K_{r(\sigma)}\|\|K_{r(\tau)}\|\langle v_\sigma, v_\tau\rangle| \leq \mathcal{K}(\sigma,\tau)$, and the Bessel argument of Theorem 2.1 concludes. Polynomial density, continuity of point evaluation and Fatou's lemma extend the estimate to every $F \in H^p(\mathbb{T}^\infty)$. □

The factorization is not analytic in the remaining variables, so the argument cannot be repeated in a second coordinate.

THEOREM 2.9 (The local constant). *$\|\mathcal{K}_2\| = \Gamma(1/4)^2/\sqrt{2\pi}$. Hence, for every $0 < p < \infty$ and $f \in \mathscr{H}^p$,*

$$\int_0^2 |f(1/2+\sigma)|^p\,\mathrm{d}\sigma \leq \frac{\Gamma(1/4)^2}{\sqrt{2\pi}}\,\|f\|_{\mathscr{H}^p}^p.$$

PROOF. *Lower bound.* As $\sigma, \tau \to 0$, $\mathcal{K}$ is asymptotic to

$$L_0(\sigma,\tau) = 2^{-1/2}(\sigma\tau)^{-1/4}(\sigma+\tau)^{-1/2},$$

homogeneous of degree -1. Take the unit vector $f_T(\sigma) = T^{-1/2}\sigma^{-1/2}\mathbf{1}_{(e^{-2T},e^{-T})}(\sigma)$. The substitution $x = -\log\sigma$, with the unitary factor $\sqrt{\sigma}$, turns L_0 into convolution by $k(x) = 1/(2\sqrt{\cosh(x/2)})$. Thus

$$\langle L_0 f_T, f_T\rangle = \int_{\mathbb{R}} k(x)(1-|x|/T)_+\,\mathrm{d}x \longrightarrow \int_{\mathbb{R}} k(x)\,\mathrm{d}x = \frac{\Gamma(1/4)^2}{\sqrt{2\pi}}.$$

Since $\mathcal{K}/L_0 \to 1$ uniformly on the support of $f_T \otimes f_T$, the same limit holds for $\mathcal{K}$.

Upper bound. Schur's test with $w(\tau) = \tau^{-1/2}e^{3\tau/10}$. For $\sigma \le 1/100$ it is analytic: $\zeta(1+x) \le x^{-1}(1 + 7x/12 + x^2/12)$ gives $\mathcal{K} \le L_0$ times explicit factors $1 + O(\sigma+\tau)$, and the saving $\int_2^\infty L_0(\sigma,\tau)\tau^{-1/2}\,\mathrm{d}\tau \asymp \sigma^{1/4}$ beats the loss $O(\sigma^{1/4})$ from these factors, the margin being 0.0926 at $\sigma = 1/100$. On $[1/100, 2]$ the row sums were bounded in ball arithmetic (`arb`) on 268 cells of ratio at most 1.02, with smallest margin 0.177. □

This replaces $5 + 2\sqrt{2}$ on $(0,1)$ by the exact near-boundary constant of the method on $(0,2)$. Numerically $\|\mathcal{K}_\delta\|$ stays equal to the same value up to some $\delta_* \in (2,3)$ and then grows.

2.4. Measures on curves

A set R of primes has *density* $\theta \in [0,1]$ *with constant* B if

$$\Big|\sum_{p\in R} p^{-1-t} - \theta\log\frac{1}{t}\Big| \le B \qquad (0 < t \le 2). \tag{2.1}$$

All primes have density 1. Unions of residue classes have rational densities. Sets with $\sum_{p\in R} 1/p < \infty$ have density 0. Let $r_R(\sigma) = (p^{-1/2-\sigma})_{p\in R}$.

LEMMA 2.10 (Energy and kernel growth). *Let R have density $\theta \in (0,1]$ with constant B, and $u, v \in (0,1]$. In each geometry,*

$$E\big(r_R(u), r_R(v)\big) \ge \theta|\log u - \log v| - \theta\log 4 - 4B.$$

The kernel norm $\|K_{r_R(\sigma)}\|_2^2$ is at most $e^{1+B}(2\sigma)^{-\theta}$, $e^{\beta(1+B)}(2\sigma)^{-\beta\theta}$ and $e^B(2\sigma)^{-\theta}$ in the Hardy, Bergman and Gauss cases. We write κ for the exponent: θ, $\beta\theta$ and θ respectively.

PROOF. Since $\rho(a,b) \ge |a-b|$ for $a, b \in [0,1)$, every energy is at least

$$\sum_{p\in R}(p^{-1/2-u} - p^{-1/2-v})^2.$$

Expanding gives three values of $\sum_{p\in R} p^{-1-t}$, with $t \in \{2u, 2v, u+v\} \subset (0,2]$. By (2.1) the sum is at least $\theta\log((u+v)^2/(4uv)) - 4B$, and $(u+v)^2/(4uv) \ge e^{|\log u - \log v|}/4$. For the kernels, use $-\log(1-x) \le x + x^2$ for $x \le 1/2$, $\sum_p p^{-2} < 1$, and (2.1) at $t = 2\sigma$. □

THEOREM 2.11 (Curves in three geometries). *Let R have density $\theta \in (0,1]$ with constant B, and let κ be the kernel exponent. There is $C = C(\theta, B, \text{geometry})$ such that, for every $0 < p < \infty$ and every F in the coordinates of R:*

(1) (block maximum) for every choice of $\tau_j \in [e^{-j-1}, e^{-j}]$,

$$\sum_{j\ge 0} e^{-j\kappa} |F(r_R(\tau_j))|^p \le C \|F\|_p^p;$$

(2) (measures) for every positive ν on $(0,1]$ with $[\nu]_\kappa = \sup_t \nu((0,t])/t^\kappa < \infty$,

$$\int |F(r_R(\sigma))|^p \,\mathrm{d}\nu(\sigma) \le C[\nu]_\kappa \|F\|_p^p.$$

For all primes in the Hardy geometry, (2) holds with $C < 21$.

PROOF. In each geometry $H(\omega_r, \omega_s) \le e^{-cE(r,s)}$, with $c = 1/4$ (Hardy and Gauss, Proposition 1.4) or $c = \beta/4$ (Bergman, Proposition 1.4).

(1) Apply Theorem 2.1 to $\mu = \sum_j e^{-j\kappa}\delta_{r_R(\tau_j)}$ with the weight $\|K\|$. Since $\|K_{r_R(\tau_k)}\|_2^2 \le C_K e^{(k+1)\kappa}$, the row sums are at most

$$C_K e^{\kappa} \sum_k \min\bigl(1, e^{-c(\theta(|j-k|-1)-\theta\log 4 - 4B)}\bigr),$$

which is bounded uniformly in j by Lemma 2.10.

(2) Apply Theorem 2.1 to ν itself with the weight $\|K\|$, and split $(0,1]$ into the blocks $I_k = (e^{-k-1}, e^{-k}]$. Then $\int_{I_k} \|K\|^2 \,\mathrm{d}\nu \le C_K e^{(k+1)\kappa}[\nu]_\kappa e^{-k\kappa}$, while for $\sigma \in I_j$ and $\tau \in I_k$ the affinity is at most $\min(1, e^{-c(\theta(|j-k|-1)-C_2)})$, with $C_2 = \theta\log 4 + 4B$, by Lemma 2.10. Sum over k.

For all primes in the Hardy geometry, Perfekt's estimate $\sum_p (p^{-1/2-u} - p^{-1/2-v})^2 > |\log u - \log v| - 7$ and $\zeta(1+2\sigma) \le 3/(2\sigma)$ give the explicit bound $(3/2)e\bigl(17 + 2e^{-1/4}/(1-e^{-1/4})\bigr) < 99$ in both parts. For (2) the Gram kernel does better. By Lemma 2.4, the Schur row at σ with the weight $\|K_\tau\| = \zeta(1+2\tau)^{1/2}$ is at most $\int g_\sigma \,\mathrm{d}\nu$ with $g_\sigma(\tau) = \zeta(1+2\sigma)^{-1/4}\zeta(1+2\tau)^{3/4}\zeta(1+\sigma+\tau)^{1/2}$, which decreases in τ. Since $\nu((0,t]) \le [\nu]t$, the layer-cake formula gives $\int g_\sigma \,\mathrm{d}\nu \le [\nu]\int_0^1 g_\sigma \,\mathrm{d}\tau$, and $1/x \le \zeta(1+x) \le 3/x$ for $0 < x \le 2$ gives

$$\int_0^1 g_\sigma \,\mathrm{d}\tau \le 3^{5/4} \int_0^\infty (2\sigma)^{1/4}(2\tau)^{-3/4}(\sigma+\tau)^{-1/2} \,\mathrm{d}\tau = 3^{5/4}\frac{\Gamma(1/4)^2}{\sqrt{2\pi}} < 20.71.$$

□

In the block maximum the points can be chosen to maximize $|F|$ in each block, which is what atomic measures need (chapter 4). Averaging over the position in the block recovers $\int_0^1 |F(r_R(\sigma))|^p \sigma^{\kappa-1} \,\mathrm{d}\sigma \lesssim \|F\|_p^p$. For prime sets this is the Fejér–Riesz inequality with weight $\sigma^{\theta-1}$; for the Bergman geometry with all primes it is $\int_0^1 |f(1/2+\sigma)|^p \sigma^{\beta-1} \,\mathrm{d}\sigma \lesssim \|f\|_{\mathscr{A}^p_\beta}^p$.

On the curves themselves the condition of Theorem 2.11 is also necessary, so there the Hellinger–Schur bound loses nothing.

COROLLARY 2.12 (Carleson measures on curves). *Let R have density $\theta \in (0,1]$ with constant B, let κ be the kernel exponent, and let ν be a positive measure on $(0,1]$. If $\int |F(r_R(\sigma))|^p \,\mathrm{d}\nu(\sigma) \le M\|F\|_p^p$ for some $0 < p < \infty$, then $[\nu]_\kappa \le CM$. Hence the following are equivalent: the embedding holds for one p; it holds for every p; $[\nu]_\kappa < \infty$; and the Gram operator $\mathcal{G}_\nu$ is bounded.*

PROOF. Test with $F_t = K_{r_R(t)}^{2/p} \|K_{r_R(t)}\|_2^{-2/p}$, of norm one. All coefficients of the reproducing kernels are nonnegative. Since $r_R(\sigma) \ge r_R(t)$ coordinatewise for $\sigma \le t$,

$$|F_t(r_R(\sigma))|^p = \frac{K_{r_R(t)}(r_R(\sigma))^2}{\|K_{r_R(t)}\|_2^2} \ge \|K_{r_R(t)}\|_2^2 \ge ct^{-\kappa}.$$

The last inequality follows from (2.1): one can take $c = e^{-B}2^{-\theta}$ in Hardy and Gauss geometry, and $c = e^{-\beta B}2^{-\beta\theta}$ in Bergman geometry. Hence $\nu((0,t]) \le CMt^\kappa$. Part (2) of the proof of Theorem 2.11 bounds $\|\mathcal{G}_\nu\|$ by $C[\nu]_\kappa$, and Theorem 2.1 closes the circle. □

2.5. Further consequences

Four notions of separation appear in this chapter, and they are not interchangeable. Ordered by the way the argument uses them:

(1) *Energy.* Boundedly many near neighbours, and $\sum_{\text{far } k} e^{-E(r^j,r^k)/4} < \infty$ uniformly in j.
(2) *Summable rows.* $\sup_j \sum_k H(\omega_{r^j}, \omega_{r^k}) < \infty$.
(3) *Bounded affinity matrix.* $(H(\omega_{r^j}, \omega_{r^k}))$ is bounded on ℓ^2; this is all that Theorem 2.1 needs.
(4) *Geometric separation.* Pseudo-hyperbolic distances bounded below.

Here (1) implies (2) by Proposition 1.4, and (2) implies (3) by Schur's test. The converse of the second step fails: in the tree family of Proposition 2.14 every row diverges while the matrix is bounded. And (4) controls only nearest neighbours, not how many points lie at each distance: the separated grids of Proposition 2.18 carry Carleson measures whose Gram operators are unbounded, because the long tails of the affinities accumulate. The prime curve works because overlap decays exponentially in the hyperbolic distance (Theorem 1.6) while the number of points within distance d along the curve grows only linearly in d, so the rows of the affinity matrix are geometric series.

COROLLARY 2.13 (Energy-separated families). *Let (r^j) be points with*

$$S = \sup_j \sum_k H(\omega_{r^j}, \omega_{r^k}) < \infty,$$

and let $c_j \ge 0$. The following are equivalent:

(1) $\sum c_j |F(r^j)|^p \lesssim \|F\|_p^p$ for one, equivalently every, $p > 0$;
(2) $\sum c_j |F(r^j)|^2 \lesssim \|F\|_2^2$;
(3) $c_j \lesssim \|K_{r^j}\|_2^{-2}$.

PROOF. The one-point condition and Theorem 2.1 with the weight $\|K\|$ give every p. Testing (1) with $K_{r^j}^{2/p}/\|K_{r^j}\|_2^{2/p}$ gives the one-point condition (3), and so does

testing (2) with K_{r^j}. The summability of the affinities holds, for instance, when there are boundedly many near neighbours and $\sum_{\text{far } k} e^{-E_{jk}/4} < \infty$. □

Summability is used only to bound the affinity matrix on ℓ^2, and that bound is all the argument needs.

PROPOSITION 2.14 (Bounded affinity matrices). *(1) Let (r^j) be points whose affinity matrix $\mathbf{H} = (H(\omega_{r^j}, \omega_{r^k}))_{j,k}$ is bounded on ℓ^2, and let $c_j \geq 0$. Then $\sum c_j |F(r^j)|^p \lesssim \|F\|_p^p$ for one $p > 0$ if and only if $c_j \lesssim \|K_{r^j}\|_2^{-2}$, and then the Gram operator is bounded and the embedding holds for every $p > 0$.*

(2) In Hardy and Gauss geometry there are families with pairwise energies bounded below by a positive constant, for which $\mathbf{H}$ is bounded on ℓ^2 but $\sum_k H(\omega_{r^j}, \omega_{r^k}) = \infty$ for every j.

PROOF. (1) Testing with $K_{r^j}^{2/p}/\|K_{r^j}\|_2^{2/p}$, as in Corollary 2.12, gives $c_j \|K_{r^j}\|_2^2 \leq C_p$. For the measure $\mu = \sum_j c_j \delta_{r^j}$, the map $f \mapsto (\sqrt{c_j} f(r^j))_j$ identifies $L^2(\mu)$ with ℓ^2 and turns $\mathcal{G}_\mu$ into $D\mathbf{H}D$, with D diagonal and $D_{jj} = (c_j \|K_{r^j}\|_2^2)^{1/2}$. So $\|\mathcal{G}_\mu\| \leq C_p \|\mathbf{H}\|$, and Theorem 2.1 gives every p.

(2) Let T be the tree in which every vertex has degree 3, with adjacency operator A, and let $c = 3/5$. The weighted Schur test with weight $2^{-|v|/2}$, where $|v|$ is the distance to a fixed root, gives $\|A\| \leq 2\sqrt{2}$. For $v \neq w$ exactly one neighbour of v is closer to w, so $h_w(v) = c^{d(v,w)}$ satisfies $((1 + 2c^2)I - cA)h_w = (1 - c^2)e_w$, and $h_w \in \ell^2(T)$ since $2c^2 < 1$. As $(1 + 2c^2)I - cA \geq (1 - \sqrt{2}c)^2 I$, the matrix $(c^{d(v,w)})$ equals $(1 - c^2)((1 + 2c^2)I - cA)^{-1}$ and is bounded, while its row sums $1 + 3c \sum_{n \geq 0} (2c)^n$ diverge.

Index the coordinates by the edges of T. To the vertex v assign the point r^v with coordinate t on each edge of the path from the root to v and 0 elsewhere. In the Hardy geometry choose t with $h(t) = c$; it exists because h is continuous, $h(0) = 1$, and $h(t) \leq (1 - t^2)^{1/4}$ (Lemma 2.3). In the Gauss geometry choose t with $e^{-t^2/4} = c$. Shared edges contribute affinity 1 and the $d(v, w)$ edges of the symmetric difference contribute c each, so $H(\omega_{r^v}, \omega_{r^w}) = c^{d(v,w)}$ by Proposition 1.4, and $E(r^v, r^w) = t^2 d(v, w)$. □

With $c_v = \|K_{r^v}\|_2^{-2}$ the Gram operator of the tree family is $\mathbf{H}$ itself, so it is bounded and every embedding holds. The Hellinger–Schur rows with the weight $\|K\|$ are the divergent row sums of $\mathbf{H}$, so a different weight is needed to show boundedness.

COROLLARY 2.15 (Independent general frequencies). *In Hardy geometry, let $b_j > 0$ be linearly independent over $\mathbb{Q}$, let $0 < q_j < 1$ with $\sum_j [-\log(1 - q_j^2) - q_j^2] < \infty$, put $r_j(\sigma) = q_j e^{-b_j \sigma}$, and suppose that $A(\sigma) = \sum_j q_j^2 e^{-2b_j \sigma}$ satisfies $|A(\sigma) - \theta \log(1/\sigma)| \leq M$ on $(0, 1]$ for some $\theta > 0$ and $M < \infty$. Then for every $p > 0$,*

$$\int_0^1 |F(r(\sigma))|^p \, \sigma^{\theta - 1} \, \mathrm{d}\sigma \leq C \|F\|_p^p.$$

Proof. Put $D=\sum_j[-\log(1-q_j^2)-q_j^2]<\infty$. Then

$$\log\|K_{r(\sigma)}\|_2^2 \le A(\sigma)+D \le \theta\log(1/\sigma)+M+D.$$

For $0<u,v\le 1$, the energy satisfies

$$\begin{aligned} E(r(u),r(v)) &\ge \sum_j q_j^2(e^{-b_j u}-e^{-b_j v})^2 \\ &= A(u)+A(v)-2A((u+v)/2) \\ &\ge \theta\log\frac{(u+v)^2}{4uv}-4M. \end{aligned}$$

These are the kernel-growth and energy bounds used in the proof of Theorem 2.11, so the same Schur estimate applies. For general Dirichlet series whose frequencies are generated by the b_j, rational independence gives the identification with $H^p(\mathbb{T}^\infty)$ of Defant and Schoolmann [32]. □

Remark 2.16. Two further consequences need no new argument.

- *Vector-valued H^p.* For X-valued F, the function $\|F\|_X^{p/2}$ is plurisubharmonic. So stability, and with it every result above, holds for $H^p(\mathbb{T}^\infty;X)$ and every Banach space X.
- *Translation-average Bergman spaces.* Averaging Theorem 2.5 over horizontal translations gives $\int_0^1|f(1/2+t)|^p\mu([0,t])\,\mathrm{d}t\le(5+2\sqrt{2})\|f\|_{\mathscr{A}^p_\mu}^p$ for the spaces $\mathscr{A}^p_\mu$ of Bailleul and Lefèvre [2].

2.6. What the Gram bound sees

The inequality replaces each evaluation by the positive function $|K_r|$, and so forgets the phases of the kernels. The cleanest way to see what this costs is to compare, in one coordinate, the exact Carleson constant $C(\mu)=\sup\int|F|^p\,\mathrm{d}\mu/\|F\|_p^p$ with $\|\mathcal{G}_\mu\|$ for measures that interpolate between a point and a circle.

Lemma 2.17. *For $k\ge 2$, $0<p<\infty$, $x\in\mathbb{D}$ and $g\in H^p(\mathbb{T})$,*

$$\frac{1}{k}\sum_{\omega^k=1}|g(\omega x)|^p \le \frac{\|g\|_{H^p}^p}{1-|x|^{2k}},$$

with equality for $g=(1-\overline{x}^k z^k)^{-2/p}$. For $k=2$ this is $(|g(x)|^p+|g(-x)|^p)/2\le\|g\|_p^p/(1-|x|^4)$.

Proof. Write $g=Ih^{2/p}$ with I inner and h outer, so $|g(\omega x)|^p\le|h(\omega x)|^2$ and $\|h\|_2^2=\|g\|_p^p$. Split h by residues of the exponent, $h(z)=\sum_{r<k}z^r h_r(z^k)$, so that $\|h\|_2^2=\sum_r\|h_r\|_2^2$. Averaging over the roots of unity removes the cross terms:

$$\frac{1}{k}\sum_{\omega^k=1}|h(\omega x)|^2=\sum_{r<k}|x|^{2r}|h_r(x^k)|^2\le\sum_{r<k}\frac{\|h_r\|_2^2}{1-|x|^{2k}}=\frac{\|h\|_2^2}{1-|x|^{2k}},$$

by the point estimate at x^k. Equality holds when only h_0 is present and is the reproducing kernel at x^k, and $I=1$. □

PROPOSITION 2.18. *In the Hardy geometry, in one coordinate and for $0 < r < 1$, let μ_k be the average of the point masses at the k points ωr, $\omega^k = 1$, and let m_r be normalized arc length on $\{|z| = r\}$. For every $0 < p < \infty$:*

(1) $C(\mu_k) = 1/(1-r^{2k})$ and $C(\mu_k) \le \|\mathcal{G}_{\mu_k}\| \le kC(\mu_k)$. For fixed k the ratio tends to 1 as $r \to 1$ and as $r \to 0$.
(2) $C(m_r) = 1$, while $\|\mathcal{G}_{m_r}\| \ge c\log^2\big(1/(1-r)\big)$.
(3) For $\mu = (6/\pi^2)\sum_j j^{-2} m_{1-e^{-j}}$, $C(\mu) \le 1$ while $\mathcal{G}_\mu$ is unbounded.
(4) Separation does not help. Every probability measure ν on $\{|z| = r\}$ has $\|\mathcal{G}_\nu\| \ge c\log^2\big(1/(1-r)\big)$. For every $\delta < 1$ there are radii $r_j \to 1$ and grids ν_j, each the normalized counting measure on n_j equally spaced points of $\{|z| = r_j\}$, such that any two points of the union are at pseudo-hyperbolic distance at least δ, and $\mu = \sum_j j^{-3/2}\nu_j$ satisfies $C(\mu) < \infty$ for every p while $\mathcal{G}_\mu$ is unbounded.

PROOF. (1) The constant is Lemma 2.17, sharp. The Gram matrix is circulant with positive entries $(1-r^2)^{-1}h(\rho(r,\omega r))/k$, so its norm is the row sum $(1-r^2)^{-1}k^{-1}\sum_\omega h(\rho(r,\omega r))$. This lies between $(1-r^2)^{-1}/k$ and $(1-r^2)^{-1}$, and $k(1-r^2) \ge 1-r^{2k} \ge 1-r^2$. As $r \to 1$, $\rho(r,\omega r) \to 1$ for $\omega \ne 1$, so $\sum_\omega h(\rho(r,\omega r)) \to 1$ and the ratio tends to $\lim(1-r^{2k})/(k(1-r^2)) = 1$. As $r \to 0$, both sides tend to 1. (2) Subharmonicity gives $\int |F|^p \,\mathrm{d}m_r \le \|F\|_p^p$, with equality for constants. With V as in the proof of Theorem 2.1, $\|\mathcal{G}\| = \|V^*\|^2 \ge \int \langle 1, |K_z|\rangle^2 \,\mathrm{d}m_r(z)$, and $\langle 1, |K_z|\rangle = \int_{\mathbb{T}} |1-\overline{z}\zeta|^{-1}\,\mathrm{d}m(\zeta) \asymp \log(1/(1-r))$. (3) Subharmonicity on each circle gives $C(\mu) \le 1$. The lower bound $\|\mathcal{G}_\mu\| \ge \int \langle 1, |K_z|\rangle^2 \,\mathrm{d}\mu(z)$ from (2) is linear in μ, and here it is $\asymp \sum_j j^{-2} \cdot j^2 = \infty$. (4) The bound in (2) used only that $\langle 1, |K_z|\rangle$ depends on $|z|$ alone, so it holds for every probability measure on the circle. Let $\lambda, \eta > 0$, put $a_j = e^{-\lambda j}$, $r_j = 1 - a_j$ and $n_j = \lceil \eta/a_j \rceil$, and keep $j \ge j_0$ with $\eta/a_{j_0} \ge 2$. Since $|F|^p$ is subharmonic, averaging the Poisson kernel over the grid gives $\int |F|^p \,\mathrm{d}\nu_j \le (1+r_j^{n_j})(1-r_j^{n_j})^{-1}\|F\|_p^p$, and $r_j^{n_j} \le e^{-\eta}$; the weights $j^{-3/2}$ are summable. For points on different circles, $\rho(z,w) \ge \rho(|z|,|w|) \ge (a_j - a_k)/(a_j + a_k) \ge \tanh(\lambda/2)$ when $j < k$. On one circle, two distinct points differ by an angle θ with $|\sin(\theta/2)| \ge 2/n_j \ge a_j/\eta$, and $1 - r_j^2 \le 2a_j$, so

$$\rho^2 = \frac{4r_j^2 \sin^2(\theta/2)}{(1-r_j^2)^2 + 4r_j^2\sin^2(\theta/2)} \ge \frac{r_j^2/\eta^2}{1 + r_j^2/\eta^2}.$$

Both lower bounds exceed δ once λ is large and η small. Finally $\|\mathcal{G}_{\nu_j}\| \gtrsim \lambda^2 j^2$, so the compression of $\mathcal{G}_\mu$ to the jth grid has norm $\gtrsim \lambda^2 j^{1/2}$. □

So the Gram bound loses even when each kernel's support, of angular width about $1-r$, contains at most one point of the measure. The loss is in the tails: the exact formula of Proposition 1.4 gives $h(\rho) \sim (2/\pi)(1-\rho^2)^{1/2}\log\big(4(1-\rho^2)^{-1/2}\big)$ as $\rho \to 1$, so on a grid the affinity to the mth neighbour is about $m^{-1}\log m$, and the rows of the (circulant) affinity matrix sum to about $\log^2(1/(1-r))$. Thus the affinity matrix is unbounded on ℓ^2, which is exactly the hypothesis of Proposition 2.14 that fails. The loss comes from directions in which the kernels rotate: on a circle, $K_{\omega r}$ and K_r differ by a phase, and the cancellation that makes $\int |F|^p \,\mathrm{d}m_r \le \|F\|_p^p$ trivial

is invisible to $|K|$. Along the prime curve the kernels at different heights have no relative rotation, every coordinate being real and positive, and Theorem 2.11 is efficient there. Rotational averages are handled instead by the symmetry mechanism of section 4.4.1, which uses analyticity in the rotated variable.

2.7. Controlled evaluation superpositions

Let $X = \mathbb{D}^\infty \cap \ell^2$, $K_r(z) = \prod_j (1 - \overline{r_j} z_j)^{-1}$, $N_r = \|K_r\|_2$, $P_r = |K_r|^2 / N_r^2$ and $H(r,s) = \int_{\mathbb{T}^\infty} \sqrt{P_r P_s}\, dm$. Write $\delta_r(F) = F(r)$. Then $\|\delta_r\|_{(H^1)^*} = N_r^2$ by Proposition 1.3.

Call $L \in (H^1(\mathbb{T}^\infty))^*$ *controlled* if $L(F) = \int_X F(r) w(r)\, d\mu(r)$ for all $F \in H^1(\mathbb{T}^\infty)$, where μ is a positive Borel measure on X, w is μ-measurable with $|w(r)| \le N_r^{-2}$, and

$$S(\mu) = \operatorname*{ess\,sup}_{r \in \operatorname{supp}\mu} \int_X H(r,s) N_s^2\, d\mu(s) < \infty.$$

Write $\mathcal{R}$ for this class and $\mathcal{E} = \operatorname{span}\{\delta_r : r \in X\}$.

LEMMA 2.19. *If $L \in \mathcal{R}$, then there is $C < \infty$ such that for every $F \in H^1(\mathbb{T}^\infty)$*

$$|L(F)| \le C \sup_{s \in X} |F(s)| N_s^{-3}.$$

PROOF. Fix a representation (w, μ) with $\mu \ne 0$ (if $\mu = 0$ then $L = 0$). The row bound holds μ-almost everywhere and $\operatorname{supp}\mu$ has full measure, X being separable, so we may choose $r_0 \in \operatorname{supp}\mu$ with $\int H(r_0, s) N_s^2\, d\mu(s) \le S(\mu)$. The product $K_{r_0} K_s$ is analytic with constant coefficient one, so

$$H(r_0, s) = \frac{1}{N_{r_0} N_s} \int |K_{r_0} K_s|\, dm \ge \frac{1}{N_{r_0} N_s}.$$

Hence $\int N_s\, d\mu(s) \le N_{r_0} S(\mu)$, and

$$|L(F)| \le \int |F(s)| N_s^{-2}\, d\mu(s) \le \sup_s \bigl(|F(s)| N_s^{-3}\bigr) \int N_s\, d\mu(s).$$

Take $C = N_{r_0} S(\mu)$. □

THEOREM 2.20. *For the controlled class and evaluation span,*

$$\mathcal{E} \subset \mathcal{R} \subset \overline{\mathcal{E}}^{\,\|\cdot\|} \subsetneq (H^1(\mathbb{T}^\infty))^*.$$

In particular, $\overline{\mathcal{R}}^{\,\|\cdot\|} = \overline{\mathcal{E}}^{\,\|\cdot\|}$. Every element of $\mathcal{R}$, when restricted to the copy of $H^1(\mathbb{T})$ in the first coordinate, has a representing analytic symbol in VMOA. *In particular, the bounded functional*

$$L_\Theta(F) = \int_{\mathbb{T}^\infty} F(\zeta) \overline{\Theta(\zeta_1)}\, dm(\zeta), \qquad \Theta(z) = \exp\!\left(-\frac{1+z}{1-z}\right),$$

belongs to $(H^1(\mathbb{T}^\infty))^*$ but does not belong to $\overline{\mathcal{E}}^{\,\|\cdot\|}$, hence has no controlled superposition representation.

PROOF. Every controlling measure is finite by Lemma 2.19, and $\|w(r)\delta_r\| \le 1$ by the point-evaluation bound.

The map $r \mapsto \delta_r$ is norm continuous on X with its ℓ^2 topology. Indeed, a small ℓ^2 ball about any fixed r remains in X and has uniformly bounded N_s^2; the point-evaluation bound and Cauchy's estimate on each complex line give a local Lipschitz bound for δ_s. Since X is separable, $r \mapsto w(r)\delta_r$ is strongly measurable. It is Bochner integrable because $\mu(X) < \infty$ and its norm is at most one. Approximation by simple functions proves $L = \int w(r)\delta_r \, \mathrm{d}\mu(r) \in \overline{\mathcal{E}}^{\,\|\cdot\|}$.

Conversely, for a finite sum $L = \sum_{j=1}^{J} c_j \delta_{r_j}$, combine repeated points and choose atomic masses $A_j \ge |c_j| N_{r_j}^2$. The measure $\mu = \sum_j A_j \delta_{r_j}$ has finite Schur rows, and $w(r_j) = c_j / A_j$ obeys the pointwise cap. Thus $\mathcal{E} \subset \mathcal{R}$.

Let $J : H^1(\mathbb{T}) \to H^1(\mathbb{T}^\infty)$ place a function in the first coordinate. Restriction J^* maps δ_r to ordinary disc evaluation at r_1. Under the classical H^1–BMOA duality [36, Theorem 7.1], that evaluation has the symbol $(1 - \overline{r_1} z)^{-1}$, analytic across the closed disc. By the closure characterization [36, Theorem 5.5], VMOA is a closed subspace of BMOA containing these symbols. Hence $J^*(\overline{\mathcal{E}}^{\,\|\cdot\|}) \subset \mathrm{VMOA}$.

The function Θ is inner, so $|\Theta| \le 1$ in $\mathbb{D}$ and $\|L_\Theta\| \le 1$. Its boundary values away from 1 satisfy $\Theta(e^{i\theta}) = \exp(-i \cot(\theta/2))$. For $I_h = (0, h)$, integration by parts after the substitution $u = \cot(\theta/2)$ gives

$$\Theta_{I_h} := \frac{1}{h} \int_0^h \Theta(e^{i\theta}) \, \mathrm{d}\theta \longrightarrow 0 \qquad (h \downarrow 0).$$

Indeed, the phase derivative is comparable to θ^{-2} and the endpoint and derivative terms are $O(h^2)$. Since $|\Theta| = 1$ almost everywhere, the mean absolute oscillation satisfies

$$\frac{1}{h} \int_0^h |\Theta(e^{i\theta}) - \Theta_{I_h}| \, \mathrm{d}\theta \ge 1 - |\Theta_{I_h}| \longrightarrow 1.$$

Thus $\Theta \notin \mathrm{VMOA}$. Since $J^* L_\Theta$ has symbol Θ, the asserted properness follows.

□

THEOREM 2.21. *Let $a_k = 1 - 2^{-k}$, $r_k = (a_k, 0, 0, \dots) \in X$ and*

$$L = \sum_{k \ge 1} \frac{2^{-k}}{k^2} \, \delta_{r_k}.$$

The series converges absolutely in $(H^1(\mathbb{T}^\infty))^*$, so L lies in the norm-closed span of point evaluations, but L is not controlled.

The theorem holds equally if the supremum in $S(\mu)$ runs over all $r \in X$, since that only shrinks $\mathcal{R}$.

PROOF OF THEOREM 2.21. Since $N_{r_k}^2 = (1 - a_k^2)^{-1} \le 2^k$, we have

$$\sum_k 2^{-k} k^{-2} \|\delta_{r_k}\| \le \sum_k k^{-2} < \infty.$$

So the series converges absolutely in norm, and $L(F) = \sum_k 2^{-k} k^{-2} F(r_k)$ for every $F \in H^1(\mathbb{T}^\infty)$.

For $0 < \rho < 1$ put $F_\rho(z) = (1 - \rho z_1)^{-3/2}$, a bounded analytic function of the first variable, so $F_\rho \in H^1(\mathbb{T}^\infty)$. For $s \in X$,

$$|F_\rho(s)| \le (1 - |s_1|)^{-3/2} \le 2^{3/2}(1 - |s_1|^2)^{-3/2} \le 2^{3/2} N_s^3,$$

because $|1 - \rho s_1| \ge 1 - |s_1|$ and $N_s^2 = \prod_j (1 - |s_j|^2)^{-1} \ge (1 - |s_1|^2)^{-1}$. Thus $\sup_s |F_\rho(s)| N_s^{-3} \le 2^{3/2}$ for all ρ. On the other hand every term of

$$L(F_\rho) = \sum_{k \ge 1} \frac{2^{-k}}{k^2} (1 - \rho a_k)^{-3/2}$$

is positive and increases to $2^{-k} k^{-2} (1 - a_k)^{-3/2} = 2^{k/2} k^{-2}$ as $\rho \uparrow 1$. By monotone convergence $L(F_\rho) \to \infty$. By Lemma 2.19, $L \notin \mathcal{R}$. □

PROPOSITION 2.22 (Weak-$*$ density). *The span of the evaluations at $\gamma(\sigma) = (p_j^{-\sigma})_j$, $\sigma > 1$, is weak-$*$ dense in $(H^1(\mathbb{T}^\infty))^*$. Hence so is the controlled class $\mathcal{R}$, although it is not norm dense.*

PROOF. Write $\gamma(\sigma) = (p_j^{-\sigma})_j$ for the prime curve, $\sigma > 1$. These evaluations lie in $\mathcal{E}$. If $F \in H^1(\mathbb{T}^\infty)$ is annihilated by all of them, its Dirichlet series $\sum_{n \ge 1} a_n n^{-\sigma}$ vanishes for every $\sigma > 1$. Since $|a_n| \le \|F\|_1$, the series is absolutely convergent there. Isolating the least nonzero coefficient and letting $\sigma \to \infty$ contradicts vanishing. Hence $F = 0$, and the preannihilator criterion gives the stated weak-$*$ density. □

Open

QUESTION 2. On prime curves, and for families whose affinity matrix is bounded on ℓ^2, boundedness of the modulus Gram operator is necessary (Corollary 2.12 and Proposition 2.14); for merely separated families it is not (part (4) of Proposition 2.18). Is there a geometric condition on the family, between the two, under which it is necessary? What exactly does the passage from K_r to $|K_r|$ lose?

QUESTION 3. What is the sharp constant in Theorem 2.5? Gram methods give exactly $\Gamma(1/4)^2/\sqrt{2\pi}$ near the boundary (Theorem 2.9), and exactness in finitely many primes does not lower this. For $p = 2$ the sharp constant is π, including the constant coefficient (Proposition 2.7). Determine the sharp constant for general $p > 0$.

CHAPTER 3

Carleson measures near the critical line

Question. Which measures in $\mathbb{D}^\infty \cap \ell^2$, or in the half-plane $\mathbb{C}_{1/2}$, are Carleson measures for H^p, and when does the answer depend on p?

The Hellinger–Bessel inequality (chapter 2) gives a sufficient condition that does not see p, and it settles discrete families separated in energy completely (Proposition 2.14). This chapter tests how far it reaches. It handles twisted points, and hence complex measures on the half-plane, at no extra cost. Its role for the rest of the book is Theorem 3.1, which is used in chapter 10. On vertical lines, where factorization is lost in a different way, the Gram condition is not the right tool: an L^r argument with the Poisson kernel, the positive representative of section 1.2, gives a rate below every power of ε, for every p. The chapter ends with the dual question on the line itself: $(\mathscr{H}^1)^*$ is not locally in L^q for $q > 7/2$ (Theorem 3.9). Apart from the embedding theorems quoted in Proposition 3.3, the results of the chapter are, to our knowledge, new.

3.1. Twists and the half-plane

For a Dirichlet series f and a completely multiplicative $\chi : \mathbb{N} \to \mathbb{T}$ write $f_\chi = \sum a_n \chi(n) n^{-s}$. Vertical translation is the case $\chi(n) = n^{-iy}$. The point is that twisting the evaluation points costs nothing: kernel norms depend only on moduli, and energies only increase, since $|a_p(r,s)| \geq \big||r_p| - |s_p|\big|$.

THEOREM 3.1 (Twisted block maximum). *Let R have density $\theta \in (0,1]$ with constant B. For every $0 < p < \infty$ and every $f \in \mathscr{H}^p$ supported on the R-integers,*

$$\sum_{j \geq 0} e^{-j\theta} \max_{\tau \in \overline{I_j}} \sup_\chi |f_\chi(1/2+\tau)|^p \leq C \|f\|_{\mathscr{H}^p}^p,$$

$$\int_0^1 \sup_\chi |f_\chi(1/2+\sigma)|^p \sigma^{\theta-1} \, \mathrm{d}\sigma \leq C \|f\|_{\mathscr{H}^p}^p,$$

with C as in Theorem 2.11.

PROOF. For a polynomial the supremum over χ is a maximum over a finite torus. Choose maximizing (τ_j, χ_j) and the points $r^j = (\chi_j(p) p^{-1/2-\tau_j})_p$. The proof of Theorem 2.11 (1) uses the points only through $\|K_{r^j}\|_2$ and lower bounds on the energies by moduli. The second estimate follows from the first, since $\int_{I_j} \sigma^{\theta-1} \, \mathrm{d}\sigma = e^{-j\theta}(1 - e^{-\theta})/\theta \leq e^{-j\theta}$. Polynomial approximation and Fatou's lemma extend both estimates to every $f \in \mathscr{H}^p$. □

COROLLARY 3.2 (Complex measures on the half-plane). *Let μ be a complex measure on $\mathbb{C}_{1/2}$ with*

$$[\mu] = \sup_{0<t\le 1} \frac{|\mu|(\{1/2 < \Re s \le 1/2+t\})}{t} < \infty.$$

Then $\int |f|^p \,\mathrm{d}|\mu| \le (C[\mu] + \zeta(3)|\mu|(\mathbb{C}_{1/2}))\|f\|^p_{\mathscr{H}^p}$ for every $0 < p < \infty$, and the Helson matrix with moments $\alpha(n) = \int n^{-s}\,\mathrm{d}\mu(s)$ has a bounded symbol.

PROOF. On the strip $\Re s - 1/2 \in I_j$, $|f(s)| \le \max_\tau \sup_\chi |f_\chi(1/2+\tau)|$. Apply Theorem 3.1; for $\Re s > 3/2$ use $|f(s)|^p \le \zeta(2\Re s)\|f\|_p^p$. With $p = 1$, Hahn–Banach gives the symbol. □

The strip condition is sufficient but not necessary.

PROPOSITION 3.3 (Necessity on the half-plane). *(1) An $\mathscr{H}^p$ Carleson measure satisfies the box condition $|\mu|(\{\Re s - 1/2 \le \delta,\ |\Im s - y| \le \delta\}) \lesssim \delta$. For $p = 2$ and μ supported in a bounded box above the line, the box condition is also sufficient, by the local embedding theorem and Carleson's theorem [67]. For $2 < p < \infty$, the same conclusion is conditional on a theorem of Chen, Fang, Guo, Hou, Shao and Zhou [28, Theorem 1.1], together with the local Carleson-measure transfer of Olsen and Saksman.*

(2) The strip condition is not necessary for $\mathscr{H}^2$: with λ_ε Lebesgue measure on $\{1/2+\varepsilon+it : 0 \le t \le 1\}$ and $\varepsilon_k = e^{-k^2}$, the measure $\sum_k 2^{-k}\lambda_{\varepsilon_k}$ is $\mathscr{H}^2$-Carleson.

(3) There is an explicit $\mathscr{H}^2$- but not $\mathscr{H}^1$-Carleson measure of the form $\sum_k 2^{-k}\lambda_{\varepsilon_k}$.

PROOF. (1) Test with $\zeta(s+1/2+\delta-iy)^{2/p}$ truncated to finitely many primes, using $|\zeta(1+w)| \ge 1/|w| - 3$, and let the truncation grow. (2) The local embedding for $\mathscr{H}^2$ and contractivity of horizontal translation make each λ_ε uniformly Carleson. (3) The norms $N_1(\varepsilon) = \sup \int |f|\,\mathrm{d}\lambda_\varepsilon / \|f\|_{\mathscr{H}^1}$ are finite but unbounded, for otherwise $\int_0^1 |f(1/2+it)|\,\mathrm{d}t \lesssim \|f\|_{\mathscr{H}^1}$, which Harper's theorem [40] rules out. Choose ε_k with $N_1(\varepsilon_k) \ge 8^k$. □

The measure in part (3) lies on complex points. This distinction matters: its construction uses vertical translation, not real evaluation points.

3.2. Vertical lines

The most prominent open problem about Carleson measures for Dirichlet series concerns a single segment of the critical line.

PROBLEM 1 (The embedding problem; Saksman and Seip [80, Problem 2.1]). *For which $0 < p < \infty$ is there a constant C_p with $\int_0^1 |f(1/2+it)|^p\,\mathrm{d}t \le C_p\|f\|^p_{\mathscr{H}^p}$ for every Dirichlet polynomial f?*

The answer is yes for $p = 2$ and hence for $p \in 2\mathbb{N}$, by applying the case $p = 2$ to $f^{p/2}$. It is no for every $0 < p < 2$: Harper's low-moment estimates [40] show that for $f = \sum_{n\le x} n^{-s}$ the pth power of the $\mathscr{H}^p$ norm is smaller than $x^{p/2}$ by a power of $\log\log x$, while $\int_0^1 |f(1/2+it)|^p\,\mathrm{d}t \asymp x^{p/2}$. Chen, Fang, Guo, Hou, Shao

and Zhou [28] have announced an affirmative answer for every $p > 2$, by reducing the dual problem to a log-correlated Gaussian field and a critical branching random walk; if their proof stands, the answer is exactly $p \geq 2$. We study the problem at a fixed distance from the line. For $0 < \varepsilon \leq 1/2$ let

$$N_p(\varepsilon) = \sup_{f \neq 0} \frac{\int_0^1 |f(1/2 + \varepsilon + it)|^p \, \mathrm{d}t}{\|f\|_{\mathscr{H}^p}^p}.$$

For $p \in 2\mathbb{N}$ these are bounded, and for $p = 1$ they are unbounded (Proposition 3.3). Throughout, q denotes a prime and $L = \log(1/\varepsilon)$. Pointwise evaluation gives only $N_p(\varepsilon) \leq \zeta(1 + 2\varepsilon) \approx (2\varepsilon)^{-1}$. The following theorem shows that, uniformly in p, the growth is slower than every power of ε; it is our contribution to Problem 1.

THEOREM 3.4. *There is an absolute constant C such that for all $0 < p < \infty$, $0 < \varepsilon \leq 1/2$ and $0 < \delta \leq 1/2$,*

$$N_p(\varepsilon) \leq \frac{C}{\delta} \big(C(2 + L)\big)^{1/\delta} \varepsilon^{-\delta/(1-\delta)}.$$

Consequently, uniformly in p, $N_p(\varepsilon) \leq \exp\big(C\sqrt{(1 + L)\log(2 + L)}\big)$, and $N_p(\varepsilon) \leq C_\eta \varepsilon^{-\eta}$ for every $\eta > 0$.

The Gram condition of Theorem 2.1 works in L^2. Applied to $\zeta(1 + 2\varepsilon)^{-1} \, \mathrm{d}t$ on $[0, 1]$, with $r_t = \gamma(1/2 + \varepsilon + it)$ and the affinity bound $H(\omega_{r_t}, \omega_{r_u}) \leq C_0(1 + (t - u)^2/(4\varepsilon^2))^{-1/4}$ of Theorem 1.6 and Proposition 1.4, it gives $N_p(\varepsilon) \leq C\varepsilon^{-1/2}$: one point evaluation per unit hyperbolic length, with the affinity decaying like $e^{-d/4}$. Its normalized kernel powers $|K_s|^{1+\alpha}$ reach exponents down to $3 - 2\sqrt{2}$ and, heuristically, no lower. The proof below replaces $|K_s|$ by the Poisson kernel, the positive representative of section 1.2, and L^2 by L^r with r large. The Poisson kernel has L^1 norm one, and the price is a moment of order $1 + 1/(r - 1)$ of a critical multiplicative chaos, which we bound shell by shell. The method is related to that of Chen, Fang, Guo, Hou, Shao and Zhou [28], who also bound a dual fractional moment of a log-correlated field, but on the line itself, at $\varepsilon = 0$.

PROOF. It suffices to bound $\int_0^1 |f(s_t)|^p \, \mathrm{d}t$, with $s_t = 1/2 + \varepsilon + it$, for Dirichlet polynomials f whose Bohr lift F depends on the first n primes, uniformly in n. For $|a| < 1$ let $P(a, \cdot)$ be the Poisson kernel of the disc at a, and let $P_t(z) = \prod_q P(q^{-s_t}, z_q)$, the product over the first n primes, a function of n variables with $\int P_t \, \mathrm{d}m = 1$; it is the Poisson kernel of section 1.2 at $\gamma(s_t)$, restricted to the first n primes. Since $|F|^\theta$ is subharmonic in each variable, $|f(s_t)|^\theta \leq \int |F|^\theta P_t \, \mathrm{d}m$ for every $\theta > 0$.

Duality. Put $\varrho = 1 + \delta$ and let $r = \varrho/\delta$ be the conjugate exponent. With $\theta = p/r$ and $v = |F|^{p/r}$, so that $\|v\|_r^r = \|F\|_p^p$, we get $\int_0^1 |f(s_t)|^p \, \mathrm{d}t \leq \|T\|^r \|F\|_p^p$ for the operator $Tv(t) = \int v P_t \, \mathrm{d}m$ from $L^r(m)$ to $L^r(0, 1)$. Its adjoint is $g \mapsto M = \int_0^1 g(u) P_u \, \mathrm{d}u$. Since $r/\varrho = 1/\delta$, it suffices to show for $g \geq 0$

$$\int M^\varrho \, \mathrm{d}m \leq C(2 + L)\delta^{-\delta} \varepsilon^{-\delta^2/(1-\delta)} \|g\|_\varrho^\varrho; \tag{3.1}$$

then $N_p(\varepsilon) \le \delta^{-1}(C(2+L))^{1/\delta}\varepsilon^{-\delta/(1-\delta)}$. The choice $\delta = \sqrt{\log(2+L)/(1+L)}$ (or $\delta = 1/2$ if this exceeds $1/2$) gives the second bound, and $\delta = \min(\eta/2, 1/2)$ the third.

Shells. Pointwise $M^\varrho = \int_0^1 g(t) P_t M^\delta \,\mathrm{d}t$. Fix t, let $K = \lceil L \rceil$, and split $[0,1]$ into the shells U_k where $e^{-k-1} < |u-t| \le e^{-k}$ for $k < K$, and U_K where $|u-t| \le e^{-K}$. With $M_k = \int_{U_k} g P_u \,\mathrm{d}u$ we have $M^\delta \le \sum_k M_k^\delta$. The phase difference in coordinate q is $(t-u)\log q$. On U_k, primes with $\log q \le e^k$ have phase difference at most one; we put them in the coarse factor P_u^c. The remaining primes form P_u^f. Their overlap is controlled by an oscillatory prime sum, uniformly in the shell. In the last shell, the radial displacement 2ε supplies the lower bound that the angular displacement no longer gives.

Fine primes. For fixed coarse variables $P_t^f \,\mathrm{d}m_f$ is a probability measure, so Jensen's inequality gives $\int P_t^f M_k^\delta \,\mathrm{d}m_f \le (\Gamma X_k)^\delta$, where $X_k = \int_{U_k} g P_u^c \,\mathrm{d}u$ and $\Gamma = \sup_{u \in U_k} \int P_t^f P_u^f \,\mathrm{d}m$. In one variable $\int P(a,\cdot)P(b,\cdot)\,\mathrm{d}m = (1-|ab|^2)/|1-a\bar b|^2$, so $\log\Gamma \le 2|\sum_f q^{-1-w}| + 2$, with the sum over the fine primes and $w = 2\varepsilon + i(t-u)$. Here $\Re w > 0$, $|w| \le 3$, and $e^k|w| \ge e^{-1}$ on U_k (for $k = K$ because $|w| \ge 2\varepsilon$). The prime number theorem with the de la Vallée Poussin error term [62, Theorem 6.9] and one integration by parts give, for $1 \le X < X' \le \infty$, $\Re w > 0$ and $|w| \le 3$,

$$\Big|\sum_{e^X < q \le e^{X'}} q^{-1-w}\Big| \le C + \frac{4}{X|w|}.$$

With $X = e^k$, Γ is bounded by an absolute constant.

Coarse primes. Hölder's inequality with exponents $\lambda = 1/(1-\delta)$ and $1/\delta$, together with $\int P_u^c \,\mathrm{d}m = 1$, gives

$$\int P_t^c X_k^\delta \,\mathrm{d}m \le \Big(\int (P_t^c)^\lambda \,\mathrm{d}m\Big)^{1/\lambda} \Big(\int_{U_k} g\Big)^\delta.$$

In one variable put $x = |a|^2 \le 1/2$ and $\alpha = \lambda - 1 \in (0,1]$. Euler's transformation gives

$$\int P(a,\cdot)^\lambda \,\mathrm{d}m = (1-x)^{-\alpha} \sum_{n \ge 0} \frac{(-\alpha)_n^2}{(n!)^2} x^n.$$

For $n \ge 1$, $|(-\alpha)_n/n!| \le \alpha/n$, so the sum is at most $1 + \alpha^2 x + x^2/2$. Since $-\log(1-x) \le x + x^2$, taking logarithms gives

$$\log \int P(a,\cdot)^\lambda \,\mathrm{d}m \le \lambda(\lambda-1)x + 3x^2/2.$$

With $\sum_{\log q \le y} 1/q \le \log y + 2$ for $y \ge 1$, from [79, (3.18)], the first factor is at most $e^{(\lambda-1)(k+2)+1}$. The second is at most $(2e^{-k}\mathcal{M}g(t))^\delta$, where $\mathcal{M}$ is the Hardy–Littlewood maximal function.

Since $\lambda - 1 - \delta = \delta^2/(1-\delta)$, summing over $k \le K$ gives $\int P_t M^\delta \,\mathrm{d}m \le C(2+L)\varepsilon^{-\delta^2/(1-\delta)}\mathcal{M}g(t)^\delta$. Integrate against $g(t)\,\mathrm{d}t$, and use Hölder's inequality and the maximal theorem on L^ϱ, whose constant is $O(1/\delta)$ [88, Chapter I, Theorem 1]; this proves (3.1). □

The growth is not bounded either, and for $p<1$ the lower bound improves on the order $L^{p/4}$ that the partial sums of ζ give directly.

PROPOSITION 3.5 (Lower bounds). *(1) For $0<p<2$ let k be the largest integer with $kp<2$. Then $N_p(\varepsilon)\ge c_p L^{kp/4}$ for $\varepsilon\le\varepsilon_p$.*
(2) For $\varepsilon\le\varepsilon_0$ some $p\in(1,2)$ has $N_p(\varepsilon)\ge c\sqrt{L}/\log L$. In particular $\sup_p N_p(\varepsilon)\ge c\sqrt{L}/\log L$.

PROOF. Let $X=\lfloor e^{1/\varepsilon}\rfloor$ and $P_X(s)=\sum_{n\le X}n^{-s}$. By the approximate formula of Titchmarsh [92, Theorem 4.11], $P_X(s)=\zeta(s)+X^{1-s}/(1-s)+O(X^{-1/2})$ uniformly for $s=1/2+\varepsilon+it$, $0\le t\le 1$. Since ζ is bounded there, $|1-s|\le 2$ and $X^{1/2-\varepsilon}\ge X^{1/2}/e$, we get $|P_X(s)|\ge X^{1/2}/8$ for small ε, and so $\int_0^1|P_X(1/2+\varepsilon+it)|^{2\kappa}\,\mathrm{d}t\ge X^{\kappa}/64$ for $0<\kappa<1$. The Bohr lift of P_X is $\sum_{n\le X}f(n)$ with f a Steinhaus random multiplicative function, and Harper's theorem [40, Theorem 1], which is uniform in the exponent $0\le\kappa\le 1$, gives

$$\|P_X\|_{\mathscr{H}^{2\kappa}}^{2\kappa}\le C\Big(\frac{X}{1+(1-\kappa)\sqrt{\log\log X}}\Big)^{\kappa},\qquad \log\log X=L+o(1).$$

So the quotient defining $N_{2\kappa}(\varepsilon)$ is at least $c((1-\kappa)\sqrt{L/2})^{\kappa}$ at P_X. (1) Apply this to P_X^k with $\kappa=kp/2$, since $|P_X^k|^p=|P_X|^{kp}$. (2) Take $\kappa=1-1/\log L$ and $p=2\kappa$. □

So for $0<p<2$, $N_p(\varepsilon)$ is unbounded but $\varepsilon^{-o(1)}$, and $\sup_p N_p(\varepsilon)$ lies between $\sqrt{L}/\log L$ and $\exp(C\sqrt{L\log L})$. For $p\in 2\mathbb{N}$, $N_p(\varepsilon)$ is bounded, and for $p>2$ it is bounded if the proof of Chen et al. [28] stands. For fixed $p<2$ the order is open; Harper's work suggests $(\log(1/\varepsilon))^{1/4}$ at $p=1$.

Translating vertically, the theorem answers the remaining question about the strip condition: it is not necessary, for any p.

COROLLARY 3.6. *Let $0<\beta<1$. The measure $\mathrm{d}\mu=\varepsilon^{-\beta}e^{-|t|}\,\mathrm{d}\varepsilon\,\mathrm{d}t$ on $\{1/2+\varepsilon+it:0<\varepsilon<1/4,\ t\in\mathbb{R}\}$ is an $\mathscr{H}^p$ Carleson measure for every $0<p<\infty$, but $\mu(\{1/2<\Re s\le 1/2+\delta\})=2\delta^{1-\beta}/(1-\beta)$ for $\delta\le 1/4$, so it violates the strip condition.*

PROOF. Vertical translation preserves the $\mathscr{H}^p$ norm, so Theorem 3.4 bounds $\int_k^{k+1}|f(1/2+\varepsilon+it)|^p\,\mathrm{d}t$ by $C_\eta\varepsilon^{-\eta}\|f\|_{\mathscr{H}^p}^p$ for every integer k. Take $\eta=(1-\beta)/2$, sum against $e^{-|t|}$ and integrate $\varepsilon^{-\beta-\eta}=\varepsilon^{-(1+\beta)/2}$ over $(0,1/4)$. □

For $\beta=1/4$ the measure of Corollary 3.6 also satisfies the Hellinger–Schur condition of Theorem 2.1, though with a weight other than $\|K\|$. Let $r=\gamma(1/2+a+iy)$ and $s=\gamma(1/2+\varepsilon+it)$, where $0<a,\varepsilon<1/4$ and $\gamma(z)=(q^{-z})_q$. By Proposition 1.4,

$$\mathcal{G}(r,s)\le\big(\zeta(1+2a)\zeta(1+2\varepsilon)\big)^{1/4}|\zeta(1+a+\varepsilon+i(t-y))|^{1/2}.$$

Using $\zeta(1+x)\le 1+1/x$ for $x>0$ and $|\zeta(1+w)|\le C/|w|$ for $|w|\le 3$, and $|\zeta(1+w)|\le C\log(2+|w|)$ otherwise, we obtain $\mathcal{G}(r,s)\le C(a\varepsilon)^{-1/4}k(t-y)$, where $k(\tau)=|\tau|^{-1/2}$ for $|\tau|\le 2$ and $k(\tau)=\log^{1/2}(2+|\tau|)$ otherwise. Schur's test with

the weight $\varepsilon^{-1/4}e^{|t|/2}$ bounds every row by

$$C\int_0^{1/4}\varepsilon^{-3/4}\,\mathrm{d}\varepsilon\cdot\sup_y e^{-|y|/2}\int_{\mathbb{R}}e^{-|t|/2}k(t-y)\,\mathrm{d}t<\infty.$$

So for $\beta=1/4$ the Gram condition alone already proves Corollary 3.6. It cannot do so for $\beta\ge 1/2$. Let μ_δ be the restriction to $\varepsilon\ge\delta$, and define $B_\delta g=\int g(s)|K_s|\,\mathrm{d}\mu_\delta(s)$. Then $\mathcal{G}_{\mu_\delta}=B_\delta^*B_\delta$ and $B_\delta^*1(s)=\|K_s\|_{\mathscr{H}^1}$. Since the constant function has norm one on the Bohr torus, we need only estimate $\|K_s\|_1$. The one-variable expansion $\log\int_{\mathbb{T}}|1-az|^{-1}\,\mathrm{d}m=|a|^2/4+O(|a|^4)$, uniformly for $|a|\le 2^{-1/2}$, gives $\log\|K_s\|_1=(1/4)\sum_q q^{-1-2\varepsilon}+O(1)=(1/4)\log(1/\varepsilon)+O(1)$. Thus $\|K_s\|_1\asymp\varepsilon^{-1/4}$, independently of t, and

$$\|\mathcal{G}_{\mu_\delta}\|\ge\|B_\delta^*1\|^2_{L^2(\mu_\delta)}\ge c\int_\delta^{1/4}\varepsilon^{-1/2-\beta}\,\mathrm{d}\varepsilon.$$

This tends to infinity as $\delta\to 0$.

For $p=2$ the limit of $N_2(\varepsilon)$ is the constant of the local embedding on the line itself. The author's thesis [82, Section 4.4] suggested the value 2; in fact the constant lies just above 2π, and it can be located quite precisely.

PROPOSITION 3.7 (The constant at $p=2$). *Let C_0 be the least constant with $\int_0^1|P(1/2+it)|^2\,\mathrm{d}t\le C_0\|P\|^2_{\mathscr{H}^2}$ for all Dirichlet polynomials P. Then $N_2(\varepsilon)\le C_0$ for every ε, $N_2(\varepsilon)\to C_0$ as $\varepsilon\to 0$, and*

$$2\pi\le C_0<2\pi+1/8.$$

PROOF. Horizontal translation is a contraction on $\mathscr{H}^2$, so $N_2(\varepsilon)\le C_0$; conversely, for a fixed polynomial P the integral over $1/2+\varepsilon+[0,1]i$ tends to the one over the line, so $\liminf N_2(\varepsilon)\ge C_0$.

For the lower bound, write $\zeta(1+w)=1/w+\psi(1+w)$ with ψ bounded on $\{|w|\le 2\}$, and test $N_2(\varepsilon)$ with $f(s)=\zeta(1/2+\eta+s-i/2)$, for which $\|f\|^2_{\mathscr{H}^2}=\zeta(1+2\eta)$. With $\delta=\eta+\varepsilon$,

$$\int_0^1|f(1/2+\varepsilon+it)|^2\,\mathrm{d}t=\int_{-1/2}^{1/2}\frac{\mathrm{d}v}{\delta^2+v^2}+O\big(\log(1/\delta)\big)=\frac{\pi}{\delta}+O\big(\log(1/\delta)\big),$$

and $\zeta(1+2\eta)=1/(2\eta)+O(1)$. Taking $\eta=\sqrt{\varepsilon}$, the quotient tends to 2π as $\varepsilon\to 0$.

For the upper bound, let $T_\varepsilon a(t)=\sum_n a_n n^{-1/2-\varepsilon-it}$ on $[0,1]$, so that $N_2(\varepsilon)=\|T_\varepsilon\|^2$. Then $T_\varepsilon T_\varepsilon^*$ has kernel $\zeta(1+2\varepsilon+i(t-u))=k(t-u)+\psi(1+2\varepsilon+i(t-u))$, where $k(x)=1/(2\varepsilon+ix)=\int_0^\infty e^{-(2\varepsilon+ix)y}\,\mathrm{d}y$ and $\psi(s)=\zeta(s)-1/(s-1)$. Convolution with k on $L^2(\mathbb{R})$ is the Fourier multiplier $2\pi e^{2\varepsilon\xi}\mathbf{1}_{\xi<0}$, which lies below $2\pi\mathbf{1}_{\xi<0}$. Compressing to $[0,1]$, $T_\varepsilon T_\varepsilon^*\le S+\Psi_\varepsilon$, where S is the compression of $2\pi\mathbf{1}_{\xi<0}$ and Ψ_ε has kernel $\psi(1+2\varepsilon+i(t-u))$. In terms of the finite Hilbert transform $\mathcal{H}f(t)=\mathrm{PV}\int_0^1 f(u)(t-u)^{-1}\,\mathrm{d}u$, $S=\pi I-i\mathcal{H}$.

The operator S can be diagonalized. The substitution $t=e^x/(1+e^x)$ gives a unitary map $Uf(x)=f(t)(\mathrm{d}t/\,\mathrm{d}x)^{1/2}$ from $L^2(0,1)$ onto $L^2(\mathbb{R})$. Since $t-u=\sinh((x-y)/2)/(2\cosh(x/2)\cosh(y/2))$, it carries $\mathcal{H}$ to convolution with $\mathrm{PV}\,1/(2\sinh(x/2))$, the Fourier multiplier $-i\pi\tanh(\pi\xi)$. So S becomes multiplication by $\pi(1-\tanh(\pi\xi))$,

and $0 \le S \le 2\pi$. The constant function goes to $1/(2\cosh(x/2))$, whose Fourier transform is $\pi\operatorname{sech}(\pi\xi)$. Substituting $s = \pi(1 - \tanh(\pi\xi))$,

$$\langle \varphi(S)1, 1\rangle = \frac{1}{2\pi}\int_0^{2\pi} \varphi(s)\,\mathrm{d}s$$

for bounded Borel φ: the spectral measure of 1 is uniform on $[0, 2\pi]$.

Let $J = 1 \otimes 1$ and $c > 0$. For $\lambda > 2\pi$ the rank-one operator $c(\lambda - S)^{-1/2}J(\lambda - S)^{-1/2}$ has norm $c\langle(\lambda - S)^{-1}1, 1\rangle = (c/2\pi)\log(\lambda/(\lambda - 2\pi))$. When this is less than 1, $\lambda - S - cJ > 0$. Hence

$$S + cJ \le \lambda_c = \frac{2\pi}{1 - e^{-2\pi/c}}.$$

Take $c = \psi(1) = \gamma$, Euler's constant. Then $\lambda_\gamma < 2\pi + 2\cdot 10^{-4}$ and $T_\varepsilon T_\varepsilon^* \le \lambda_\gamma + R_\varepsilon$, where $R_\varepsilon = \Psi_\varepsilon - \gamma J$ has kernel $\psi(1 + 2\varepsilon + i(t-u)) - \psi(1)$. By the Euler–Maclaurin formula,

$$\psi(s) = \frac{1}{2} + \frac{s}{12} - \frac{s(s+1)}{2}\int_1^\infty \widetilde{B}_2(x)x^{-s-2}\,\mathrm{d}x,$$

where $\widetilde{B}_2$ is the periodic Bernoulli function, with $|\widetilde{B}_2| \le 1/6$. Differentiating, and using $\int_1^\infty x^{-3}\,\mathrm{d}x = 1/2$ and $\int_1^\infty x^{-3}\log x\,\mathrm{d}x = 1/4$, for $s = 1 + iv$ with $|v| \le 1$

$$|\psi'(s)| \le \frac{1}{12} + \frac{\sqrt{13}}{24} + \frac{\sqrt{10}}{48} < \frac{3}{10},$$

and the same bound holds up to $O(\varepsilon)$ for $1 \le \Re s \le 1 + 2\varepsilon$. So the kernel of R_ε is at most $(3/10 + O(\varepsilon))|t - u| + O(\varepsilon)$. Its Hilbert–Schmidt norm is at most $(3/10)(\int_0^1\int_0^1 (t-u)^2\,\mathrm{d}t\,\mathrm{d}u)^{1/2} + O(\varepsilon) = 3/(10\sqrt{6}) + O(\varepsilon) < 0.1225 + O(\varepsilon)$. Hence $N_2(\varepsilon) < 2\pi + 0.0002 + 0.1225 + O(\varepsilon)$, and letting $\varepsilon \to 0$ gives the upper bound. □

The upper bound improves the constant $5\pi/2$ that follows from Brevig's weighted inequality [16] $\pi^{-1}\int_{\mathbb{R}} |f(1/2 + it)|^2(1 + t^2)^{-1}\,\mathrm{d}t \le 2\|f\|^2_{\mathscr{H}^2}$.

The proof shows more. The edge 2π is the top of the spectrum of S, and the constant function sits at the edge with a logarithmically divergent resolvent. So any positive multiple of J pushes an eigenvalue above 2π, by the exponentially small amount $2\pi/(e^{2\pi/c} - 1)$. The true remainder Ψ is not a multiple of J, so this heuristic does not decide whether $C_0 > 2\pi$. A computation does.

PROPOSITION 3.8 (Above the edge). *Let $A\colon \mathscr{H}^2 \to L^2(0,1)$ be the bounded extension of $P \mapsto P(1/2 + it)$.*

(1) $AA^* = S + \Psi$, where Ψ has the continuous kernel $\psi(1 + i(t-u))$.
(2) If $C_0 > 2\pi$, then C_0 is an isolated eigenvalue of AA^* of finite multiplicity, and there is $f \in \mathscr{H}^2$ with $\|Af\|^2 = C_0\|f\|^2_{\mathscr{H}^2}$.
(3) $2\pi + 2.8\cdot 10^{-7} < C_0 < 2\pi + 10^{-3}$.

PROOF. (1) With D_ε the diagonal operator $n^{-\varepsilon}$, $T_\varepsilon = AD_\varepsilon$ and $D_\varepsilon^2 \to I$ strongly, so $T_\varepsilon T_\varepsilon^* \to AA^*$ strongly. In the proof of Proposition 3.7, the compressed multipliers converge strongly to S, and $\Psi_\varepsilon \to \Psi$ uniformly.

(2) Since Ψ is compact and $S \le 2\pi$, the essential spectrum of AA^* lies in $[0, 2\pi]$. Hence the spectrum above 2π consists of isolated eigenvalues of finite multiplicity,

and $C_0 = \|AA^*\|$ is one of them. If $AA^*g = C_0 g$ with $g \neq 0$, then $f = A^*g$ is nonzero and $\|Af\|^2 = \langle AA^*g, AA^*g\rangle = C_0\|f\|^2$.

(3) For the lower bound, take $g = \sum_{n<192} a_n P_n(2t-1)$ with rational a_n, real for even n and imaginary for odd n, found by an eigenvalue computation. By (1), $\langle AA^*g, g\rangle = \pi\|g\|^2 + \langle -iHg, g\rangle + \langle \Psi g, g\rangle$. The Hilbert form is exact: $\int_0^1 P_n(2t-1)\,(HP_m(2\cdot -1))(t)\,\mathrm{d}t$ equals $2/((n-m)(n+m+1))$ when $n+m$ is odd, and 0 otherwise. On $|w| \le 1$, $\psi(1+w)$ is replaced by the degree-96 Taylor polynomial of an Euler–Maclaurin approximant with cutoff 64. The error is below $2\cdot 10^{-18}$ by the remainder formula and Cauchy's estimate, and the moments of g are exact. Exact rational arithmetic then gives $\langle AA^*g, g\rangle > (2\pi + 2.8\cdot 10^{-7})\|g\|^2$.

For the upper bound, repeat the proof of Proposition 3.7 with the rank-two model $c + ia(t-u)$ in place of γJ, where c and a are rationals close to γ and $-\gamma_1$ and $\psi(1+w) = \gamma - \gamma_1 w + \cdots$. Its positive part is $\mu\, q\otimes q$ with q in the span of 1 and $t-1/2$. After applying U and the Fourier transform, $t-1/2$ goes to a multiple of ξ times the image of 1, so $\langle(\lambda - S)^{-1}q, q\rangle$ is an explicit integral over $[0, 2\pi]$ with logarithmic and dilogarithmic terms. At $\lambda = 2\pi + 1/340000$, $\mu\langle(\lambda-S)^{-1}q, q\rangle < 0.999$, in exact arithmetic. The remainder kernel has Hilbert–Schmidt norm below $9.7\cdot 10^{-4}$, by exact integration of the Taylor polynomial above. □

3.3. The dual of $\mathscr{H}^1$ on the critical line

Let $X = (\mathscr{H}^1)^*$. Every functional on $\mathscr{H}^1$ has a bounded symbol, whose analytic projection lies in $\mathscr{H}^2$, so we may regard X as a subspace of $\mathscr{H}^2$ under the coefficient pairing. Elements of $\mathscr{H}^2$ have boundary values in L^2_{loc} on the critical line, by the local embedding of Hedenmalm, Lindqvist and Seip [42, Theorem 4.11]. Konyagin, Queffélec, Saksman and Seip [54, Section 3.1] showed, by an argument of Brevig with power singularities $|t|^{\beta-1}$ and Helson's inequality, that some $\varphi \in X$ has $t \mapsto \varphi(1/2+it)$ outside $L^q(-1,1)$ when $q > 4$, and asked whether this happens for every $q > 2$. Their test family cannot reach $q \le 4$. The next theorem uses many translated kernels at one scale instead.

THEOREM 3.9. *For every finite $q > 7/2$ there is $\varphi \in (\mathscr{H}^1)^*$ such that $t \mapsto \varphi(1/2+it)$ is not in $L^q(-1,1)$.*

For $\delta > 0$ and $u \in \mathbb{R}$ let $K_{\delta,u}(s) = \zeta(1/2+\delta+iu+s)$, and let $b(\delta) = \|K_{\delta,u}\|_{\mathscr{H}^1}$, which does not depend on u. Write $D_{\delta,u} = |K_{\delta,u}|/b(\delta)$ on $\mathbb{T}^\infty$; these are the densities of the first powers of the kernels, not of the harmonic measures of chapter 1.

LEMMA 3.10. *There are absolute constants $c, C, \delta_0, h_0 > 0$ such that $c\delta^{-1/4} \le b(\delta) \le C\delta^{-1/4}$ for $0 < \delta \le \delta_0$, and*

$$\int \sqrt{D_{\delta,t} D_{\tau,u}}\,\mathrm{d}m \le C\Big(\frac{\delta\tau}{(\delta+\tau)^2 + (t-u)^2}\Big)^{1/16} \qquad (0 < \delta, \tau \le \delta_0,\ |t-u| \le h_0).$$

PROOF. In one variable, for $|a|, |b| \le 2^{-1/2}$, put $I(a) = \int_{\mathbb{T}} |1-az|^{-1}\,\mathrm{d}m$ and $I(a,b) = \int_{\mathbb{T}} |1-az|^{-1/2}|1-bz|^{-1/2}\,\mathrm{d}m$. Expanding $|1-az|^{-1/2} = (1-az)^{-1/4}(1-\overline{az})^{-1/4}$ gives

$$\log I(a) = \frac{|a|^2}{4} + O(|a|^4), \qquad \log \frac{I(a,b)}{\sqrt{I(a)I(b)}} = -\frac{|a-b|^2}{16} + O(|a|^4 + |b|^4).$$

Take $a_p = p^{-1/2-\delta-it}$ and $b_p = p^{-1/2-\tau-iu}$ and multiply over the primes; the products pass to the limit since finite-prime kernels converge in $\mathscr{H}^2$. The first expansion and $\sum_p p^{-1-2\delta} = \log(1/\delta) + O(1)$ give the bounds for $b(\delta)$. The second gives the affinity bound, since, as in the proof of Theorem 1.6, $\sum_p |a_p - b_p|^2 = \log\big(((\delta+\tau)^2 + (t-u)^2)/(4\delta\tau)\big) + O(1)$. $\square$

LEMMA 3.11. *Suppose that $F_1, \dots, F_M$ have norm one in $L^1(m)$, that $0 \le \eta < 1$, and that*

$$\sum_{k \neq j} \int |F_j F_k|^{1/2} \, \mathrm{d}m \le \eta \qquad (1 \le j \le M).$$

Then $\|\sum_j a_j F_j\|_1 \ge (1-\eta) \sum_j |a_j|$ for all complex numbers a_j.

PROOF. For complex numbers z_j, comparing with the term of largest modulus gives

$$\Big|\sum_j z_j\Big| \ge \sum_j |z_j| - 2\sum_{j<k} \min(|z_j|, |z_k|).$$

Apply this to $z_j = a_j F_j$ and bound each minimum by $|a_j a_k|^{1/2} |F_j F_k|^{1/2}$. Then use the inequality $2|a_j a_k|^{1/2} \le |a_j| + |a_k|$ and integrate. $\square$

Let $E\varphi(t) = \varphi(1/2 - it)$ on $(-1,1)$; the reflection does not change any norm. Its adjoint $A = E^* : L^2(-1,1) \to \mathscr{H}^2$ is $Ag(s) = \sum_n \widehat{g}(\log n) n^{-1/2-s}$, with $\widehat{g}(\xi) = \int_{-1}^{1} g(t) e^{-it\xi} \, \mathrm{d}t$. For $0 < \delta \le \delta_0$ and $|u| \le 1/2$ put $g_{\delta,u}(t) = \delta^{1/4}/(\delta - i(t-u))$ on $(-1,1)$. On the whole line, $\int_{\mathbb{R}} e^{-it\xi}(\delta - i(t-u))^{-1} \, \mathrm{d}t = 2\pi e^{-\delta\xi - iu\xi}$ for $\xi > 0$, while the tails $|t| > 1$ contribute $O(1/\xi)$ after an integration by parts, and the coefficient at $n = 1$ is bounded. Hence

$$Ag_{\delta,u} = 2\pi\delta^{1/4} K_{\delta,u} + R_{\delta,u}, \qquad \|R_{\delta,u}\|_{\mathscr{H}^2} \le C\delta^{1/4}, \tag{3.2}$$

because for $n \ge 2$ the coefficients of $R_{\delta,u}$ are of order $\delta^{1/4}/(\sqrt{n}\log n)$, and $\sum_{n \ge 2} 1/(n \log^2 n) < \infty$.

PROOF OF THEOREM 3.9. Let $p = q/(q-1)$. If $E(X) \subset L^q(-1,1)$, the closed graph theorem makes $E : X \to L^q(-1,1)$ bounded, since convergence in X and in $L^q(-1,1)$ both imply convergence in $L^2(-1,1)$. Then, for $g \in L^2(-1,1)$,

$$\|Ag\|_{\mathscr{H}^1} = \sup_{\|\varphi\|_X \le 1} |\langle \varphi, Ag\rangle| = \sup_{\|\varphi\|_X \le 1} |\langle E\varphi, g\rangle| \le \|E\| \, \|g\|_p.$$

We show that this fails. Fix $0 < \ell < \min(h_0, 1)$, let $\delta \to 0$, put $M = \lfloor \varepsilon \delta^{-1/8} \rfloor$ with $\varepsilon > 0$ small, and let $u_1, \dots, u_M$ be equally spaced in $[-\ell/2, \ell/2]$ with spacing ℓ/M. By Lemma 3.10, the affinity row sums of the D_{δ,u_j} are at most $C\sum_{r \ge 1} (\delta M/r)^{1/8} \le C\delta^{1/8} M \le 1/2$, so Lemma 3.11 gives $\|\delta^{1/4} \sum_j K_{\delta,u_j}\|_{\mathscr{H}^1} \ge M\delta^{1/4} b(\delta)/2 \ge cM$. With $g = \sum_j g_{\delta,u_j}$ and (3.2), $\|Ag\|_{\mathscr{H}^1} \ge cM - CM\delta^{1/4} \ge cM/2$.

On the other hand, at each t the nearest u_j contributes at most $\delta^{1/4}(\delta^2 + (t-u_j)^2)^{-1/2}$, and the others at most $C\delta^{1/4} M \log(M+1)$ in total. Integrating over the cells of the grid,

$$\|g\|_p^p \le C\big(M\delta^{1-3p/4} + \delta^{p/4} M^p \log^p(M+1)\big) \le CM\delta^{1-3p/4},$$

since $(M\delta)^{p-1}\log^p(M+1)\to 0$. So

$$\frac{\|Ag\|_{\mathscr{H}^1}}{\|g\|_p}\geq cM^{1/q}\delta^{3/4-1/p}\geq c\,\delta^{-1/4+7/(8q)},$$

which tends to infinity exactly when $q>7/2$. □

The same kernels at geometrically separated scales, $\delta_j=e^{-aj}$ with a large and $u=0$, give $\|Ag_M\|_{\mathscr{H}^1}\geq cM$ and $\|g_M\|_{4/3}\leq CM^{3/4}$, a second proof at $q=4$ that does not use the reciprocal-divisor sum of [54]. To our knowledge, Theorem 3.9 is new. The range $2<q\leq 7/2$ is open; $7/2$ is where pairwise affinity row sums stop, not a known threshold.

Open

(1) The exact value of C_0. By Proposition 3.8 it exceeds 2π by between $2.8\cdot 10^{-7}$ and 10^{-3}.
(2) The order of $N_p(\varepsilon)$ for fixed $p<2$, between $L^{kp/4}$ (Proposition 3.5) and $\exp(C\sqrt{L\log L})$ (Theorem 3.4); is it $(\log(1/\varepsilon))^{1/4}$ at $p=1$?
(3) Is $\sup_p N_p(\varepsilon)$ polylogarithmic in $1/\varepsilon$?
(4) Are the $\mathscr{H}^1$ and $\mathscr{H}^{1/2}$ Carleson classes on $\mathbb{C}_{1/2}$ the same? By Theorem 4.24 the classes are not the same for all $p<2$.
(5) Does $(\mathscr{H}^1)^*$ fail to embed into $L^q(-1,1)$ on the critical line for $2<q\leq 7/2$ (Konyagin, Queffélec, Saksman and Seip)? Theorem 3.9 covers $q>7/2$.
(6) A version of Proposition 2.14 for continuous families with many near neighbours, such as uncountably many sign patterns at one scale, beyond the product structure of Proposition 4.6.

Part 2

Where Nehari's theorem survives: structured Helson matrices

CHAPTER 4

Positive Helson matrices on the half-line

In one variable a non-negative Hankel matrix is a moment matrix. Widom characterized the bounded ones by a Carleson condition on the moment measure [96], and Nehari's theorem then supplies a bounded symbol for free. In infinitely many variables Nehari's theorem fails, but the known counterexamples, built from tensor powers of small symbols such as $z_1 + z_2$, have indefinite Hankel forms [69]. This chapter asks whether positivity rescues the theorem. In general it does not: some bounded non-negative Helson matrices, with moment measures on real points of rotated copies of the prime curve, have no bounded symbol (Theorem 4.21). For structured families it does.

Two mechanisms produce bounded symbols for positive Helson matrices, and they behave very differently.

- *Separation.* On the curve $\gamma(s) = (2^{-s}, 3^{-s}, 5^{-s}, \dots)$, points at different scales have nearly mutually singular harmonic measures. The Hellinger–Bessel inequality of chapter 2, which replaces Perfekt's localization argument [70], turns this into an H^1 Carleson embedding. It requires the moment measure to satisfy $\nu((1/2, 1/2 + t]) \lesssim t$, exactly the condition for boundedness.
- *Symmetry.* Averaging a point over independent signs in each coordinate kills every pair (m, n) except those with mn a square, in H^1 as well as in H^2. This needs no condition at all near $s = 1/2$.

The main result of the chapter, Theorem 4.2, runs both mechanisms at once. Keep a set R of primes on the curve and average the signs of the others. If R has density θ among the primes, the condition becomes $\nu((1/2, 1/2 + t]) \lesssim t^\theta$. Within the whole family, boundedness, the H^1 embedding and the existence of a bounded symbol are equivalent, with a constant depending only on R. The curve ($\theta = 1$) and the full sign average ($\theta = 0$) are the two endpoints. The same interpolation describes the compact members (Theorem 4.10), which is a Hartman-type theorem whose condition moves from $o(t)$ at one endpoint to no condition at the other. On the curve the essential norm is comparable to the essential size of the symbol (Theorem 4.11); for measures with a density it equals $\pi w(1/2)$, and the essential spectrum is $[0, \pi w(1/2)]$ (Corollary 4.13). To our knowledge the interpolating family, the main theorem with its uniform constant, the sign-average mechanism and the Hartman-type theorem are new.

Section 4.1 sets up the dictionary between symbols, functionals and Carleson measures. Section 4.2 defines the family and states the results. Sections 4.3 and 4.4 prove the two mechanisms separately; the second, averaging over roots of unity,

is developed there for every order $k \geq 2$ and every p, with exact spectra and decimation. Then section 4.5 combines them. Section 4.6 treats compactness, section 4.7 collects consequences, and section 4.8 shows that the general question has a negative answer.

4.1. Positive Helson matrices and Carleson measures

Throughout, $M(\alpha) = (\alpha(mn))_{m,n\geq 1}$ is a Helson matrix, and $z^{\kappa(n)}$ is the Bohr lift of n^{-s}. For a Dirichlet polynomial $h(s) = \sum_k h_k k^{-s}$ with Bohr lift H, put

$$\Lambda_\alpha(h) = \sum_k h_k\, \alpha(k).$$

For finitely supported $a, b \in \ell^2$ with Dirichlet polynomials f, g, the form of $M(\alpha)$ is $\sum_{m,n} \alpha(mn) a_m b_n = \Lambda_\alpha(fg)$.

Definition. A *bounded symbol* for $M(\alpha)$ is a function $g \in L^\infty(\mathbb{T}^\infty)$ with $\int_{\mathbb{T}^\infty} z^{\kappa(n)} g \, \mathrm{d}m = \alpha(n)$ for every $n \geq 1$.

For real α this agrees, after complex conjugation, with the convention of [71], where a symbol satisfies $\langle A, z^{\kappa(n)}\rangle = \alpha(n)$. The bilinear form is more convenient here.

Lemma 4.1. *$M(\alpha)$ has a bounded symbol if and only if Λ_α extends to a bounded functional on $\mathscr{H}^1$. In that case*

$$\|M(\alpha)\| \leq \inf\{\|g\|_\infty : g \text{ a bounded symbol}\} = \|\Lambda_\alpha\|_{(\mathscr{H}^1)^*}.$$

Proof. If g is a bounded symbol, then $\Lambda_\alpha(h) = \int H g \, \mathrm{d}m$ on polynomials, so $|\Lambda_\alpha(h)| \leq \|g\|_\infty \|h\|_{\mathscr{H}^1}$. Conversely, a bounded functional on $\mathscr{H}^1$, which the Bohr lift identifies with a closed subspace of $L^1(\mathbb{T}^\infty)$, extends by the Hahn–Banach theorem to L^1 with the same norm, and is represented by some $g \in L^\infty$ whose moments are $\Lambda_\alpha(n^{-s}) = \alpha(n)$. Finally, $\|fg\|_{\mathscr{H}^1} \leq \|f\|_{\mathscr{H}^2}\|g\|_{\mathscr{H}^2}$ gives $\|M(\alpha)\| \leq \|\Lambda_\alpha\|$. □

Positivity

Perfekt and Pushnitski [71] show that, under a mild growth condition, $M(\alpha) \geq 0$ exactly when α is a moment sequence $\alpha(n) = \int t^{\kappa(n)} \, \mathrm{d}\mu(t)$ of a positive measure μ on $\mathbb{R}^\infty$. Then $M(\alpha) = \mathcal{N}^*\mathcal{N}$ with $\mathcal{N} : \ell^2 \to L^2(\mu)$, $(a_n) \mapsto \sum_n a_n t^{\kappa(n)}$, so $M(\alpha)$ is bounded exactly when μ is a Carleson measure for $H^2(\mathbb{T}^\infty)$:

$$\int |F|^2 \, \mathrm{d}\mu \leq C \|F\|_{H^2}^2.$$

On the other side, if μ is a Carleson measure for $H^1(\mathbb{T}^\infty)$, then $|\Lambda_\alpha(h)| = |\int H \, \mathrm{d}\mu| \leq C\|h\|_{\mathscr{H}^1}$, and Lemma 4.1 gives a bounded symbol. So for non-negative Helson matrices, Nehari's theorem follows from the inclusion

$$\{H^2\text{-Carleson measures}\} \subseteq \{H^1\text{-Carleson measures}\} \tag{4.1}$$

for the moment measures in question. The reverse inclusion always holds: apply an H^1 embedding to F^2, using $\|F^2\|_{H^1} = \|F\|_{H^2}^2$. In one variable (4.1) is trivial, because every H^1 function is a product of two H^2 functions with $\|F\|_1 = \|G\|_2\|H\|_2$.

On $\mathbb{T}^\infty$ that factorization fails, and (4.1) is false for general measures: arc length on a vertical segment of the critical line is a counterexample, by Harper's work [40, 70], and chapter 3 gives one in the interior of $\mathbb{D}^\infty$. Whether this can happen for measures on real points is the following problem.

PROBLEM 2 (Positive Nehari; Perfekt and Pushnitski [71, Section 4]). *Does every bounded non-negative Helson matrix have a bounded symbol?*

The H^1 Carleson inclusion (4.1) would imply a positive answer, but boundedness of $\int F\,\mathrm{d}\mu$ is a priori weaker than boundedness of $\int|F|\,\mathrm{d}\mu$. The counterexample below makes both fail. The answer is no (Theorem 4.21), but most of this chapter shows that it is yes for large structured families.

4.2. Partially averaged curve measures

The real half-line $s > 1/2$ lifts to the curve

$$\gamma(s) = (p^{-s})_p \in (0,1)^\infty \cap \ell^2,$$

and $F(\gamma(s)) = f(s)$ for every Dirichlet series f with Bohr lift F. Split the primes into two sets: R, the primes kept on the curve, and Q, the primes whose signs are averaged. Let $\varepsilon = (\varepsilon_p)_{p\in Q}$ be independent uniformly distributed signs, and write $\varepsilon\gamma(s)$ for the point with coordinates $\varepsilon_p p^{-s}$ for $p \in Q$ and p^{-s} for $p \in R$.

For a finite positive Borel measure ν on $(1/2,\infty)$, let $\mu_{\nu,R}$ be the image of $\nu \otimes \mathbb{P}_\varepsilon$ under $(s,\varepsilon) \mapsto \varepsilon\gamma(s)$. Its moments are

$$\alpha_{\nu,R}(n) = \int \mathbb{E}_\varepsilon\big[\varepsilon^{\kappa(n)}\big] n^{-s}\,\mathrm{d}\nu(s) = \mathbf{1}\big[n_Q \text{ is a square}\big] \int n^{-s}\,\mathrm{d}\nu(s),$$

where n_Q is the part of n composed of primes in Q. So $M(\alpha_{\nu,R}) \geq 0$, and $\int|F|\,\mathrm{d}\mu_{\nu,R} = \int \mathbb{E}_\varepsilon|F(\varepsilon\gamma(s))|\,\mathrm{d}\nu(s)$.

The size of R is measured by its density among the primes. Say that R has *density* $\theta \in [0,1]$ *with constant* B if

$$\Big|\sum_{p\in R} p^{-1-t} - \theta\log\frac{1}{t}\Big| \leq B, \qquad 0 < t \leq 2. \tag{4.2}$$

All primes have density 1, since $\sum_p p^{-1-t} = \log(1/t) + O(1)$. Finite sets, and more generally sets with $\sum_{p\in R} 1/p < \infty$, have density 0. The primes in a union of a residue classes modulo q coprime to q have density $a/\varphi(q)$, by the prime number theorem in arithmetic progressions, and the odd-indexed primes have density $1/2$. Put

$$[\nu]_\theta = \sup_{0<t\leq 1} \frac{\nu((1/2,1/2+t])}{t^\theta},$$

which for $\theta = 0$ is just $\nu((1/2,3/2])$.

THEOREM 4.2. *Let R have density $\theta \in [0,1]$ with constant B, and let ν be a finite positive Borel measure on $(1/2,\infty)$. The following are equivalent.*

(1) $M(\alpha_{\nu,R})$ is bounded on ℓ^2.
(2) $[\nu]_\theta < \infty$.
(3) $\mu_{\nu,R}$ is a Carleson measure for $H^1(\mathbb{T}^\infty)$.
(4) $M(\alpha_{\nu,R})$ has a bounded symbol.

Moreover there is $C = C(\theta, B)$ such that

$$\|M(\alpha_{\nu,R})\| \le \inf\|g\|_\infty \le C\,\|M(\alpha_{\nu,R})\|,$$

the infimum being over all bounded symbols. If $\theta = 0$, all four conditions hold for every finite ν.

The two endpoints are the motivating cases.

COROLLARY 4.3 (The curve). *For R the set of all primes, $\alpha_\nu(n) = \int n^{-s}\,\mathrm{d}\nu(s)$, and $[\nu] = [\nu]_1$: the matrix $M(\alpha_\nu)$ is bounded if and only if $[\nu] < \infty$, if and only if $\int |f|\,\mathrm{d}\nu \lesssim \|f\|_{\mathscr{H}^1}$, if and only if it has a bounded symbol. One may take $\inf\|g\|_\infty \le (2 \cdot 20.71 + \zeta(3))\|M(\alpha_\nu)\| < 43\,\|M(\alpha_\nu)\|$.*

The equivalence of the first two conditions is due to Perfekt and Pushnitski [71, Theorem 5.1]. The multiplicative Hilbert matrix [22] comes from Lebesgue measure, which is not finite but differs from a finite measure by a Hilbert–Schmidt term (Corollary 4.18). The constant does not depend on ν; by contrast, the best constant in Nehari's theorem on $\mathbb{T}^d$ is at least $(\pi^2/8)^{d/4}$ [69].

COROLLARY 4.4 (Full sign averages). *For $R = \emptyset$ and every finite positive ν, the Helson matrix with $\alpha(n) = \mathbf{1}_\square(n) \int n^{-s}\,\mathrm{d}\nu(s)$, where $\mathbf{1}_\square$ is the indicator of the squares, is bounded and has a bounded symbol with*

$$\inf\|g\|_\infty \le \zeta(2)\,\nu((1/2,\infty)) \le \zeta(2)\,\|M(\alpha)\|.$$

So the full sign averages satisfy Nehari's theorem with constant at most $\zeta(2) \approx 1.645$, uniformly in ν. That is far better than the constant for the curve. Only the single orbits of Proposition 4.7, where the constant is exactly 1, do better.

4.3. Mechanism I: separation

The difficulty on the curve is a logarithm. Point evaluation at $\gamma(1/2+\sigma)$ has norm $\zeta(1+2\sigma) \asymp 1/\sigma$ on $\mathscr{H}^1$, by the Cole–Gamelin estimate [30]. If $[\nu] < \infty$, then ν puts mass $O(e^{-j})$ on the scale $1/2 + (e^{-j-1}, e^{-j}]$, so each scale contributes $O(\|f\|_{\mathscr{H}^1})$ to $\int |f|\,\mathrm{d}\nu$, and there are $\log(1/\delta)$ scales above height δ. One function cannot be near-extremal at many scales at once: the harmonic measures at different scales have exponentially small Hellinger affinity (chapter 1), and the Hellinger–Bessel inequality turns this into a Carleson estimate (chapter 2). For the Hardy geometry the kernel exponent equals the density, $\kappa = \theta$, and Theorem 2.11 with $p = 1$ reads as follows.

LEMMA 4.5 (Block-maximal evaluation on prime sets). *Let R have density $\theta \in (0,1]$ with constant B. There is $C_1 = C_1(\theta, B)$ such that every $f \in \mathscr{H}^1$ whose coefficients are supported on integers composed of primes in R satisfies*

$$\sum_{j\ge 0} e^{-j\theta} \max_{e^{-j-1} \le t \le e^{-j}} |f(1/2+t)| \le C_1 \|f\|_{\mathscr{H}^1},$$

and $\int_{(1/2,3/2]} |f|\,\mathrm{d}\nu \le C_1 [\nu]_\theta \|f\|_{\mathscr{H}^1}$ for every positive ν. For R the set of all primes one may take $C_1 = 99$ in the first bound and $C_1 < 20.71$ in the second (Theorem 2.11).

PROOF. Both are Theorem 2.11 for $p = 1$, applied to the Bohr lift; the maximum is attained since f is continuous on each closed block, and Fatou's lemma passes from polynomials to $\mathscr{H}^1$. □

For all primes, Perfekt's localization argument [70, Section 2], with Hoeffding's inequality [46] controlling the overlaps, gives the lemma with $C_1 = 3eL$, $L = 10^5$. The Gram route of chapter 2 needs no disjointness of the localized sets, which is why its constant is so much smaller.

Taking R to be all primes and ν Lebesgue measure on $(1/2, 3/2]$ recovers Perfekt's inequality $\int_0^1 |f(1/2+\sigma)|\,\mathrm{d}\sigma \lesssim \|f\|_{\mathscr{H}^1}$. The block maximum is the strengthening a general measure needs: ν may be atomic, and an average says nothing about any particular point.

4.4. Mechanism II: symmetry

Averaging a point over roots of unity in some of its coordinates is a second way of bounding evaluations, independent of separation. It works for every p, and it reaches points outside ℓ^2.

4.4.1. Averages over roots of unity

The root-of-unity estimate of Lemma 2.17 interpolates between a single evaluation and the full circle average. Its constant $1/(1-|x|^{2k})$ is the Cole–Gamelin bound $1/(1-|x|^2)$ for a single point ($k = 1$), and it tends to 1, the trivial bound for the full circle average, as $k \to \infty$.

PROPOSITION 4.6 (Root-of-unity averages in every H^p). *Let $0 < p < \infty$, let Q be a set of coordinates with orders $k_q \geq 2$ and points $x_q \in \mathbb{D}$, and fix the remaining coordinates at w. With independent ω_q uniform on the k_qth roots of unity, every analytic polynomial F satisfies*

$$\mathbb{E}_\omega |F((\omega_q x_q)_{q\in Q}, w)|^p \leq \prod_{q\in Q} \frac{1}{1-|x_q|^{2k_q}} \int_{\mathbb{T}^Q} |F(\zeta, w)|^p \,\mathrm{d}m(\zeta).$$

For $x = \gamma(s)$ with $\Re s \geq 1/2$ and all $k_q \geq k$, the constant is at most $\zeta(k)$; for signs it is $\zeta(4\Re s) \leq \zeta(2)$.

PROOF. Induct on the coordinates, applying Lemma 2.17 in one and Fubini in the rest; $\prod_q (1-q^{-2k_q\Re s})^{-1} \leq \prod_q (1-q^{-k})^{-1}$. □

So averaging over kth roots in each coordinate replaces the domain ℓ^2 of point evaluations by ℓ^{2k}: the averaged evaluation is bounded on every H^p at points where no single evaluation is, and the higher the order, the more of $\mathbb{D}^\infty$ it reaches. The analytic average is decimation in disguise: $\mathbb{E}_\omega F(\omega x) = (D_{k,Q}F)(x^k)$ in the notation of section 4.4.2; the proposition controls instead the average of $|F|^p$, which decimation does not.

A single orbit can be computed exactly. For $k \geq 2$ and $s \geq 1/2$ let $M_{k,s}(m,n) = \mathbf{1}[mn \text{ is a } k\text{th power}](mn)^{-s}$.

PROPOSITION 4.7 (Root-of-unity orbits). *For $0 < p < \infty$,*

$$\|M_{k,s}\|_{S_p}^p = \zeta(2ks)^p \prod_\ell \bigl(1 + (k-1)\ell^{-ksp}\bigr),$$

the product over primes, so $M_{k,s} \in S_p$ exactly when $ksp > 1$. Moreover $\|M_{k,s}\| = \zeta(2ks)$, and $M_{k,s}$ has a bounded symbol of norm exactly $\zeta(2ks)$. For $k \geq 3$ the matrix is indefinite, and every finite positive mixture $\int M_{k,s}\,\mathrm{d}\nu(s)$ over $s \geq 1/2$ is trace class.

PROOF. The matrix is the tensor product over primes ℓ of the matrices $(\mathbf{1}[k \mid a+b]x^{a+b})_{a,b\geq 0}$ with $x = \ell^{-s}$. Split the exponents by their residues mod k. Residue 0 against 0 gives the rank-one block $(x^{k(\alpha+\beta)})$ with singular value $1/(1-x^{2k})$; residue r against $k-r$, $1 \leq r < k$, gives $x^k w \otimes w$ with $w = (x^{k\alpha})_\alpha$. So the singular values are $1/(1-x^{2k})$ once and $x^k/(1-x^{2k})$ with multiplicity $k-1$, and for $k \geq 3$ the pairs $r \neq k-r$ give eigenvalues of both signs. Tensoring gives the S_p norm and the operator norm. For the symbol, the functional is $F \mapsto \mathbb{E}_\omega F(\omega x) = (D_k F)(x^k)$, and decimation is contractive on H^1, being an average of rotations followed by the measure-preserving map $z \mapsto z^k$, so its norm is at most $\prod_\ell (1-\ell^{-2ks})^{-1} = \zeta(2ks)$. For $k \geq 3$ and $s \geq 1/2$, the trace norm is at most $\zeta(3)\prod_\ell(1+(k-1)\ell^{-3/2})$. □

So an individual orbit has Nehari constant exactly one, even when it is indefinite, and at $s = 1/2$ it lies in S_p exactly for $p > 2/k$.

4.4.2. Decimation

For $k \geq 2$ and a set Q of coordinates let $D_{k,Q}$ replace each exponent ν_q, $q \in Q$, by $k\nu_q$ in the coefficient index: $\widehat{D_{k,Q}F}(\nu) = \widehat{F}(\tilde{\nu})$. For Q all primes, $D_k \sum a_n n^{-s} = \sum a_{n^k} n^{-s}$.

PROPOSITION 4.8. *$\|D_{k,Q}\|_{H^p \to H^p} = 1$ for $1 \leq p \leq \infty$. For $0 < p < 1$ and $|Q| = d$, $\|D_{k,Q}\| = k^{d(1/p-1)}$; so $D_{k,Q}$ is unbounded for infinite Q. If $\alpha(n^k) = \beta(n)$ and $\alpha = 0$ off kth powers, the functionals Λ_α and Λ_β on H^1 have equal norms, hence equal optimal symbol norms.*

PROOF. $(D_{k,Q}F) \circ \pi_k$ is the average of F over kth roots of unity in the Q-coordinates, where π_k raises those coordinates to the kth power and preserves Haar measure. For $p \geq 1$ this is a conditional expectation. For $p < 1$ the upper bound is p-subadditivity; the lower bound uses outer functions of modulus 1 on an arc whose k rotations are disjoint and δ elsewhere, tensored d times. For the functionals, $\Lambda_\alpha = \Lambda_\beta \circ D_k$ and $D_k J_k = I$ with the isometry $J_k G = G \circ \pi_k$. □

Decimation identifies analytic sign averages exactly,

$$\mathbb{E}_\varepsilon F((\varepsilon_q q^{-s})_Q, (p^{-s})_R) = (D_{2,Q}F)((q^{-2s})_Q, (p^{-s})_R),$$

but it bounds $|\mathbb{E}_\varepsilon F|$, not $\mathbb{E}_\varepsilon |F|$ (already $F = z_q$ has $D_2 F = 0$), so it does not give Proposition 4.6.

For the functional, and hence for the symbol, this already suffices. Since $\Lambda_{\alpha_{\nu,R}}$ is linear, $\Lambda_{\alpha_{\nu,R}}(h) = \int (D_{2,Q}H)((q^{-2s})_Q, (p^{-s})_R)\,\mathrm{d}\nu(s)$, and $D_{2,Q}$ is contractive on H^1. The bent curve $s \mapsto ((q^{-2s})_Q, (p^{-s})_R)$ lies in ℓ^2, its kernel norms exceed those

of γ_R by at most the factor $\prod_{q\in Q}(1-q^{-4s})^{-1} \le \zeta(2)$, and its Hellinger affinities are at most those of γ_R, being products with factors at most 1. So the Schur bound in the proof of Theorem 2.11 bounds $\Lambda_{\alpha_{\nu,R}}$ on $\mathscr{H}^1$ directly: averaging the signs moves the Q-coordinates from $\gamma(s)$ to $\gamma(2s)$, and only the singularity of the coordinates in R is left. The two-point lemma is needed only for the stronger Carleson statement, which bounds $\mathbb{E}_\varepsilon|F|$ rather than $|\mathbb{E}_\varepsilon F|$.

4.4.3. Signs

The second mechanism is the case $k_q = 2$ of the root-of-unity averages of section 4.4.1. For sign averages Lemma 2.17 reads $(|g(x)| + |g(-x)|)/2 \le \|g\|_{H^1}/(1-|x|^4)$, with equality for $g = (1-\overline{x}^2 z^2)^{-2}$, and Proposition 4.6 gives, for an analytic polynomial F, a set Q of coordinates with points $x_p \in \mathbb{D}$, and the remaining coordinates fixed at w,

$$\mathbb{E}_\varepsilon\big|F\big((\varepsilon_p x_p)_{p\in Q}, w\big)\big| \le \prod_{p\in Q} \frac{1}{1-|x_p|^4} \int_{\mathbb{T}^Q} \big|F(\zeta, w)\big|\, \mathrm{d}m(\zeta). \tag{4.3}$$

For $x = \gamma(s)$ the constant is at most $\zeta(4s) \le \zeta(2)$. Two features matter here. The constant does not blow up as $s \to 1/2$, and $\sum|x_p|^4 < \infty$ is weaker than $x \in \ell^2$: the sign-averaged evaluation is bounded on H^1 at points where no single point evaluation is.

4.5. Proof of the main theorem

The two mechanisms fit together because they act on different coordinates. Point evaluation at $\gamma(1/2+\sigma)$ costs $\zeta(1+2\sigma) \asymp 1/\sigma$ on $\mathscr{H}^1$, and the cost is a product over primes. Averaging the signs of the primes in Q removes their share of this singularity at the bounded price $\prod_{q\in Q}(1-q^{-4\Re s})^{-1} \le \zeta(2)$, since sign averages need only ℓ^4. What survives is the share of R, which is $\zeta_R(1+2\sigma) \asymp \sigma^{-\theta}$. Separation along the curve then handles exactly this residual singularity, and the condition $\nu((1/2, 1/2+t]) = O(t^\theta)$ matches it. So θ measures how much of the evaluation singularity survives averaging, and the two endpoints are visible before any estimate: at $\theta = 1$ nothing is averaged, and at $\theta = 0$ nothing survives.

(1) $\Rightarrow$ (2). Assume $\theta > 0$; for $\theta = 0$ there is nothing to prove. Test with $a_n = n^{-1/2-\delta}$, $0 < \delta \le 1$, on the integers composed of primes in R. For such m, n the Q-part of mn is 1, so $\alpha_{\nu,R}(mn) = \int (mn)^{-s}\, \mathrm{d}\nu(s)$. By Tonelli's theorem, with $\zeta_R(w) = \prod_{p\in R}(1-p^{-w})^{-1}$,

$$\sum_{m,n} \alpha_{\nu,R}(mn) a_m a_n = \int \zeta_R(1/2+\delta+s)^2\, \mathrm{d}\nu(s) \ge \zeta_R(1+2\delta)^2\, \nu((1/2, 1/2+\delta]).$$

The left side is at most $\|M(\alpha_{\nu,R})\|\zeta_R(1+2\delta)$, and by (4.2) $\zeta_R(1+2\delta) \ge e^{-B}(2\delta)^{-\theta}$. Hence $[\nu]_\theta \le 2e^B \|M(\alpha_{\nu,R})\|$. We also record $\nu((1/2,\infty)) = \alpha_{\nu,R}(1) \le \|M(\alpha_{\nu,R})\|$.

(2) $\Rightarrow$ (3). Let F be an analytic polynomial. For fixed s, (4.3) with the R-coordinates fixed at $\gamma_R(s)$ gives

$$\mathbb{E}_\varepsilon|F(\varepsilon\gamma(s))| \le \zeta(2) \int_{\mathbb{T}^Q} |F_\zeta(\gamma_R(s))|\, \mathrm{d}m(\zeta),$$

where $F_\zeta = F(\zeta,\cdot)$ is a polynomial in the R-coordinates. It is the Bohr lift of a Dirichlet polynomial f_ζ supported on integers composed of primes in R, with $f_\zeta(s) = F_\zeta(\gamma_R(s))$. Integrate in ν and use Tonelli:

$$\int |F|\,\mathrm{d}\mu_{\nu,R} \le \zeta(2)\int_{\mathbb{T}^Q}\Big(\int |f_\zeta|\,\mathrm{d}\nu\Big)\,\mathrm{d}m(\zeta).$$

If $\theta > 0$, the measure estimate in Lemma 4.5 gives

$$\int_{(1/2,3/2]} |f_\zeta|\,\mathrm{d}\nu \le C_1[\nu]_\theta \|F_\zeta\|_{L^1(\mathbb{T}^R)}.$$

If $\theta = 0$, the point estimate gives $|f_\zeta(s)| \le \zeta_R(2\Re s)\|F_\zeta\|_1 \le e^{1+B}\|F_\zeta\|_1$ for every $s > 1/2$. On $(3/2,\infty)$, $|f_\zeta| \le \zeta(3)\|F_\zeta\|_1$. Since $\int_{\mathbb{T}^Q}\|F_\zeta\|_{L^1(\mathbb{T}^R)}\,\mathrm{d}m(\zeta) = \|F\|_{L^1}$, we obtain

$$\int |F|\,\mathrm{d}\mu_{\nu,R} \le \zeta(2)\big(C_1[\nu]_\theta + \zeta(3)\,\nu((1/2,\infty))\big)\|F\|_{H^1} \tag{4.4}$$

for $\theta > 0$, and $\int|F|\,\mathrm{d}\mu_{\nu,R} \le \zeta(2)e^{1+B}\nu((1/2,\infty))\|F\|_{H^1}$ for $\theta = 0$. Fatou's lemma extends both to $H^1(\mathbb{T}^\infty)$.

(3) $\Rightarrow$ (4). For polynomials, $\Lambda_{\alpha_{\nu,R}}(h) = \int H\,\mathrm{d}\mu_{\nu,R}$. Apply Lemma 4.1.

(4) $\Rightarrow$ (1). Lemma 4.1.

The constant. Combine Lemma 4.1, (4.4) and the bounds from the first step. Corollary 4.3 follows with $\zeta(2)$ omitted, since $Q = \emptyset$, and with $C_1 < 20.71$, $[\nu] \le 2\|M\|$. Corollary 4.4 follows from (4.3) directly, since then $R = \emptyset$ and $\int|F|\,\mathrm{d}\mu_{\nu,\emptyset} \le \int \zeta(4s)\,\mathrm{d}\nu(s)\,\|F\|_{H^1}$. □

4.6. Compactness

In one variable, Hartman's theorem [41] says that a Hankel operator is compact exactly when its symbol can be taken in $H^\infty + C$, and Sarason's theorem [81] says that this space is closed. Let

$$N = \Big\{ g \in L^\infty(\mathbb{T}^\infty) : \int z^\kappa g\,\mathrm{d}m = 0 \text{ for every multi-index } \kappa \ge 0\Big\},$$

so that the bounded symbols of a given $M(\alpha)$ form a coset $g + N$. Call a functional on $\mathscr{H}^1$ a *polynomial functional* if it has the form $h \mapsto \sum_{k\le K} h_k\beta(k)$.

PROPOSITION 4.9. *$C(\mathbb{T}^\infty) + N$ is closed in $L^\infty(\mathbb{T}^\infty)$.*

PROOF. For $k \ge 1$ let E_k be the conditional expectation onto the first k coordinates and F_k convolution on $\mathbb{T}^k$ with the product of Fejér kernels of order k, and put $T_k = F_kE_k$. Then $\|T_k\|_{L^\infty\to L^\infty} \le 1$, T_k maps L^∞ into trigonometric polynomials, and $T_kc \to c$ uniformly for $c \in C(\mathbb{T}^\infty)$, by density of finite-coordinate polynomials. Moreover $T_kN \subseteq N$: E_k is self-adjoint and sends z^κ either to itself or to 0, and F_k multiplies Fourier coefficients. So for $c \in C$ and $n \in N$, $\operatorname{dist}(c, C\cap N) \le \|c - T_kn\|_\infty \le \|c - T_kc\|_\infty + \|c - n\|_\infty$, and letting $k\to\infty$ gives $\operatorname{dist}(c, C\cap N) = \operatorname{dist}(c, N)$. Thus $C/(C\cap N) \to L^\infty/N$ is isometric, its image $(C+N)/N$ is complete, hence closed, and $C + N$ is its preimage. This is Rudin's argument for $H^\infty + C$, which needs only the maps T_k, not an algebra structure. □

THEOREM 4.10. *Let R have density $\theta \in [0,1]$ with constant B, and let ν be a finite positive Borel measure on $(1/2,\infty)$ with $[\nu]_\theta < \infty$. The following are equivalent.*

(1) $M(\alpha_{\nu,R})$ is compact.
(2) $\nu((1/2, 1/2+t]) = o(t^\theta)$ as $t \to 0^+$.
(3) $\Lambda_{\alpha_{\nu,R}}$ lies in the $(\mathscr{H}^1)^*$-closure of the polynomial functionals.
(4) Some, equivalently every, bounded symbol lies in $C(\mathbb{T}^\infty) + N$, which is closed (Proposition 4.9).

In particular, if $\theta = 0$, then $M(\alpha_{\nu,R})$ is compact for every finite ν.

The condition interpolates in the same way as in Theorem 4.2. At the curve endpoint it is $o(t)$; at the sign-average endpoint it is vacuous, because $\nu((1/2, 1/2+t]) \to 0$ for every finite measure. The full sign averages of Corollary 4.4 are therefore all compact, while the multiplicative Hilbert matrix, on the curve, is not.

PROOF. (1) $\Rightarrow$ (2). Assume $\theta > 0$. The unit vectors k_δ proportional to $(n^{-1/2-\delta})$ on the R-integers tend to 0 weakly as $\delta \to 0^+$, because $\zeta_R(1+2\delta) \to \infty$. A compact operator maps them to a norm-null family, and the computation in the proof of (1) $\Rightarrow$ (2) above gives $\nu((1/2, 1/2+\delta]) \le e^B (2\delta)^\theta \langle M(\alpha_{\nu,R}) k_\delta, k_\delta \rangle$.

(2) $\Rightarrow$ (3). Fix $0 < \varepsilon \le 1$ and write $\nu = \nu_\varepsilon + \nu^\varepsilon$, with ν_ε the restriction to $(1/2, 1/2+\varepsilon]$. Then $[\nu_\varepsilon]_\theta \le \eta(\varepsilon) := \sup_{0<u\le\varepsilon} \nu((1/2,1/2+u])/u^\theta$, which tends to 0 by (2), and the estimates in the proof of Theorem 4.2 give $\|\Lambda_{\alpha_{\nu_\varepsilon,R}}\| \le \zeta(2) C_1 \eta(\varepsilon)$, or $\le \zeta(2) e^{1+B} \nu((1/2,1/2+\varepsilon])$ when $\theta = 0$. The measure ν^ε lives on $[1/2+\varepsilon, \infty)$, so $0 \le \alpha_{\nu^\varepsilon,R}(n) \le \nu((1/2,\infty))\, n^{-1/2-\varepsilon}$. By Cauchy–Schwarz and Helson's inequality $\sum_k |h_k|^2/d(k) \le \|h\|^2_{\mathscr{H}^1}$ [44],

$$\Big| \Lambda_{\alpha_{\nu^\varepsilon,R}}(h) - \sum_{k \le K} h_k \alpha_{\nu^\varepsilon,R}(k) \Big| \le \Big(\sum_{k>K} d(k) \alpha_{\nu^\varepsilon,R}(k)^2 \Big)^{1/2} \|h\|_{\mathscr{H}^1},$$

which tends to 0 as $K \to \infty$ because $\sum_k d(k) k^{-1-2\varepsilon} = \zeta(1+2\varepsilon)^2$.

(3) $\Rightarrow$ (1). A polynomial functional with coefficients $\beta(k)$, $k \le K$, gives the Helson matrix $(\beta(mn))$, which vanishes unless $m, n \le K$ and so has finite rank. By Lemma 4.1, $\|M(\alpha) - M(\beta)\| \le \|\Lambda_\alpha - \Lambda_\beta\|$.

(3) $\Leftrightarrow$ (4). Let $q : L^\infty \to L^\infty/N \cong (\mathscr{H}^1)^*$ be the quotient map. The image $q(C(\mathbb{T}^\infty))$ is closed by Proposition 4.9. Trigonometric polynomials are uniformly dense in $C(\mathbb{T}^\infty)$, and their images under q are precisely the polynomial functionals: a monomial z^μ contributes zero unless $\mu \le 0$ coordinatewise. Thus the closure of the polynomial functionals is $q(C(\mathbb{T}^\infty))$. Its preimage is $C(\mathbb{T}^\infty) + N$, proving the equivalence for every symbol in the coset. □

Essential norms

Compactness is the qualitative statement; the quantitative one compares the essential norm $\|M\|_{\mathrm{e}}$ with the essential size of the symbol,

$$\mathcal{E}_{\mathrm{sym}}(M) = \operatorname{dist}_{L^\infty}\big(g, C(\mathbb{T}^\infty) + N\big),$$

which does not depend on the bounded symbol g chosen. A continuous symbol gives a compact matrix, being a uniform limit of trigonometric polynomials, and

N gives the zero matrix, so $\|M\|_{\mathrm{e}} \le \mathcal{E}_{\mathrm{sym}}(M)$ always. On the curve the converse holds up to an absolute constant.

THEOREM 4.11 (Essential Nehari on the curve). *Let R be all primes, let ν be finite and positive with $[\nu] < \infty$, and put $\ell_\nu = \limsup_{t\to 0^+} \nu((1/2, 1/2+t])/t$. Then*

$$\frac{1}{2}\ell_\nu \le \|M(\alpha_\nu)\|_{\mathrm{e}} \le \mathcal{E}_{\mathrm{sym}}(M(\alpha_\nu)) \le \frac{\Gamma(1/4)^2}{\sqrt{2\pi}}\,\ell_\nu \le \frac{2\Gamma(1/4)^2}{\sqrt{2\pi}}\,\|M(\alpha_\nu)\|_{\mathrm{e}}.$$

PROOF. The lower bound is the test in (1) ⇒ (2) of Theorem 4.10, with $\delta\zeta(1+2\delta) \to 1/2$ and the weak nullity of k_δ. For the upper bound, split $\nu = \nu_\varepsilon + \nu^\varepsilon$ as in (2) ⇒ (3), and put $A_\varepsilon = \sup_{t\le\varepsilon} \nu((1/2, 1/2+t])/t$. The functional of ν^ε is a norm limit of polynomial functionals. For ν_ε apply Theorem 2.1 with $p = 1$ and the weight $\|K_\tau\|$: by Lemma 2.4 the row at σ is at most $\int g_\sigma \,\mathrm{d}\nu_\varepsilon$ with $g_\sigma(\tau) = \zeta(1+2\sigma)^{-1/4}\zeta(1+2\tau)^{3/4}\zeta(1+\sigma+\tau)^{1/2}$, which decreases in τ, so the layer-cake formula bounds it by $A_\varepsilon \int_0^\varepsilon g_\sigma \,\mathrm{d}\tau$. Since $x^{-1} \le \zeta(1+x) \le (1+x)/x$, the integral is at most $(1+2\varepsilon)^{5/4}\Gamma(1/4)^2/\sqrt{2\pi}$, as in the proof of Theorem 2.11. Through the quotient map q, $\mathcal{E}_{\mathrm{sym}}$ is the distance of $q(g)$ to the closure of the polynomial functionals, which is at most this bound. Let $\varepsilon \to 0$. □

The same proof works on every prime set.

COROLLARY 4.12. *Let R have density $\theta \in (0,1]$ with constant B, let $[\nu]_\theta < \infty$, and put $\ell_{\nu,\theta} = \limsup_{t\to 0^+} \nu((1/2, 1/2+t])/t^\theta$. Then*

$$e^{-B}2^{-\theta}\ell_{\nu,\theta} \le \|M(\alpha_{\nu,R})\|_{\mathrm{e}} \le \mathcal{E}_{\mathrm{sym}}(M(\alpha_{\nu,R})) \le \zeta(2)C_1\ell_{\nu,\theta},$$

with $C_1 = C_1(\theta, B)$ from Lemma 4.5.

PROOF. For the lower bound, test with the normalized kernels k_δ of $\gamma_R(1/2+\delta)$ in the coordinates of R, which are weakly null since $\zeta_R(1+2\delta) \to \infty$: $\langle Mk_\delta, k_\delta\rangle \ge \zeta_R(1+2\delta)\nu((1/2, 1/2+\delta])$ and $\zeta_R(1+2\delta) \ge e^{-B}(2\delta)^{-\theta}$. For the upper bound, split ν as in the proof above and use (4.4) for ν_ε, which has no mass on $(3/2, \infty)$. □

COROLLARY 4.13. *If $\mathrm{d}\nu = w\,\mathrm{d}s$ near $1/2$, with w right-continuous at $1/2$, then $M(\alpha_\nu)$ has essential spectrum $[0, \pi w(1/2)]$; in particular $\|M(\alpha_\nu)\|_{\mathrm{e}} = \pi w(1/2)$.*

PROOF. Let $\nu_0 = w(1/2)\mathbf{1}_{(1/2,3/2]}\,\mathrm{d}s$. The positive and negative parts of $\nu - \nu_0$ have $o(t)$ mass near $1/2$, so their matrices are compact by Theorem 4.10. Extending ν_0 to $(1/2, \infty)$ changes the matrix, after deleting the first row and column, by a Hilbert–Schmidt term, and leaves $w(1/2)$ times the multiplicative Hilbert matrix, whose essential spectrum is $[0, \pi]$ [72]. □

The same decomposition compares symbols. Let M be the multiplicative Hilbert matrix.

COROLLARY 4.14. *Under the hypotheses of Corollary 4.13, $\mathcal{E}_{\mathrm{sym}}(M(\alpha_\nu)) = w(1/2)\,\mathcal{E}_{\mathrm{sym}}(M)$.*

PROOF. Put $w_0 = w(1/2)$ and $\nu_0 = w_0\mathbf{1}_{(1/2,3/2]}\,\mathrm{d}s$. The positive and negative parts of $\nu - \nu_0$ satisfy the compactness condition, so their functionals have symbols in $C(\mathbb{T}^\infty) + N$ by Theorem 4.10. The remaining comparison with $w_0 M$ uses the identity $\int_{1/2}^{3/2} n^{-s}\,\mathrm{d}s = 1/(\sqrt{n}\log n) - n^{-3/2}/\log n$ for $n \ge 2$, together with a

possible constant coefficient. Put $c_n = n^{-3/2}/\log n$. Helson's inequality makes this correction a norm limit of polynomial functionals, since $\sum_{n\ge 2} d(n)c_n^2 < \infty$. Hence the two symbol classes differ by an element of $C(\mathbb{T}^\infty)+N$, which proves the equality of their distances. □

By Corollary 11.7, the essential size of the symbol is at least $w(1/2)C_{\mathrm{Gauss}}$, while the essential norm is $\pi w(1/2)$. So an excess $C_{\mathrm{Gauss}} > \pi$ would give a strict essential Nehari gap for every curve measure with a density continuous and positive at $1/2$. For the multiplicative Hilbert matrix $\mathscr{M}$ itself, the same splitting of the functional $\int_0^\infty (f(1/2+\sigma)-a_1)\,\mathrm{d}\sigma$ gives $\pi \le \mathcal{E}_{\mathrm{sym}}(\mathscr{M}) \le \Gamma(1/4)^2/\sqrt{2\pi} \approx 5.244$: the essential Nehari gap lies in $[1, 1.67]$, and its upper end is exactly the price of discarding phases in section 2.2.

So for a curve measure with a density the essential norm is read off from a single number, and the symbol cannot be made essentially smaller than $\pi w(1/2)$ by any choice in the coset. For the multiplicative Hilbert matrix this gives $\mathcal{E}_{\mathrm{sym}} \ge \pi$: *no* bounded symbol is within distance less than π of $C(\mathbb{T}^\infty)+N$, a statement about the whole coset that the sharp-constant problem (chapter 11) only asks about at the point 0.

4.7. Consequences

Let R have density $\theta \in (0,1]$, Q its complement, orders $k_q \ge 2$ on Q, and ν a finite positive measure on $(1/2,\infty)$; if Q is empty, read $\zeta(k)$ below as 1. Let $\mu_{\nu,R}$ be the law of $\big((\omega_q q^{-s})_{q\in Q}, (p^{-s})_{p\in R}\big)$ under $\mathrm{d}\omega \times \mathrm{d}\nu(s)$. Its moments are $\alpha(n) = \mathbf{1}[k_q \mid v_q(n)\ \forall q \in Q]\int n^{-s}\,\mathrm{d}\nu(s) \ge 0$. For signs, $k_q = 2$, these are the measures of Theorem 4.2, and then the Carleson measures of the family are the same for every p.

Corollary 4.15 (One Carleson class for all p). *For every $0 < p < \infty$, $\mu_{\nu,R}$ is a Carleson measure for $H^p(\mathbb{T}^\infty)$ if and only if $[\nu]_\theta = \sup_t \nu((1/2,1/2+t])/t^\theta < \infty$, and then, with $k = \min_q k_q$,*

$$\int |F|^p\,\mathrm{d}\mu_{\nu,R} \le \zeta(k)\big(C[\nu]_\theta + \zeta(3)\nu((1/2,\infty))\big)\|F\|_p^p.$$

In particular the Helson matrix $(\alpha(mn))$ has a bounded symbol of norm at most the constant for $p = 1$.

Proof. Sufficiency: Proposition 4.6 in the Q-coordinates, then Theorem 2.11 (2) on each slice $F(\zeta,\cdot) \in H^p(\mathbb{T}^R)$, then Fubini. For necessity, let C_p be the assumed H^p embedding constant. Take $f_\delta = K_\delta^{2/p}/\|K_\delta\|_2^{2/p}$, where K_δ is the kernel at $\gamma_R(1/2+\delta)$ in the R-coordinates. Then $\|f_\delta\|_p = 1$ and $|f_\delta|^p = |K_\delta|^2/\|K_\delta\|_2^2$. These functions do not see the averaging, so the kernel test in section 4.5 gives $C_p \ge e^{-B}(2\delta)^{-\theta}\nu((1/2,1/2+\delta])$. Hence $[\nu]_\theta \le 2e^B C_p$. The symbol comes from Hahn–Banach. □

Compactness needs a Montel lemma in infinitely many variables.

LEMMA 4.16. *Let $0 < p < \infty$. Every bounded sequence in $H^p(\mathbb{T}^\infty)$ has a subsequence that converges uniformly on every compact subset of $\mathbb{D}^\infty \cap \ell^2$, compact in the ℓ^2 topology.*

PROOF. Point evaluation satisfies $|F(r)| \le \|F\|_p \prod_j (1-|r_j|^2)^{-1/p}$, which is at most $\|F\|_p \exp(M^2/(p(1-\rho^2)))$ when $\|r\|_2 \le M$ and $\|r\|_\infty \le \rho < 1$. A compact set K has $\sup_K \|r\|_2 < \infty$ and $\sup_K \|r\|_\infty < 1$, and the same bounds hold, with room to spare, on a fixed ℓ^2-neighbourhood of K and of its truncations $r^{(D)} = (r_1, \dots, r_D, 0, \dots)$. So the sequence is uniformly bounded there, and Cauchy's estimate along lines bounds its derivatives uniformly. Since $\sup_K \|r - r^{(D)}\|_2 \to 0$, we get $|F_n(r) - F_n(r^{(D)})| \to 0$ uniformly in n and $r \in K$. The coefficients of a bounded sequence are bounded, so a diagonal subsequence converges coefficientwise; on each $\mathbb{D}^D$ the finite-dimensional Montel theorem then gives uniform convergence on compact sets. Combining the two statements, the subsequence is uniformly Cauchy on K. □

PROPOSITION 4.17 (Compact embeddings). *In the setting of Corollary 4.15, suppose $[\nu]_\theta < \infty$.*

(1) If the embedding $H^p \to L^p(\mu_{\nu,R})$ is compact for some $0 < p < \infty$, then $\nu((1/2, 1/2+t]) = o(t^\theta)$.
(2) If $\nu((1/2, 1/2+t]) = o(t^\theta)$, the embedding is compact for every $0 < p < \infty$.

PROOF. (1) Let K_δ be the kernel at $\gamma_R(1/2+\delta)$ and $f_\delta = K_\delta^{2/p}/\|K_\delta\|_2^{2/p}$, of norm one. At every point of the support, $f_\delta \to 0$ as $\delta \to 0$, because $K_\delta(z)$ stays bounded there while $\|K_\delta\|_2 \to \infty$. If the embedding is compact, every sequence $\delta_n \to 0$ has a subsequence along which f_{δ_n} converges in $L^p(\mu)$, necessarily to 0, the pointwise limit. But $\int |f_\delta|^p \, d\mu \ge \zeta_R(1+2\delta)\nu((1/2, 1/2+\delta]) \gtrsim \delta^{-\theta}\nu((1/2, 1/2+\delta])$, as in section 4.5. (2) Split ν at $1/2+\delta$. By Corollary 4.15, the part near $1/2$ has embedding norm at most a constant times $\sup_{t \le \delta} \nu((1/2, 1/2+t])t^{-\theta} + \nu((1/2, 1/2+\delta])$, which tends to 0 with δ. The far part is the image of a compact subset K_δ of $\mathbb{D}^\infty \cap \ell^2$: its points have coordinates of modulus at most $q^{-1/2-\delta}$, whose squares have uniformly summable tails, and the root phases range over a compact product. Given a bounded sequence in H^p, Lemma 4.16 provides a subsequence converging uniformly on K_δ, hence in L^p of the finite far measure. Since the near parts are uniformly small, the images form a Cauchy sequence; for $p < 1$ use the p-subadditivity of $\int |\cdot|^p$. □

For density $\theta = 0$, that is $\sum_{p \in R} 1/p < \infty$, the root-of-unity average is bounded uniformly in $s > 1/2$:

$$\mathbb{E}_\omega |F((\omega_q q^{-s})_q, (p^{-s})_p)|^p \le \zeta(k) \prod_{p \in R} (1 - 1/p)^{-1} \|F\|_p^p.$$

So every finite ν gives a Carleson measure for every p, and the splitting in part (2) shows that the embedding is compact, the near part now being controlled by $\nu((1/2, 1/2+\delta]) \to 0$.

For signs the Helson matrix is positive and the rest of this chapter develops its full theory. For some $k_q \ge 3$ it is not: the measure lives on complex points, and

already the one-variable Hankel matrix of $\mathbf{1}[3 \mid n]r^n$ has the indefinite block $\left(\begin{smallmatrix} 0 & r^3 \\ r^3 & 0 \end{smallmatrix}\right)$. Boundedness of the matrix and the H^2 Carleson condition then need not coincide, and the corollary gives the bounded symbol directly from the H^1 embedding.

So the inclusion (4.1) holds throughout the family, and Problem 2 has a positive answer for every measure of the form $\mu_{\nu,R}$.

The best-known special case is the multiplicative Hilbert matrix, with entries $1/(\sqrt{mn}\log mn)$ for $m, n \geq 2$.

PROBLEM 3 (Saksman and Seip [80, Problem 3.2]). *Does the multiplicative Hilbert matrix have a bounded symbol?*

Perfekt [70] answered Problem 3 in the affirmative. The family of Theorem 4.2 contains the matrix, up to a harmless correction, and so gives a second proof.

COROLLARY 4.18 (Perfekt [70, Corollary 1.2]). *The multiplicative Hilbert matrix has a bounded symbol: there is $g \in L^\infty(\mathbb{T}^\infty)$ with $\int g \,\mathrm{d}m = 1$ and $\int z^{\kappa(n)} g \,\mathrm{d}m = 1/(\sqrt{n}\log n)$ for $n \geq 2$.*

PROOF. Take R to be all primes and $\nu = \mathbf{1}_{(1/2,3/2]}\,\mathrm{d}s$. Then $[\nu] = 1$, $\alpha_\nu(1) = 1$, and

$$\alpha_\nu(n) = \frac{1}{\sqrt{n}\log n} - c_n, \qquad c_n = \frac{n^{-3/2}}{\log n} \qquad (n \geq 2).$$

Corollary 4.3 gives a bounded symbol for α_ν. The functional $h \mapsto \sum_{n\geq 2} h_n c_n$ is bounded on $\mathscr{H}^1$ by Helson's inequality, since $\sum_n d(n)n^{-3} < \infty$, so it too has a bounded symbol by Lemma 4.1. The sum of the two symbols works. □

The coefficient series also converges in its natural order, by Perfekt's Tauberian argument (Corollary 2.6). Both proofs take the symbol from the Hahn–Banach theorem; Theorem 9.18 writes one down, with norm at most $10^5 + 7$.

The same subtraction applied to the odd-indexed primes ($\theta = 1/2$) gives a bounded symbol for the matrix $\big(\mathbf{1}[(mn)_Q \in \square]/(\sqrt{mn}\log mn)\big)_{m,n\geq 2}$, the multiplicative Hilbert matrix with the signs of the even-indexed primes averaged out.

4.7.1. Passing from p to q

Corollary 4.15 gives one Carleson class for all p in this family. Bayart, Brevig, Haimi, Ortega-Cerdà and Perfekt asked whether this is always so in one direction. For a positive measure μ on $\mathbb{D}^\infty$ and $0 < r < \infty$ let $C_r(\mu)$ be the least constant in $\int |F|^r \,\mathrm{d}\mu \leq C_r(\mu)\|F\|_r^r$ over analytic polynomials F in finitely many variables, so that μ is a Carleson measure for $H^r(\mathbb{T}^\infty)$ exactly when $C_r(\mu) < \infty$.

PROBLEM 4 (Bayart, Brevig, Haimi, Ortega-Cerdà and Perfekt [12, Section 5]). *If μ is a Carleson measure for $H^p(\mathbb{T}^\infty)$ for some $0 < p < \infty$, is it a Carleson measure for $H^q(\mathbb{T}^\infty)$ for every $q > p$?*

For $q = kp$ with k a positive integer the answer is yes, with $C_q \leq C_p$: apply the H^p inequality to F^k. On a finite polydisc the Carleson class does not depend on p, by Chang's theorem [26], as recalled in [12, Section 5], and Chalmoukis and Dayan [24, Lemma 3.4] give an interpolation proof of the upward inclusion for $1 < p < q < \infty$; in both the constants depend on the dimension. The next theorem

shows that Problem 4 is a question about sharp constants of finitely many point masses.

THEOREM 4.19. *Suppose that $0 < p < q < \infty$. The following are equivalent.*

1. *Every Carleson measure for $H^p(\mathbb{T}^\infty)$ is a Carleson measure for $H^q(\mathbb{T}^\infty)$.*
2. *$C_q(\nu) \le C_p(\nu)$ for every d and every finite positive atomic measure ν on $\mathbb{D}^d$.*
3. *$C_q(\mu) \le C_p(\mu)$ for every finite positive measure μ on $\mathbb{D}^\infty$.*

If (2) fails, there is a countably atomic measure μ on points with finitely many nonzero coordinates such that $\mu(\mathbb{D}^\infty) \le 1$, $C_p(\mu) \le 1$ and $C_q(\mu) = \infty$.

PROOF. For measures ν_1, ν_2 on disjoint blocks of coordinates, applying the two inequalities to slices and using Fubini's theorem gives $C_r(\nu_1 \otimes \nu_2) \le C_r(\nu_1) C_r(\nu_2)$, and products $F_1(z)F_2(w)$ give the reverse inequality; this holds for every $r > 0$. A measure placed on a block of coordinates, with the others zero, keeps its constant, by the sub-mean value property in the unused coordinates.

Suppose that (2) fails for ν on $\mathbb{D}^d$, put $a = C_p(\nu) \in (0, \infty)$, and take a polynomial F whose q-ratio $\int |F|^q \, \mathrm{d}\nu / \|F\|_q^q$ is ρa with $\rho > 1$. Choose k_j with $\rho^{k_j} \ge 4^j$, place $\eta_j = a^{-k_j} \nu^{\otimes k_j}$ on pairwise disjoint blocks, and put $\mu = \sum_{j \ge 1} 2^{-j} \eta_j$. Then $C_p(\eta_j) = 1$, and $\eta_j(\mathbb{D}^\infty) \le 1$ since $\nu(\mathbb{D}^d) \le a$, so μ has mass and constant at most 1. The tensor power of F on the jth block has q-ratio at least $2^{-j} \rho^{k_j} \ge 2^j$ for μ. This proves the last statement and that (1) implies (2).

Suppose (2), and let ν have compact support $K \subset \mathbb{D}^d$. The functions $|F|^t$ with $\|F\|_t \le 1$ are uniformly bounded and equicontinuous on K, by the Poisson bound in each variable and Cauchy's estimate. Moving the mass of each cell of a fine Borel partition of K to one of its points therefore changes C_p and C_q by arbitrarily little, so $C_q(\nu) \le C_p(\nu)$. For general μ with $C_p(\mu) < \infty$, let $\nu_{d,r}$ be the image of μ under $z \mapsto (rz_1, \dots, rz_d)$, $0 < r < 1$. Dilation contracts H^p norms, so $C_p(\nu_{d,r}) \le C_p(\mu)$, and $\nu_{d,r}$ has compact support. Hence $\int |F(rz)|^q \, \mathrm{d}\mu \le C_p(\mu) \|F\|_q^q$ for polynomials F in d variables; let $r \to 1$ by dominated convergence. This gives (3), and (3) implies (1). □

So a negative answer needs one finite atomic measure and one polynomial whose q-ratio exceeds the whole constant C_p; two polynomials with different ratios do not suffice. At $p = 2$ that constant is explicit: if $\nu = \sum_\ell w_\ell \delta_{z_\ell}$, then $C_2(\nu)$ is the norm of the Gram matrix $G_{\ell m} = \sqrt{w_\ell w_m}\, K_{z_m}(z_\ell)$, since the restriction map $F \mapsto (\sqrt{w_\ell} F(z_\ell))_\ell$ composed with its adjoint is G. The next proposition excludes two natural classes of test functions.

PROPOSITION 4.20. *Let ν be a finite positive atomic measure on $\mathbb{D}^d$, and $A = C_2(\nu)$.*

1. *If F is a zero-free analytic polynomial on $\mathbb{D}^d$, then $\int |F|^q \, \mathrm{d}\nu \le A \|F\|_q^q$ for every $q > 0$. If some extremal U for A is zero-free on $\mathbb{D}^d$, then $C_q(\nu) \ge A$ for every $q > 0$, with equality when q is an even integer.*

(2) *Let $M \geq 2$, $0 < r < 1$, $t = r^2$, $\zeta = e^{2\pi i/M}$ and $\nu = M^{-1}\sum_{j<M} \delta_{(r\zeta^j, r\zeta^j)}$ on $\mathbb{D}^2$. Then*

$$A = \max_{0 \leq a < M} t^a \Big(\frac{a+1}{1-t^M} + \frac{Mt^M}{(1-t^M)^2} \Big),$$

and every homogeneous polynomial F of degree one satisfies $\int |F|^3 \, \mathrm{d}\nu \leq (3\pi/8)\sqrt{2/3}\, A \|F\|_3^3$, where $(3\pi/8)\sqrt{2/3} < 0.962$.

Proof. (1) Apply the H^2 inequality to $F^{q/2}$, which is bounded and holomorphic on $\mathbb{D}^d$, after dilation and polynomial approximation. The extremal U is a combination of reproducing kernels, so $U^{2/q}$ is bounded and has q-ratio A; the power trick gives $C_{2k} \leq A$.

(2) The Gram matrix is $M^{-1}(1 - t\zeta^{j-k})^{-2} = M^{-1}\sum_{n \geq 0}(n+1)t^n\zeta^{n(j-k)}$. It is circulant, with eigenvalues $\sum_{n \equiv a \pmod M}(n+1)t^n$, $0 \leq a < M$, and summing gives the formula; in particular $A \geq \max(1, 2t, 3t^2)$. A homogeneous linear F is $F(z,w) = w\,P(z/w)$ with P of degree at most one, so its 3-ratio is $t^{3/2}|P(1)|^3/\|P\|^3_{H^3(\mathbb{T})}$. With $P(1) = 1$, averaging P with its coefficient reversal and convexity show that $\|P\|_3$ is smallest for $P = (1+z)/2$, where $\|P\|_3^3 = \int |\cos(\theta/2)|^3 \, \mathrm{d}m(\theta) = 4/(3\pi)$. So the ratio is at most $(3\pi/4)t^{3/2}$, and $(3\pi/4)t^{3/2}/\max(1, 2t, 3t^2)$ is largest at $t = 2/3$. □

In (1) the branch through zero-free extremals is flat: if U is zero-free near the closed polydisc, the ratio $\int |U|^t \, \mathrm{d}\nu / \|U\|_t^t$ has derivative zero in t at $t = 2$, by the extremal equation tested with $U \log U$. So a counterexample must use functions with zeros, and (2) shows that the simplest such class, linear forms on cyclic diagonal measures, stays below A by a uniform factor. Product measures cannot be counterexamples either: by inner-outer factorization in one variable, $C_p = C_q$ for measures on $\mathbb{D}$, and the constants are multiplicative. The amplification in Theorem 4.19 is the principle behind the even-integer case of [12, Theorem 22]; the reduction to finite atomic measures with sharp constants is, to our knowledge, new. Problem 4 remains open.

4.8. Positive Nehari fails

Problem 2 has a positive answer for every measure $\mu_{\nu,R}$, and for discrete measures on energy-separated families with weights bounded by the inverse kernel norm (chapter 3). The separation mechanism controls the coordinates kept on a curve. The symmetry mechanism controls the coordinates whose signs are averaged, and costs only the factor $\zeta(2)$. Rotations $z \mapsto \varepsilon z$ of $\mathbb{T}^\infty$ preserve every H^p norm, so each statement also holds for a fixed rotated copy of the family. Since a positive measure dominates each of its parts, finite sums of such measures are covered as well.

Explicit compressions to $\mathcal{S}_R$ at every borderline are listed in section 5.5, and Proposition 5.34 compares them with the averaged matrices.

The first case not covered is a countable superposition of measures on rotated copies of the curve, with mass that is neither separated in energy nor symmetric under sign changes in any fixed set of coordinates. There the answer is negative.

For a positive measure μ on real points write, as in section 4.7.1, $C_2(\mu)$ for its H^2 Carleson constant, which is the norm of its Helson matrix, and

$$D(\mu) = \sup_{F \neq 0} \frac{|\int F \, \mathrm{d}\mu|}{\|F\|_1},$$

the norm of Λ_α on $\mathscr{H}^1$; by Lemma 4.1 the matrix has a bounded symbol exactly when $D(\mu) < \infty$, and $D(\mu) \le C_1(\mu)$.

THEOREM 4.21. *There is a countably atomic positive measure μ with $C_2(\mu) \le 1$ and $D(\mu) = C_1(\mu) = \infty$, supported on points $\varepsilon\gamma(s)$ with $s > 1/2$ and $\varepsilon \in \{-1,1\}^{\mathbb{P}}$ having finitely many negative entries. Consequently the Helson matrix of μ is bounded and non-negative and has no bounded symbol, and Problem 2 has a negative answer.*

In particular there is an H^2 Carleson measure on $(-1,1)^\infty \cap \ell^2$ that is not an H^1 Carleson measure, which answers the question of Perfekt and Pushnitski [71, after Remark 4.11], and the countable superpositions just described are not always H^1 Carleson measures.

The proof rests on one local gain. For a finite vector x with entries in $(0,1)$ let $\sigma_x = (\delta_x + \delta_{-x})/2$. Its Gram matrix has diagonal $\prod_j (1-x_j^2)^{-1}/2$ and off-diagonal $\prod_j (1+x_j^2)^{-1}/2$, so

$$C_2(\sigma_x) = A_x = \frac{1}{2}\Big(\prod_j (1-x_j^2)^{-1} + \prod_j (1+x_j^2)^{-1}\Big),$$

with extremal $U_x = \prod_j (1-x_j z_j)^{-1} + \prod_j (1+x_j z_j)^{-1}$. Then $U_x(\pm x) = 2A_x$ and $\|U_x\|_2^2 = 4A_x$, so U_x^2 attains $\int U_x^2 \, \mathrm{d}\sigma_x = A_x \|U_x^2\|_1$. Subtracting a small positive constant decreases its values at the atoms at rate one; its H^1 norm decreases at the average phase of U_x. A gain occurs when the latter decrease, multiplied by A_x, exceeds one.

LEMMA 4.22. *There are $\eta > 0$ and $\rho > 1$ such that, whenever $\max_j x_j < \eta$ and $|\sum_j x_j^2 - 3| < \eta$, some polynomial f_x with real coefficients, positive on $[-1,1]^d$, satisfies $\int f_x \, \mathrm{d}\sigma_x \ge \rho A_x \|f_x\|_1$.*

PROOF. Let $\max_j x_j \to 0$ and $\sum_j x_j^2 \to 3$. Then $\sum_j x_j z_j$ converges in law to a complex Gaussian $W = X + iY$ with X, Y independent of variance $3/2$, the higher terms of $\log \prod_j (1 \mp x_j z_j)^{-1}$ tend to 0 in L^2, and $\mathbb{E}|\prod_j (1 \mp x_j z_j)^{-1}|^4 = \prod_j (1+x_j^2)(1-x_j^2)^{-3}$ stays bounded. Thus $U_x \to U_0 = 2\cosh W$ in law with uniformly bounded fourth moments. In particular, $|U_x^2 - \delta|$ is uniformly integrable for every fixed $\delta \ge 0$. Also $A_x \to A_0 = \cosh 3$, so $\|U_x^2 - \delta\|_1 \to \mathbb{E}|U_0^2 - \delta|$.

Put $c_0 = \mathbb{E}\Re(\overline{\cosh W}/\cosh W)$. Given X, let $a = e^{-2|X|}$; the phase is $(a + e^{\mp 2iY})/(1 + a e^{\mp 2iY})$, and expanding in powers of a with $\mathbb{E}e^{-2ikY} = e^{-3k^2}$ gives

$$c_0 = \mathbb{E}\Big[a + (1-a^2)\sum_{k \ge 1} (-a)^{k-1} e^{-3k^2}\Big] \ge \mathbb{E}a,$$

since the alternating series has decreasing terms. Using $e^{-u} \geq 1-u$,

$$\mathbb{E}a = \frac{2}{\sqrt{3\pi}} \int_0^\infty e^{-2v-v^2/3}\,\mathrm{d}v \geq \frac{2}{\sqrt{3\pi}} \int_0^\infty e^{-2v}(1-v^2/3)\,\mathrm{d}v = \frac{5}{6\sqrt{3\pi}} > \frac{5}{24}.$$

As $\cosh 3 > 11/2$, we get $A_0 c_0 > 55/48 > 1$. The difference quotients of $\delta \mapsto |U_0^2 - \delta|$ are bounded by 1, so dominated convergence gives

$$\frac{\mathrm{d}}{\mathrm{d}\delta}\Big(4A_0^2 - \delta - A_0 \mathbb{E}|U_0^2 - \delta|\Big)\Big|_{\delta=0^+} = A_0 c_0 - 1 > 0.$$

Here $4A_0^2 - \delta$ is the limit of $\int (U_x^2 - \delta)\,\mathrm{d}\sigma_x$, so some fixed $0 < \delta < 1$ gives a ratio above 1 in the limit. By the convergence above, the same holds, with a fixed $\rho > 1$, for all x with $\max_j x_j$ and $|\sum_j x_j^2 - 3|$ small, uniformly in d; otherwise a sequence would contradict the limit. On the real cube $U_x \geq 2$, since the product of its two terms is $\prod_j (1 - x_j^2 z_j^2)^{-1} \geq 1$; so $U_x^2 - \delta \geq 3$ there, and a Taylor polynomial of $U_x^2 - \delta$ keeps both the positivity and the gain. □

Proof of Theorem 4.21. Shrinking η if necessary, assume $0 < \eta < 1$. Choose P_0 with $p^{-1/2} < \eta/2$ for $p > P_0$. Since $\sum_{p>P_0} p^{-2s} \to \infty$ as $s \to 1/2^+$, for each m there is $s_m > 1/2$ and disjoint finite sets $B_1, \ldots, B_m$ of primes above P_0 with $3 \leq \sum_{p \in B_i} p^{-2s_m} < 3 + \eta^2/4$. The vectors $x^{(i)} = (p^{-s_m})_{p \in B_i}$ satisfy Lemma 4.22. Let ν_m be the uniform probability measure on the 2^m points $\varepsilon\gamma(s_m)$ where ε is constant on each B_i and equal to 1 off $\bigcup_i B_i$. On the complement C of the blocks, ν_m is a point mass at a real point r, of cost $B_C = \|K_r\|_2^2$ both in H^2 and in H^1 (Proposition 1.3), attained by K_r^2, which is positive on the real cube. Slice testing and product tests give $C_2(\nu_m) = B_C \prod_i A_{x^{(i)}}$. Choose finitely many complementary coordinates J with $B_J = \prod_{j \in J}(1 - r_j^2)^{-1} > B_C/2$. Taylor polynomials approximate $K_{r_J}^2$ uniformly on the closed polydisc. Since $K_{r_J}^2$ is positive and bounded away from zero on the real cube, sufficiently close approximants remain positive there and have evaluation ratio greater than $B_C/2$. Multiplying one of these by the $f_{x^{(i)}}$ gives a polynomial F, positive on the real cube, with $\int F\,\mathrm{d}\nu_m \geq \rho^m C_2(\nu_m)\|F\|_1/2$.

Choose m_j with $\rho^{m_j} \geq 2 \cdot 4^j$ and put $\mu = \sum_{j \geq 1} 2^{-j} \nu_{m_j}/C_2(\nu_{m_j})$. Then $C_2(\mu) \leq 1$. For the polynomial F_j chosen for ν_{m_j} with $\|F_j\|_1 = 1$, the other components contribute non-negative amounts, so $\int F_j\,\mathrm{d}\mu \geq 2^j$. Hence $D(\mu) = \infty$, and $C_1(\mu) \geq D(\mu)$. Every atom lies in ℓ^2 since $s_m > 1/2$. □

The gain needs a function with zeros. Suppose an extremal U for a finite atomic measure ν on real points has real coefficients and no zeros near the closed polydisc. For a polynomial h, put $g = (h/U)^\sharp$, where $a^\sharp(z) = \overline{a(\overline{z})}$. This function is analytic near the closed polydisc, so polynomial approximation makes it an admissible test in $\int U\overline{g}\,\mathrm{d}\nu = C_2 \int U\overline{g}\,\mathrm{d}m$. On real points $U\overline{g} = h$. Conjugating all coordinates on the torus and using the real coefficients of U gives $\int h\,\mathrm{d}\nu = C_2 \int h\,\overline{U}/U\,\mathrm{d}m$. Thus $C_2\overline{U}/U$ is a bounded symbol and $D = C_2$. For two opposite atoms in one or two coordinates, U has numerator 2 or $2(1 + r^2 z_1 z_2)$, which is zero-free. Three coordinates already suffice for a gain, with a computer-assisted first variation.

PROPOSITION 4.23. *Let $r = 4/5$ and $\sigma = \sigma_{(r,r,r)}$. Then*

$$C_2(\sigma) = A = \frac{544140625}{50243409}, \qquad D(\sigma) > A.$$

PROOF. With $U = U_{(r,r,r)}$, the function $F_\varepsilon = U^2 - \varepsilon$ has $\int F_\varepsilon \, \mathrm{d}\sigma = 4A^2 - \varepsilon$ and, by dominated convergence, $\|F_\varepsilon\|_1 = 4A - \varepsilon c_0 + o(\varepsilon)$, where $c_0 = \int_{\mathbb{T}^3} \Re(\overline{U}/U) \, \mathrm{d}m$. So $D(\sigma) > A$ as soon as $Ac_0 > 1$, after replacing F_ε by a Taylor polynomial. The numerator $2(1 + r^2(z_1z_2 + z_1z_3 + z_2z_3))$ of U vanishes at $z_j = i/(\sqrt{3}r)$, so the zero-free argument does not apply. Split $\mathbb{T}^3$ into 256^3 cells of half-width $h = \pi/256$ in each angle. On the cell with centre ξ, the derivatives of U give $|U - U(\xi)| \le \delta(\xi)$ with an explicit $\delta(\xi)$, and then $\Re(\overline{U}/U) \ge \max\{-1, \Re(\overline{U(\xi)}/U(\xi)) - 2\delta(\xi)/|U(\xi)|\}$. Summing these lower bounds in interval arithmetic, with outward rounding and an exact final summation, gives

$$c_0 \ge \frac{409587562197037715}{2^{61}} > \frac{17}{100} > \frac{1}{A}. \qquad \square$$

Amplifying σ by tensor powers on disjoint blocks, as in the proof of Theorem 4.19, gives a second counterexample, on real points that are not on rotated copies of the curve. Theorem 4.21 needs no computation: its gain comes from the Gaussian limit of many small coordinates. To our knowledge both are new. They leave the positive results of this chapter as they are, but they show that some structure, such as a single curve or a fixed set of averaged signs, is needed.

4.8.1. Different Carleson classes below two

Below $p = 2$ the half-plane Carleson class also depends on p. For a positive measure μ on $\mathbb{C}_{1/2}$ and $0 < r < \infty$ let $A_r(\mu)$ be the least constant in $\int |f|^r \, \mathrm{d}\mu \le A_r(\mu)\|f\|^r_{\mathscr{H}^r}$ over Dirichlet polynomials. Taking powers gives $A_{kp} \le A_p$ for positive integers k, so one might expect a single class for all $p < 2$; it is not so.

THEOREM 4.24. *There are $q \in (1,2)$ and a finite, countably atomic positive measure μ on $\mathbb{C}_{1/2}$ with $A_q(\mu) \le 1$ and $A_1(\mu) = \infty$.*

PROOF. We use the blocks of Lemma 4.22, with its η and ρ: for x with entries below η and $|\sum_j x_j^2 - 3| < \eta$, the two-point measure σ_x has H^2 constant $a(x) = A_x$ and H^1 constant at least $\rho a(x)$, attained up to that factor by a polynomial positive on the real cube. For the H^q constant, let $\theta = \sqrt{q/2}$. Weissler's hypercontractivity, in the form $\|F(\theta\cdot)\|_{H^2} \le \|F\|_{H^q}$ of [15, Lemma 4.2], holds in every dimension by Minkowski's inequality, since $q \le 2$. Applied to $F(\pm x) = G(\pm x/\theta)$, $G = F(\theta\cdot)$, and with Hölder's inequality on the probability measure σ_x, it gives $\int |F|^q \, \mathrm{d}\sigma_x \le a(x/\theta)\|F\|_q^q$. Write $a(x) = (B_x + D_x)/2$ with $B_x = \prod_j (1 - x_j^2)^{-1}$ and $D_x = \prod_j (1 + x_j^2)^{-1}$; dilation raises B and lowers D, so

$$\frac{a(x/\theta)}{a(x)} \le \frac{B_{x/\theta}}{B_x} \le \exp\Big(\frac{(\theta^{-2} - 1)\sum_j x_j^2}{1 - \eta^2/\theta^2}\Big) < \sqrt{\rho}$$

when $\sum_j x_j^2 \le 4$ and $\theta < 1$ is close enough to 1. Fix such θ and $q = 2\theta^2 \in (1,2)$, and put $\kappa = \sqrt{\rho}$.

Pack blocks $B_1, \dots, B_m$ at $s_m > 1/2$ and form ν_m on the points $\varepsilon\gamma(s_m)$ as in the proof of Theorem 4.21, with complement point cost B_C. Slice testing gives an H^q constant at most $D_m = B_C \prod_i a(x^{(i)}/\theta)$, and the product tests give an H^1 constant at least $B_C \prod_i \rho a(x^{(i)}) \geq \kappa^m D_m$.

These points are real, not on the half-plane curve. But for $\|F\|_{\mathscr{H}^r} = 1$ the functions $|F|^r$ are uniformly equicontinuous in the ℓ^2 metric near each point $\varepsilon\gamma(s)$, by the point bound and Cauchy's estimate. By Kronecker's theorem, $(p^{-it})_{p \leq P}$ comes arbitrarily close to any $(\varepsilon_p)_{p \leq P}$, since the $\log p$ are linearly independent over $\mathbb{Q}$, and the tail $\sum_{p>P} p^{-2s_m}$ is small. So ν_m is approximated by atomic measures λ_m on the line $\Re s = s_m$ with $A_q(\lambda_m) \leq 2D_m$ and $A_1(\lambda_m) \geq \kappa^m D_m/4$. Choose m_j with $\kappa^{m_j} \geq 8 \cdot 4^j$ and put $\mu = \sum_j 2^{-j} \lambda_{m_j}/A_q(\lambda_{m_j})$. Then $A_q(\mu) \leq 1$, and $A_1(\mu) \geq 2^{-j} A_1(\lambda_{m_j})/A_q(\lambda_{m_j}) \geq 2^j$ for every j. □

So no single Carleson class serves all $p < 2$. Whether the $\mathscr{H}^1$ and $\mathscr{H}^{1/2}$ classes coincide, and whether any two distinct exponents below 2 give different classes, is open.

Notes

The counterexample in Theorem 4.21, including the Gaussian block gain and its amplification, is presented as a standalone paper in [85]. The curve and sign-average theorems of this chapter identify structured classes in which bounded symbols still exist.

CHAPTER 5

Spectral, Schatten and rank structure

Question. What do positive Helson matrices look like spectrally, and how far can the Nehari constant grow among matrices of bounded rank?

The answer to the first question is a dictionary. For a curve matrix the dyadic profile $a_j = e^{j\theta}\nu(I_j)$ of the measure determines everything: boundedness is $a \in \ell^\infty$, compactness is $a \in c_0$ (chapter 4), and membership in S_p is $a \in \ell^p$ for every $p > 0$, and likewise for every Lorentz class (Theorems 5.31 and 5.38). For the compression of the multiplicative Hilbert matrix to a prime set of density $\theta < 1$ the spectrum decays root-exponentially; for the full sign average $s_n \sim 1/(2n \log n)$. The second question has a short answer at the level of S_2, where Helson's inequality is exactly a weak-factorization identity, and a gap between the exponents 0.174 and 1/2.

Four different spectral regimes appear below, and it helps to keep them apart.

- *Scale invariance.* When the companion kernel is asymptotically homogeneous at the critical line, the Mellin transform diagonalizes the model and the essential spectrum is an interval (Theorem 5.4).
- *Periodicity in logarithmic scale.* An oscillating prime density makes the model periodic, and Floquet theory gives bands (Proposition 5.7); for the explicit sets R_P the bands are absolutely continuous spectrum (Theorem 5.18).
- *A weaker singularity.* The Hilbert matrix of a prime set of density $\theta < 1$ has a kernel with singularity $t^{-\theta}$, which is integrable against $t\,\mathrm{d}t$; Taylor approximation then gives root-exponential decay (Theorem 5.44).
- *Many blocks.* The full sign average splits into one block for each squarefree q, each of rank one up to a trace-class error, and the asymptotics count the leading eigenvalues of the blocks (Theorem 5.45 and Proposition 5.46).

Restriction to a sparse prime set and sign averaging both simplify the matrix, but in different ways. Restriction removes primes and so weakens the singularity of a single kernel: few singular values survive. Averaging keeps every prime but cuts the matrix into infinitely many blocks, and then the number of blocks, not the smoothness of one kernel, sets the decay, which is only polynomial.

Chavan, Sahu and Sinha [27] study Helson matrices induced by measures and give Schatten criteria in terms of the entries $\alpha(mn)$; the criteria below are in terms of the measure. To our knowledge the following are new: the dictionary between the profile and the Schatten–Lorentz classes for every $p > 0$, with its pointwise corollaries and the separated-atom theorem, which extends to unbounded Helson operators; the exact asymptotics of the full sign average; the critical spectrum for general prime sets, including the band structure for oscillating densities, the spectral

type of these bands for the explicit sets R_P (Theorem 5.18), and the absolutely continuous spectrum for Chebotarev sets. The S_2 endpoint in the last section, where Helson's inequality gives bounded symbols exactly, is where chapter 7 starts.

5.1. The companion kernel

Throughout, R is a set of primes, $\mathcal{S}_R$ the multiplicative semigroup it generates, and $\zeta_R(w) = \prod_{p\in R}(1-p^{-w})^{-1}$. In this chapter a curve measure is a finite positive measure ν on $(0,\infty)$, shifted so that $u=0$ is the critical line:

$$\alpha_\nu(n) = \int_0^\infty n^{-1/2-u}\,\mathrm{d}\nu(u), \qquad M_{\nu,R} = \big(\alpha_\nu(mn)\big)_{m,n\in\mathcal{S}_R}.$$

For R all primes we write M_ν. With this shift, $M_{\nu,R}$ is the compression of the matrix $M(\alpha_{\nu,R})$ of chapter 4 to $\ell^2(\mathcal{S}_R)$: for $m,n\in\mathcal{S}_R$ the Q-part of mn is 1. Density of R is defined as in (2.1). Put $I_j = (e^{-j-1}, e^{-j}]$ for $j\ge 0$.

LEMMA 5.1. *Let $\mathcal{N}:\ell^2(\mathcal{S}_R)\to L^2(\nu)$, $\mathcal{N}e_n(u) = n^{-1/2-u}$, on finitely supported vectors. Then $M_{\nu,R} = \mathcal{N}^*\mathcal{N}$ as forms, and whenever either side is bounded, $M_{\nu,R}$ and the integral operator on $L^2(\nu)$ with kernel $\zeta_R(1+u+v)$ have the same nonzero spectrum with multiplicities.*

PROOF. $\langle \mathcal{N}e_n, \mathcal{N}e_m\rangle_{L^2(\nu)} = \int (mn)^{-1/2-u}\,\mathrm{d}\nu(u) = \alpha_\nu(mn)$, and $\mathcal{N}\mathcal{N}^*$ has kernel $\sum_{n\in\mathcal{S}_R} n^{-1-u-v} = \zeta_R(1+u+v)$. The nonzero spectra of $\mathcal{N}^*\mathcal{N}$ and $\mathcal{N}\mathcal{N}^*$ coincide. □

The kernel $\zeta_R(1+u+v)$ is a Hankel kernel in the additive variable $u+v$, singular like $(u+v)^{-\theta}$ at the origin. Every result in the first half of this chapter is a statement about this one kernel restricted to ν.

PROPOSITION 5.2 (Traces). *For every finite positive ν and integer $k\ge 1$,*

$$\operatorname{Tr} M_{\nu,R} = \int \zeta_R(1+2u)\,\mathrm{d}\nu(u),$$

$$\operatorname{Tr} M_{\nu,R}^k = \int \prod_{i=1}^{k} \zeta_R(1+u_i+u_{i+1})\,\mathrm{d}\nu(u_1)\cdots\mathrm{d}\nu(u_k),$$

with $u_{k+1}=u_1$, both sides possibly infinite. In particular $\|M_{\nu,R}\|_{S_2}^2 = \iint \zeta_R(1+u+v)^2\,\mathrm{d}\nu(u)\,\mathrm{d}\nu(v)$.

PROOF. Restrict ν to $[\varepsilon,\infty)$. Then $\mathcal{N}$ is Hilbert–Schmidt, the formula is the trace of the kth power of the companion, and the compressions to $n\le N$ increase to it. Monotone convergence as $N\to\infty$ and $\varepsilon\to 0$ gives the general case. □

COROLLARY 5.3. *Let R have density $\theta\in[0,1)$. The multiplicative Hilbert matrix restricted to $\mathcal{S}_R\setminus\{1\}$ is positive and trace class, with*

$$\operatorname{Tr} = \frac{1}{2}\int_0^\infty \big(\zeta_R(1+t)-1\big)\,\mathrm{d}t.$$

More generally, $\big(\Gamma(\kappa)(mn)^{-1/2}(\log mn)^{-\kappa}\big)$ on $\mathcal{S}_R\setminus\{1\}$ is trace class for $\kappa>\theta$, bounded and not compact for $\kappa=\theta>0$, and unbounded for $\kappa<\theta$.

PROOF. The matrix is $M_{\nu,R}$ for Lebesgue measure ν on $(0,\infty)$ with the index 1 removed, and Proposition 5.2 with $\zeta_R(1+2u)-1$ in place of $\zeta_R(1+2u)$ gives the trace, which is finite because $\zeta_R(1+t) \lesssim t^{-\theta}$ near 0 and $\zeta_R(1+t)-1 \le 2^{1-t}(\zeta(2)-1)$ for $t \ge 1$. The second family is $\nu = u^{\kappa-1}\,\mathrm{d}u$. Its block profile is $a_j \asymp e^{-j(\kappa-\theta)}$. The part of ν on $(1,\infty)$ has infinite mass, but once the index 1 is removed its companion kernel $\zeta_R(1+u+v)-1$ decays exponentially, so it contributes a trace-class term; for the part on $(0,1]$, the trace integral is bounded above by a constant times $\int_0^1 u^{\kappa-\theta-1}\,\mathrm{d}u$, finite when $\kappa > \theta$. At equality the block profile is bounded and does not tend to zero; below it the profile is unbounded. The remaining conclusions follow from Theorems 4.2 and 4.10. □

At the critical exponent the matrix is bounded but not compact. For the multiplicative Hilbert matrix the essential spectrum is $[0,\pi]$ [72], and the Mellin structure of the companion kernel suggests a universal answer.

PROBLEM 5 (A universal spectral constant). *Let the companion kernel of a positive Helson matrix grow like $c\,x^{-\kappa}$ at the boundary, with the fixed critical weight $u^{\kappa-1}\,\mathrm{d}u$. Is its critical essential spectrum always $[0, c\,\Gamma(\kappa/2)^2/\Gamma(\kappa)]$?*

The answer is yes under a literal asymptotic, by Theorem 5.4. Comparability alone does not imply this formula; Corollary 5.29 gives the precise upper-limit criterion.

THEOREM 5.4 (Critical essential spectrum). *Let $\kappa > 0$, let $h \ge 0$ be continuous on $(0,\infty)$ with $x^{\kappa}h(x) \to c$ as $x \to 0$, and let $\nu = g(u)\,\mathrm{d}u$ be a nonzero positive measure on $(0,\infty)$ with $u^{1-\kappa}g(u) \to b$ as $u \to 0$. Suppose that $\iint_{\max(u,v)\ge\delta} h(u+v)^2\,\mathrm{d}\nu(u)\,\mathrm{d}\nu(v) < \infty$ for every $\delta > 0$. Then the integral operator T on $L^2(\nu)$ with kernel $h(u+v)$ is bounded, and*

$$\sigma_{\mathrm{ess}}(T) = \Big[0,\ bc\,\frac{\Gamma(\kappa/2)^2}{\Gamma(\kappa)}\Big].$$

PROOF. Multiplication by $g^{1/2}$ identifies $L^2(\nu)$ with $L^2(\{g>0\},\mathrm{d}u)$. Extend the transformed operator by zero to $L^2(0,\infty)$; its kernel is $K(u,v) = g(u)^{1/2}h(u+v)g(v)^{1/2}$. Compare it with $K_0(u,v) = (uv)^{(\kappa-1)/2}(u+v)^{-\kappa}$, which is homogeneous of degree -1. The unitary map $(Uf)(t) = e^{t/2}f(e^t)$ from $L^2(0,\infty)$ to $L^2(\mathbb{R})$ carries K_0 to convolution with $C_\kappa(t) = (2\cosh(t/2))^{-\kappa}$, since

$$e^{(t+s)/2}K_0(e^t,e^s) = (2\cosh((t-s)/2))^{-\kappa}.$$

The Fourier transform therefore carries this operator to multiplication by

$$m(\xi) = \int_0^\infty t^{\kappa/2-1+i\xi}(1+t)^{-\kappa}\,\mathrm{d}t = \frac{|\Gamma(\kappa/2+i\xi)|^2}{\Gamma(\kappa)},$$

a continuous even function, decreasing in $|\xi|$, with maximum $m_0 = \Gamma(\kappa/2)^2/\Gamma(\kappa)$ at $\xi = 0$ and limit 0 at infinity. So K_0 is bounded with $\sigma(K_0) = \sigma_{\mathrm{ess}}(K_0) = [0,m_0]$. Let P_δ be multiplication by the indicator of $(0,\delta)$. The functions $f_n(u) = n^{-1/2}u^{-1/2+i\xi}\mathbf{1}_{(e^{-2n},e^{-n})}(u)$ are, up to $o(1)$ in norm, approximate eigenvectors of K_0 with eigenvalue $m(\xi)$. Indeed, Uf_n is a normalized plane wave on $(-2n,-n)$; for each fixed translation its boundary error tends to zero. Integrating against

$C_\kappa \in L^1(\mathbb{R})$ proves the assertion. These vectors tend weakly to zero and are eventually supported in $(0,\delta)$. Hence $\sigma_{\mathrm{ess}}(P_\delta K_0 P_\delta) = [0, m_0]$ for every $\delta > 0$.

Now $K = P_\delta K P_\delta + R_\delta$, where R_δ has kernel supported in $\{\max(u,v) \geq \delta\}$ and is Hilbert–Schmidt by hypothesis. On $(0,\delta)^2$ the hypotheses give $|K - bcK_0| \leq \varepsilon(\delta) K_0$ pointwise, with $\varepsilon(\delta) \to 0$. A kernel dominated by a positive kernel has at most its norm, so $\|P_\delta(K - bcK_0)P_\delta\| \leq \varepsilon(\delta) m_0$. In particular T is bounded. Now put $D = K - bcP_1K_0P_1$. For $\delta \leq 1$, the operator $D - P_\delta D P_\delta$ is Hilbert–Schmidt: its K part by hypothesis, and its model part because K_0 is square integrable on $(0,1)^2 \setminus (0,\delta)^2$. Since $\|P_\delta D P_\delta\| \leq \varepsilon(\delta) m_0 \to 0$, D is a norm limit of compact operators, hence compact: the operator is the boundary model $bcP_1K_0P_1$ modulo compacts. Hence its essential spectrum is $[0, bcm_0]$. Zero already belongs to the essential spectrum on $L^2(\nu)$: choose a set E of finite positive ν-measure bounded away from zero. The restriction of T to inputs supported in E is Hilbert–Schmidt by hypothesis. An orthonormal sequence in $L^2(E,\nu)$ is therefore a singular Weyl sequence at zero. The zero extension adds no essential spectrum, and $\sigma_{\mathrm{ess}}(T) = [0, bcm_0]$. □

For a prime set R with

$$c_R = \lim_{x \to 0} x^\theta \zeta_R(1+x) \quad \text{existing},$$

for instance when $\sum_{p \in R} p^{-1-t} = \theta \log(1/t) + C + o(1)$, take $\nu = u^{\theta-1}\,\mathrm{d}u$ and $h(x) = \zeta_R(1+x) - 1$, which decays exponentially. By Lemma 5.1 the matrix $\big(\Gamma(\theta)(mn)^{-1/2}(\log mn)^{-\theta}\big)$ on $\mathcal{S}_R \setminus \{1\}$ and this operator have the same nonzero spectrum, so

$$\sigma_{\mathrm{ess}}\Big(\big(\Gamma(\theta)(mn)^{-1/2}(\log mn)^{-\theta}\big)_{m,n \in \mathcal{S}_R \setminus \{1\}}\Big) = \Big[0,\, c_R \frac{\Gamma(\theta/2)^2}{\Gamma(\theta)}\Big].$$

For R all primes, $c_R = 1$, $\theta = 1$, and this is the interval $[0,\pi]$ of the multiplicative Hilbert matrix found by Perfekt and Pushnitski [72]; the proof above is the standard Mellin argument for kernels homogeneous of degree -1. The theorem applies verbatim to any positive family whose companion kernel and weight vary regularly at the boundary, and so answers Problem 5 in that setting.

Ma, Lu and Zu [58, Theorem 3.3] prove a sharper statement under stronger hypotheses. For kernels $w(x)b(x+y)w(y)$ whose normalized profiles $t^\kappa b(t)$ and $t^{(1-\kappa)/2}w(t)$ are C^2, have limits at 0 and at ∞, and satisfy $|(t\partial_t)^2 f(t)| \lesssim (1 + |\log t|)^{-2}$, they find the same essential spectrum, with a contribution from each end, and by a Mourre estimate they exclude singular continuous spectrum and show that each compact window inside the open bands contains only finitely many eigenvalues. For the Bergman forms with weights d_α on all primes this gives essential and absolutely continuous spectrum $[0, \Gamma(\alpha/2)^2/\Gamma(\alpha)]$, the case $\kappa = \alpha$ of the theorem above, together with finitely many simple eigenvalues above it, which occur exactly when α exceeds a threshold $\alpha_* \in (1,2)$ [58, Theorem 1.1]. Theorem 5.4 asks only for limits, which is what lets it reach every prime set for which c_R exists, where ζ_R need not be smooth in the required sense; by itself it gives no information on the finer spectral type. For the prime sets of algebraic number theory the boundary behaviour is smooth, and the spectral type can be determined.

Let $K/\mathbb{Q}$ be a finite Galois extension with group G and let $C \subseteq G$ be a conjugacy class. The *Chebotarev set* of (K, C) is $R = \{p \text{ unramified in } K : \mathrm{Frob}_p \in C\}$; it has density $\theta = |C|/|G|$. The primes $p \equiv a \bmod q$ with $\gcd(a, q) = 1$ are the case $K = \mathbb{Q}(e^{2\pi i/q})$, $C = \{a \bmod q\}$, $\theta = 1/\varphi(q)$. Write

$$A_R = \big(\Gamma(\theta)(mn)^{-1/2}(\log mn)^{-\theta}\big)_{m,n\in\mathcal{S}_R\setminus\{1\}}, \qquad m_\theta = \frac{\Gamma(\theta/2)^2}{\Gamma(\theta)}$$

for the critical matrix and the constant of Theorem 5.4 at $\kappa = \theta$.

Proposition 5.5 (Spectral type for Chebotarev sets). *Let R be a Chebotarev set of density θ, for instance the primes in an arithmetic progression. Then $c_R \in (0,\infty)$ exists, A_R is bounded with $\ker A_R = \{0\}$, $\sigma_{\mathrm{ess}}(A_R) = [0, c_R m_\theta]$, and the absolutely continuous part of A_R is unitarily equivalent to multiplication by λ on $L^2([0, c_R m_\theta], \mathrm{d}\lambda)$: its absolutely continuous spectrum is $[0, c_R m_\theta]$, of multiplicity one.*

Proof. *Smoothness at the boundary.* We show that $\log \zeta_R(1+x) = -\theta \log x + h(x)$ with h real analytic near $x = 0$; then $c_R = e^{h(0)}$. By column orthogonality of the irreducible characters χ of G, $\mathbf{1}_C = \theta \sum_\chi \overline{\chi(C)}\chi$. For each χ the Euler product of the Artin L-function gives $\log L(s,\chi) = \sum_{p \text{ unram.}} \chi(\mathrm{Frob}_p) p^{-s} + E_\chi(s)$, where E_χ collects the prime powers and the finitely many ramified factors and is analytic in $\mathrm{Re}\, s > 1/2$; likewise $\log \zeta_R(s)$ differs from $\sum_{p\in R} p^{-s}$ by a function analytic there. Hence $\log \zeta_R(s) = \theta \sum_\chi \overline{\chi(C)} \log L(s,\chi) + E(s)$ with E analytic in $\mathrm{Re}\, s > 1/2$. For $\chi = 1$, $\log \zeta(s) = -\log(s-1)$ plus a function analytic near $s = 1$. For $\chi \neq 1$, Brauer's induction theorem writes $\chi = \sum_i n_i \operatorname{Ind}_{H_i}^G \psi_i$ with $n_i \in \mathbb{Z}$ and one-dimensional ψ_i, so $L(s,\chi) = \prod_i L(s,\psi_i)^{n_i}$ with Hecke L-functions, which are holomorphic and nonzero at $s = 1$ except for a simple pole when ψ_i is trivial [63, Chapter VII]. By Frobenius reciprocity the order of $L(s,\chi)$ at $s = 1$ is $-\sum_i n_i \langle \psi_i, 1\rangle_{H_i} = -\langle \chi, 1\rangle_G = 0$. As $L(\sigma,\chi) \neq 0$ for $\sigma > 1$, $\log L(s,\chi)$ is analytic near $[1,\infty)$. For progressions this is just $L(1,\chi) \neq 0$ for the nonprincipal Dirichlet characters. Putting $s = 1 + x$ gives the claim.

Factorization. Let $\nu = u^{\theta-1}\,\mathrm{d}u$ on $(0,\infty)$ and $b(t) = \zeta_R(1+t) - 1$, so that $b(t) \le Ct^{-\theta}$ and, with $p_0 = \min R$, b and its derivatives are $O(p_0^{-t})$ as $t \to \infty$. As in the proof of Corollary 5.3, we use Lemma 5.1 for this infinite measure with the index 1 removed. The companion T with kernel $b(u+v)$ is bounded by Theorem 5.4; its off-corner Hilbert–Schmidt condition follows from the exponential decay of b. For $g \in L^2(\nu)$ supported in a compact subset of $(0,\infty)$, expansion of the kernel gives

$$\sum_{n\in\mathcal{S}_R\setminus\{1\}} \left| \int_0^\infty n^{-1/2-u} g(u)\,\mathrm{d}\nu(u) \right|^2 = \langle Tg, g\rangle \le \|T\|\,\|g\|^2_{L^2(\nu)}.$$

Thus the adjoint evaluation map extends boundedly from this dense subspace, and its adjoint $\mathcal{N} : \ell^2(\mathcal{S}_R \setminus \{1\}) \to L^2(\nu)$ is bounded. Moreover, $A_R = \mathcal{N}^*\mathcal{N}$ because $\int (mn)^{-1/2-u} u^{\theta-1}\,\mathrm{d}u = \Gamma(\theta)(mn)^{-1/2}(\log mn)^{-\theta}$, and $\mathcal{N}\mathcal{N}^*$ has kernel $b(u+v)$. If $\mathcal{N}c = 0$, the Dirichlet series $\sum_n c_n n^{-s}$ vanishes for $s > 1/2$, so $c = 0$; thus $\ker A_R = \{0\}$, and the polar decomposition of $\mathcal{N}$ makes A_R unitarily equivalent to the restriction of $\mathcal{N}\mathcal{N}^*$ to $(\ker \mathcal{N}\mathcal{N}^*)^\perp$. In particular the two operators have unitarily

equivalent absolutely continuous and singular continuous parts. Multiplication by $u^{(\theta-1)/2}$ carries $\mathcal{N}\mathcal{N}^*$ to the weighted Hankel operator on $L^2(0,\infty)$ with kernel $w(u)b(u+v)w(v)$, $w(t)=t^{(\theta-1)/2}$.

The spectral type. Fedele and Pushnitski [33, Theorem 1.3] treat exactly these operators, with $w(t)=t^{\alpha}$, $\alpha=(\theta-1)/2>-1/2$. If $\Phi(t)=t^{1+2\alpha}b(t)$ satisfies $(\Phi-a_0)^{(m)}=O(t^{\varepsilon-m})$ as $t\to 0$ and $(\Phi-a_\infty)^{(m)}=O(t^{-\varepsilon-m})$ as $t\to\infty$ for $m\le 2$ and some $\varepsilon>0$, the absolutely continuous part is that of multiplication by λ on $[0,\pi_\alpha a_0]$ and on $[0,\pi_\alpha a_\infty]$, with $\pi_\alpha=\Gamma(1/2+\alpha)^2/\Gamma(1+2\alpha)=m_\theta$; an interval of length zero drops out. Here $\Phi(t)=t^\theta b(t)=e^{h(t)}-t^\theta$, so $(\Phi-c_R)^{(m)}=O(t^{\theta-m})$ as $t\to 0$ for $m\le 2$, since $\theta\le 1$ and h is smooth, while Φ and its derivatives decay exponentially at infinity, so $a_\infty=0$. This gives the absolutely continuous part. The essential spectrum is Theorem 5.4, applied as in the paragraph after it: the Hilbert–Schmidt hypothesis holds because $b(u+v)^2\lesssim p_0^{-2u-2v}$ when $\max(u,v)\ge\delta$. □

The rest of the spectral type depends on the Mourre estimate of Ma, Lu and Zu, which has not yet been refereed.

Corollary 5.6 (Conditional on [58, Theorem 3.3]). *In Proposition 5.5, A_R has no singular continuous spectrum, and every compact subinterval of $(0,c_Rm_\theta)$ contains only finitely many eigenvalues of A_R, each of finite multiplicity.*

Proof. Apply [58, Theorem 3.3], quoted above, to the weighted Hankel operator of the proof of Proposition 5.5, with normalized profiles Φ and $t^{(1-\theta)/2}w(t)\equiv 1$. Both are C^2, with limits c_R and 0 for Φ. With $D=t\,\mathrm{d}/\,\mathrm{d}t$, $D^2e^h=(th'+t^2h''+t^2h'^2)e^h=O(t)$ and $D^2t^\theta=\theta^2t^\theta$, so $D^2\Phi=O(t^\theta)$ at 0, and $D^2\Phi$ decays exponentially at infinity; hence $|D^2\Phi(t)|\lesssim(1+|\log t|)^{-2}$. The conclusions transfer to A_R by the unitary equivalence in that proof. □

For R all primes this is the multiplicative Hilbert matrix on $n\ge 2$, and Proposition 5.5 recovers the absolutely continuous spectrum $[0,\pi]$ of Perfekt and Pushnitski [72]. Absence of embedded eigenvalues in the band follows under the explicit smoothness hypothesis of Theorem 5.22. Above-band eigenvalues depend on the prime set (Theorems 5.20 and 5.23). If only c_R exists, the derivative bounds on Φ can fail, and neither tool applies. If only the bounded remainder (2.1) is known, $\psi(L)=\log\bigl(x^\theta\zeta_R(1+x)\bigr)$, $L=\log(1/x)$, may oscillate without converging, and then the answer changes.

Proposition 5.7 (Oscillating densities). *Let R have density $\theta\in(0,1]$, and suppose that $\psi(L)-\psi_P(L)\to 0$ as $L\to\infty$ for a continuous P-periodic function ψ_P. Let T_P be the operator on $L^2(\mathbb{R})$ with kernel*

$$\Bigl(2\cosh\frac{s-t}{2}\Bigr)^{-\theta}e^{\psi_P(\Lambda(s,t))},\qquad \Lambda(s,t)=-\log(e^{-s}+e^{-t}).$$

Then the essential spectrum of the critical matrix

$$\bigl(\Gamma(\theta)(mn)^{-1/2}(\log mn)^{-\theta}\bigr)_{m,n\in\mathcal{S}_R\setminus\{1\}}$$

is $\sigma(T_P)$, the closure of the union over $k\in[0,2\pi)$ of the spectra of the compact fibre operators on $L^2(0,P)$ with kernels $\sum_{n\in\mathbb{Z}}K_P(s,t+nP)e^{ikn}$, where K_P is the kernel of T_P.

PROOF. In the variable $u = e^{-s}$ the companion operator of Lemma 5.1, with $\nu = u^{\theta-1}\,\mathrm{d}u$, becomes the operator on $L^2(\mathrm{d}s)$ with kernel $(2\cosh((s-t)/2))^{-\theta} e^{\tilde\psi(\Lambda(s,t))}$, where $x^\theta(\zeta_R(1+x)-1) = e^{\tilde\psi(\log(1/x))}$ and $\tilde\psi - \psi \to 0$. Since $\Lambda(s+P, t+P) = \Lambda(s,t)+P$, the operator T_P commutes with translation by P, and the Floquet–Bloch decomposition gives its spectrum as stated; the fibres are Hilbert–Schmidt because the kernel decays like $e^{-\theta|s-t|/2}$. As in the proof of Theorem 5.4, the part of the companion operator with $\min(s,t) \le s_0$ is Hilbert–Schmidt, and on $\{s, t > s_0\}$ its kernel differs from that of T_P by at most $\varepsilon(s_0)$ times the kernel of T_P, with $\varepsilon(s_0) \to 0$. So the essential spectrum is that of the compression of T_P to a half-line. The pieces of T_P coupling $(-\infty, s_0)$ with (s_0, ∞) are Hilbert–Schmidt, by the exponential decay, so the compressions to the two half-lines together have essential spectrum $\sigma(T_P)$; and Weyl sequences for T_P, truncated Bloch waves translated by multiples of P, can be placed in (s_0, ∞). Hence the compression to (s_0, ∞) has essential spectrum $\sigma(T_P)$ as well. □

When ψ_P is constant, the fibres are diagonalized by the Mellin transform, the bands of T_P fill $[0, c_R\Gamma(\theta/2)^2/\Gamma(\theta)]$, and Theorem 5.4 is recovered. Genuine oscillations occur for prime sets. If $R_P = \{p : \{\log\log p/P\} < \theta\}$, then by the prime number theorem ψ is asymptotically the P-periodic smoothing of the indicator of $\{y : \{y/P\} < \theta\}$ by the kernel e^{-e^z}, whose kth Fourier coefficient carries the factor $\Gamma(2\pi ik/P)$, and R_P has density θ with bounded remainder. Numerically, for $\theta = 1/2$ and $P = 8$, where ψ oscillates between ± 0.53, the two largest bands of T_P are $[3.137, 8.236]$ and $[0.359, 0.790]$, stable to four digits under refinement of the discretization; for $P = 12$ they are $[7.43, 10.80]$ and $[1.03, 1.46]$. The reduction to T_P is proved, and for $P = 8$ the next remark certifies a gap, so for densities with bounded remainder the critical essential spectrum need not be an interval. If "grow like" in Problem 5 is read as comparability, this answers it in the negative.

REMARK 5.8 (A certified gap). For $\theta = 1/2$ and $P = 8$ one gap can be certified. Let $c_R = \lim_{L\to\infty}(\log\zeta_R(1+e^{-L}) - L/2 - \psi_0(L))$, with ψ_0 the periodic profile above, normalized to mean zero. Truncating the Floquet fibres to 128×128 matrices on a mesh of 129 Bloch parameters, enclosing their entries in ball arithmetic, and bounding the truncation and mesh errors gives

$$\sigma_{\mathrm{ess}} \cap \Big(\frac{189}{100}e^{c_R}, \frac{211}{100}e^{c_R}\Big) = \varnothing$$

for the critical matrix of R_8. Since the essential spectrum contains 0 and points above this interval, it is not an interval. The certificate is published with the book's numerics (chapter A). It does not locate the band edges or exclude eigenvalues in the gap.

5.2. Spectral type for oscillating prime sets

For the explicit sets R_P, the bands just constructed are absolutely continuous spectrum: the matrix has no singular continuous spectrum, and its absolutely continuous multiplicity is half that of the full-line periodic model (Theorem 5.18). The compact comparison in Proposition 5.7 alone does not give these conclusions.

We need a rate for the arithmetic error, a trace-class comparison, and a resolvent estimate that controls singular spectrum.

5.2.1. The periodic model

Write K_P for the kernel of T_P in Proposition 5.7.

LEMMA 5.9 (Periodic model and decoupling). *Let T_P be the operator with kernel K_P on $L^2(\mathbb{R})$. Its singular continuous spectrum is empty. Its nonzero spectral measures are sums of absolutely continuous measures from nonconstant local analytic Floquet eigenvalue branches and atoms from constant branches. A possible spectral contribution at zero is an atom.*

Suppose, in addition, that

$$h_P(x) = x^{-\theta} e^{\psi_P(-\log x)} = \int_0^\infty e^{-x\lambda}\, \mathrm{d}\mu(\lambda)$$

for a positive locally finite measure μ. Write T_+ and T_- for the compressions of T_P to $(0,\infty)$ and $(-\infty,0)$, respectively. Then

$$T_P - (T_+ \oplus T_-) \in S_1.$$

Consequently their absolutely continuous parts satisfy

$$(T_P)_{\mathrm{ac}} \simeq (T_+)_{\mathrm{ac}} \oplus (T_-)_{\mathrm{ac}}, \qquad m_{\mathrm{ac}}(T_P;\lambda) = m_{\mathrm{ac}}(T_+;\lambda) + m_{\mathrm{ac}}(T_-;\lambda)$$

for almost every λ.

The Laplace representation hypothesis holds for every periodic profile obtained from the prime-set scaling limit in Proposition 5.7.

PROOF. The bounded periodic factor and $\big(2\cosh((s-t)/2)\big)^{-\theta} \le e^{-\theta|s-t|/2}$ give boundedness by the Schur test. Simultaneous translation by P preserves the kernel. Under the usual cell Floquet transform the fibres on $L^2(0,P)$ have kernels

$$F(k;s,t) = \sum_{n\in\mathbb{Z}} K_P(s,t+nP)e^{ikn}, \qquad 0 < s,t < P.$$

For s,t in this cell,

$$|K_P(s,t+nP)| \le Ce^{-\theta P(|n|-1)/2}.$$

The series therefore converges locally uniformly in Hilbert–Schmidt norm for $|\operatorname{Im} k| < \theta P/2$, as do all its k derivatives on smaller strips. Thus $F(k)$ is a compact analytic operator family and is self-adjoint for real k.

Every nonzero eigenvalue of a real fibre is isolated and has finite multiplicity. Local analytic spectral projections reduce its cluster to a finite-dimensional self-adjoint analytic family. The local Rellich theorem, in the form of [76, Theorem XII.13], gives analytic eigenvalue branches and analytic orthonormal eigenvectors on a sufficiently small real interval. The argument is local away from zero; no assertion that the branches continue through zero is needed.

On such an interval, the corresponding direct-integral subspace reduces to an orthogonal sum of multiplication operators by these analytic branches. For a nonconstant real analytic branch E, its critical points are discrete. Away from them it has a locally Lipschitz inverse, so the inverse image under E of every Lebesgue null set is null. Hence its multiplication operator has absolutely continuous spectral

measures. A constant branch contributes only an atom. A countable cover of the nonzero fibre spectrum by these local charts proves the assertion away from zero: such a cover exists since the parameter-energy plane is second countable. The fibre kernel contributes only to the spectral projection at zero, which is an atom. Equivalently, a continuous spectral measure supported on the singleton $\{0\}$ vanishes. This proves $\sigma_{\mathrm{sc}}(T_P) = \varnothing$.

We next prove trace-class decoupling. The unitary change of variables $u = e^{-s}$ carries T_P to the weighted Hankel kernel

$$(uv)^{(\theta-1)/2} h_P(u+v)$$

on $L^2(0,\infty;\mathrm{d}u)$. The cut $s = 0$ becomes $u = 1$. Periodicity and boundedness give $h_P(x) \le Cx^{-\theta}$, so its positive Laplace measure satisfies

$$\mu([0,L]) \le e\, h_P(1/L) \le C_1 L^{\theta} \qquad (L > 0).$$

In particular the measure has no atom at zero.

Fix $0 < \varepsilon < \theta$. Define operators into $L^2(\mu)$ by

$$(B_0 f)(\lambda) = \lambda^{-\varepsilon/2} \int_0^1 u^{(\theta-1)/2} e^{-\lambda u} f(u)\, \mathrm{d}u,$$
$$(B_\infty g)(\lambda) = \lambda^{\varepsilon/2} \int_1^\infty v^{(\theta-1)/2} e^{-\lambda v} g(v)\, \mathrm{d}v.$$

Their squared Hilbert–Schmidt norms are

$$\|B_0\|_{S_2}^2 = \int_0^\infty \lambda^{-\varepsilon} \int_0^1 u^{\theta-1} e^{-2\lambda u}\, \mathrm{d}u\, \mathrm{d}\mu(\lambda),$$
$$\|B_\infty\|_{S_2}^2 = \int_0^\infty \lambda^{\varepsilon} \int_1^\infty v^{\theta-1} e^{-2\lambda v}\, \mathrm{d}v\, \mathrm{d}\mu(\lambda).$$

For the first integral, the inner integral is bounded by a constant when $\lambda \le 1$ and by $C_\theta \lambda^{-\theta}$ when $\lambda \ge 1$. On the dyadic intervals $(2^{-j-1}, 2^{-j}]$ the resulting integrals are bounded by $C 2^{-j(\theta-\varepsilon)}$; on $[2^j, 2^{j+1})$ they are bounded by $C 2^{-j\varepsilon}$. Both sums converge.

For the second integral, when $\lambda \le 1$ the inner integral is bounded by $C_\theta \lambda^{-\theta}$, giving the summable bounds $C 2^{-j\varepsilon}$ on the small dyadic intervals. When $\lambda \ge 1$, the inequality $2\lambda v \ge \lambda + v$ for $\lambda, v \ge 1$ gives an inner bound $C_\theta e^{-\lambda}$. The measure growth estimate then makes the large dyadic sum converge as well.

Thus B_0 and B_∞ are Hilbert–Schmidt. On compactly supported test functions the Laplace representation and Fubini give

$$B_0^* B_\infty(u,v) = (uv)^{(\theta-1)/2} h_P(u+v), \qquad 0 < u < 1 < v.$$

By density this is the cross-cut block; products of two Hilbert–Schmidt operators are trace class. The opposite block is its adjoint, proving the stated trace-class difference. The unitary equivalence of absolutely continuous parts follows from the Kato–Rosenblum theorem [77, Theorem XI.8]; multiplicities add under orthogonal sums.

Finally, let R be a prime set from Proposition 5.7 and put $H(x) = \zeta_R(1+x) - 1$. It is the Laplace transform of the positive measure

$$\sum_{n \in \mathcal{S}_R \setminus \{1\}} n^{-1} \delta_{\log n}.$$

Set $a_j = e^{-jP}$ and $H_j(x) = a_j^\theta H(a_j x)$. The hypothesis $\psi(L) - \psi_P(L) \to 0$, with $\psi(L) = \log(x^\theta \zeta_R(1+x))$ and $L = \log(1/x)$, gives $H_j(x) \to h_P(x)$ locally uniformly for $x > 0$. Each H_j is the Laplace transform of a positive measure μ_j.

For every fixed $L > 0$, $\mu_j([0, L]) \le eH_j(1/L)$ is bounded uniformly in j. Diagonal compactness on bounded intervals gives a vaguely convergent subsequence with positive locally finite limit μ on $[0, \infty)$. For fixed $x > 0$ its Laplace tails satisfy

$$\int_L^\infty e^{-x\lambda} \, \mathrm{d}\mu_j(\lambda) \le e^{-xL/2} H_j(x/2),$$

uniformly in j. Passing to the limit first on bounded intervals and then letting $L \to \infty$ proves $h_P(x) = \int_{[0,\infty)} e^{-x\lambda} \, \mathrm{d}\mu(\lambda)$. The bound $h_P(x) \le Cx^{-\theta}$ as $x \to \infty$ excludes an atom at zero. This supplies the required representation and completes the proof. □

Remark 5.10. Lemma 5.9 concerns the periodic model alone. The trace-class identity gives only the *sum* of the two half-line multiplicities, and Kato–Rosenblum transfers no singular continuous information. For the explicit prime sets R_P, Theorem 5.18 below supplies the missing steps: no flat bands, a trace-class arithmetic comparison, the split of the multiplicity between the two half-lines, and a limiting absorption principle for the actual matrix.

5.2.2. The arithmetic comparison

Fix $P > 0$ and $0 < \theta < 1$. Put $\mathcal{A} = \{y \in \mathbb{R} : \{y/P\} < \theta\}$ and $R_P = \{p : \log\log p \in \mathcal{A}\}$, the prime set introduced after Proposition 5.7. For $\theta = 1$, R_P is the set of all primes, whose matrix is covered by Proposition 5.5. Write $R = R_P$, $\mathcal{S}_R$ for its multiplicative semigroup, $b(x) = \zeta_R(1+x) - 1$, and, with $x = e^{-L}$,

$$\psi(L) = \log\big(x^\theta \zeta_R(1+x)\big), \qquad \tilde{\psi}(L) = \log\big(x^\theta b(x)\big).$$

Let

$$G(L) = \int_{-\infty}^0 \mathbf{1}_{\mathcal{A}}(y)\big(e^{-e^{y-L}} - 1\big) \, \mathrm{d}y + \int_0^\infty \mathbf{1}_{\mathcal{A}}(y) e^{-e^{y-L}} \, \mathrm{d}y.$$

Then $G'(L) = \int_{\mathbb{R}} \mathbf{1}_{\mathcal{A}}(y) e^{y-L} e^{-e^{y-L}} \, \mathrm{d}y$ is the convolution of the P-periodic function $\mathbf{1}_{\mathcal{A}}$ with a probability density. It is smooth, bounded, P-periodic, and has mean θ. Hence $G(L) - \theta L$ is smooth and P-periodic. Throughout, $K_0(r) = (2\cosh(r/2))^{-\theta}$, $\Lambda(s,t) = -\log(e^{-s} + e^{-t})$, and C denotes constants that may change from line to line.

Lemma 5.11. *There is a constant c_2 such that $\psi_P = G - \theta L + c_2$ satisfies*

$$|\tilde{\psi}(L) - \psi_P(L)| + |\tilde{\psi}'(L) - \psi_P'(L)| \le Ce^{-\theta L} \qquad (L \ge 0).$$

In particular ψ_P is the continuous periodic profile of Proposition 5.7, and $h_P(x) = x^{-\theta} e^{\psi_P(\log(1/x))} = e^{c_2} e^{G(\log(1/x))}$.

PROOF. By the prime number theorem with the de la Vallée Poussin error term [62, Theorem 6.9], $E(t) = \pi(t) - \mathrm{li}(t)$ satisfies $|E(t)| \le Cte^{-c\sqrt{\log t}}$ for $t \ge 2$. The primes $p > e$ of R are those in the windows $[a_j, b_j)$, $a_j = \exp(e^{jP})$, $b_j = \exp(e^{jP+\theta P})$, $j \ge 0$. For a C^1 function f, Stieltjes integration by parts gives

$$\sum_{a_j \le p < b_j} f(p) - \int_{a_j}^{b_j} \frac{f(t)\,\mathrm{d}t}{\log t} = f(b_j)E(b_j^-) - f(a_j)E(a_j^-) - \int_{a_j}^{b_j} E(t)f'(t)\,\mathrm{d}t.$$

Take first $f_x(t) = t^{-1-x}$, $0 \le x \le 1$. Since $|f_x - f_0| \le t^{-1}\min(1, x\log t)$ and $|f_x' - f_0'| \le 2xt^{-2}(1+\log t)$, the sum over j of the right-hand sides is a function $\mathrm{Er}(x)$ with $|\mathrm{Er}(x) - \mathrm{Er}(0)| \le Cx$: the bound is Cx times

$$\sum_{j \ge 0} e^{jP+\theta P} e^{-ce^{jP/2}} + \int_e^\infty (1+\log t) e^{-c\sqrt{\log t}} \frac{\mathrm{d}t}{t} < \infty.$$

The substitution $t = \exp(e^y)$ turns the main term into

$$\begin{aligned}\int_e^\infty \mathbf{1}_{\mathcal{A}}(\log\log t)\frac{t^{-1-x}\,\mathrm{d}t}{\log t} &= \int_0^\infty \mathbf{1}_{\mathcal{A}}(y) e^{-e^{y-L}}\,\mathrm{d}y \\ &= G(L) - \int_{-\infty}^0 \mathbf{1}_{\mathcal{A}}(y)\bigl(e^{-e^{y-L}} - 1\bigr)\,\mathrm{d}y,\end{aligned}$$

and the last integral is $O(e^{-L})$ because $|e^{-a} - 1| \le a$. The primes $p < e$ and the prime powers contribute to $\log\zeta_R(1+x)$ a function that is C^1 on $[0,1]$. Hence $\log\zeta_R(1+x) = G(L) + c_2 + r(x)$ with $|r(x)| \le Cx$.

For the derivative, $\mathrm{d}/\,\mathrm{d}L = -x\,\mathrm{d}/\,\mathrm{d}x$. The same computation with $f(t) = x(\log t)t^{-1-x}$, for which $|f| \le xt^{-1}\log t$ and $|f'| \le 3xt^{-2}(1+\log t)$, gives error terms bounded by Cx and the main term $\int_0^\infty \mathbf{1}_{\mathcal{A}}(y)e^{y-L}e^{-e^{y-L}}\,\mathrm{d}y = G'(L) + O(e^{-L})$. The remaining terms are x times derivatives of C^1 functions. So $|xr'(x)| \le Cx$. This proves the estimate for $\psi = -\theta L + G + c_2 + r$ in place of $\tilde\psi$.

Finally $\tilde\psi - \psi = \log(1 - 1/\zeta_R(1+x))$. Since ψ is bounded on $L \ge 0$, $1/\zeta_R(1+x) = x^\theta e^{-\psi} \le Cx^\theta$, and $1/\zeta_R(1+x) < 1$ for $0 < x \le 1$. Moreover $\mathrm{d}(1/\zeta_R)/\,\mathrm{d}L = -(\psi' + \theta)/\zeta_R$ with ψ' bounded. Thus $\tilde\psi - \psi$ and its derivative are $O(x^\theta) = O(e^{-\theta L})$. □

From now on ψ_P is this profile, T_P is the operator of Lemma 5.9 built from it, and $F(k)$, $k \in \mathbb{R}$, are its Floquet fibres on $L^2(0,P)$. We use the Floquet transform $(Uf)(k,s) = \sum_n e^{-ikn} f(s+nP)$, $0 < s < P$, which is unitary from $L^2(\mathbb{R})$ onto $L^2([0,2\pi)\times(0,P);\mathrm{d}k\,\mathrm{d}s/2\pi)$ and carries T_P to multiplication by $F(k)$. By Lemma 5.9, F extends to a Hilbert–Schmidt analytic family on $|\mathrm{Im}\,k| < \theta P/2$, $F(k+2\pi) = F(k)$, and $F(-k) = \overline{F(k)}$ for real k because K_P is real.

LEMMA 5.12. *The Laplace measure μ of h_P from Lemma 5.9 satisfies $\mu(e^P B) = e^{\theta P}\mu(B)$ for Borel sets B, and $\mathrm{d}\mu(\lambda) = w(\lambda)\,\mathrm{d}\lambda$ with $0 \le w(\lambda) \le C\lambda^{\theta-1}$.*

PROOF. Since ψ_P is P-periodic, $h_P(e^{-P}x) = e^{\theta P}h_P(x)$. Comparing Laplace transforms, the image of μ under $\lambda \mapsto e^{-P}\lambda$ is $e^{\theta P}\mu$, which is the first claim. By Lemma 5.11,

$$-h_P'(x) = h_P(x)\frac{G'(\log(1/x))}{x} = h_P(x)\int_0^\infty e^{-x\rho}\mathbf{1}_{\mathcal{A}}(\log\rho)\,\mathrm{d}\rho.$$

The left side is the Laplace transform of the positive measure $\lambda\,\mathrm{d}\mu$, the right side that of the convolution of μ with $\mathbf{1}_{\mathcal{A}}(\log\rho)\,\mathrm{d}\rho$. By uniqueness of Laplace transforms these measures agree. The convolution has the density $\int_{[0,\lambda)}\mathbf{1}_{\mathcal{A}}(\log(\lambda-\lambda'))\,\mathrm{d}\mu(\lambda')\le\mu([0,\lambda])\le C\lambda^{\theta}$, by the growth bound in the proof of Lemma 5.9. Since μ has no atom at zero, the second claim follows. □

LEMMA 5.13. *T_P has no eigenvalue in $(0,\infty)$, and 0 is an accumulation point of $\sigma(T_P)\setminus\{0\}$.*

PROOF. Define $L:L^2(\mathbb{R})\to L^2(\mu)$ by $(Lf)(\lambda)=\int e^{-\theta s/2}e^{-\lambda e^{-s}}f(s)\,\mathrm{d}s$ on compactly supported f. The Laplace representation gives $\|Lf\|^2=\langle T_Pf,f\rangle$, since $e^{-\theta(s+t)/2}h_P(e^{-s}+e^{-t})=K_P(s,t)$. So L is bounded and $T_P=L^*L$. The operator LL^* has kernel $\int e^{-\theta s}e^{-(\lambda+\lambda')e^{-s}}\,\mathrm{d}s=\Gamma(\theta)(\lambda+\lambda')^{-\theta}$ on $L^2(\mu)$.

Let m be the image of μ under $\lambda\mapsto\sigma=\log\lambda$ and $\mathrm{d}\tilde\nu=e^{-\theta\sigma}\,\mathrm{d}m$. By Lemma 5.12, $\tilde\nu$ is P-periodic and has a bounded density $q(\sigma)=e^{(1-\theta)\sigma}w(e^{\sigma})$, which is not almost everywhere zero. The map $f\mapsto e^{\theta\sigma/2}f(e^{\sigma})$ is unitary from $L^2(\mu)$ onto $L^2(\tilde\nu)$ and carries LL^* to the operator with kernel $\Gamma(\theta)K_0(\sigma-\sigma')$. Put $g(y)=e^{\theta y/2}e^{-e^y}$ and $(Af)(\tau)=\int g(\tau-\sigma)f(\sigma)\,\mathrm{d}\tilde\nu(\sigma)$. Since $\int g(\tau-\sigma)g(\tau-\sigma')\,\mathrm{d}\tau=\Gamma(\theta)K_0(\sigma-\sigma')$, A^*A is that operator. So $A:L^2(\tilde\nu)\to L^2(\mathbb{R})$ is bounded. The operators L^*L, LL^*, A^*A, AA^* have the same nonzero eigenvalues and, off zero, unitarily equivalent parts.

The operator AA^* has the P-periodic kernel $\mathcal{K}(\tau,\tau')=\int g(\tau-\sigma)g(\tau'-\sigma)\,\mathrm{d}\tilde\nu(\sigma)$, which is at most $\|q\|_\infty\Gamma(\theta)K_0(\tau-\tau')$. Its Floquet fibres act on the basis $P^{-1/2}e^{i\xi_n\tau}$, $\xi_n=\xi_n(k)=(2\pi n+k)/P$, of the quasi-periodic functions. Since $\hat g(\eta)=\int e^{-i\eta y}g(y)\,\mathrm{d}y=\Gamma(\theta/2-i\eta)$, unfolding the periodic measure gives the matrix

$$M(k)_{n'n}=\frac{1}{P}\Gamma(\theta/2-i\xi_{n'})\hat\nu_{n'-n}\Gamma(\theta/2+i\xi_n),\qquad \hat\nu_j=\int_{[0,P)}e^{-2\pi ij\sigma/P}\,\mathrm{d}\tilde\nu(\sigma).$$

Here $|\hat\nu_j|\le\tilde\nu([0,P))$.

Suppose $AA^*f=\lambda_0 f$ with $\lambda_0>0$ and $f\neq0$. Then, on a set of real k of positive measure, there is $0\neq c\in\ell^2(\mathbb{Z})$ with $M(k)c=\lambda_0 c$. Put $d_n=\Gamma(\theta/2+i\xi_n)c_n$. Since $\Gamma(\theta/2+i\xi_n)$ decays exponentially in n, $d\in\ell^1$, and $d\neq0$ because $\lambda_0 c$ is a function of d. Writing $\mathfrak{m}(\xi)=\Gamma(\theta/2+i\xi)\Gamma(\theta/2-i\xi)$, the eigenvalue equation becomes

$$d_{n'}=\frac{\mathfrak{m}(\xi_{n'})}{\lambda_0 P}\sum_n\hat\nu_{n'-n}d_n.$$

Let $\omega_n=e^{-\pi^2|n|/P}$ and $e_n=d_n/\omega_n$. Define $X(k)$ by

$$X(k)_{n'n}=\frac{\mathfrak{m}(\xi_{n'}(k))\hat\nu_{n'-n}\omega_n}{\lambda_0 P\omega_{n'}}.$$

Then $e=X(k)e$, and $e\in\ell^2$ because $|d_{n'}|\le C|\mathfrak{m}(\xi_{n'})|$.

The function $\mathfrak{m}$ is meromorphic with poles only at $\xi=\pm i(\theta/2+j)$, $j\ge0$. Hence every entry of $X(k)$ is analytic on $\Omega=\mathbb{C}\setminus\mathcal{E}$, where $\mathcal{E}$ is the discrete set of k for which some $\xi_n(k)$ is a pole of $\mathfrak{m}$. Stirling's formula gives $|\mathfrak{m}(x+iy)|\le C|x|^{\theta-1}e^{-\pi|x|}$ for $|x|\ge1$, uniformly for y in compact sets. So $\sum_{n'}|\mathfrak{m}(\xi_{n'})/\omega_{n'}|^2$ and $\sum_n\omega_n^2$ are

locally bounded on Ω. Thus X is an analytic Hilbert–Schmidt-valued function on the connected open set Ω. By the analytic Fredholm theorem [78, Theorem VI.14], $1 - X(k)$ is either noninvertible for every $k \in \Omega$ or invertible off a discrete set. The first alternative holds, because $1 - X(k)$ has a kernel on a set of real k of positive measure.

Now take $Y = \theta/2+1/2+N$ with $N \in \mathbb{N}$ and $k_* = iPY$, so that $\xi_n(k_*) = x_n+iY$, $x_n = 2\pi n/P$, and $k_* \in \Omega$. By the reflection formula,

$$\mathfrak{m}(x + iY) = \frac{\pi}{\sin(\pi(\theta/2 - Y + ix))} \cdot \frac{\Gamma(w)}{\Gamma(w + 1 - \theta)}, \qquad w = \theta/2 + Y - ix.$$

Here $|\sin(\pi(\theta/2-Y+ix))|^2 = \sin^2(\pi(\theta/2-Y))+\sinh^2(\pi x) = \cosh^2(\pi x)$. Moreover $\Gamma(w)/\Gamma(w + 1 - \theta) = B(w, 1 - \theta)/\Gamma(1 - \theta)$, and $|B(w, 1 - \theta)| \le B(\operatorname{Re} w, 1 - \theta)$. Hence

$$|\mathfrak{m}(x + iY)| \le \frac{\pi r_Y}{\cosh(\pi x)}, \qquad r_Y = \frac{\Gamma(\theta/2 + Y)}{\Gamma(1 - \theta/2 + Y)} \longrightarrow 0 \quad (N \to \infty),$$

because $\theta < 1$. Consequently

$$\|X(k_*)\|_{S_2}^2 \le \frac{\tilde\nu([0, P))^2}{\lambda_0^2 P^2} \sum_{n'} \frac{4\pi^2 r_Y^2 e^{-4\pi^2|n'|/P}}{\omega_{n'}^2} \sum_n \omega_n^2 \le C_P\, r_Y^2,$$

which is less than one for large N. Then $1 - X(k_*)$ is invertible, a contradiction. So AA^*, and therefore T_P, has no eigenvalue in $(0, \infty)$.

For the second claim, let $c \in \ell^2$ be finitely supported and d as above. For real k, $\langle M(k)c, c\rangle = P^{-1} \int_{[0,P)} |\sum_n d_n e^{2\pi i n\sigma/P}|^2 q(\sigma)\, d\sigma$. A nonzero trigonometric polynomial vanishes only on a finite subset of a period, and q is positive on a set of positive measure. So $M(k)$ is injective on the infinite-dimensional space of finitely supported vectors, hence has infinite rank. Each $M(k)$ is compact and positive, so its eigenvalues accumulate at 0. Therefore 0 is an accumulation point of $\sigma(AA^*) \setminus \{0\} = \sigma(T_P) \setminus \{0\}$. □

5.2.3. Bands and propagation

Let $k_0 \in \mathbb{R}$ and $\lambda_0 > 0$. Choose $\varepsilon > 0$ so that the circle $|E - \lambda_0| = \varepsilon$ misses $\sigma(F(k_0))$ and lies in $\operatorname{Re} E > 0$. For k in a complex neighbourhood of k_0 its Riesz projection $\Pi(k)$ is analytic and of constant finite rank r. By Rellich's theorem [76, Theorem XII.13] there are a real interval $I \ni k_0$ and functions $E_1, \dots, E_r$ and $\varphi_1, \dots, \varphi_r$, analytic on a complex neighbourhood of I, such that for real $k \in I$ the $\varphi_j(k)$ are orthonormal, span $\operatorname{Ran}\Pi(k)$, and $F(k)\varphi_j = E_j\varphi_j$. Since $\det(F(k)|_{\operatorname{Ran}\Pi(k)} - E) = \prod_j (E_j(k) - E)$ on I, the identity persists for complex k near I, so the $E_j(k)$ are then all the eigenvalues of $F(k)$ inside the circle. We call (I, E_j, φ_j) a *chart* and the E_j with $E_j(k) = \lambda$ the *branches through* (k, λ). A branch constant on I would give T_P the eigenvalue λ_0, through the vectors with Floquet transform $a(k)\varphi_j(k)$; by Lemma 5.13, every branch is nonconstant.

Let $\mathcal{B}$ be the set of $\lambda > 0$ for which some real k has a branch through (k, λ) with zero derivative, or two branches through (k, λ) whose derivatives have opposite signs. Put $\mathcal{G} = (0, \infty) \setminus \mathcal{B}$. For $\lambda \in \mathcal{G}$ every branch through a point (k, λ) has a

nonzero derivative, and all branches through the same point have derivatives of one sign, the *velocity sign*. Let

$$N(\lambda) = \sum_{k\in[0,2\pi)} \dim\ker(F(k)-\lambda),$$

and let $N_\pm(\lambda)$ count only the points k of velocity sign $\pm$.

LEMMA 5.14. *(1) $\mathcal{B}\cap[\Lambda,\infty)$ is finite for every $\Lambda>0$.*

(2) Let $\lambda_0\in\mathcal{G}$. There are finitely many points $k_1,\dots,k_q\in[0,2\pi)$ with $\lambda_0\in\sigma(F(k_i))$, and charts $(I_i,E_{ij},\varphi_{ij})$, $I_i=(k_i-r,k_i+r)$, $j\le r_i$, with $E_{ij}(k_i)=\lambda_0$, $\varsigma_i E'_{ij}\ge c>0$ on I_i for a sign ς_i, and an open interval $\Delta\ni\lambda_0$, such that for real k

$$\mathbf{1}_\Delta(F(k)) = \sum_{i,j} \mathbf{1}_{I_i}(k)\mathbf{1}_\Delta(E_{ij}(k))\,|\varphi_{ij}(k)\rangle\langle\varphi_{ij}(k)|.$$

Consequently $\Delta\subset\mathcal{G}$, $N=\sum_i r_i$ and $N_\pm=\sum_{\varsigma_i=\pm} r_i$ on Δ, and $E_\Delta(T_P)$ is unitarily equivalent to the orthogonal sum of $N(\lambda_0)$ copies of multiplication by λ on $L^2(\Delta)$.

(3) $N_+=N_-=N/2$ on $\mathcal{G}$.

PROOF. (1) Fix $\Lambda'<\Lambda$ with $\Lambda'\notin\sigma(F(k))$ for k near a given k_0. A Riesz projection onto $\sigma(F(k))\cap[\Lambda',\infty)$ and Rellich's theorem give one chart on a neighbourhood of k_0 containing all eigenvalues $\ge\Lambda'$. Finitely many such charts, restricted to compact subintervals, cover $[0,2\pi)$. On each, finitely many nonconstant analytic branches have finitely many critical points, and two branches that are not identical meet at finitely many points; identical branches have equal derivatives. Hence $\mathcal{B}\cap[\Lambda,\infty)$ is contained in a finite set of values.

(2) Every point (k,λ_0) lies on nonconstant branches, so the k with $\lambda_0\in\sigma(F(k))$ are isolated, hence finitely many on the circle. Take charts at them. Choose r so small that every E_{ij} is strictly monotone on I_i with $|E'_{ij}|\ge c$, and the other eigenvalues of $F(k)$, $k\in I_i$, stay at distance η_0 from λ_0. By continuity of F, the spectra $\sigma(F(k))$ for k outside $\bigcup_i I_i$ also stay at distance $\eta_1>0$. Take $\Delta=(\lambda_0-\eta,\lambda_0+\eta)$ with $\eta<\min(\eta_0,\eta_1,cr)$. The displayed identity follows, and with it the formulas for N and $N_\pm$. In the Floquet representation $E_\Delta(T_P)$ is the orthogonal sum over (i,j) of multiplication by E_{ij} on $L^2(E_{ij}^{-1}(\Delta)\cap I_i)$. The change of variables $\lambda=E_{ij}(k)$ makes each summand multiplication by λ on $L^2(\Delta)$.

(3) For real k, $F(-k)=\overline{F(k)}$. So complex conjugation of eigenvectors maps the branches through (k,λ) to branches through $(-k,\lambda)$ and reverses the signs of their derivatives. □

By Lemma 5.9 and Lemma 5.13, T_P has purely absolutely continuous spectrum on $(0,\infty)$, and $\mathcal{B}$ carries no spectral measure. Hence the absolutely continuous subspace of T_P is $E_\mathcal{G}(T_P)L^2(\mathbb{R})=\mathcal{H}_\rightarrow\oplus\mathcal{H}_\leftarrow$, where $\mathcal{H}_\rightarrow$ (respectively $\mathcal{H}_\leftarrow$) consists of the f with $(Uf)(k)$ in the range of the spectral projection of $F(k)$ onto the points of $\sigma(F(k))\cap\mathcal{G}$ of velocity sign + (respectively −). These subspaces are orthogonal, since eigenspaces of $F(k)$ for distinct eigenvalues are, and they reduce T_P. By part (2) of Lemma 5.14, $T_P|_{\mathcal{H}_\rightarrow}$ has multiplicity $N_+(\lambda)$ on $\mathcal{G}$, and the *packets*, the

vectors with $(Uf)(k) = a(k)\varphi_{ij}(k)$, $a \in C_c^\infty$ supported in $I_i \cap E_{ij}^{-1}(\Delta)$, $\varsigma_i = +$, span a dense subspace of $\mathcal{H}_\to$, because $\mathcal{G}$ is covered by countably many of the intervals Δ.

LEMMA 5.15 (Limiting absorption). *Let $\lambda_0 \in \mathcal{G}$, $a > 0$ and $\rho(s) = e^{-a|s|}$. There is a complex neighbourhood $\mathcal{U}$ of λ_0 such that the function $z \mapsto \rho(T_P - z)^{-1}\rho$ on $\mathcal{U} \cap \{\operatorname{Im} z > 0\}$ extends analytically to $\mathcal{U}$.*

PROOF. In the Floquet representation the resolvent has the block kernel

$$\mathbf{1}_n (T_P - z)^{-1} \mathbf{1}_m \simeq G_{n-m}(z), \qquad G_\ell(z) = \frac{1}{2\pi}\int_0^{2\pi} e^{ik\ell}(F(k) - z)^{-1}\,\mathrm{d}k,$$

where $\mathbf{1}_n$ is the indicator of $[nP, (n+1)P)$, identified with $(0, P)$. The integrand is analytic in k near the real axis wherever $z \notin \sigma(F(k))$, and 2π-periodic. Use the data of part (2) of Lemma 5.14. Replace each segment $[k_i - r/2, k_i + r/2]$ of the circle by the three sides of the rectangle with that base and depth δ, lying in $\operatorname{Im} k < 0$ if $\varsigma_i = +$ and in $\operatorname{Im} k > 0$ if $\varsigma_i = -$. Call the new contour Γ.

For complex k near I_i the eigenvalues of $F(k)$ near λ_0 are the $E_{ij}(k)$, and $\operatorname{Im} E_{ij}(x \mp iy) = -y|E'_{ij}(x)| + O(y^2)$ when $\varsigma_i = \pm$. Fix δ so small that $\operatorname{Im} E_{ij}(k) \le -cy/2$ on the rectangles, $|E_{ij}(k) - E_{ij}(\operatorname{Re} k)| \le cr/8$ there, and $\delta < aP$. Let $\mathcal{U} = \{|\operatorname{Re} z - \lambda_0| < cr/8,\ |\operatorname{Im} z| < c\delta/4\}$, shrunk if needed below $\min(\eta_0, \eta_1)/2$. Then no $z \in \mathcal{U}$ is an eigenvalue of $F(k)$ for $k \in \Gamma$. On the real part of Γ this holds by the choice of η_0, η_1 and since $|E_{ij}(k) - \lambda_0| \ge cr/2$ for $r/2 \le |k - k_i| < r$. On the vertical sides $|E_{ij}(k) - \lambda_0| \ge cr/2 - cr/8$. On the bottom, or top, $\operatorname{Im} E_{ij}(k) \le -c\delta/2$. For $\operatorname{Im} z > 0$ no $z \in \mathcal{U}$ is an eigenvalue for k between the real axis and Γ, since there $\operatorname{Im} E_{ij}(k) \le 0$. By Cauchy's theorem $G_\ell(z)$ equals the same integral over Γ for $z \in \mathcal{U}$, $\operatorname{Im} z > 0$.

The integral over Γ is analytic in $z \in \mathcal{U}$. Its integrand is bounded: split $(F(k) - z)^{-1} = (F(k) - z)^{-1}\Pi(k) + (F(k) - z)^{-1}(1 - \Pi(k))$. The second part is jointly continuous, and the first is the resolvent of an $r_i \times r_i$ analytic matrix whose eigenvalues avoid z. Since $|e^{ik\ell}| \le e^{\delta|\ell|}$ on Γ, we get $\|G_\ell(z)\| \le Ce^{\delta|\ell|}$ uniformly on compact subsets of $\mathcal{U}$. As $\rho \le e^{aP}e^{-aP|n|}$ on the nth cell,

$$\sum_{n,m} \|\mathbf{1}_n \rho\, G_{n-m}(z)\, \rho \mathbf{1}_m\|^2 \le C \sum_{n,m} e^{-2aP(|n|+|m|)+2\delta|n-m|} < \infty,$$

because $\delta < aP$. So the block operator with these entries is bounded, depends analytically on $z \in \mathcal{U}$, and equals $\rho(T_P - z)^{-1}\rho$ for $\operatorname{Im} z > 0$. □

Let $\mathcal{A}_R$ be the operator on $L^2(\mathbb{R})$ with kernel $K_0(s-t)e^{\tilde\psi(\Lambda(s,t))}$; the function $e^{\tilde\psi}$ is bounded on $\mathbb{R}$. Let $\Pi_\pm$ be multiplication by $\mathbf{1}_{\mathbb{R}_\pm}$ and $T_\pm = \Pi_\pm T_P \Pi_\pm$ on $L^2(\mathbb{R}_\pm)$.

LEMMA 5.16. *(1) The matrix A_R is unitarily equivalent to $\mathcal{A}_R$ restricted to $(\ker \mathcal{A}_R)^\perp$.*

(2) $\mathcal{A}_R - (T_+ \oplus 0)$ is trace class.

(3) The kernels of $\Pi_+(\mathcal{A}_R - T_P)\Pi_+$, $\Pi_\pm \mathcal{A}_R \Pi_\mp$, $\Pi_- \mathcal{A}_R \Pi_-$ and $\Pi_\pm T_P \Pi_\mp$ are bounded by $Ce^{-\theta(|s|+|t|)/2}$.

PROOF. (1) For finitely supported c let

$$(\mathcal{N}c)(s) = \sum_{n\in\mathcal{S}_R\setminus\{1\}} c_n e^{-\theta s/2} n^{-1/2-e^{-s}}.$$

Then $\mathcal{N}^*\mathcal{N} = A_R$, because $\int (mn)^{-1/2-u} u^{\theta-1}\,\mathrm{d}u = \Gamma(\theta)(mn)^{-1/2}(\log mn)^{-\theta}$, and $\mathcal{N}\mathcal{N}^*$ has the kernel

$$e^{-\theta(s+t)/2} b(e^{-s}+e^{-t}) = K_0(s-t)e^{\tilde\psi(\Lambda)}.$$

The Schur test with $K_0 \le e^{-\theta|s-t|/2}$ bounds this kernel, so $\mathcal{N}$ is bounded and $\mathcal{N}\mathcal{N}^* = \mathcal{A}_R$. If $\mathcal{N}c = 0$, the Dirichlet series $\sum c_n n^{-s}$ vanishes for $s > 1/2$, so $c = 0$. Polar decomposition gives the claim.

(3) On $s,t>0$ the kernel is $K_0(s-t)\varepsilon(\Lambda)$ with $\varepsilon = e^{\tilde\psi} - e^{\psi_P}$. As $\Lambda \ge \min(s,t) - \log 2$, Lemma 5.11 gives $|\varepsilon(\Lambda)| + |\varepsilon'(\Lambda)| \le Ce^{-\theta\min(s,t)}$, and $\theta|s-t|/2 + \theta\min(s,t) = \theta(s+t)/2$. For $s>0>t$, $K_0(s-t) \le e^{-\theta(s+|t|)/2}$, and $e^{\tilde\psi}$, e^{ψ_P} are bounded. For $s,t<0$, $\Lambda \le \min(s,t)$, so $x = e^{-\Lambda} \ge e^{\max(|s|,|t|)}$ and $x^\theta b(x) \le Cx^{-2} \le Ce^{-|s|-|t|}$.

(2) Write $\mathcal{A}_R - (T_+ \oplus 0) = \Pi_+(\mathcal{A}_R - T_P)\Pi_+ + D$, where D collects the blocks of $\mathcal{A}_R$ meeting $\mathbb{R}_-$. Let ν_b be the measure $\sum_{n\in\mathcal{S}_R\setminus\{1\}} n^{-1}\delta_{\log n}$, so that $b(x) = \int e^{-x\lambda}\,\mathrm{d}\nu_b(\lambda)$, and $\nu_b([0,L]) \le eb(1/L) \le CL^\theta$; it vanishes for $L < \log 2$. The proof of trace-class decoupling in Lemma 5.9 used only such a growth bound. With ν_b in place of μ it shows that $\Pi_+\mathcal{A}_R\Pi_-$ and its adjoint are trace class. The operator $\Pi_-\mathcal{A}_R\Pi_-$ is B^*B with $B : L^2(\mathbb{R}_-) \to L^2(\nu_b)$, $(Bf)(\lambda) = \int_{-\infty}^0 e^{-\theta s/2} e^{-\lambda e^{-s}} f(s)\,\mathrm{d}s$, and $\|B\|_{S_2}^2 = \int_1^\infty u^{\theta-1} b(2u)\,\mathrm{d}u < \infty$. So $D \in S_1$.

For the first term, cover $(0,\infty)^2$ by the unit squares $Q_{nm} = [n,n+1)\times[m,m+1)$. A kernel κ on Q_{nm} that is C^1 in t satisfies $\kappa(s,t) = \kappa(s,m) + \int_m^t \partial_t\kappa(s,r)\,\mathrm{d}r$. This writes the operator as a rank-one operator plus the product of an operator with kernel $\partial_t\kappa$ and the Volterra operator with kernel $\mathbf{1}_{r<t}$. Therefore its trace norm is at most $\sup|\kappa| + \sup|\partial_t\kappa|$. Here $|K_0'| \le \theta K_0/2$ and $0 < \partial_t\Lambda < 1$. By (3) the sum over Q_{nm} is bounded by $C\sum_{n,m\ge0} e^{-\theta|n-m|/2-\theta\min(n,m)} < \infty$. □

LEMMA 5.17 (Half-line splitting). *$(T_+)_{\mathrm{ac}}$ is unitarily equivalent to $T_P|_{\mathcal{H}_\rightarrow}$ and $(T_-)_{\mathrm{ac}}$ to $T_P|_{\mathcal{H}_\leftarrow}$.*

PROOF. Let $T_D = T_+ \oplus T_-$. By Lemma 5.9, $T_D - T_P \in S_1$, so the wave operator $W = \lim_{t\to+\infty} e^{itT_D}e^{-itT_P}P_{\mathrm{ac}}(T_P)$, a strong limit, exists, is complete, and intertwines [77, Theorem XI.8]. Thus W maps $\mathcal{H}_\rightarrow \oplus \mathcal{H}_\leftarrow$ unitarily onto $\mathcal{H}_{\mathrm{ac}}(T_+) \oplus \mathcal{H}_{\mathrm{ac}}(T_-)$.

Let f be a packet in $\mathcal{H}_\rightarrow$, with $E = E_{ij}$, $\varphi = \varphi_{ij}$. On the nth cell,

$$(e^{-itT_P}f)(s+nP) = \frac{1}{2\pi}\int e^{i\Phi(k)} a(k)\varphi(k)(s)\,\mathrm{d}k, \qquad \Phi(k) = kn - tE(k).$$

For $n \le -1$ and $t \ge 0$, $|\Phi'| \ge |n| + ct$ and $|\Phi^{(2)}| + |\Phi^{(3)}| \le Ct$ on the support of a. Two integrations by parts bound the $L^2(0,P)$ norm by $C(1+|n|+t)^{-2}$. Hence $\|\Pi_- e^{-itT_P}f\|^2 \le C\sum_{n\ge1}(n+t)^{-4} \to 0$. As Π_- commutes with T_D,

$$\Pi_- W f = \lim_{t\to\infty} e^{itT_-}\Pi_- e^{-itT_P} f = 0.$$

By density $W\mathcal{H}_{\to} \subset L^2(\mathbb{R}_+)$, and likewise, with $n \geq 0$, $W\mathcal{H}_{\leftarrow} \subset L^2(\mathbb{R}_-)$. Since W is onto the orthogonal sum, $W\mathcal{H}_{\to} = \mathcal{H}_{\mathrm{ac}}(T_+)$ and $W\mathcal{H}_{\leftarrow} = \mathcal{H}_{\mathrm{ac}}(T_-)$. □

THEOREM 5.18 (Spectral type for the explicit prime sets). *Let $0 < \theta < 1$, $P > 0$ and $R = R_P$, and let A_R be the critical matrix*

$$A_R = \left(\frac{\Gamma(\theta)}{\sqrt{mn}(\log mn)^{\theta}} \right)_{m,n \in \mathcal{S}_R \setminus \{1\}}.$$

(1) The singular continuous spectrum of A_R is empty. Its eigenvalues in $(0,\infty)$ can accumulate only at 0 and at points of the set $\mathcal{B}$, which is finite on every $[\Lambda,\infty)$, $\Lambda > 0$.
(2) The absolutely continuous part of A_R is unitarily equivalent to $T_P|_{\mathcal{H}_{\to}}$. Hence $\sigma_{\mathrm{ac}}(A_R) = \sigma(T_P) = \sigma_{\mathrm{ess}}(A_R)$: every band of T_P is absolutely continuous spectrum, and there are no flat bands.
(3) For $\lambda \in (0,\infty) \setminus \mathcal{B}$ the number $N(\lambda)$ is finite and even, and the absolutely continuous multiplicity of A_R is $N(\lambda)/2$, half the multiplicity of T_P.

PROOF. (2) and (3). By parts (1) and (2) of Lemma 5.16 and the Kato–Rosenblum theorem [77, Theorem XI.8], $(A_R)_{\mathrm{ac}} \simeq (\mathcal{A}_R)_{\mathrm{ac}} \simeq (T_+)_{\mathrm{ac}}$, which is $T_P|_{\mathcal{H}_{\to}}$ by Lemma 5.17. Its multiplicity is $N_+ = N/2$ on $\mathcal{G}$ by Lemma 5.14. Every positive point of $\sigma(T_P)$ lies on a nonconstant branch, so it is a limit of points of $\mathcal{G}$ with $N > 0$; together with Lemma 5.13 this gives $\sigma_{\mathrm{ac}}(A_R) = \sigma(T_P)$. The equality with $\sigma_{\mathrm{ess}}(A_R)$ is Proposition 5.7.

(1) Let $\mathcal{H} = L^2(\mathbb{R}) \oplus L^2(\mathbb{R}_-)$ and $H = \mathcal{A}_R \oplus T_-$. Regroup $\mathcal{H} = L^2(\mathbb{R}_+) \oplus L^2(\mathbb{R}_-) \oplus L^2(\mathbb{R}_-)'$, where the prime marks the second summand, and identify $L^2(\mathbb{R}_+) \oplus L^2(\mathbb{R}_-)'$ with $L^2(\mathbb{R})$. Let H_0 act as T_P on this copy of $L^2(\mathbb{R})$ and as 0 on the remaining $L^2(\mathbb{R}_-)$. The blocks of $V = H - H_0$ are $\Pi_+(\mathcal{A}_R - T_P)\Pi_+$, the blocks of $\mathcal{A}_R$ meeting $\mathbb{R}_-$, and minus the blocks $\Pi_\pm T_P \Pi_\mp$ coupling $L^2(\mathbb{R}_+)$ with $L^2(\mathbb{R}_-)'$. By part (3) of Lemma 5.16, with $\rho(s) = e^{-\theta|s|/8}$ on each summand, $V = \rho V_1 \rho$, where V_1 has kernel bounded by $Ce^{-3\theta(|s|+|t|)/8}$ and so is Hilbert–Schmidt.

Let $\lambda_0 \in \mathcal{G}$ and let $\mathcal{U}$ be given by Lemma 5.15 with $a = \theta/8$. Then $Q_0(z) = \rho(H_0 - z)^{-1}\rho$, which is $\rho(T_P - z)^{-1}\rho \oplus (-z^{-1}\rho^2)$, is analytic on $\Omega = \{\operatorname{Im} z > 0\} \cup \mathcal{U}$. The resolvent identity gives $(1 + Q_0(z)V_1)\rho(H - z)^{-1}\rho = Q_0(z)$. For $\operatorname{Im} z > 0$ the operator $1 + Q_0V_1$ is injective: if $\chi = -\rho(H_0 - z)^{-1}\rho V_1 \chi$, then $u = (H_0 - z)^{-1}\rho V_1 \chi$ satisfies $(H - z)u = 0$, so $u = 0$ and $\chi = 0$. As Q_0V_1 is compact, the analytic Fredholm theorem shows that $(1 + Q_0V_1)^{-1}$ is meromorphic on Ω, with a discrete set $\mathcal{N}$ of poles. On a compact interval $J \subset (\mathcal{U} \cap \mathbb{R}) \setminus \mathcal{N}$, $\rho(H - \lambda - i\eta)^{-1}\rho$ is bounded for $\lambda \in J$ and $0 < \eta < 1$. By Stone's formula, every $g = \rho f$ has a spectral measure with bounded density on the interior of J, and it has no atoms there. Since ρL^2 is dense, H is purely absolutely continuous on $(\mathcal{U} \cap \mathbb{R}) \setminus \mathcal{N}$. On a smaller closed interval around λ_0, $\mathcal{N}$ is finite.

Thus every point of $\mathcal{G}$ has a neighbourhood free of singular continuous spectrum of H and containing finitely many eigenvalues. Since $H \geq 0$ and $\{0\} \cup \mathcal{B}$ is countable, H has no singular continuous spectrum. Its eigenvalues in $(0,\infty)$ accumulate only at $\{0\} \cup \mathcal{B}$. The same holds for the summand $\mathcal{A}_R$, and by part (1) of Lemma 5.16 for A_R. □

REMARK 5.19. The conclusion applies to the explicit sets R_P. The proof uses two properties of these sets: the exponential rate of Lemma 5.11, which gives both the trace-class comparison and the exponential localisation needed in Lemma 5.15, and the absolutely continuous Laplace measure of Lemma 5.12. For a prime set satisfying only the $o(1)$ hypothesis of Proposition 5.7, neither step is available, and the question remains open for that class.

5.3. Eigenvalues of critical prime-set matrices

For a prime set R, write $\mathcal{S}_R$ for its multiplicative semigroup and $\zeta_R(s) = \prod_{p\in R}(1-p^{-s})^{-1}$. Assume

$$t^\theta \zeta_R(1+t) \longrightarrow c_R \in (0,\infty) \qquad (t \downarrow 0), \qquad 0<\theta<1. \tag{5.1}$$

Consider the positive critical matrix

$$A_R(m,n) = \frac{\Gamma(\theta)}{\sqrt{mn}(\log(mn))^\theta}, \qquad m,n \in \mathcal{S}_R \setminus \{1\},$$

with band edge $L_R = c_R\Gamma(\theta/2)^2/\Gamma(\theta)$.

Eigenvalues above the band depend on R, not on θ or on the smoothness of ζ_R: for the primes $p \equiv 1 \pmod 4$ there are none (Theorem 5.23), while finite deletions from any regular set produce any prescribed number (Theorem 5.20). Under the smoothness hypothesis (5.3) below, which holds for every arithmetic progression and its finite modifications, the band contains no eigenvalues at all (Theorem 5.22).

Put $a = \theta/2$, $\nu = u^{\theta-1}\,\mathrm{d}u$ on $(0,\infty)$, $b(t) = \zeta_R(1+t)-1$ and, on finite sequences,

$$(Nc)(u) = \sum_{n\in\mathcal{S}_R\setminus\{1\}} c_n n^{-1/2-u}.$$

Gamma integration gives $N^*N = A_R$ as forms, and $T = NN^*$ is the integral operator on $L^2(\nu)$ with kernel $b(u+v)$. Theorem 5.4, applied with $\kappa = \theta$ and the fixed critical weight, shows that N is bounded and

$$\sigma_{\mathrm{ess}}(A_R) = [0, L_R]. \tag{5.2}$$

Here $b(t) \le C2^{-t}$ for $t \ge 1$, since every $n \ge 2$ has $n^{-1-t} \le 2^{1-t}n^{-2}$. If $Nc = 0$, then $\sum c_n n^{-s}$ vanishes for real $s > 1/2$, so $c = 0$; hence $\ker A_R = \{0\}$, and $E > 0$ is an eigenvalue of A_R if and only if it is one of T, with $c \mapsto Nc$ mapping eigenvectors to eigenfunctions.

We will use the following regularity condition on $\Phi(t) = t^\theta\zeta_R(1+t)$:

$$\Phi \text{ extends to a } C^2 \text{ function on } [0,1] \text{ with } \Phi(0) = c_R > 0. \tag{5.3}$$

THEOREM 5.20. *Let R_0 satisfy (5.1). For every integer $k \ge 1$ and every prime $p \in R_0$, there is a finite set $F \subset R_0\setminus\{p\}$ such that $R = R_0\setminus F$ has the same exponent θ and A_R has at least k eigenvalues, counted with multiplicity, strictly above L_R. Finite deletions preserve hypothesis (5.3). More explicitly, any positive $k\times k$ principal block B that remains after deletion gives at least k above-band eigenvalues whenever $\lambda_{\min}(B) > L_R$; for one retained prime it suffices that $\Gamma(\theta)/(p(2\log p)^\theta) > L_R$.*

PROOF. For a finite $F \subset R_0$,

$$\zeta_{R_0\setminus F}(1+t) = \zeta_{R_0}(1+t)\prod_{q\in F}(1-q^{-1-t}), \qquad c_{R_0\setminus F} = c_{R_0}\prod_{q\in F}(1-1/q). \quad (5.4)$$

Each finite Euler-factor product is analytic and nonzero near $[0,1]$, which proves preservation of θ and of (5.3). Moreover, $\sum_{q\in R_0} 1/q = \infty$. Otherwise

$$\log\zeta_{R_0}(1+t) = \sum_{q\in R_0} q^{-1-t} + O\Big(\sum_{q\in R_0} q^{-2}\Big)$$

would stay bounded for $t > 0$, contradicting (5.1). Consequently the products in (5.4) can be made arbitrarily small while avoiding the one prime p.

Keep the indices $p, p^2, \dots, p^k$. Their principal block B_k has entries

$$B_k(i,j) = \frac{\Gamma(\theta)}{p^{(i+j)/2}((i+j)\log p)^\theta}, \qquad 1 \le i,j \le k.$$

This block is independent of every deletion avoiding p and is strictly positive definite. Indeed, for $c \in \mathbb{C}^k$,

$$\langle B_k c, c\rangle = \int_0^\infty u^{\theta-1}\Big|\sum_{j=1}^k c_j p^{-j/2} e^{-j(\log p)u}\Big|^2 \,\mathrm{d}u.$$

If this vanishes, the polynomial $\sum_{j=1}^k c_j p^{-j/2}x^j$ vanishes on $0 < x < 1$, so all coefficients vanish. Put $\eta_k = \lambda_{\min}(B_k) > 0$. Choose F avoiding p so that

$$c_{R_0}\prod_{q\in F}(1-1/q)\frac{\Gamma(\theta/2)^2}{\Gamma(\theta)} < \eta_k.$$

For $R = R_0 \setminus F$, the k-dimensional coordinate space of these retained powers has every nonzero Rayleigh quotient at least $\eta_k > L_R$. Hence the spectral projection $\mathbf{1}_{(L_R,\infty)}(A_R)$ has dimension at least k: otherwise this coordinate space would contain a nonzero vector in its orthogonal complement, whose Rayleigh quotient would be at most L_R. By (5.2), spectrum above L_R consists of isolated eigenvalues of finite multiplicity. The same argument proves the block criterion, and the single index p gives the last condition. □

For the primes in a progression $p \equiv r \pmod q$, $\gcd(r,q) = 1$, one has $\theta = 1/\varphi(q)$ and $\log\zeta_R(s) = \theta\sum_\chi \overline{\chi(r)}\log L(s,\chi) + E(s)$ with E analytic in $\operatorname{Re} s > 1/2$; since $(s-1)\zeta(s)$ and $L(s,\chi)$, $\chi \ne \chi_0$, are analytic and nonzero near $[1,2]$, Φ is real analytic on a neighbourhood of $[0,1]$.

LEMMA 5.21. *Let $0 < \theta \le 1$, $a = \theta/2$ and $0 < \lambda \le \Gamma(a)^2/\Gamma(\theta)$. If $-1 < \operatorname{Re} x < \theta$, $x \ne 0$, and $\Gamma(x)\Gamma(\theta-x)/\Gamma(\theta) = \lambda$, then $\operatorname{Re} x = a$.*

PROOF. Put $w = x - a$ and $G(w) = \Gamma(a+w)\Gamma(a-w)/\Gamma(\theta)$, an even function of w. The Weierstrass product for $1/\Gamma$ gives, with $m = \Gamma(a)^2/\Gamma(\theta)$,

$$G(w) = m\prod_{k\ge 0}\frac{(a+k)^2}{(a+k-w)(a+k+w)}.$$

For real w with $0 < |w| < a$ every factor exceeds 1, so $G(w) > m \geq \lambda$; for $a < |w| < 1 + a$ only the factor $k = 0$ is negative, so $G(w) < 0$. Now let $w = \alpha + i\beta$ with $\beta > 0$ and, by evenness and conjugation symmetry, $0 < \alpha < 1 + a$. Writing $\arg(y + i\beta) \in (0, \pi)$, the kth factor has argument

$$\delta_k = \arg(a + k - \alpha + i\beta) - \arg(a + k + \alpha + i\beta) = \int_{a+k-\alpha}^{a+k+\alpha} \frac{\beta \, \mathrm{d}y}{y^2 + \beta^2} \in (0, \pi).$$

Since $\delta_k = O(k^{-2})$, the sum $S = \sum_{k \geq 0} \delta_k$ is an argument of $G(w)$. Put $\varphi(y) = \arg(y + i\beta)$, so φ decreases from π to 0. Enlarging each integration interval from radius $\alpha < 1 + a$ to radius $1 + a$ gives

$$\begin{aligned} 0 < S &\leq \sum_{k \geq 0} \big(\varphi(k - 1) - \varphi(k + 1 + 2a)\big) \\ &= \varphi(-1) + \varphi(0) + \sum_{k \geq 0} \big(\varphi(k + 1) - \varphi(k + 1 + 2a)\big). \end{aligned}$$

The first part telescopes. The intervals $[k + 1, k + 1 + 2a]$ have disjoint interiors because $2a = \theta \leq 1$. Their contribution is at most $\varphi(1)$. As $\varphi(-1) + \varphi(1) = \pi$ and $\varphi(0) = \pi/2$, we obtain $0 < S \leq 3\pi/2 < 2\pi$. Hence $G(w)$ is not positive real, so $G(w) = \lambda$ forces $\operatorname{Re} w = 0$. □

THEOREM 5.22. *Let R satisfy (5.1) and (5.3). Then A_R has no eigenvalue in $[0, L_R]$. In particular the band contains no embedded eigenvalues, and every eigenvalue of A_R is one of the isolated eigenvalues above L_R.*

PROOF. The argument extends that of Perfekt and Pushnitski [72, Theorem 2.4], which treats $\theta = 1$, to the weight $u^{\theta-1}$; the new ingredients are Lemma 5.21, a bootstrap in place of their C^2 hypothesis on the kernel (here $b(t) - c_R t^{-\theta} = -1 + O(t^{1-\theta})$ is not C^2 at 0), and a weighted Hardy inequality in the last step. Zero is not an eigenvalue. Suppose that $Tf = Ef$ with $0 < E \leq L_R$, $f \in L^2(\nu)$ and $f \neq 0$. As the kernel of T is real, we may take f real. Constants C may change from line to line.

Kernel bounds. Let $\tilde{b}(t) = b(t) - c_R t^{-\theta} = t^{-\theta}(\Phi(t) - c_R) - 1$ on $(0, 1]$. By (5.3), the derivatives of $t^{-\theta}$, and the exponential bounds $|b^{(j)}(t)| \leq C 2^{-t}$ for $t \geq 1$, $j \leq 2$,

$$\begin{gathered} b(t) \leq C t^{-\theta} 2^{-t}, \qquad |b'(t)| \leq C t^{-1-\theta}, \\ |\tilde{b}(t)| \leq \frac{C}{1 + t^{\theta}}, \qquad |\tilde{b}'(t)| \leq \frac{C t^{-\theta}}{1 + t}, \qquad |\tilde{b}''(t)| \leq \frac{C t^{-1-\theta}}{1 + t} \end{gathered} \tag{5.5}$$

for all $t > 0$.

A priori bounds. The integral of $(t + s)^{-2\theta}$ against ν equals $C t^{-\theta}$, so the Cauchy–Schwarz inequality gives

$$|f(t)| \leq E^{-1} \|b(t + \cdot)\|_{L^2(\nu)} \|f\| \leq C t^{-a} 2^{-t}. \tag{5.6}$$

In particular $f \in L^1(\nu)$. Differentiation under the integral sign is legitimate for $t > 0$, so f is smooth and $Ef'(t) = \int b'(t + s) f(s) \, \mathrm{d}\nu(s)$.

The Mellin equation. Let T_0 have kernel $c_R(u+v)^{-\theta}$ on $L^2(\nu)$ and put

$$g(t)=\int_0^\infty \tilde b(t+s)f(s)\,\mathrm{d}\nu(s),\qquad \text{so that}\qquad T_0f-Ef=-g.$$

By (5.5) and (5.6), g is bounded, $g=O(t^{-\theta})$, $g'=O(t^{-a})$ as $t\to0$, $g'=O(t^{-1-\theta})$ as $t\to\infty$, and $g''=O(t^{-2-\theta})$ as $t\to\infty$. For the Mellin transform $\mathcal{M}f(x)=\int_0^\infty s^{x-1}f(s)\,\mathrm{d}s$, analytic in $S_+=\{a<\operatorname{Re}x<\theta\}$ by (5.6), Tonelli's theorem and $\int_0^\infty t^{x-1}(t+s)^{-\theta}\,\mathrm{d}t=s^{x-\theta}B(x,\theta-x)$ give

$$\big(c_RB(x,\theta-x)-E\big)\mathcal{M}f(x)=-\mathcal{M}g(x),\qquad x\in S_+. \tag{5.7}$$

Integrating by parts twice, with vanishing boundary terms by the bounds on g,

$$\mathcal{M}g(x)=\frac{1}{x(x+1)}\int_0^\infty s^{x+1}g''(s)\,\mathrm{d}s,\qquad 0<\operatorname{Re}x<\theta. \tag{5.8}$$

Suppose that $|g''(t)|\le Ct^{-\gamma}$ on $(0,1]$. Then the integral in (5.8) converges absolutely, locally uniformly, for $\gamma-2<\operatorname{Re}x<\theta$, so $\mathcal{M}g$ continues meromorphically to that strip with at most simple poles at 0 and -1, and $|\mathcal{M}g(\sigma+i\tau)|\le C(1+\tau^2)^{-1}$ for $|\tau|\ge1$, uniformly for σ in compact subsets.

Let $\Sigma=\{\max(\gamma-2,-1)<\operatorname{Re}x<\theta\}$ and $H=-\mathcal{M}g/(c_RB(x,\theta-x)-E)$, meromorphic in Σ. By Lemma 5.21 with $\lambda=E/c_R\le\Gamma(a)^2/\Gamma(\theta)$, the denominator vanishes in Σ only on the line $\operatorname{Re}x=a$; at $x=0$ the simple pole of $B(x,\theta-x)$ cancels the possible pole of $\mathcal{M}g$. So $H=\mathcal{M}f$ on S_+ by (5.7), and the only possible poles of H in Σ lie on $\operatorname{Re}x=a$. By Plancherel, $\int|\mathcal{M}f(\sigma+i\tau)|^2\,\mathrm{d}\tau=2\pi\int_0^\infty|f(s)|^2s^{2\sigma-1}\,\mathrm{d}s$ is bounded uniformly for $a<\sigma<\theta$, using $s^{2\sigma-1}\le s^{\theta-1}$ on $(0,1]$ and (5.6) on $[1,\infty)$. A pole of order $k\ge1$ at $a+i\tau_0$ would make $\int_{|\tau-\tau_0|<\eta}|H(a+\eta+i\tau)|^2\,\mathrm{d}\tau\ge c\eta^{-1}$ for small $\eta>0$, which is impossible. Thus H is analytic in Σ. As $|\tau|\to\infty$, Stirling's formula gives $B(\sigma+i\tau,\theta-\sigma-i\tau)\to0$ locally uniformly in σ, so $|H(\sigma+i\tau)|\le C(1+\tau^2)^{-1}$ uniformly on closed substrips of Σ. Mellin inversion on a line $\operatorname{Re}x=\sigma_1\in(a,\theta)$ and a shift of contour therefore give

$$|f(s)|\le C_\sigma s^{-\sigma}\quad(0<s\le1),\qquad \text{for every }\sigma\in\Sigma\cap\mathbb{R}. \tag{5.9}$$

Bootstrap. By (5.5) and (5.6), $|g''(t)|\le C\int_0^\infty(t+s)^{-1-\theta}s^{a-1}\,\mathrm{d}s+C\le Ct^{-1-a}$ on $(0,1]$. With $\gamma=1+a$, (5.9) holds for all $\sigma>a-1$. Fix $\sigma_0\in(a-1,0)$; then $|g''(t)|\le C\int_0^\infty(t+s)^{-1-\theta}s^{\theta-\sigma_0-1}\,\mathrm{d}s+C\le Ct^{-1-\sigma_0}$ on $(0,1]$. With $\gamma=1+\sigma_0<1$, $\Sigma=\{-1<\operatorname{Re}x<\theta\}$. Taking $\sigma=a/2-1$ in (5.9), and then using $Ef'(t)=\int b'(t+s)f(s)\,\mathrm{d}\nu(s)$ with (5.5),

$$f(s)=O(s^{1-a/2}),\qquad f'(s)=O(s^{-a/2})\qquad(s\to0), \tag{5.10}$$

while f and f' decay exponentially as $s\to\infty$.

The commutator argument. Put

$$G(s)=f'(s)+(\theta-1)f(s)/s,\qquad (f(s)s^{\theta-1})'=G(s)s^{\theta-1}.$$

By (5.10), f' and G lie in $L^2(\nu)$ and $f(s)s^{\theta-1}\to0$ as $s\to0$. Integrating by parts in s,

$$Ef'(t)=\int_0^\infty\partial_s\big(b(t+s)\big)f(s)s^{\theta-1}\,\mathrm{d}s=-\int_0^\infty b(t+s)G(s)\,\mathrm{d}\nu(s)=-(TG)(t).$$

Since $T = NN^* \ge 0$, this gives $E\langle f', G\rangle_{L^2(\nu)} = -\langle TG, G\rangle \le 0$. On the other hand, put $J = \int_0^\infty f(s)^2 s^{\theta-3}\,\mathrm{d}s$, finite by (5.10). Integration by parts, whose boundary terms vanish by (5.10), gives

$$\int_0^\infty f(s)f'(s)s^{\theta-2}\,\mathrm{d}s = \frac{2-\theta}{2}J, \qquad \langle f', G\rangle = \|f'\|^2 - \frac{(1-\theta)(2-\theta)}{2}J.$$

By Cauchy–Schwarz in the first identity, $(2-\theta)J/2 \le J^{1/2}\|f'\|$, so $J \le 4(2-\theta)^{-2}\|f'\|^2$, and

$$0 \ge E\langle f', G\rangle \ge E\Big(1 - \frac{2(1-\theta)}{2-\theta}\Big)\|f'\|^2 = \frac{E\theta}{2-\theta}\|f'\|^2.$$

Hence $f' = 0$, so f is constant; as ν has infinite mass, $f = 0$, a contradiction. □

THEOREM 5.23. *Let $R = \{p \equiv 1 \pmod 4\}$, so $\theta = 1/2$. Then*

$$\zeta_R(1+t) - 1 \le c_R t^{-1/2} \qquad (t > 0), \tag{5.11}$$

$\|A_R\| = L_R = c_R\Gamma(1/4)^2/\sqrt{\pi}$, and A_R has no eigenvalues. Here $c_R = (\pi P/8)^{1/2}$ with $P = \prod_{p\equiv 3\,(4)}(1-p^{-2})$, and $c_R \in [0.579821, 0.579822]$, $L_R \in [4.30012, 4.30014]$.

PROOF. Comparing Euler factors, with χ_4 the nonprincipal character modulo 4 and $P(s) = \prod_{p\equiv 3\,(4)}(1-p^{-2s})$,

$$\zeta_R(s)^2 = \zeta(s)L(s,\chi_4)(1-2^{-s})P(s).$$

As $L(1,\chi_4) = \pi/4$, this gives $c_R^2 = \pi P/8$, and (5.3) holds because $t\zeta_R(1+t)^2$ is analytic and positive near $[0,1]$. We use, for $s = 1+t$, $t > 0$: $\zeta(s) \le 1 + 1/t$; $L(s,\chi_4) < 1$ and $0 < L'(s,\chi_4) < (\log 3)/3$, since both series alternate with terms decreasing in absolute value from $n = 3$ on; $1 - 2^{-s} \le 1/2 + t(\log 2)/2$; and $P(1+t) \le Pe^{Kt}$ with $K = \sum_{p\equiv 3\,(4)} 2\log p/(p^2-1)$.

For $t \ge 1/2$, $\zeta_R(1+t)^2 < \zeta(1+t) \le 1 + 1/t$, which is at most $(1 + c_R t^{-1/2})^2$ as soon as $1 - c_R^2 \le 2c_R t^{1/2}$; it suffices that $1 - c_R^2 \le \sqrt{2}c_R$.

For $0 < t \le 1/2$, put $Q(t) = (1+t)(\pi/4 + t(\log 3)/3)(1/2 + t(\log 2)/2)$. Then $t\zeta_R(1+t)^2 \le PQ(t)e^{Kt}$, and (5.11) follows from $f_L(t) = P(Q(t)e^{Kt} - \pi/8) \le 2c_R t^{1/2}$. As $Q(0) = \pi/8$ and Qe^{Kt} is a product of positive, increasing, convex functions, f_L is convex with $f_L(0) = 0$, so $f_L(t) \le 2tf_L(1/2)$, and it suffices that $f_L(1/2) \le \sqrt{2}c_R$.

Interval arithmetic in `eigenvalues-mod4/zeta_mod4.py` (chapter A) (primes up to 10^6, with explicit tail bounds) gives $P \in [0.856108, 0.856110]$ and $K \le 0.45751$. Hence $1 - c_R^2 \le 0.66382 < 0.81997 \le \sqrt{2}c_R$ and $f_L(1/2) \le 0.71642 < \sqrt{2}c_R$. This proves (5.11).

The companion T is unitarily equivalent, through multiplication by $u^{(\theta-1)/2}$, to the operator on $L^2(0,\infty)$ with kernel $(uv)^{-1/4}b(u+v)$. By (5.11) this positive kernel is dominated pointwise by $c_R(uv)^{-1/4}(u+v)^{-1/2}$, whose operator norm is $c_R\Gamma(1/4)^2/\Gamma(1/2) = L_R$ by the Mellin transform. Thus $\|A_R\| = \|T\| \le L_R$, with equality by (5.2). Hence A_R has no eigenvalues above L_R, and none in $[0, L_R]$ by Theorem 5.22. □

COROLLARY 5.24. *For $R = \{p \equiv 1 \pmod 4\}$ and $F(s) = \sum_{n\in\mathcal{S}_R} a_n n^{-s}$, the sharp constant in $\int_0^\infty |F(1/2+\sigma) - a_1|^2 \sigma^{-1/2}\,\mathrm{d}\sigma \le C\sum_{n\neq 1}|a_n|^2$ is $C = L_R = c_R\Gamma(1/4)^2/\sqrt{\pi}$, and it is not attained.*

PROOF. The left side is $\|Na\|^2_{L^2(\nu)} = \langle A_R a, a\rangle$, so the sharp constant is $\|A_R\| = L_R$. It is not attained, since L_R is not an eigenvalue. □

REMARK 5.25. Without (5.3), when only (5.1) holds, Theorem 5.22 is not claimed. The pointwise criterion $\zeta_R(1+t) - 1 \le c_R t^{-\theta}$ for all $t > 0$ forces $\|A_R\| = L_R$ for any regular R; which other progressions satisfy it, or have eigenvalues above the band, is not decided here. Near $t = 0$ the criterion holds with room whenever (5.3) holds and $\theta < 1$, since then $\zeta_R(1+t) - 1 - c_R t^{-\theta} \to -1$. The Ma–Lu–Zu threshold concerns different Bergman coefficients and is the source of the analogy only; none of its unrefereed spectral results is used.

5.4. Limits of translated kernels

We now allow the normalized kernel and weight to oscillate. Fix $\kappa > 0$, put $K_0(r) = (2\cosh(r/2))^{-\kappa}$ and

$$m(\xi) = \int_{\mathbb{R}} K_0(r) e^{i\xi r}\,\mathrm{d}r = \frac{|\Gamma(\kappa/2 + i\xi)|^2}{\Gamma(\kappa)}, \qquad m_\kappa = m(0) = \frac{\Gamma(\kappa/2)^2}{\Gamma(\kappa)},$$

so m is even and strictly decreasing in $|\xi|$ (proof of Theorem 5.4). Let h be real-valued and continuous on $(0,\infty)$ and let $\mathrm{d}\nu = u^{\kappa-1}\gamma(u)\,\mathrm{d}u$ be nonzero, with $\gamma \ge 0$ continuous. Assume, as in Theorem 5.4, that $\iint_{\max(u,v)\ge\delta} |h(u+v)|^2\,\mathrm{d}\nu(u)\,\mathrm{d}\nu(v) < \infty$ for every $\delta > 0$, and let T be the integral operator with kernel $h(u+v)$ on $L^2(\nu)$. Put $\Lambda(s,t) = -\log(e^{-s} + e^{-t})$ and

$$\varphi(L) = e^{-\kappa L} h(e^{-L}), \qquad w(s) = \gamma(e^{-s})^{1/2}.$$

The isometry $f \mapsto e^{-\kappa s/2}(\gamma^{1/2} f)(e^{-s})$ from $L^2(\nu)$ into $L^2(\mathbb{R})$ (onto when $\gamma > 0$) carries T, extended by zero on the orthogonal complement of its image, to the operator with kernel

$$K_0(s-t)\,\varphi(\Lambda(s,t))\,w(s)w(t), \tag{5.12}$$

since $(uv)^{\kappa/2}(u+v)^{-\kappa} = K_0(s-t)$ and $u + v = e^{-\Lambda(s,t)}$ for $u = e^{-s}$, $v = e^{-t}$. We keep the name T for this operator.

Hypothesis (U). For some L_0, φ and w are bounded and uniformly continuous on $[L_0,\infty)$. Extend them to $\mathbb{R}$ by their values at L_0. The *hull* Ω is the set of pairs (α,β) of continuous functions on $\mathbb{R}$ such that $\varphi(\cdot + a_n) \to \alpha$ and $w(\cdot + a_n) \to \beta$ locally uniformly for some $a_n \to \infty$. For $(\alpha,\beta) \in \Omega$ let $T_{\alpha,\beta}$ be the operator on $L^2(\mathbb{R})$ with kernel $K_0(s-t)\alpha(\Lambda(s,t))\beta(s)\beta(t)$.

By the theorem of Arzelà and Ascoli, every sequence $a_n \to \infty$ has a subsequence along which the translates converge, so Ω is nonempty and compact for local uniform convergence. Since $\Lambda(s+b, t+b) = \Lambda(s,t) + b$, it is invariant under translation. Every α in the hull has the modulus of continuity of φ on $[L_0,\infty)$, and $\alpha \le \limsup_{L\to\infty}\varphi(L)$. By the Schur test and $\int K_0 = m_\kappa$, every $T_{\alpha,\beta}$ is bounded with norm at most $m_\kappa\|\alpha\|_\infty\|\beta\|_\infty^2$.

THEOREM 5.26 (Hull formula). *Under (U), T is bounded and*

$$\sigma_{\mathrm{ess}}(T) = \overline{\bigcup_{(\alpha,\beta)\in\Omega} \sigma(T_{\alpha,\beta})}.$$

THEOREM 5.27 (Upper-limit reading). *Assume (U), $\gamma(u) \to b > 0$ as $u \to 0$, and that $h \geq 0$ and the kernel $h(u+v)$ is positive semi-definite, as for the companion kernel of every positive Helson matrix (Lemma 5.1). Put $\varphi^* = \limsup_{L\to\infty} \varphi(L)$ and*

$$M^+ = \limsup_{N\to\infty} \limsup_{a\to\infty} \frac{1}{N} \int_a^{a+N} \varphi(L)\,\mathrm{d}L.$$

Then $bm_\kappa M^+ \leq \max \sigma_{\mathrm{ess}}(T) \leq bm_\kappa \varphi^*$, and the following are equivalent:

(1) $\sigma_{\mathrm{ess}}(T) = [0, b\varphi^* m_\kappa]$;
(2) $\max \sigma_{\mathrm{ess}}(T) = b\varphi^* m_\kappa$;
(3) for every $N, \varepsilon > 0$ there are arbitrarily large L with $\varphi \geq \varphi^* - \varepsilon$ on $[L, L+N]$.

They hold in particular when $\varphi(L+a) - \varphi(L) \to 0$ as $L \to \infty$, locally uniformly in a, for instance when $x^\kappa h(x)$ is bounded below by a positive constant and slowly varying at 0.

LEMMA 5.28. *Let $h(x) = \int_{[0,\infty)} e^{-x\lambda}\,\mathrm{d}\mu(\lambda)$ with $\mu \geq 0$, and suppose $x^\kappa h(x) \leq B$ for $0 < x \leq x_0$. Then φ is Lipschitz on $[\log(2/x_0), \infty)$ with constant $(\kappa + 2^{\kappa+1}/e)B$.*

PROOF. With $x = e^{-L}$, $\varphi'(L) = -\kappa\varphi(L) - x^\kappa \cdot xh'(x)$. Since $ye^{-y/2} \leq 2/e$ for $y \geq 0$, $|xh'(x)| = \int x\lambda e^{-x\lambda}\,\mathrm{d}\mu \leq (2/e)h(x/2) \leq (2/e)2^\kappa B x^{-\kappa}$ when $x \leq x_0/2$. □

COROLLARY 5.29. *Let R be a set of primes of density $\theta \in (0,1]$ with $x^\theta \zeta_R(1+x)$ bounded on $(0,1]$, and let A_R be the critical matrix $\big(\Gamma(\theta)(mn)^{-1/2}(\log mn)^{-\theta}\big)$ on $\mathcal{S}_R \setminus \{1\}$. Put $\varphi(L) = x^\theta \zeta_R(1+x)$, $x = e^{-L}$, and $c^+ = \limsup_{x\to 0} x^\theta \zeta_R(1+x)$. Then $\sigma_{\mathrm{ess}}(A_R) = \overline{\bigcup_\alpha \sigma(T_{\alpha,1})}$ over the hull of φ, and $\sigma_{\mathrm{ess}}(A_R) = [0, c^+ m_\theta]$ if and only if φ satisfies condition (3) of Theorem 5.27 with $\varphi^* = c^+$. If $\log\varphi(L) - \psi_P(L) \to 0$ for a continuous, nonconstant, P-periodic ψ_P, as in Proposition 5.7, then*

$$m_\theta \frac{1}{P} \int_0^P e^{\psi_P} \leq \max \sigma_{\mathrm{ess}}(A_R) < m_\theta \max e^{\psi_P}.$$

PROOF OF THEOREM 5.26. *Reduction.* Fix $s_0 \geq L_0 + \log 2$ and let P be multiplication by $\mathbf{1}_{(s_0,\infty)}$. In the variable u, the region $\{\min(s,t) \leq s_0\}$ is $\{\max(u,v) \geq e^{-s_0}\}$, so $T - PTP$ is Hilbert–Schmidt by hypothesis. On $(s_0,\infty)^2$ we have $\Lambda \geq s_0 - \log 2 \geq L_0$, so the kernel of PTP is that of $PT_\# P$, where $T_\#$ is the operator on $L^2(\mathbb{R})$ with kernel (5.12) built from the extended φ, w. $T_\#$ is bounded by the Schur test, hence so is T. For $S \geq s_0$ the parts of $T_\#$ coupling (S,∞) with $(-\infty, S)$ are Hilbert–Schmidt, because $|T_\#(s,t)| \leq CK_0(s-t)$ and $\int_{-\infty}^S \int_S^\infty K_0(s-t)^2\,\mathrm{d}t\,\mathrm{d}s < \infty$.

Zero. As in the proof of Theorem 5.4, choose a set E of finite positive ν-measure bounded away from zero. The restriction to inputs in $L^2(E,\nu)$ is Hilbert–Schmidt, so an orthonormal sequence there shows that zero belongs to the essential spectrum of the original operator on $L^2(\nu)$. Thus zero extension leaves its essential spectrum

unchanged. For $(\alpha,\beta)\in\Omega$, $\chi\in C_c(\mathbb{R})$ and $f_\xi = e^{i\xi\cdot}\chi$, Riemann–Lebesgue gives $(T_{\alpha,\beta}f_\xi)(s)\to 0$ for each s as $\xi\to\infty$, with domination by $C\int K_0(s-t)|\chi(t)|\,\mathrm{d}t\in L^2$; so $0\in\sigma(T_{\alpha,\beta})$.

Local norm convergence. Let $(\varphi,w)(\cdot+a_n)\to(\alpha,\beta)$ locally uniformly, and let T_n be $T_\#$ conjugated by translation by a_n; its kernel is $K_0(s-t)\varphi(\Lambda(s,t)+a_n)w(s+a_n)w(t+a_n)$. For $W>0$ and $Q_W=\mathbf{1}_{[-W,W]}$ we claim $\|(T_n-T_{\alpha,\beta})Q_W\|\to 0$. The kernel difference is bounded by $CK_0(s-t)$; on $[-W-M,W+M]\times[-W,W]$ it tends to 0 uniformly, since there Λ ranges over a compact set; and the Hilbert–Schmidt norm of the rest is at most $C(2W)^{1/2}\big(\int_{|r|>M}K_0^2\big)^{1/2}$, small for large M.

Inclusion $\supseteq$. Fix $\lambda\in\sigma(T_{\alpha,\beta})$, $\lambda\neq 0$, and let $a_n\to\infty$ define this hull element. Choose unit vectors v_j supported in bounded intervals with $\|(T_{\alpha,\beta}-\lambda)v_j\|<1/j$. By local norm convergence, we may choose a subsequence a_{n_j} so far to the right that the translated supports are disjoint, lie in (s_0,∞) and escape to infinity, and $\|(T_{n_j}-T_{\alpha,\beta})v_j\|<1/j$. Then $u_j=v_j(\cdot-a_{n_j})$ is orthonormal and $\|(T_\#-\lambda)u_j\|<2/j$. The compact comparison and cross-cut operators from the reduction send u_j to zero in norm. Hence $\|(T-\lambda)u_j\|\to 0$, and Weyl's criterion gives $\lambda\in\sigma_{\mathrm{ess}}(T)$.

Inclusion $\subseteq$. Let $\lambda\in\sigma_{\mathrm{ess}}(T)$, $\lambda\neq 0$, and let $S\geq s_0$. Then $\lambda\in\sigma_{\mathrm{ess}}(P_ST_\#P_S)$ with $P_S=\mathbf{1}_{(S,\infty)}$, by the reduction, and a Weyl sequence for it, being weakly null, is moved only by $o(1)$ under the Hilbert–Schmidt coupling terms. Hence there is a unit g supported in (S,∞) with $\|(T_\#-\lambda)g\|\leq 1/S$. Let $\chi\in C^\infty(\mathbb{R})$, $0\leq\chi\leq 1$, supported in $[-1,1]$, with $\sum_{j\in\mathbb{Z}}\chi(x-j)^2=1$, and put $\chi_j(s)=\chi(s/W-j)$. Then $\sum_j\chi_j^2=1$, at most two χ_j are nonzero at a point, and $|\chi_j(s)-\chi_j(t)|\leq C|s-t|/W$. By Cauchy–Schwarz and $|T_\#(s,t)|\leq CK_0(s-t)$,

$$\sum_j\|[T_\#,\chi_j]g\|^2\leq C\int K_0\cdot\sup_t\int K_0(s-t)\sum_j|\chi_j(s)-\chi_j(t)|^2\,\mathrm{d}s\,\|g\|^2\leq\frac{C_1}{W^2},$$

since $\int K_0(r)r^2\,\mathrm{d}r<\infty$. Therefore

$$\sum_j\|(T_\#-\lambda)\chi_jg\|^2\leq 2\sum_j\|\chi_j(T_\#-\lambda)g\|^2+\frac{2C_1}{W^2}\leq\Big(\frac{2}{S^2}+\frac{2C_1}{W^2}\Big)\sum_j\|\chi_jg\|^2,$$

and some j has $\chi_jg\neq 0$ and $\|(T_\#-\lambda)\chi_jg\|\leq(2/S^2+2C_1/W^2)^{1/2}\|\chi_jg\|$. The support of χ_jg lies in an interval $[a,a+2W]$ with $a\geq S-2W$. Translating by a gives $v=\chi_jg(\cdot+a)/\|\chi_jg\|$, supported in $[0,2W]$, with $\|(T^{(a)}-\lambda)v\|\leq(2/S^2+2C_1/W^2)^{1/2}$, where $T^{(a)}$ is $T_\#$ conjugated by translation by a. Keep W fixed and let $S\to\infty$ along a sequence; then $a\to\infty$, and along a subsequence the translates converge to some $(\alpha,\beta)\in\Omega$. By local norm convergence with Q_{2W}, $\|(T_{\alpha,\beta}-\lambda)v\|\leq(2C_1)^{1/2}/W+o(1)$ for the corresponding unit vectors, so $\operatorname{dist}(\lambda,\sigma(T_{\alpha,\beta}))\leq(2C_1)^{1/2}/W$, since $T_{\alpha,\beta}$ is self-adjoint. Letting $W\to\infty$ proves the inclusion. □

Proof of Theorem 5.27. Here every β in the hull is the constant $b^{1/2}$, so $T_{\alpha,\beta}=bT_\alpha$, where T_α has kernel $K_0(s-t)\alpha(\Lambda(s,t))$. We take $b=1$ and write Ω_φ for the profile hull. By Theorem 5.26, $\max\sigma_{\mathrm{ess}}(T)=\sup_{\alpha\in\Omega_\varphi}\max\sigma(T_\alpha)$. Since $0\leq\alpha\leq\varphi^*$, Schur's test gives the upper bound $m_\kappa\varphi^*$.

Lower bound. Let $f_a = N^{-1/2}\mathbf{1}_{[a,a+N]}$. With $r = s - t$, $\Lambda(t + r, t) = t + \ell(r)$, where $\ell(r) = \min(r, 0) - \log(1 + e^{-|r|})$, so $|\ell(r)| \le |r| + \log 2$, and

$$\langle T_\# f_a, f_a\rangle = \frac{1}{N}\int_{\mathbb{R}} K_0(r)\int_{J_r}\varphi(t + \ell(r))\,\mathrm{d}t\,\mathrm{d}r + o(1)$$

as $a \to \infty$, because $w(s)w(t) \to 1$, with J_r an interval of length $(N - |r|)_+$ in $[a, a + N]$. The interval $J_r + \ell(r)$ differs from $[a, a + N]$ by a set of measure at most $4|r| + 2\log 2$, so $\langle T_\# f_a, f_a\rangle = m_\kappa N^{-1}\int_a^{a+N}\varphi + O(1/N)$. The f_a are weakly null as $a \to \infty$, and for f supported in (s_0, ∞), $\langle Tf, f\rangle - \langle T_\# f, f\rangle = \langle (T - PTP)f, f\rangle$ with $T - PTP$ compact. For a self-adjoint operator and a weakly null unit sequence, $\limsup\langle Tf_a, f_a\rangle \le \max\sigma_{\mathrm{ess}}(T)$, because the spectral projection above $\max\sigma_{\mathrm{ess}}(T) + \varepsilon$ has finite rank. Letting $a \to \infty$ and then $N \to \infty$ gives $\max\sigma_{\mathrm{ess}}(T) \ge m_\kappa M^+$.

Equivalences. (1) implies (2) is trivial. Condition (3) says exactly that the constant φ^* lies in Ω_φ: given (3), take $a_n \to \infty$ with $\varphi \ge \varphi^* - 1/n$ on $[a_n - n, a_n + n]$ and use $\limsup\varphi = \varphi^*$; the converse is immediate. Then $\sigma(T_{\varphi^*}) = \varphi^* m(\mathbb{R}) = [0, \varphi^* m_\kappa]$, which lies in $\sigma_{\mathrm{ess}}(T)$ by Theorem 5.26. Since $T \ge 0$ and $\max\sigma_{\mathrm{ess}}(T) \le \varphi^* m_\kappa$, this gives (3) implies (1). It remains to show (2) implies (3); if $\varphi^* = 0$, (3) holds trivially.

Suppose that (3) fails and $\varphi^* > 0$. Its negation gives $N, \varepsilon > 0$ such that every sufficiently far-right interval of length N contains a point where $\varphi < \varphi^* - 2\varepsilon$. Local uniform convergence gives a point with $\alpha \le \varphi^* - 2\varepsilon$ in every interval of length N, for every $\alpha \in \Omega_\varphi$. By their common modulus of continuity, there is $\delta \in (0, 1/2]$ such that every interval of length N contains a point x with $\alpha \le \varphi^* - \varepsilon$ on $[x, x + \delta]$.

Fix $s \in \mathbb{R}$ and choose such an x in $[s - N - 2, s - 2]$. For $\sigma \in [x, x + \delta]$, the inverse of $\sigma = \Lambda(s, t)$ is $t = s - \log(e^{s-\sigma} - 1)$. Hence $|s - t| \le N + 2$ and $\mathrm{d}t/\,\mathrm{d}\sigma = (1 - e^{\sigma - s})^{-1} \ge 1$. Therefore

$$\int_{\mathbb{R}} K_0(s - t)\big(\varphi^* - \alpha(\Lambda(s, t))\big)\,\mathrm{d}t \ge \varepsilon\delta K_0(N + 2) = d > 0.$$

Since $0 \le \alpha \le \varphi^*$, every row integral of the kernel of T_α is at most $\varphi^* m_\kappa - d$. The kernel is symmetric, so Schur's test gives $\|T_\alpha\| \le \varphi^* m_\kappa - d$, uniformly over the hull. By Theorem 5.26, condition (2) cannot hold. Thus (2) implies (3).

If $\varphi(L + a) - \varphi(L) \to 0$ locally uniformly in a, every α in the hull is constant, and the constants in the hull are the limit points of φ at infinity, which include φ^*; so (3) holds. □

Proof of Corollary 5.29. Take $\kappa = \theta$, $\gamma \equiv 1$ and $h(x) = \zeta_R(1 + x) - 1 = \int e^{-x\lambda}\,\mathrm{d}\mu$ with $\mu = \sum_{n \in \mathcal{S}_R \setminus \{1\}} n^{-1}\delta_{\log n}$. The Hilbert–Schmidt hypothesis holds as in the proof of Proposition 5.5, and Lemma 5.28 gives (U); $x^\theta h(x)$ and $x^\theta\zeta_R(1 + x)$ differ by $x^\theta \to 0$, so they have the same hull and upper limit. The kernel $h(u + v)$ is positive semi-definite. By Lemma 5.1 and polar decomposition, A_R and T are unitarily equivalent on the orthogonal complements of their kernels, so their essential spectra agree away from 0. Both contain 0: for T by the proof of Theorem 5.26, for A_R because its diagonal entries tend to 0, so $\|A_R e_n\| \le \|A_R\|^{1/2}\langle A_R e_n, e_n\rangle^{1/2} \to 0$. This gives the first two assertions. In the periodic case the hull is $\{e^{\psi_P(\cdot + y)}\}$,

M^+ is the mean of e^{ψ_P}, and (3) fails as ψ_P is nonconstant; so $\max\sigma_{\mathrm{ess}}(A_R) < m_\theta \max e^{\psi_P}$. □

THEOREM 5.30 (Literal kernel asymptotic, oscillating weight). *Assume the setting, with $x^\kappa h(x) \to c > 0$ as $x \to 0$, and suppose that $\gamma(e^{-s}) - q(s) \to 0$ as $s \to \infty$, where q is continuous, positive, nonconstant and P-periodic, with mean $\bar q = P^{-1}\int_0^P q$ and maximum Q. Let $S_q = M_{\sqrt q} K_0 M_{\sqrt q}$ on $L^2(\mathbb{R})$, with K_0 acting by convolution. Then $\sigma_{\mathrm{ess}}(T) = c\,\sigma(S_q)$, and $\tau = \max\sigma_{\mathrm{ess}}(T)$ satisfies*

$$c\,m_\kappa \bar q < \tau < c\,m_\kappa Q.$$

In particular, let $0 < \varepsilon < 1$, $P > 0$, let χ be continuous on $(0,\infty)$ with $0 \le \chi \le 1$, $\chi = 1$ on $(0, 1/2]$ and $\chi = 0$ on $[1,\infty)$, and let

$$\mathrm{d}\nu(u) = \big(1 + \varepsilon\cos(2\pi\log(1/u)/P)\big)\chi(u)\,\mathrm{d}u,$$
$$M_\nu = \Big(\int_0^\infty (mn)^{-1/2-u}\,\mathrm{d}\nu(u)\Big)_{m,n\ge 1}.$$

The companion kernel $\zeta(1+x)$ of M_ν satisfies $x\zeta(1+x) \to 1$, and the weight has density 1 on average in $\log(1/u)$, but M_ν is bounded and $\sigma_{\mathrm{ess}}(M_\nu)$ contains a point $\tau > \pi$, so $\sigma_{\mathrm{ess}}(M_\nu) \ne [0,\pi]$.

PROOF. *Hull.* The profile φ is continuous with limit c, hence bounded and uniformly continuous on $[L_0,\infty)$; and $w = \sqrt q + o(1)$ is bounded and uniformly continuous there, since $\sqrt{\cdot}$ is uniformly continuous on $[0,\infty)$. So (U) holds. Every limit of translates $\varphi(\cdot + a_n)$ is the constant c, and, passing to a subsequence along which a_n modulo P converges to some y, every limit of $w(\cdot + a_n)$ is $\sqrt q(\cdot + y)$. Thus $\Omega = \{(c, \sqrt q(\cdot + y)) : y \in \mathbb{R}\}$, and each $T_{c,\sqrt q(\cdot+y)}$ is unitarily equivalent to cS_q by translation. By Theorem 5.26, $\sigma_{\mathrm{ess}}(T) = c\,\sigma(S_q)$. Since $m \ge 0$, $S_q \ge 0$.

Lower bound. Let $f_N = \sqrt q\,\mathbf{1}_{[0,NP]}$, so $\|f_N\|^2 = NP\bar q$. Since K_0 is even,

$$\langle S_q f_N, f_N\rangle = \int_0^{NP} q(t)\int_{\mathbb{R}} K_0(r)q(t+r)\,\mathrm{d}r\,\mathrm{d}t - E_N = N\int_0^P q\,(K_0 * q) - E_N,$$

with $0 \le E_N \le 2Q^2\int_{\mathbb{R}} |r|K_0(r)\,\mathrm{d}r$, the contribution of $t + r \notin [0, NP]$. With $q = \sum_k \hat q_k e^{2\pi i k t/P}$, the periodic function $K_0 * q$ has coefficients $m(2\pi k/P)\hat q_k$, and Parseval gives $P^{-1}\int_0^P q\,(K_0 * q) = \sum_k m(2\pi k/P)|\hat q_k|^2$. Hence

$$\max\sigma(S_q) \ge \lim_{N\to\infty} \frac{\langle S_q f_N, f_N\rangle}{\|f_N\|^2} = \frac{1}{\bar q}\sum_{k\in\mathbb{Z}} m(2\pi k/P)|\hat q_k|^2 > m_\kappa \bar q,$$

because $\hat q_0 = \bar q$, $m > 0$ everywhere, and $\hat q_k \ne 0$ for some $k \ne 0$ as q is nonconstant.

Upper bound. Apply the weighted Schur test to the kernel $\sqrt{q(s)}K_0(s-t)\sqrt{q(t)}$ with weight $\sqrt q$. Its weighted row integral is

$$\frac{1}{\sqrt{q(s)}}\int_{\mathbb{R}} \sqrt{q(s)}K_0(s-t)\sqrt{q(t)}\sqrt{q(t)}\,\mathrm{d}t = (K_0 * q)(s).$$

Thus $\|S_q\| \le \max_{s\in[0,P]}(K_0 * q)(s)$. Since $K_0 > 0$ and q is continuous and nonconstant, $(K_0 * q)(s) < m_\kappa Q$ for every s. The convolution is continuous and P-periodic, so its maximum is strictly less than $m_\kappa Q$. This proves the upper bound.

Example. Take $\kappa = 1$, so $m_1 = \pi$, $h(x) = \zeta(1+x)$, $c = 1$, $\gamma(u) = (1 + \varepsilon\cos(2\pi\log(1/u)/P))\chi(u)$ and $q(s) = 1 + \varepsilon\cos(2\pi s/P)$, so that $\gamma(e^{-s}) = q(s)$ for $s \geq \log 2$ and $\bar{q} = 1$. The kernel $\zeta(1+u+v) = \sum_n n^{-1-u-v}$ is real, and $\zeta(1+x) \leq 1 + 1/x$ makes it bounded on $\{\max(u,v) \geq \delta\}$; as ν is finite, the Hilbert–Schmidt hypothesis holds. By what was proved, T is bounded and $\tau = \max \sigma_{\mathrm{ess}}(T) > \pi$. By Lemma 5.1, $M_\nu = \mathcal{N}^*\mathcal{N}$ is bounded, and M_ν and $T = \mathcal{N}\mathcal{N}^*$ are unitarily equivalent on the orthogonal complements of their kernels, so their essential spectra agree away from 0 and $\tau \in \sigma_{\mathrm{ess}}(M_\nu)$. □

The hull formula is a limit-operator theorem for this class of kernels; compare Rabinovich, Roch and Silbermann [75] and Chandler-Wilde and Lindner [25, Theorem 8.4]. The proof above supplies the needed localization argument directly. For the fixed critical weight, Problem 5 therefore has a positive answer under a literal asymptotic and a negative answer under comparability. Oscillating weights give a further failure even when the kernel has a limit.

5.5. Schatten classes along the curve

Put

$$a_j(\nu) = e^{j\theta}\nu(I_j), \qquad j \geq 0.$$

Chapter 4 shows that $M_{\nu,R}$ is bounded exactly when a is bounded, and compact exactly when $a_j \to 0$. The trace formula shows that $M_{\nu,R}$ is trace class exactly when $\int_0^1 u^{-\theta}\,\mathrm{d}\nu < \infty$, that is, when $a \in \ell^1$. The intermediate classes interpolate.

The profile is the natural quantity. On the block I_j one evaluation along the curve costs $\zeta_R(1+2u) \asymp e^{j\theta}$, so a_j is the mass of the block times the size of an evaluation there; it is, up to constants, the trace of the piece M_j of the matrix that comes from I_j. Two facts turn this into a spectral classification. Within one block, the kernel $k(u)$ depends analytically on u across a region of fixed hyperbolic size, so M_j is of rank one up to geometrically small corrections (5.15): one block contributes one principal singular value, of size about a_j. Across blocks, normalized kernels at distant scales are nearly orthogonal (Lemma 2.4), so these principal values interact only weakly. The singular values of $M_{\nu,R}$ are therefore, up to constants and geometric errors, the numbers a_j. Nothing in the profile records how the mass is arranged inside a block, which is why it determines the distribution of the spectrum (Theorem 5.38) but not individual singular values (Proposition 5.35).

THEOREM 5.31 (Schatten classes of curve matrices). *Let R have density $\theta \in (0,1]$ with constant B, and let $1 \leq p < \infty$. There are $c, C > 0$ depending only on θ, B and p such that, for every finite positive ν on $(0,\infty)$,*

$$c\sum_{j\geq 0} a_j(\nu)^p \leq \|M_{\nu,R}\|_{S_p}^p \leq C\Big(\sum_{j\geq 0} a_j(\nu)^p + \nu((1,\infty))^p\Big).$$

In particular $M_{\nu,R} \in S_p$ if and only if $a(\nu) \in \ell^p$. The same holds for every $0 < p < \infty$, and for every Lorentz class $S_{p,q}$, by Theorem 5.38 below.

PROOF. The part of ν on $(1,\infty)$ gives a positive matrix whose trace is at most $\zeta(3)\,\nu((1,\infty))$. So assume that ν lives on $(0,1]$.

Upper bound. Let $\nu_j = \nu|_{I_j}$ and, for those j with $a_j > 0$, let $B_j = M_{\nu_j,R}/a_j$. For a finitely supported complex sequence c put $Tc = \sum_j c_j B_j$, so that $M_{\nu,R} = Ta$. Since $\zeta_R(1+2u) \le e^{1+B}(2u)^{-\theta}$,

$$\|B_j\|_{S_1} = \operatorname{Tr} B_j \le e^{1+B} 2^{-\theta} e^{(j+1)\theta} \nu(I_j)/a_j \le e^{2+B},$$

so $\|Tc\|_{S_1} \le e^{2+B}\|c\|_{\ell^1}$. For real c and finitely supported x, positivity gives $|\langle Tc\,x, x\rangle| \le \|c\|_\infty \langle M_{\nu',R}\,x, x\rangle$ with $\nu' = \sum_j \nu_j/a_j$. Every block of ν' has $a_j(\nu') = 1$, so $\nu'((0,t]) \le (e^\theta/(1-e^{-\theta}))\,t^\theta$ and Theorem 2.11 with $p = 2$ bounds $\|M_{\nu',R}\|$ by a constant. Splitting complex c into real and imaginary parts, $\|Tc\| \le C\|c\|_{\ell^\infty}$. Complex interpolation, with $[\ell^1, \ell^\infty]_{1-1/p} = \ell^p$ and $[S_1, B(H)]_{1-1/p} = S_p$, gives $\|Tc\|_{S_p} \le C\|c\|_{\ell^p}$. Apply this to $c = a$.

Lower bound. For $0 < \delta \le 1$ let $k_\delta(n) = n^{-1/2-\delta}\zeta_R(1+2\delta)^{-1/2}$, a unit vector in $\ell^2(\mathcal{S}_R)$. For $u \le \delta$ one has $\zeta_R(1+\delta+u) \ge \zeta_R(1+2\delta)$, so

$$\langle M_{\nu,R}k_\delta, k_\delta\rangle = \frac{\int \zeta_R(1+\delta+u)^2\,\mathrm{d}\nu(u)}{\zeta_R(1+2\delta)} \ge \zeta_R(1+2\delta)\,\nu((0,\delta]) \ge e^{-B}(2\delta)^{-\theta}\nu((0,\delta]).$$

With $\delta_j = e^{-j}$ this is at least $e^{-B}2^{-\theta}a_j$. The density bounds, used as in Lemma 2.10, give

$$\langle k_{\delta_j}, k_{\delta_k}\rangle = \frac{\zeta_R(1+\delta_j+\delta_k)}{\sqrt{\zeta_R(1+2\delta_j)\zeta_R(1+2\delta_k)}} \le e^{2B+1}\Big(\frac{2\sqrt{\delta_j\delta_k}}{\delta_j+\delta_k}\Big)^\theta \le 2e^{2B+1}e^{-\theta|j-k|/2}.$$

By Schur's test, the normalized kernels form a Bessel sequence:

$$\sum_{j\ge 0}|\langle h, k_{\delta_j}\rangle|^2 \le C_V\|h\|^2, \qquad C_V = 2e^{2B+1}\frac{1+e^{-\theta/2}}{1-e^{-\theta/2}}.$$

Write $M = M_{\nu,R}$ and assume $M \in S_p$; otherwise the lower estimate is immediate. Since k_{δ_j} is a unit vector, Jensen's inequality for its spectral probability measure gives $\langle Mk_{\delta_j}, k_{\delta_j}\rangle^p \le \langle M^p k_{\delta_j}, k_{\delta_j}\rangle$. Expanding M^p in an orthonormal eigenbasis and applying the Bessel bound, we obtain

$$\begin{aligned} e^{-Bp}2^{-\theta p}\sum_j a_j^p &\le \sum_j \langle Mk_{\delta_j}, k_{\delta_j}\rangle^p \\ &\le \sum_j \langle M^p k_{\delta_j}, k_{\delta_j}\rangle \le C_V \operatorname{Tr} M^p. \end{aligned}$$

This lower bound needs only that the kernels form a Bessel sequence; the separated-atom theorem below needs a Gram matrix bounded from below, and therefore a stronger separation. □

COROLLARY 5.32. *With $F(t) = \nu((0,t])$ and $0 < \theta \le 1$:*

1. *$M_{\nu,R} \in S_1$ if and only if $\int_0^1 u^{-\theta}\,\mathrm{d}\nu(u) < \infty$;*
2. *$M_{\nu,R} \in S_2$ if and only if $\int_0^1 F(t)^2 t^{-1-2\theta}\,\mathrm{d}t < \infty$;*
3. *if $\theta = 0$, then $M_{\nu,R}$ is trace class for every finite ν, with $\operatorname{Tr} M_{\nu,R} \le e^{1+B}\nu((0,\infty))$.*

PROOF. (1) and (2) restate $a \in \ell^1$ and $a \in \ell^2$; for (2), note $e^{j\theta}F(e^{-j}) = \sum_{k\geq j} e^{-(k-j)\theta}a_k$, a convolution of a with a summable kernel, so $(e^{j\theta}F(e^{-j})) \in \ell^2$ exactly when $a \in \ell^2$. (3) is the trace formula with $\zeta_R(1+2u) \leq e^{1+B}$. □

Part (1) is the trace formula. For R all primes the boundedness criterion that the profile refines is due to Perfekt and Pushnitski [71, Theorem 5.1].

Part (3) sharpens the $\theta = 0$ case of Theorem 4.10, which gave only compactness.

REMARK 5.33 (Positive matrices in every Schatten class have symbols). Take $\nu = \sum_j j^{-1/p} e^{-j\theta}\delta_{e^{-j}}$. Then $M_{\nu,R}$ lies in S_q exactly for $q > p$, and it has a bounded symbol: Theorem 4.2 gives one for $M(\alpha_{\nu,R})$, and its conditional expectation onto the coordinates in R has moments $\alpha_\nu(n)\mathbf{1}_{\mathcal{S}_R}(n)$ and no larger norm. So positive curve matrices occupy every Schatten class, each with a bounded symbol. The counterexamples of chapter 7, which exist in S_p for every $p > 4$, are therefore necessarily not of this form. Inside the positive family the S_p condition is a pure size condition on ν; the S_p failure of Nehari's theorem is a phenomenon of cancellation.

The matrices $M_{\nu,R}$ of this chapter are compressions to $\mathcal{S}_R$. The sign-averaged matrices $M(\alpha_{\nu,R})$ of chapter 4, on all of $\ell^2(\mathbb{N})$, have the same Schatten classes for $p > 1$ but not at $p \leq 1$.

PROPOSITION 5.34. *Let R have density $\theta \in [0,1]$, let ν be a finite positive measure on $(0,\infty)$, and put $W = M(\alpha_{\nu,R})$, with the shift of this chapter.*

(1) For $p > 1$, $\|M_{\nu,R}\|_{S_p} \leq \|W\|_{S_p} \leq \zeta(2)\zeta(p)^{1/p}\|M_{\nu,R}\|_{S_p}$.
(2) $\operatorname{Tr} W = \int \zeta(1+2u)\,\mathrm{d}\nu(u)$, so $W \in S_1$ exactly when $\int_0^1 u^{-1}\,\mathrm{d}\nu < \infty$, whatever R is.
(3) If $\theta < 1$ and $\nu((0,1]) > 0$, then $W \notin S_p$ for $p \leq 1/3$.

PROOF. Let Q be the complement of R and write $m = m_R q a^2$ with q squarefree and $q, a \in \mathcal{S}_Q$. The entry $\alpha_{\nu,R}(mn)$ vanishes unless m and n have the same q, so $W = \bigoplus_q W_q$, and $W_q = \mathcal{N}_q^*\mathcal{N}_q$ with $\mathcal{N}_q e_{(m_R,a)}(u) = m_R^{-1/2-u}(qa^2)^{-1/2-u}$. Hence $\mathcal{N}_q\mathcal{N}_q^* = \sum_a w_{qa^2} K w_{qa^2}$, where K is the companion operator of Lemma 5.1 and w_n is multiplication by $n^{-1/2-u}$, of norm at most $n^{-1/2}$. For $p \geq 1$ the triangle inequality gives $\|W_q\|_{S_p} \leq \zeta(2)q^{-1}\|K\|_{S_p}$, and summing pth powers over q gives (1) for $p > 1$; the block $q = a = 1$ gives the lower bound. For (2), every n^2 has a square Q-part. For (3), testing $\mathcal{N}_q\mathcal{N}_q^*$ with $\mathbf{1}_{(0,1]}$ gives an eigenvalue at least $q^{-3}\nu((0,1])$ in each block, and $\sum_q q^{-1} = \prod_{p\in Q}(1+1/p) = \infty$ when $\theta < 1$. □

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The criteria of chapter 4 and Theorem 5.31 turn every borderline into an explicit matrix. Let R have density $\theta \in (0,1]$.

(1) *Power weights.* For $\kappa > 0$, $\mathrm{d}\nu = u^{\kappa-1}\,\mathrm{d}u$ gives, on $\mathcal{S}_R \setminus \{1\}$, $\alpha(n) = \Gamma(\kappa)n^{-1/2}(\log n)^{-\kappa}$ and $a_j \asymp e^{-j(\kappa-\theta)}$: in every S_p for $\kappa > \theta$, bounded but not compact for $\kappa = \theta$, unbounded for $\kappa < \theta$ (Corollary 5.3). For R all primes and $\kappa = 1$ this is the multiplicative Hilbert matrix.

(2) *Logarithmic weights.* $\mathrm{d}\nu = u^{\theta-1}(\log(e/u))^{-\alpha}\,\mathrm{d}u$ on $(0,1]$ has $a_j \asymp (1+j)^{-\alpha}$. It is bounded exactly for $\alpha \geq 0$, compact exactly for $\alpha > 0$, and in S_p, $p > 0$, exactly for $\alpha p > 1$. For R all primes, Miheisi and Pushnitski [61] found the exact eigenvalue asymptotics $\lambda_n \sim \kappa(\alpha) n^{-\alpha}$ for the closely related weights $1/(\sqrt{n}\log n(\log\log n)^{\alpha})$.

(3) *Iterated logarithms.* With an extra factor $(\log(e\log(e/u)))^{-\beta}$ at $\alpha = 1$, $a_j \asymp (1+j)^{-1}\log^{-\beta}(e+j)$: compact and in every S_p, $p > 1$, and trace class exactly for $\beta > 1$.

(4) *Sparse atoms.* For a sufficiently large integer L, $\nu = \sum_j c_j \delta_{e^{-Lj}}$ has $a_{Lj} = e^{Lj\theta}c_j$ and $a = 0$ elsewhere. With $c_j = e^{-Lj\theta}j^{-1/q}$ the singular values are exactly of order $n^{-1/q}$ (Theorem 5.36 below), so $M \in S_p$ iff $p > q$, for every $p > 0$.

(5) *Same profile, different spectra.* One atom in a block gives a rank-one matrix; two atoms in the same block, with the same total mass, give rank two (Proposition 5.35). So the profile cannot see individual singular values, and Theorem 5.36 needs its separation hypothesis.

Each example has a bounded symbol whenever it is bounded, with the constant of Theorem 4.2.

5.6. What the block profile determines

The profile $a(\nu)$ determines the operator norm, compactness and the S_p norms for $p \geq 1$, up to constants. It does not determine finer spectral information.

PROPOSITION 5.35. *Let R have density $\theta \in (0,1]$ with constant B.*

(1) $c\sup_j a_j(\nu) \leq \|M_{\nu,R}\| \leq C\sup_j a_j(\nu) + C\nu((1,\infty))$.

(2) If ν consists of N atoms at distinct points of $(0,\infty)$, then $\operatorname{rank} M_{\nu,R} = N$. In particular measures with the same profile can have different ranks, and no statement about individual singular values depends on $a(\nu)$ alone.

PROOF. (1) The lower bound is the test in the proof of Theorem 5.31; the upper bound is Theorem 2.11 with $[\nu]_\theta \lesssim \sup_j a_j$. (2) $M = \mathcal{N}^*\mathcal{N}$ with $\mathcal{N}$ mapping into $L^2(\nu) \cong \mathbb{C}^N$, and the vectors $(n^{-1/2-u_i})_{n\in\mathcal{S}_R}$ are linearly independent. Indeed, fix a prime $p \in R$ and restrict them to $n = p^k$, $0 \leq k < N$. Removing the common factors $p^{-k/2}$ leaves the Vandermonde matrix $((p^{-u_i})^k)_{k,i}$, whose nodes are distinct. Thus $\mathcal{N}$ is onto. □

Individual singular values are controlled once each atom has its own scale.

THEOREM 5.36 (Separated atoms). *Let R have density $\theta \in (0,1]$ with constant B. There is $L = L(\theta, B)$ such that, if $\nu = \sum_i c_i\delta_{u_i}$ with $u_i \in (0,1]$ and $|\log u_i - \log u_j| \geq L$ for $i \neq j$, and the sequence b below is bounded, then*

$$\frac{1}{2}b_n^* \leq s_n(M_{\nu,R}) \leq \frac{3}{2}b_n^* \qquad (n \geq 1),$$

where $b_i = c_i\zeta_R(1+2u_i) \asymp c_i u_i^{-\theta}$ and b^* is its decreasing rearrangement. In particular $M_{\nu,R} \in S_p$ if and only if $b \in \ell^p$, for every $0 < p < \infty$.

PROOF. By Lemma 5.1 the nonzero spectrum of $M_{\nu,R}$ is that of the matrix $\big(\sqrt{c_ic_j}\,\zeta_R(1+u_i+u_j)\big)_{i,j} = D^{1/2}\widetilde{K}D^{1/2}$, with $D=\operatorname{diag}(b_i)$ and $\widetilde{K}$ the Gram matrix of the normalized kernels k_{u_i}. As in the proof of Theorem 5.31, $\widetilde{K}_{ij} \le 2e^{2B+1}(2\sqrt{u_iu_j}/(u_i+u_j))^{\theta} \le 2^{1+\theta}e^{2B+1}e^{-\theta|\log u_i-\log u_j|/2}$. At most two indices j have $|\log u_i-\log u_j|\in[mL,(m+1)L)$, so for L large Schur's test gives $\|\widetilde{K}-I\|\le 1/2$. Hence $D/2 \le D^{1/2}\widetilde{K}D^{1/2} \le 3D/2$, and the min-max principle compares eigenvalues. □

The comparison survives when b is unbounded and the Helson form defines an unbounded operator. Take R to be all primes and ν a finite positive measure on $(0,\infty)$. On $\ell^2(\mathbb{N})$ let $\mathcal{N}_{\max}a(u)=\sum_n a_n n^{-1/2-u}$, a series that converges absolutely for $u>0$, with domain $\{a:\mathcal{N}_{\max}a\in L^2(\nu)\}$, and put $H_\nu=\mathcal{N}_{\max}^*\mathcal{N}_{\max}$. Ma, Lu and Zu [59, Theorem 1.2] show that $\mathcal{N}_{\max}$ is the closure of $\mathcal{N}$ on finitely supported vectors, so that H_ν is the positive self-adjoint operator of the closure of the Helson form $\sum_{m,n}\alpha_\nu(mn)a_n\overline{a_m}$. The next theorem is about H_ν and does not use that identification.

THEOREM 5.37 (Unbounded separated atoms). *Let L be as in Theorem 5.36 for R all primes, with $L>\log 2$, and let $\nu=\sum_{i\in I}c_i\delta_{u_i}$ with I infinite, $c_i>0$, $\sum_i c_i<\infty$, $u_i\in(0,1]$ and $|\log u_i-\log u_j|\ge L$ for $i\ne j$. Put $b_i=c_i\zeta(1+2u_i)$ and $D=\operatorname{diag}(b_i)$ on $\ell^2(I)$.*

(1) $\ker H_\nu$ is infinite-dimensional, and on its orthogonal complement H_ν is unitarily equivalent to a positive self-adjoint operator A on $\ell^2(I)$ with form domain $\operatorname{Dom}D^{1/2}$ and $D/2\le A\le 3D/2$ as forms.

(2) If $b_i\to\infty$, the nonzero spectrum of H_ν consists of eigenvalues $\lambda_1\le\lambda_2\le\cdots$ of finite multiplicity, 0 is isolated, and $b_n^{\uparrow}/2\le\lambda_n\le 3b_n^{\uparrow}/2$, where $b^{\uparrow}$ is the increasing rearrangement of b.

(3) $\sigma_{\mathrm{ess}}(H_\nu)=\{0\}$ if and only if $\#\{i:\varepsilon\le b_i\le K\}<\infty$ for all $0<\varepsilon<K<\infty$.

In particular, unboundedness of b is not enough: for $u_i=e^{-Li}$, $c_i=2^{-i}$ for even i and $c_i=1/\zeta(1+2u_i)$ for odd i, the operator H_ν is unbounded and has essential spectrum in $(0,\infty)$.

PROOF. Let $\zeta_i=\zeta(1+2u_i)$, $e_i=\zeta_i^{-1/2}(n^{-1/2-u_i})_{n\ge1}$, and let $U:L^2(\nu)\to\ell^2(I)$, $(Uf)_i=\sqrt{c_i}f(u_i)$, which is unitary. The proof of Theorem 5.36 bounds the Gram matrix of the e_i by $\|\widetilde{K}-I\|\le1/2$, and uses neither the c_i nor finiteness of I. So $Sx=\sum_i x_ie_i$ is bounded from $\ell^2(I)$ to $\ell^2(\mathbb{N})$, with $\|x\|^2/2\le\|Sx\|^2\le3\|x\|^2/2$ and $S^*a=(\langle a,e_i\rangle)_i$.

Put $T=SD^{1/2}$ on $\operatorname{Dom}D^{1/2}$. The lower Riesz bound gives $\|D^{1/2}g\|^2\le 2\|Tg\|^2$, so T is closed because $D^{1/2}$ is. Since S is bounded, $T^*=D^{1/2}S^*$ on $\{a:\sum_i b_i|\langle a,e_i\rangle|^2<\infty\}$, and $\sqrt{b_i}\langle a,e_i\rangle=\sqrt{c_i}\,\mathcal{N}_{\max}a(u_i)$; hence $T^*=U\mathcal{N}_{\max}$. Therefore $\mathcal{N}_{\max}$ is closed and densely defined, $\mathcal{N}_{\max}^*=TU$, and $\ker\mathcal{N}_{\max}^*=\{0\}$ by the lower bound. The polar decomposition makes H_ν on $(\ker\mathcal{N}_{\max})^{\perp}$ unitarily equivalent to $\mathcal{N}_{\max}\mathcal{N}_{\max}^*=U^{-1}AU$ with $A=T^*T$, the operator of the closed form $\|Tg\|^2=\langle\widetilde{K}D^{1/2}g,D^{1/2}g\rangle$ on $\operatorname{Dom}D^{1/2}$; this gives $D/2\le A\le3D/2$.

The kernel of $\mathcal{N}_{\max}$ contains $(\operatorname{ran} S)^{\perp}$. Let P be the orthogonal projection onto the closed range of S. The lower Riesz bound gives $\|S^*a\|^2 \geq \|Pa\|^2/2$. Since $1/\zeta(1+s) \leq s$ and separation gives $\sum_i u_i < \infty$, the standard unit vectors $\delta_n \in \ell^2(\mathbb{N})$ satisfy

$$\|S^*\delta_n\|^2 = \sum_i \frac{n^{-1-2u_i}}{\zeta_i} \leq \frac{2}{n}\sum_i u_i \longrightarrow 0.$$

If $(\operatorname{ran} S)^{\perp}$ were finite-dimensional, then $I-P$ would have finite rank and $\|P\delta_n\| \to 1$, a contradiction. Thus $\ker \mathcal{N}_{\max}$ is infinite-dimensional, proving (1).

(2) If $b_i \to \infty$, then D^{-1} is compact and $\min_i b_i > 0$. Inverses reverse the form order on a common form domain, so $2D^{-1}/3 \leq A^{-1} \leq 2D^{-1}$. Hence A^{-1} is compact, and the min-max principle gives the bounds on λ_n; 0 is isolated since $\lambda_1 \geq \min_i b_i/2$.

(3) If the condition holds, split $\nu = \nu_0 + \nu_1$ according as $b_i < 1$ or $b_i \geq 1$; then $b_i \to 0$ along the first family and $b_i \to \infty$ along the second. For ν_0 the operator $T_0 = S_0 D_0^{1/2}$ is compact, so $\mathcal{N}_{\max}$ for ν_0 is compact and everywhere defined, and H_{ν_0} is compact. The maximal operators for ν and ν_1 have the same domain, and the forms add, so $H_\nu = H_{\nu_1} + H_{\nu_0}$. By Weyl's theorem and (2), $\sigma_{\mathrm{ess}}(H_\nu) = \sigma_{\mathrm{ess}}(H_{\nu_1}) = \{0\}$. If the condition fails, let J be infinite with $\varepsilon \leq b_i \leq K$ on J, and suppose that $\sigma_{\mathrm{ess}}(A) \subseteq \{0\}$. Then $f(A) = A(1+A)^{-2}$ is compact, as a norm limit of finite-rank spectral truncations. With (δ_i) the standard basis of $\ell^2(I)$, the vectors $h_i = (1+A)^{1/2}\delta_i$, $i \in J$, have $\|h_i\|^2 = 1 + b_i$, and they tend weakly to 0 since $\langle h_i, y\rangle = \langle \delta_i, (1+A)^{1/2}y\rangle$ for $y \in \operatorname{Dom} A^{1/2}$. But

$$\langle f(A)h_i, h_i\rangle = 1 - \langle (1+A)^{-1}\delta_i, \delta_i\rangle \geq 1 - \frac{1}{1+b_i/2} \geq \frac{\varepsilon}{2+\varepsilon},$$

using $(1+A)^{-1} \leq (1+D/2)^{-1}$. This contradiction gives essential spectrum of A, hence of H_ν, in $(0,\infty)$; and 0 always lies in $\sigma_{\mathrm{ess}}(H_\nu)$ by (1).

In the example, $\sum_i c_i < \infty$ because $1/\zeta(1+2u) \leq 2u$; $b_i = 1$ for odd i, and $b_i \geq 2^{-i}e^{Li}/2 \to \infty$ for even i since $L > \log 2$. So H_ν is unbounded, and (3) gives essential spectrum in $(0,\infty)$. □

Returning to bounded matrices, the profile cannot determine individual singular values, but it does determine their distribution up to dilations, and with it every Schatten and Lorentz class. For a positive sequence or operator x write $n_x(t) = \#\{n : x_n^* > t\}$, with x^* the decreasing rearrangement of the sequence or the singular values of the operator, and let $\ell^{p,q}$ and $S_{p,q}$ be the Lorentz sequence spaces and the corresponding Schatten–Lorentz classes, $0 < p < \infty$, $0 < q \leq \infty$; $S_{p,p} = S_p$, and $S_{p,\infty}$ is weak S_p.

THEOREM 5.38 (Distribution of the spectrum). *Let R have density $\theta \in (0,1]$ with constant B, and let ν be a finite positive measure on $(0,1]$. There are $C, \varepsilon > 0$ depending only on θ and B such that, for all $\tau > 0$ and $0 < r \leq 1$,*

$$n_a(\tau) \leq C\, n_M(\varepsilon\tau), \tag{5.13}$$

$$n_M(C\tau) \leq C_r\, \tau^{-r} \sum_{a_j > \tau} a_j^r, \tag{5.14}$$

where $M = M_{\nu,R}$, $a = a(\nu)$, and C_r depends also on r. Consequently, for every finite positive ν on $(0,\infty)$, every $0 < p < \infty$ and every $0 < q \le \infty$,

$$\|M_{\nu,R}\|_{S_{p,q}} \asymp \|a(\nu)\|_{\ell^{p,q}} + \nu((1,\infty)),$$

with constants depending only on θ, B, p and q. In particular Theorem 5.31 holds for every $0 < p < \infty$.

The proof rests on two estimates. The first is that one block is almost of finite rank: with $\nu_j = \nu|_{I_j}$ and $M_j = M_{\nu_j,R}$, there is $\eta = \eta(\theta, B) < 1$ with

$$s_{N+1}(M_j) \le C a_j \eta^{2N} \qquad (N \ge 0), \qquad \text{hence} \qquad \operatorname{Tr} M_j^r \le C_r a_j^r. \tag{5.15}$$

Indeed, $M_j = \mathcal{N}_j^* \mathcal{N}_j$ by Lemma 5.1, and $\mathcal{N}_j^*$ is synthesis $f \mapsto \int k(u) f(u)\, \mathrm{d}\nu_j(u)$ by the $\ell^2(\mathcal{S}_R)$-valued function $k(u) = (n^{-1/2-u})_n$, which is analytic in $\Re u > 0$ with $\|k(u)\|^2 = \zeta_R(1 + 2\Re u)$. Write $u = t(c + z)$ with $t = e^{-j}$ and $c = (1 + e^{-1})/2$; then I_j corresponds to $|z| \le \delta = (1 - e^{-1})/2$, and on $|z| = 1/2$ we have $\Re u \ge t e^{-1}/2$, so $\|k\| \le C t^{-\theta/2}$ there. Truncating the Taylor series in z after N terms gives an operator of rank at most N, and by Cauchy's estimate its Hilbert–Schmidt distance to $\mathcal{N}_j^*$ is at most $C(t^{-\theta}\nu(I_j))^{1/2}(2\delta)^N = C\sqrt{a_j}\,\eta^N$ with $\eta = 2\delta$.

The second estimate is the concavity inequality $\operatorname{Tr}(A + B)^r \le \operatorname{Tr} A^r + \operatorname{Tr} B^r$ for positive A, B and $0 < r \le 1$. For $r = 1$ this is trace additivity. For $0 < r < 1$ it follows from $x^r = (\sin \pi r/\pi) \int_0^\infty t^{r-1} x (x + t)^{-1}\, \mathrm{d}t$ and $\operatorname{Tr} A(A + B + t)^{-1} \le \operatorname{Tr} A(A + t)^{-1}$.

Proof of Theorem 5.38. The idea is to count the blocks with large profile, which by the block estimate (5.15) carry essentially one singular value each, and to absorb all other blocks into an operator of small norm.

Upper estimate. Let $J = \{j : a_j > \tau\}$ and split $M = M' + M''$, where M' is the matrix of ν restricted to the blocks in J and M'' that of the rest; both are positive, since $\nu \mapsto M_{\nu,R}$ is linear and positivity-preserving. Every block of the second measure has profile at most τ, so part (1) of Proposition 5.35 gives $\|M''\| \le C\tau$. The first is the sum of the block matrices M_j, $j \in J$, so concavity and (5.15) give $\operatorname{Tr} M'^r \le \sum_{j \in J} \operatorname{Tr} M_j^r \le C_r \sum_{j \in J} a_j^r$. By Weyl's inequality and Chebyshev's inequality for the eigenvalues of M', $n_M((C + 1)\tau) \le n_{M'}(\tau) \le \tau^{-r} \operatorname{Tr} M'^r$.

Lower estimate. Let F be any finite set of N blocks with $a_j > \tau$, and let $\mu = \sum_{j \in F} (\tau/a_j)\nu_j \le \nu$, so that $Q = M_{\mu,R} \le M$ and every nonzero entry of $a(\mu)$ equals τ. Then $\|Q\| \le C\tau$, $\operatorname{Tr} Q \ge cN\tau$ by Proposition 5.2 and $\zeta_R(1 + 2u) \ge c u^{-\theta}$, and $\operatorname{Tr} Q^r \le C_r N \tau^r$ for $r = 1/2$ by (5.15) and concavity. The eigenvalues of Q that are at most $\varepsilon\tau$ contribute at most $(\varepsilon\tau)^{1/2} \operatorname{Tr} Q^{1/2} \le C\varepsilon^{1/2} N\tau$ to the trace. With ε so small that this is at most $cN\tau/2$, at least $cN/(2C)$ eigenvalues of Q exceed $\varepsilon\tau$, and by the min-max principle so do as many eigenvalues of M. Since F was arbitrary, $n_a(\tau) \le (2C/c)\, n_M(\varepsilon\tau)$.

Lorentz norms. The quasi-norm $\|x\|_{\ell^{p,q}}^q$ is comparable to $\int_0^\infty \tau^{q-1} n_x(\tau)^{q/p}\, \mathrm{d}\tau$ for $q < \infty$, and $\|x\|_{\ell^{p,\infty}} = \sup_\tau \tau n_x(\tau)^{1/p}$, the same for singular values. The lower bound for M follows at once from (5.13). For the upper bound fix $r < \min(p, 1)$

and split the sum in (5.14) into the bands $2^l\tau < a_j \le 2^{l+1}\tau$:

$$\tau^p n_M(C\tau) \le C_r \sum_{l\ge 0} 2^{-l(p-r)} (2^l\tau)^p n_a(2^l\tau).$$

Raise to the power q/p and integrate against $\mathrm{d}\tau/\tau$, which is invariant under the dilations $\tau \mapsto 2^l\tau$; use subadditivity of $x \mapsto x^{q/p}$ if $q \le p$ and Minkowski's inequality in $L^{q/p}$ if $q > p$. The geometric factor makes the series converge. For $q = \infty$ take suprema. Finally, the part of ν on $(1,\infty)$ gives a matrix in every $S_{p,q}$ with norm at most $C\nu((1,\infty))$: write $k(u) = e_1 + \psi(u)$ with $\|\psi(w)\|^2 = \zeta_R(1+2\Re w) - 1 \le C4^{-\Re w}$, apply the Taylor argument on the unit intervals $[m, m+1)$, $m \ge 1$, to ψ, and absorb the rank-one part e_1 by $s_{n+1}(X+Y) \le s_n(X)$ for $\operatorname{rank} Y \le 1$. The entry $\alpha_\nu(1) \ge \nu((1,\infty))$ gives the reverse bound. □

In the Banach range of exponents the upper half of the theorem also follows from Olsen's local embedding of $\mathscr{H}^2$ into a weighted Dirichlet space of the half-plane [68], together with Luecking's description of Schatten class Toeplitz operators there [57]. The argument above avoids interpolation, and so covers $p < 1$, $q < 1$ and weak S_1 directly.

Two pointwise consequences answer the question of what lies between the profile and individual singular values.

COROLLARY 5.39. *Let R and ν be as in Theorem 5.38.*

(1) If the profile is doubling, $a_n^* \le D a_{2n}^*$ for all n, then $s_n(M_{\nu,R}) \asymp a_n^*$ for all n, with constants depending only on θ, B and D.

(2) If ν is purely atomic with at most K atoms in each block, and b is the sequence of Theorem 5.36, then for all n

$$c\, b^*_{\lceil Cn\rceil} \le s_n(M_{\nu,R}) \le C\, b_n^*,$$

with c, C depending only on θ, B and K.

PROOF. (1) Iterating the doubling condition gives $a_m^* \le D(n/m)^A a_n^*$ for $m \le n$, with $A = \log_2 D$. Take $\tau = a_n^*$ and $0 < r \le 1$ with $Ar < 1$ in (5.14). Since $n_a(\tau) < n$, $\sum_{a_j > \tau} a_j^r \le \sum_{m<n} (D(n/m)^A\tau)^r \le C\tau^r n$, so $n_M(Ca_n^*) \le Cn$, that is $s_{Cn}(M) \le Ca_n^*$; doubling removes the index change. The lower bound follows in the same way from (5.13) with $\tau = a_n^*/2$.

(2) Let L be as in Theorem 5.36 and put $\ell = \lceil L\rceil$. Colour each atom by the residue of its block index modulo $\ell+1$ and its position, among at most K, inside the block. Atoms of the same color are L-separated, so each of the at most $K(\ell+1)$ colored parts M_c satisfies Theorem 5.36. Since $M = \sum_c M_c$ is a sum of positive operators, Weyl's inequality gives $n_M(K(\ell+1)t) \le \sum_c n_{M_c}(t) \le n_b(2t/3)$, the upper bound. Conversely $M \ge M_c$ for every c, and among the $K(\ell+1)n$ largest entries of b some color has at least n, so $s_n(M) \ge b^*_{K(\ell+1)n}/2$. □

The index change in (2) cannot be removed, even for $K = 1$: two atoms with the same normalized mass, at $e^{-L}(1 \pm \delta)$ on either side of a block boundary, give almost the same rank-one operator, so the second singular value of the pair is $O(\delta^2)$ times the first: the normalized Gram matrix has eigenvalues $1 \pm \rho_\delta$, with $1 - \rho_\delta = O(\delta^2)$. Placing such pairs at rapidly separated scales with masses 2^{-j^2} and

$\delta_j \to 0$ gives one measure with $s_{2j}/a^*_{2j} \to 0$. For profiles $a_j \asymp e^{-j}$, (5.14) permits a root-exponential upper bound, whereas separated single atoms give exponential decay. A matching lower bound requires information about the distribution of mass within each block.

5.7. Distribution functions and endpoint smoothness

For all primes, write $M_\nu = M_{\nu,R}$ and $F(t) = \nu((0,t])$. The dyadic criterion has an equivalent formulation in terms of F at 0, for every Schatten exponent. Interior jumps prevent a global smoothness criterion.

THEOREM 5.40. *Suppose that ν is a finite positive measure on $(0,\infty)$. For every $p > 0$,*

$$M_\nu \in S_p \quad \Longleftrightarrow \quad \int_0^1 \left(\frac{F(t)}{t}\right)^p \frac{\mathrm{d}t}{t} < \infty.$$

This is a weighted endpoint Hardy condition. For any $p > 0$ and any fixed $s \in \mathbb{R}$, $0 < r < \infty$, $0 < q \le \infty$, the equivalence

$$M_\nu \in S_p \quad \Longleftrightarrow \quad F|_{(0,1)} \in B^s_{r,q}(0,1)$$

fails for some finite positive ν. Here $B^s_{r,q}$ is the classical unweighted inhomogeneous Besov space, interpreted by restriction from $\mathbb{R}$.

PROOF. Let $I_j = (e^{-j-1}, e^{-j}]$, $a_j = e^j \nu(I_j)$, and $b_j = e^j F(e^{-j})$ for $j \ge 0$. Then

$$b_j = \sum_{k \ge j} e^{-(k-j)} a_k, \qquad 0 \le a_j \le b_j.$$

The convolution on the right preserves ℓ^p for $p \ge 1$ by Minkowski's inequality. For $0 < p < 1$, subadditivity of x^p gives

$$\sum_{j \ge 0} b_j^p \le \sum_{j \ge 0} \sum_{k \ge j} e^{-p(k-j)} a_k^p \le \frac{1}{1 - e^{-p}} \sum_{k \ge 0} a_k^p.$$

Thus $a \in \ell^p$ if and only if $b \in \ell^p$ for every $p > 0$. For $e^{-j-1} < t \le e^{-j}$, monotonicity of F yields

$$e^{-1} b_{j+1} \le \frac{F(t)}{t} \le e b_j.$$

Integration with $\mathrm{d}t/t$ on each I_j shows that the displayed integral is finite exactly when $b \in \ell^p$; the omitted $b_0 = F(1)$ is finite. Theorem 5.38, with all primes and density parameter 1, gives $M_\nu \in S_p$ exactly when $a \in \ell^p$. This proves the first assertion.

For the second assertion, fix $u_0 \in (0,1)$ and take $\nu = \delta_{u_0}$. Then $M_\nu = v \otimes v$ with $v_n = n^{-1/2-u_0} \in \ell^2$, so it belongs to every S_p, while $F|_{(0,1)}$ is a step function. The finite-difference definition of classical Besov spaces gives

$$F|_{(0,1)} \in B^s_{r,q}(0,1) \quad \Longleftrightarrow \quad s < 1/r \quad \text{or} \quad (s = 1/r \text{ and } q = \infty).$$

Indeed, for any integer difference order $m > \max(s, 1/r)$, the L^r norm of the mth difference of a step at displacement h is comparable to $h^{1/r}$ as $h \downarrow 0$. Smooth cutoffs extend the step from $(0,1)$ without introducing another jump.

In the remaining index range $s \le 1/r$, take $\mathrm{d}\nu(u) = \beta u^{\beta-1}\mathbf{1}_{(0,1)}(u)\,\mathrm{d}u$ with $0 < \beta < 1$. Now $F(t) = t^\beta$ on $(0,1)$. It belongs to $B^s_{r,q}(0,1)$ for every $s \le 1/r$ and every q: with a smooth cutoff beyond 1, its differences of any integer order $m > \beta + 1/r$ have L^r norm $O(h^{\beta+1/r})$, and $s < \beta + 1/r$. For $s \le 0$ use the embedding from any positive smoothness below $1/r$.

However,

$$\int_0^1 \left(\frac{F(t)}{t}\right)^p \frac{\mathrm{d}t}{t} = \int_0^1 t^{p(\beta-1)-1}\,\mathrm{d}t = \infty.$$

By the first assertion, $M_\nu \notin S_p$. In fact it is unbounded by Corollary 4.3, since $F(t)/t = t^{\beta-1}$ is unbounded at zero. Thus no choice of global classical Besov indices gives the equivalence. □

The criterion is a Besov condition localized at the endpoint 0. For $s > 0$, $0 < r \le \infty$ and $0 < q \le \infty$, say that a function f on $(0,1)$ belongs to the *endpoint Besov space* $B^s_{r,q}(0)$ if there is a polynomial P of degree less than s with

$$\int_0^1 \big(h^{-s} A_r(f-P;h)\big)^q \frac{\mathrm{d}h}{h} < \infty, \qquad A_r(g;h) = \left(\frac{1}{h}\int_0^h |g(x)|^r\,\mathrm{d}x\right)^{1/r},$$

with the usual changes for $r = \infty$ (essential supremum over $(0,h)$) and $q = \infty$ (supremum over h). For $q = \infty$ this is the Calderón–Zygmund pointwise class T^r_s [23], taken one-sidedly at the endpoint of $(0,1)$; finite q gives the corresponding Besov-type scale, with q-summation over scales.

THEOREM 5.41. *For every finite positive ν on $(0,\infty)$, every $p > 0$ and every $0 < r \le \infty$,*

$$M_\nu \in S_p \quad\Longleftrightarrow\quad F \in B^1_{r,p}(0) \quad\Longleftrightarrow\quad \int_0^1 \left(\frac{|F(h)-F(0)|}{h}\right)^p \frac{\mathrm{d}h}{h} < \infty,$$

where $F(0) = 0$. Thus the spectral condition is the one-dimensional Besov condition on F of smoothness 1 and summability p at the endpoint 0, with any integrability index r; the last form uses the first difference $\Delta_h F(0) = F(h) - F(0)$ only.

PROOF. The third condition is the integral of Theorem 5.40, since $F(0) = 0$. A polynomial of degree less than 1 is a constant c. Since ν is finite, $F(x) = \nu((0,x]) \to 0$ as $x \downarrow 0$. If $c \neq 0$, then $A_r(F-c;h) \ge |c|/2$ for all small h, and the integral defining $B^1_{r,p}(0)$ diverges like $\int_0 h^{-p-1}\,\mathrm{d}h$. So $F \in B^1_{r,p}(0)$ exactly when the integral converges with $c = 0$. As F is nondecreasing and nonnegative,

$$2^{-1/r} F(h/2) \le A_r(F;h) \le F(h),$$

the lower bound from the integral over $(h/2,h)$ (for $r = \infty$ read $2^{-1/r} = 1$). Hence $\int_0^1 (A_r(F;h)/h)^p\,\mathrm{d}h/h$ lies between $2^{-p-p/r}\int_0^{1/2}(F(t)/t)^p\,\mathrm{d}t/t$ and $\int_0^1 (F(t)/t)^p\,\mathrm{d}t/t$, and these two integrals converge together because $F(t)/t \le 2F(1)$ on $[1/2,1]$. Theorem 5.40 completes the proof. □

REMARK 5.42. The index convention matters at the integer smoothness $s = 1$. If linear polynomials were allowed in the definition of $B^1_{r,p}(0)$, or second differences used, then Lebesgue measure on $(0,1)$, with $F(t) = t$, would qualify; but its weighted integral is $\int_0^1 \mathrm{d}t/t = \infty$, so by Theorem 5.40 its matrix lies in no S_p. So membership

in S_p is a Besov condition on F, but only in the endpoint sense $F \in B^1_{r,p}(0)$ of Theorem 5.41; by Theorem 5.40, no global classical Besov space $B^s_{r,q}(0,1)$ serves.

5.8. Eigenvalue decay

Below the curve the spectrum is far smaller than Theorem 5.31 requires.

LEMMA 5.43. *Let h be completely monotone on $(0,\infty)$ with $h(t) \le Ct^{-\theta}$ on $(0,1]$ for some $\theta < 1$ and $h(t) \le Ce^{-ct}$ on $[1,\infty)$. The integral operator with kernel $h(u+v)$ on $L^2(0,\infty)$ has $s_n \le C'e^{-c'\sqrt{n}}$.*

PROOF. Bernstein's representation gives $|h^{(m)}(t)| \le m!(2/t)^m h(t/2)$. Fix m. Discard the outputs $u < e^{-m}$ and $u > m$, whose Hilbert–Schmidt norms are $O(e^{-cm})$ because $\theta < 1$ and the tail is exponential. Cover $[e^{-m}, m]$ by $O(m)$ intervals $[a,(1+\delta)a]$ with $\delta < 1/2$, and on each replace $h(u+v)$ by its Taylor polynomial of degree $m-1$ in u about a. Each piece has rank m, and the remainders have total Hilbert–Schmidt norm squared at most $C(2\delta)^{2m}\int_0^\infty t h(t)^2\,\mathrm{d}t$. A rank-$Cm^2$ operator is thus within Ce^{-cm} in norm. □

THEOREM 5.44. *If R has density $\theta < 1$, the multiplicative Hilbert matrix on $\mathcal{S}_R \setminus \{1\}$ has $s_n \le Ce^{-c\sqrt{n}}$.*

PROOF. By Lemma 5.1, with Lebesgue measure and the index 1 removed, the companion kernel is $h_R(u+v)$ with $h_R(t) = \zeta_R(1+t) - 1$, completely monotone, $O(t^{-\theta})$ at 0 and $O(2^{-t})$ at infinity. Apply Lemma 5.43. □

At the other end of the family of chapter 4, where all signs are averaged, the decay is only polynomial, and its exact asymptotics can be computed.

THEOREM 5.45 (The full sign average). *Let A be the matrix with entries $\mathbf{1}_{\square}(mn)/(\sqrt{mn}\log mn)$ for $m,n \ge 2$. Then*

$$s_n(A) \sim \frac{1}{2n\log n} \qquad (n \to \infty).$$

In particular $A \in S_p$ exactly for $p > 1$, and A lies in weak S_1 but not in S_1.

PROOF. Write $m = qa^2$, $n = qb^2$ with q squarefree. Then $A = \bigoplus_q A_q$ with $A_q(a,b) = (2qab\log(qab))^{-1}$, where for $q = 1$ only $a,b \ge 2$ occur. For $q \ge 2$ let $L = \log q$, $u_t = (a^{-1-t})_{a\ge 1}$, and let T be exponential with rate L. Then $A_q = (2q)^{-1}\int_0^\infty e^{-Lt} u_t \otimes u_t\,\mathrm{d}t = (2qL)^{-1}\mathbb{E}[u_T \otimes u_T]$. Subtract the rank-one mean, $C_q = (2qL)^{-1} v \otimes v$ with $v = \mathbb{E}u_T$, whose entries are $v(a) = a^{-1}L/(L+\log a)$. The remainder $R_q = A_q - C_q$ is a covariance, hence positive, and since $\|\partial_t u_t\|_2$ is bounded and $\mathbb{E}T^2 = 2/L^2$,

$$\operatorname{Tr} R_q = \frac{\mathbb{E}\|u_T - v\|_2^2}{2qL} \le \frac{\mathbb{E}\|u_T - u_0\|_2^2}{2qL} \le \frac{C}{q\log^3 q}.$$

The eigenvalue of C_q is $c_q = \|v\|_2^2/(2qL) = \zeta(2)/(2q\log q)\,(1 + O(1/\log q))$, and the squarefree integers up to x number $x/\zeta(2) + o(x)$, so $C = \bigoplus_q C_q$ has $n_C(s) \sim 1/(2s\log(1/s))$ as $s \to 0$.

Let R be the sum of the R_q and the block A_1. For $q \le Q = s^{-1/2}$ use $R_q \le A_q \le (2/q)A_2$ and Lemma 5.43: the companion kernel of A_2 is $e^{-(u+v)\log 2/2}\zeta(2+u+v)/4$, completely monotone and exponentially decaying, so $s_j(A_2) \le Ce^{-c\sqrt{j}}$ and $n_{R_q}(s) \le C\log^2(1/s)$; the block A_1, with kernel $(\zeta(2+u+v)-1)/2$, is the same. For $q > Q$ use $n_{R_q}(s) \le \operatorname{Tr} R_q/s$. Altogether

$$n_R(s) \le Cs^{-1/2}\log^2(1/s) + \frac{C}{s}\sum_{q>Q}\frac{1}{q\log^3 q} = o\Big(\frac{1}{s\log(1/s)}\Big).$$

Since $C \le A = C+R$, Weyl's inequality gives $n_C(s) \le n_A(s) \le n_C((1-\varepsilon)s)+n_R(\varepsilon s)$ for every $\varepsilon \in (0,1)$, so $n_A(s) \sim 1/(2s\log(1/s))$, which inverts to the claim. □

The order $1/n$ of the singular values of a full sign average is decided by a single number, the mass of ν on the critical line itself.

PROPOSITION 5.46 (Endpoint law for full sign averages). *Let ν be a finite positive measure on $[0,\infty)$ and $A_\nu(m,n) = \mathbf{1}_\square(mn)\int (mn)^{-1/2-u}\,\mathrm{d}\nu(u)$ for $m,n \ge 1$. Then A_ν is bounded,*

$$\lim_{n\to\infty} n\,s_n(A_\nu) = \nu(\{0\}),$$

and $A_\nu \in S_1$ if and only if $\int \zeta(1+2u)\,\mathrm{d}\nu(u) < \infty$. In particular $s_n(A_\nu) = o(1/n)$ when $\nu(\{0\}) = 0$.

PROOF. As in the proof of Theorem 5.45, $A_\nu = \bigoplus_q A_q$ over squarefree q, with $A_q = q^{-1}\int q^{-2u} v_u \otimes v_u\,\mathrm{d}\nu(u)$ and $v_u = (a^{-1-2u})_{a\ge1}$. Put $L_q = \int q^{-2u}\,\mathrm{d}\nu$ and let $\bar v_q$ be the mean of v_u under $q^{-2u}\,\mathrm{d}\nu/L_q$. Then $A_q = C_q + R_q$ with $C_q = q^{-1}L_q\bar v_q \otimes \bar v_q$ and $R_q \ge 0$ a covariance, and $\operatorname{Tr} R_q \le q^{-1}\int q^{-2u}\|v_u - v_0\|_2^2\,\mathrm{d}\nu$. Since $\|v_u - v_0\|_2^2 \le C\min(u^2,1)$ and $\sum_q q^{-1-2u} \le 1 + 1/(2u)$, the sum of the R_q is trace class. The eigenvalue $c_q = q^{-1}L_q\|\bar v_q\|_2^2$ of C_q is at most $\zeta(2)\nu([0,\infty))/q$, which gives boundedness. As $q\to\infty$, $L_q \to b = \nu(\{0\})$, and if $b > 0$ the weights concentrate at $u = 0$, so $\bar v_q \to v_0$ and $qc_q \to \zeta(2)b$; if $b = 0$, $qc_q \to 0$. Since the squarefree integers have density $1/\zeta(2)$, $n_C(s) \sim b/s$, or $n_C(s) = o(1/s)$ if $b = 0$. Trace class gives $n_R(\varepsilon s) = o(1/s)$, and $n_C(s) \le n_{A_\nu}(s) \le n_C((1-\varepsilon)s) + n_R(\varepsilon s)$ by Weyl's inequality. Finally $A_\nu \ge 0$ and $\operatorname{Tr} A_\nu = \sum_m \int m^{-1-2u}\,\mathrm{d}\nu$. □

So along the family $\mu_{\nu,R}$ of chapter 4, with ν Lebesgue measure, the spectrum moves from $[0,\pi]$ (all coordinates on the curve) to $1/(2n\log n)$ (all signs averaged). The compression to $\mathcal{S}_R$ of a sparse set R decays root-exponentially (Theorem 5.44), but the averaged matrices with R and its complement both infinite are not computed.

5.9. An unbounded family with simple continuous spectrum

For $0 < c < 1$, put

$$\mathrm{d}\mu_c(s) = \frac{(s-1/2)^{c-1}e^{-(s-1/2)}}{\Gamma(c)}\,\mathrm{d}s, \qquad s > 1/2.$$

Let H_c be the positive self-adjoint operator associated with the closure of the Helson form on finite sequences whose matrix entries are $\alpha_c(mn)$, where $\alpha_c(n) = n^{-1/2}(1+\log n)^{-c}$.

THEOREM 5.47. *Conditional on the closed-operator identification and adjoint-core result of Ma, Lu and Zu [59, Theorem 1.2 and Lemma 2.7],*

$$\sigma(H_c) = \sigma_{\mathrm{ess}}(H_c) = [0, \infty), \qquad 0 < c < 1.$$

Here essential spectrum means spectrum after removal of isolated eigenvalues of finite multiplicity. The spectral type is Theorem 5.48.

PROOF. Write $g(u) = u^{c-1}e^{-u}/\Gamma(c)$, $u > 0$. Let N be the closure of $N_0 a(s) = \sum_n a_n n^{-s}$, initially defined on finite sequences. The cited identification gives $H_c = N^*N$. The map

$$W : L^2(\mu_c) \longrightarrow L^2(0, \infty), \qquad (W\varphi)(u) = \sqrt{g(u)}\varphi(u + 1/2)$$

is unitary. Set $A = WNN^*W^{-1}$ and denote its closed positive form by a. Thus

$$a[f, f] = \|N^*W^{-1}f\|_{\ell^2}^2, \qquad \mathrm{Dom}(a) = W\,\mathrm{Dom}(N^*).$$

By the cited adjoint-core result, $\mathcal{D} = C_c(0, \infty)$ is a form core for a: multiplication by $\sqrt{g}$ maps compactly supported continuous functions onto compactly supported continuous functions. On $\mathcal{D}$, the adjoint formula [59, Lemma 2.2] and the Dirichlet expansion of zeta give

$$a[f, f] = \int_0^\infty \int_0^\infty \sqrt{g(u)g(v)}\zeta(1 + u + v)f(v)\overline{f(u)}\,\mathrm{d}v\,\mathrm{d}u. \tag{5.16}$$

All exchanges here are absolutely convergent because the support stays away from zero.

On $\mathcal{D}$, introduce the homogeneous positive form

$$q_0[f, f] = \frac{1}{\Gamma(c)} \int_0^\infty \left| \int_0^\infty u^{(c-1)/2}e^{-tu}f(u)\,\mathrm{d}u \right|^2 \mathrm{d}t. \tag{5.17}$$

Integrating in t shows that its kernel is

$$k(u, v) = \frac{(uv)^{(c-1)/2}}{\Gamma(c)(u + v)}.$$

Put $h_1(s) = e^{-s/2}\zeta(1+s) - 1/s$. The Laurent expansion of ζ at 1 gives $h_1(s) = O(1)$ at zero, while $h_1(s) = O(1/s)$ at infinity. The difference of the two kernels is

$$C(u, v) = \Gamma(c)^{-1}(uv)^{(c-1)/2}h_1(u + v).$$

The substitution $s = u + v$, $r = u/(u + v)$ gives

$$\|C\|_{\mathrm{HS}}^2 = \frac{\mathrm{B}(c, c)}{\Gamma(c)^2} \int_0^\infty s^{2c-1}|h_1(s)|^2\,\mathrm{d}s < \infty. \tag{5.18}$$

The integrand is $O(s^{2c-1})$ at zero and $O(s^{2c-3})$ at infinity. Thus C is compact and self-adjoint, and $a[f, f] = q_0[f, f] + \langle Cf, f\rangle$ on $\mathcal{D}$.

This identity also settles the domains. For $M > \|C\|$, the positive squared norms $a[f, f] + M\|f\|^2$ and $q_0[f, f] + M\|f\|^2$ on $\mathcal{D}$ are equivalent. Indeed, each is bounded above by the other plus $\|C\|\|f\|^2$, and each dominates $M\|f\|^2$. Since a is closable on $\mathcal{D}$, so is q_0. Their closures have exactly the same form domain. Let K

be the positive self-adjoint operator of the closure q of q_0. The identity extends to the common form domain. The representation of closed forms then gives

$$\mathrm{Dom}(A) = \mathrm{Dom}(K), \qquad A = K + C. \tag{5.19}$$

For completeness, membership of the operator domain is the condition that the functional $h \mapsto a[f,h]$, or respectively $h \mapsto q[f,h]$, is bounded in $\|h\|$ on the common form domain. The two functionals differ by $\langle Cf, h\rangle$, so the condition is identical and gives the stated operator equality.

For $\lambda > 0$, let $U_\lambda f(u) = \lambda^{1/2} f(\lambda u)$. This is unitary, preserves $\mathcal{D}$, and homogeneity gives

$$q_0[U_\lambda f, U_\lambda f] = \lambda^{1-c} q_0[f,f].$$

Equivalence of the corresponding form norms extends this covariance to the closure, and hence

$$U_\lambda^{-1} K U_\lambda = \lambda^{1-c} K.$$

The operator K is nonzero: for a nonzero nonnegative $f \in \mathcal{D}$, (5.17) is strictly positive. By positivity and the spectral theorem, there is therefore a point $x > 0$ in $\sigma(K)$. Covariance implies that $\lambda^{1-c}x \in \sigma(K)$ for every $\lambda > 0$. As $1 - c > 0$, these points exhaust $(0,\infty)$. Closure and positivity give $\sigma(K) = [0,\infty)$. Every point of this interval is nonisolated, so $\sigma_{\mathrm{ess}}(K) = [0,\infty)$ as well.

The common operator domain in (5.19) and compactness of C allow the usual compact-perturbation argument also for these unbounded operators. Explicitly, a normalized singular Weyl sequence for K at any $x \geq 0$ is in $\mathrm{Dom}(A)$; it converges weakly to zero, so C sends it to zero in norm. It is therefore a singular Weyl sequence for A at the same point. Positivity of A yields $\sigma(A) = \sigma_{\mathrm{ess}}(A) = [0,\infty)$.

Finally, polar decomposition of the closed densely defined operator N identifies N^*N and NN^* on the orthogonal complements of their kernels. In particular their positive spectra coincide. Thus every $x > 0$ belongs to $\sigma(H_c)$. Positivity and closure give $\sigma(H_c) = [0,\infty)$, and every point is nonisolated, giving the essential-spectrum equality. □

Spectral type

Throughout, $0 < c < 1$, $\delta = 1 - c$, and A, K, C are as in the proof of Theorem 5.47: $A = K + C$ on $\mathrm{Dom}(A) = \mathrm{Dom}(K)$. Put

$$h(s) = e^{-s/2}\zeta(1+s) = \sum_{n\geq 1} \frac{1}{n} e^{-s x_n}, \qquad x_n = \log n + \frac{1}{2}.$$

The kernel of A is $\Gamma(c)^{-1}(uv)^{-\delta/2} h(u+v)$. Recall that $h_1(s) = h(s) - 1/s$. Since ζ has only the simple pole at 1 with residue 1, h_1 is entire. Moreover $|h_1^{(j)}(s)| \leq C_j (1+s)^{-1-j}$ for $s \geq 0$. For $t \in \mathbb{R}$ write U_t for the dilation U_{e^t} of the proof above and D for its generator, $U_t = e^{itD}$. Then $U_t K U_t^{-1} = e^{-\delta t} K$.

THEOREM 5.48. *Conditional on [59, Theorem 1.2 and Lemma 2.7], the part of H_c in $(\ker H_c)^\perp$ is unitarily equivalent to multiplication by the independent variable in $L^2((0,\infty), \mathrm{d}\lambda)$. Thus H_c has purely absolutely continuous spectrum of multiplicity one on $(0,\infty)$; it has no embedded eigenvalues and no singular continuous spectrum.*

The proof occupies the rest of this section. By polar decomposition of N, $H_c|_{(\ker H_c)^\perp}$ is unitarily equivalent to $A|_{(\ker A)^\perp}$. It suffices to prove the statement for A.

LEMMA 5.49. *C is trace class.*

PROOF. For $y>0$ let $e_y(u)=\Gamma(c)^{-1/2}u^{-\delta/2}e^{-yu}$. Then $\|e_y\|^2=(2y)^{-c}$. Writing $1/(u+v)=\int_0^\infty e^{-(u+v)y}\,\mathrm{d}y$, the kernels of K and A are those of

$$\int_0^\infty |e_y\rangle\langle e_y|\,\mathrm{d}y \quad\text{and}\quad \sum_{n\ge1}\frac{1}{n}|e_{x_n}\rangle\langle e_{x_n}|.$$

The intervals $I_n=[x_n,x_{n+1}]$ tile $[1/2,\infty)$ and have length $\ell_n=\log(1+1/n)$. Hence C has the kernel of

$$-\int_0^{1/2}|e_y\rangle\langle e_y|\,\mathrm{d}y+\sum_{n\ge1}\Big(\Big(\frac{1}{n}-\ell_n\Big)|e_{x_n}\rangle\langle e_{x_n}|+\int_{I_n}\big(|e_{x_n}\rangle\langle e_{x_n}|-|e_y\rangle\langle e_y|\big)\,\mathrm{d}y\Big).$$

The first term has trace norm $\int_0^{1/2}(2y)^{-c}\,\mathrm{d}y=1/(2-2c)$. Next, $0\le 1/n-\ell_n\le 1/(2n^2)$ and $\|e_x\|\le1$ for $x\ge1/2$. For $1/2\le x\le y$,

$$\|e_x-e_y\|^2\le\frac{(y-x)^2}{\Gamma(c)}\int_0^\infty u^{2-\delta}e^{-2xu}\,\mathrm{d}u\le M(y-x)^2.$$

Since $\||a\rangle\langle a|-|b\rangle\langle b|\|_1\le(\|a\|+\|b\|)\|a-b\|$, the nth summand has trace norm at most $1/(2n^2)+2M^{1/2}\ell_n^2$. The series converges in trace norm, hence in Hilbert–Schmidt norm. Its kernel is therefore the almost-everywhere limit of the partial kernels, which is the kernel of C. $\square$

LEMMA 5.50. *K is unitarily equivalent to multiplication by λ in $L^2((0,\infty),\mathrm{d}\lambda)$.*

PROOF. First, $\ker K=\{0\}$. Define

$$(Lf)(t)=\Gamma(c)^{-1/2}\int_0^\infty u^{-\delta/2}e^{-tu}f(u)\,\mathrm{d}u,\qquad t>0.$$

For each $t>0$, $u\mapsto\Gamma(c)^{-1/2}u^{-\delta/2}e^{-tu}$ lies in L^2. If $f_n\in\mathcal{D}$ converges to f in form norm, then Lf_n converges in $L^2(\mathrm{d}t)$ and pointwise to Lf. An almost-everywhere convergent subsequence therefore gives $q[f,f]=\|Lf\|_2^2$. If $Kf=0$, then $Lf=0$ almost everywhere, hence everywhere by continuity. Since $e^{-tu}u^{-\delta/2}f(u)\in L^1$ for every $t>0$, uniqueness of the Laplace transform gives $u^{-\delta/2}f(u)=0$, so $f=0$.

Now $X=\delta^{-1}\log K$ is self-adjoint, and $U_t^{-1}XU_t=X+t$. Hence

$$U_t^{-1}e^{isX}U_t=e^{ist}e^{isX},\qquad s,t\in\mathbb{R}.$$

These are the Weyl relations. By the uniqueness theorem of von Neumann [64], the pair is unitarily equivalent to a direct sum of m copies of the Schrödinger pair on $L^2(\mathbb{R})$: multiplication by e^{isx} and translation by t. Then D has spectral multiplicity m. On the other hand, $f\mapsto e^{y/2}f(e^y)$ maps $L^2(0,\infty)$ onto $L^2(\mathbb{R})$ and conjugates U_t to translation by t. The generator of translations has simple spectrum, so $m=1$. Hence $K=e^{\delta X}$ is unitarily equivalent to multiplication by $e^{\delta x}$ in $L^2(\mathbb{R})$. The substitution $\lambda=e^{\delta x}$ completes the proof. $\square$

LEMMA 5.51. *Let $Af = Ef$ with $E > 0$, and put $\psi(u) = u^{-\delta/2} f(u)$ and $\kappa = E\Gamma(c)$. After correction on a null set, ψ is smooth on $(0,\infty)$, and*

$$\int_0^\infty h(u+v)\psi(v)\,\mathrm{d}v = \kappa u^{\delta}\psi(u), \qquad u > 0. \tag{5.20}$$

Moreover, $u^{\delta/2}\psi \in L^2(0,\infty)$, $\psi \in L^1(0,\infty)$, and ψ and ψ' are $O(e^{-u/2})$ as $u \to \infty$. We may take ψ real.

PROOF. Let $\chi \in C_c(0,\infty)$ and let $a(u,v)$ be the kernel in (5.16). The function $u \mapsto \int a(u,v)|\chi(v)|\,\mathrm{d}v$ is bounded by $C_\chi u^{-\delta/2}e^{-u/2}$ and therefore lies in L^2. Approximating f in form norm by elements of $\mathcal{D}$ gives $\langle Af, \chi\rangle = \iint a(u,v)f(v)\bar\chi(u)\,\mathrm{d}v\,\mathrm{d}u$. Hence $Af(u) = \int a(u,v)f(v)\,\mathrm{d}v$ almost everywhere, which is (5.20). Further, $\int_0^1 |\psi| \le (\int_0^1 u^{-\delta})^{1/2}\|f\|$ since $\delta < 1$. For $u \ge 1$, $h(u+v) \le 2e^{-(u+v)/2}$. The left side of (5.20) is therefore smooth in $u > 0$, and it and its derivative are $O(e^{-u/2})$. Real and imaginary parts of f are eigenfunctions, because A has a real kernel. □

For the next two lemmas fix such ψ and κ. Let $\sigma_0 = (1+\delta)/2 < 1$, and write $z = \sigma + i\tau$. Put

$$\Psi(z) = \int_0^\infty u^{z-1}\psi(u)\,\mathrm{d}u, \qquad R(z) = \int_0^\infty u^{z-1} r(u)\,\mathrm{d}u,$$
$$r(u) = \int_0^\infty h_1(u+v)\psi(v)\,\mathrm{d}v.$$

By Cauchy–Schwarz, $\int_0^\infty u^{\sigma-1}|\psi| < \infty$ for $\sigma > \sigma_0$. So Ψ is analytic in $\sigma > \sigma_0$ and bounded on each vertical strip $\sigma_0 + \varepsilon \le \sigma \le \sigma'$. Its boundary values on $\sigma = \sigma_0$ are the Mellin–Plancherel transform of ψ, and they lie in $L^2(\mathrm{d}\tau)$, because $\int |\psi|^2 u^{2\sigma_0 - 1} = \|f\|^2$.

The function r is entire. Indeed, for $|u| \le \rho$ and $v \ge 2\rho + 2$ one has $|h_1(u+v)| \le C/(1+v)$, and $\psi \in L^1$. On $[0,\infty)$ it satisfies $r^{(j)}(u) = O((1+u)^{-1-j})$. Integrating by parts k times,

$$R(z) = \frac{(-1)^k}{z(z+1)\cdots(z+k-1)} \int_0^\infty u^{z+k-1} r^{(k)}(u)\,\mathrm{d}u.$$

Hence R is meromorphic in $\sigma < 1$ with at most simple poles at $0, -1, -2, \ldots$, and $|R(z)| \le C_k|\tau|^{-k}$ for $|\tau| \ge 1$, uniformly on compact σ-ranges in $(-k, 1)$. Write $r(u) = \sum_n a_n u^n$, so $\sum_n |a_n| < \infty$. Splitting at $u = 1$ and integrating by parts once on $[1,\infty)$ gives a uniform bound

$$|R(z)| \le \frac{M}{|\tau|} \qquad (\sigma \le \sigma',\ |\tau| \ge 1) \tag{5.21}$$

for every $\sigma' < 1$.

Since $h = 1/s + h_1$ and $\int_0^\infty u^{z-1}(u+v)^{-1}\,\mathrm{d}u = \pi v^{z-1}/\sin \pi z$, taking Mellin transforms in (5.20) gives, for $\sigma_0 < \sigma < 1$,

$$\kappa\Psi(z+\delta) - \frac{\pi}{\sin \pi z}\Psi(z) = R(z). \tag{5.22}$$

All integrals converge absolutely. Rewrite (5.22) as $\Psi(z) = \pi^{-1}\sin(\pi z)(\kappa\Psi(z+\delta) - R(z))$. Here $\sin(\pi z)R(z)$ is analytic in $\sigma < 1$. Induction continues Ψ analytically to

$\sigma > \sigma_0 - n\delta$ for each n. So Ψ is entire, and (5.22) holds for $\operatorname{Re} z < 1$ away from the poles of R.

LEMMA 5.52. *There is a function Q, analytic and zero-free in $\{0 < \sigma < 1+\delta\}$, with $Q(z+\delta) = \pi Q(z)/(\kappa \sin \pi z)$ for $0 < \sigma < 1$. Moreover $|Q(\sigma_0 + i\tau)|$ is constant, and for every compact $J \subset (0, 1+\delta)$ there is $B_J \geq 1$ with*

$$B_J^{-1} \leq |Q(\sigma + i\tau)| e^{-\pi(\sigma_0 - \sigma)|\tau|/\delta} \leq B_J, \qquad \sigma \in J,\ |\tau| \geq 1. \tag{5.23}$$

Extended to $\sigma \leq 0$ by $Q(z) = \kappa\pi^{-1} \sin(\pi z) Q(z+\delta)$, Q is analytic. Its zeros are real, and the upper bound in (5.23) holds on compact σ-ranges.

PROOF. Let $s(t) = \sinh(t/2)\sinh(\delta t/2)$. For $|\operatorname{Re} w| < \sigma_0$ put

$$\ell(w) = \int_0^\infty \Big(\frac{\sinh wt}{2s(t)} - \frac{2we^{-t^2}}{\delta t} \Big) \frac{\mathrm{d}t}{t}.$$

The bracket is $O(t)$ at 0 and $O(e^{-(\sigma_0 - |\operatorname{Re} w|)t})$ at ∞. So ℓ is analytic, odd, and real on the real axis. For $0 < \sigma < 1$, the identity $\sinh((w-\delta)t) - \sinh(wt) = -2\cosh((w-\delta/2)t)\sinh(\delta t/2)$ with $w = \sigma_0 - z$ gives

$$\ell(w-\delta) - \ell(w) = -I(z), \qquad I(z) = \int_0^\infty \Big(\frac{\cosh((1/2 - z)t)}{\sinh(t/2)} - \frac{2e^{-t^2}}{t} \Big) \frac{\mathrm{d}t}{t}.$$

By [39, p. 3.511.2], differentiated in a, $\int_0^\infty t\cosh(at)/\sinh(t/2)\,\mathrm{d}t = \pi^2/\cos^2(\pi a)$ for $|\operatorname{Re} a| < 1/2$. Thus $I''(z) = \pi^2/\sin^2(\pi z)$, the second derivative of $\log(\pi/\sin \pi z)$. Both I and $\log(\pi/\sin \pi z)$ are invariant under $z \mapsto 1 - z$. Their difference is therefore a constant, κ_0. Set $Q(z) = \exp(-\ell(\sigma_0 - z) + \beta z)$ with $\beta\delta = -\kappa_0 - \log\kappa$. The functional equation follows.

On $\sigma = \sigma_0$, $\ell(-i\tau)$ is purely imaginary, so $|Q| = e^{\beta\sigma_0}$. For (5.23), note that $\ell''(w) = \int_0^\infty t\sinh(wt)/(2s(t))\,\mathrm{d}t$. The integrand is even, and $t/(2s(t))$ has a simple pole at 0 with residue $2/\delta$. Hence

$$\ell''(w) = \frac{1}{2}\,\mathrm{p.v.} \int_{\mathbb{R}} \frac{te^{wt}}{2s(t)}\,\mathrm{d}t.$$

Let $w = q - i\tau$ with $|q| \leq \sigma_0 - \varepsilon$ and $\tau \geq 1$. Move the contour to $\mathbb{R} - i\pi$; the strip between contains no other pole. On the new line $t = x - i\pi$, $|e^{wt}| = e^{qx - \pi\tau}$ and $|s(t)| \geq \cosh(x/2)\max\{\sin(\pi\delta/2), |\sinh(\delta x/2)|\}$. Since the principal value equals the shifted integral minus $i\pi$ times the residue, $\ell''(q - i\tau) = -i\pi/\delta + O(e^{-\pi\tau})$. By the symmetry $\ell(\bar w) = \overline{\ell(w)}$, $\operatorname{Re}\ell'' = O(e^{-\pi|\tau|})$ for $|\tau| \geq 1$. Further, $\tau \mapsto \ell'(-i\tau)$ is real and even, with derivative $-i\ell''(-i\tau) = -\pi/\delta + O(e^{-\pi\tau})$ for $\tau \geq 1$. Hence $\ell'(-i\tau) = -\pi|\tau|/\delta + c_1 + O(e^{-\pi|\tau|})$ with c_1 real. Integrating along $[0, q] - i\tau$,

$$\operatorname{Re}\ell(q - i\tau) = q\Big(c_1 - \frac{\pi|\tau|}{\delta}\Big) + O(e^{-\pi|\tau|}),$$

which is (5.23) with $q = \sigma_0 - \sigma$. The extension to $\sigma \leq 0$ multiplies by factors $\kappa\pi^{-1}\sin(\pi(z + j\delta))$. These are entire, vanish only on $\mathbb{R}$, and have modulus at most $e^{\pi|\tau|}$. □

LEMMA 5.53. *As $u \to 0^+$, $\psi(u) = O(u)$ and $\psi'(u) = O(1)$.*

PROOF. Let $\sigma_1 = -1$ and $\sigma_2 = \sigma_0 + c/4 + \delta < 1 + \delta$. Let Ω be the set of z with $\sigma_1 \le \sigma \le \sigma_2$ and $\tau > T$. Here $T \ge 1$ satisfies $\sinh \pi T \ge 2\pi/\kappa$. On Ω define

$$\Psi_*(z) = \frac{1}{\kappa} \sum_{j \ge 1} R(z - j\delta) \prod_{i=1}^{j-1} \frac{\pi}{\kappa \sin \pi(z - i\delta)}.$$

Each product is at most 2^{1-j}, and $\operatorname{Re}(z - j\delta) \le \sigma_2 - \delta < 1$. By (5.21) the series converges uniformly. Thus Ψ_* is analytic and bounded on Ω, and $|\Psi_*(z) - \kappa^{-1} R(z-\delta)| \le C e^{-\pi\tau}$. Reindexing the series shows that Ψ_* satisfies (5.22) whenever $z, z + \delta \in \Omega$.

By Lemma 5.52, $p = (\Psi - \Psi_*)/Q$ is analytic on Ω. Subtracting the two copies of (5.22) and using $\pi Q(z)/\sin \pi z = \kappa Q(z + \delta)$ gives $\kappa Q(z+\delta)(p(z+\delta) - p(z)) = 0$. Hence p is δ-periodic. Since Ω is wider than δ, p extends to a δ-periodic analytic function on $\{\tau > T\}$. So $p(z) = P(e^{2\pi i z/\delta})$ with P analytic in $0 < |\xi| < e^{-2\pi T/\delta}$. On the period strip $\sigma_0 + c/4 \le \sigma \le \sigma_2$, Ψ and Ψ_* are bounded. By (5.23), $|p| \le C e^{\pi(c/4+\delta)\tau/\delta}$. In terms of ξ, this says $|P(\xi)| \le C|\xi|^{-\theta}$ with $\theta = (c/4+\delta)/2 < 1$. Hence the singularity at $\xi = 0$ is removable, and $p(z) = P(0) + O(e^{-2\pi\tau/\delta})$ uniformly in σ.

On the line $\sigma = \sigma_0$, Ψ is in $L^2(\mathrm{d}\tau)$ and $|\Psi_*| \le C/\tau$. So $Qp = \Psi - \Psi_*$ is square integrable on $\tau > T + 1$. But $|Q|$ is a positive constant there and $p \to P(0)$. Hence $P(0) = 0$ and $|p(z)| \le C e^{-2\pi\tau/\delta}$. With (5.23), for $z \in \Omega$ and $\tau \ge T + 1$,

$$|\Psi(z)| \le \frac{|R(z-\delta)|}{\kappa} + C e^{-\pi\tau} + C e^{-\pi(\sigma - \sigma_0 + 2)\tau/\delta} \le C\tau^{-3}.$$

Here we used $\sigma - \sigma_0 + 2 \ge c/2$ on Ω and the bound on R for $k = 3$. Since ψ is real, $\Psi(\bar z) = \overline{\Psi(z)}$, and the same bound holds for $\tau \le -T - 1$.

Let $\sigma_3 = \sigma_0 + c/4$. Then Ψ is integrable on $\sigma = \sigma_3$, so Mellin inversion holds there pointwise. Since Ψ is entire and decays on the horizontal segments, we may move the contour to $\sigma_1 = -1$:

$$\psi(u) = \frac{1}{2\pi} \int_{\mathbb{R}} u^{1 - i\tau} \Psi(-1 + i\tau)\, \mathrm{d}\tau,$$

$$\psi'(u) = \frac{1}{2\pi} \int_{\mathbb{R}} (1 - i\tau) u^{-i\tau} \Psi(-1 + i\tau)\, \mathrm{d}\tau.$$

Both integrals converge absolutely, which gives the claim. □

PROOF OF THEOREM 5.48. *No eigenvalues in* $(0, \infty)$. This adapts [72, Theorem 2.4], with Lemma 5.53 in place of [72, Lemma 4.1]. Let ψ be as in Lemma 5.51. Differentiate (5.20) and integrate by parts in v. The boundary term $h(u)\psi(0^+)$ vanishes by Lemma 5.53, so

$$\int_0^\infty h(u + v)\psi'(v)\, \mathrm{d}v = -\kappa (u^\delta \psi)'(u).$$

Multiply by $\psi'(u)$ and integrate. All integrals converge absolutely by Lemmas 5.51 and 5.53, and $h(s) \le C/s$ near 0. With $\rho = \sum_n n^{-1} \delta_{x_n}$, the left side equals

$$\int \left| \int_0^\infty e^{-xv} \psi'(v)\, \mathrm{d}v \right|^2 \mathrm{d}\rho(x) \ge 0.$$

On the right, $\int \delta u^{\delta-1}\psi\psi' = -\delta(\delta-1)/2\int u^{\delta-2}\psi^2$. The boundary terms vanish since $\psi = O(u)$. Hence

$$0 \le -\kappa\Big(\int_0^\infty u^\delta \psi'(u)^2\,\mathrm{d}u + \frac{\delta c}{2}\int_0^\infty u^{\delta-2}\psi(u)^2\,\mathrm{d}u\Big).$$

As $\kappa > 0$ and $\delta c > 0$, $\psi = 0$. Thus A has no eigenvalues in $(0,\infty)$.

No singular continuous spectrum. Let $C_t = U_t C U_t^{-1}$, so $U_t A U_t^{-1} = e^{-\delta t}K + C_t$. The kernel of C_t is

$$\Gamma(c)^{-1}(uv)^{-\delta/2}(W_t h_1)(u+v),$$

where $(W_t m)(s) = e^{ct}m(e^t s)$ is a unitary group on $L^2(s^{2c-1}\,\mathrm{d}s)$ with generator $m \mapsto cm + sm'$. The map from m to the operator with kernel $\Gamma(c)^{-1}(uv)^{-\delta/2}m(u+v)$ is bounded from $L^2(s^{2c-1}\,\mathrm{d}s)$ to the Hilbert–Schmidt class, by the computation behind (5.18). Each $s^j h_1^{(j)}$ is bounded near 0 and $O(1/s)$ at infinity. So h_1 lies in the domain of every power of the generator, and $t \mapsto C_t$ is C^∞ in Hilbert–Schmidt norm. Let $R_t = (e^{-\delta t}K + C_t + 1)^{-1} = U_t(A+1)^{-1}U_t^{-1}$. Then $KR_t = e^{\delta t}(1 - (1+C_t)R_t)$ is bounded, locally uniformly in t. With the resolvent identity, this shows that $t \mapsto R_t$ is C^2 in norm. Thus $A \in C^2(D)$ in the sense of [37], and

$$[A, iD] = \delta K - \dot C_0 = \delta A + K', \qquad K' = -\delta C - \dot C_0,$$

with K' compact. Fix $E > 0$ and an interval $I \ni E$ with $\inf I > 0$. Then $E_I(A)[A,iD]E_I(A) \ge \delta \inf I\, E_I(A) + E_I(A)K'E_I(A)$. As E is not an eigenvalue, $E_J(A) \to 0$ strongly when the interval $J \ni E$ shrinks. Then $\|E_J(A)K'E_J(A)\| \to 0$. Hence a strict Mourre estimate holds on a small open interval $J \ni E$. By [37, Theorem 1.2], $\sup\|\langle D\rangle^{-1}(A-z)^{-1}\langle D\rangle^{-1}\| < \infty$ for $\operatorname{Re} z$ in a closed subinterval $J' \ni E$. By Stone's formula, each $\langle D\rangle^{-1}g$ has a spectral measure that is absolutely continuous on J'. These vectors are dense, so $E_{J'}(A)$ projects into the absolutely continuous subspace. Covering $(0,\infty)$, A has purely absolutely continuous spectrum there.

Multiplicity. By Lemma 5.49 and the Kato–Rosenblum theorem [52, Theorem X.4.4], the absolutely continuous parts of A and K are unitarily equivalent. By Lemma 5.50, K is multiplication by λ in $L^2((0,\infty),\mathrm{d}\lambda)$. By the preceding steps, the absolutely continuous part of A is $A|_{(\ker A)^\perp}$. □

Remark 5.54. The decisive step is Lemma 5.53. It is the analogue of [72, Lemma 4.1] at $u = 0$, where the a.c. channel lives. In [72] that channel appears as poles on the critical line. Here it appears as the δ-periodic factor p, and square integrability on $\sigma = \sigma_0$ removes it. The identity $\int \delta u^{\delta-1}\psi\psi' = (\delta c/2)\int u^{\delta-2}\psi^2$, with positive sign because $c < 1$, replaces the case $c = 1$ of [72, Theorem 2.4]. The spectral results are conditional on the unrefereed Ma–Lu–Zu manuscript [59]. It was checked at its latest version, v1, on 13 October 2026.

5.10. Rank and Hilbert–Schmidt structure

For a submonoid $S \subseteq \mathbb{N}$ let $d_S(k) = \#\{(m,n) \in S^2 : mn = k\}$. The space $W_2(S)$ is the completion of polynomials under

$$\|F\|_{W_2(S)} = \inf\Big\{ \Big(\sum_j \|G_j\|_2^2 \|H_j\|_2^2 \Big)^{1/2} : F = \sum_j G_j H_j \Big\},$$

with (G_j) and (H_j) orthogonal families in $H^2(S)$; equivalently the quotient of $H^2(S) \otimes_2 H^2(S)$ under multiplication.

THEOREM 5.55 (Helson [44], restated). *For $F = \sum_{k \in S} a_k \chi_k$,*

$$\|F\|_{W_2(S)}^2 = \sum_{k \in S} \frac{|a_k|^2}{d_S(k)}.$$

Hence $\sum_k |a_k|^2/d_S(k) \le C^2 \|F\|_1^2$ is equivalent, with the same constant, to the statement that every Hilbert–Schmidt S-Hankel form H_b has a bounded symbol of norm at most $C\|H_b\|_{S_2}$.

PROOF. A tensor $U = (u_{mn})$ maps to F when $\sum_{mn=k} u_{mn} = a_k$ for each k. The fibres are disjoint, and Cauchy–Schwarz in each fibre gives $\sum_{mn=k} |u_{mn}|^2 \ge |a_k|^2/d_S(k)$, with equality for the constant choice $u_{mn} = a_k/d_S(k)$. The dual norm is $\sum_k d_S(k)|b_k|^2 = \|H_b\|_{S_2}^2$, and Hahn–Banach supplies the symbol. □

For $S = \mathbb{N}$ this is Helson's inequality [44] read as $H^1(\mathbb{T}^\infty) \hookrightarrow W_2$ contractively. Classical weak factorization is the W_1 endpoint, with cost $\sum_j \|G_j\|_2 \|H_j\|_2$, and it fails on $\mathbb{T}^\infty$ [69]; Nehari's theorem in S_q corresponds to embedding into $W_{q'}$. The divisor weight therefore supplies exactly the S_2 theorem and nothing more.

COROLLARY 5.56. *(1) If $K_S = \sup_{k \in S} d(k)/d_S(k) < \infty$, the intrinsic W_2 estimate holds with constant $\sqrt{K_S}$; for $S = \{1, 2^2, 2^3, \dots\}$, $K_S = 2$.*

(2) If S is freely generated by integers with $\mathbb{Z}$-independent exponent vectors, the estimate holds with constant 1.

Write $\mathcal{N}(H)$ for the least norm of a bounded symbol of a Helson matrix H, and $C(r) = \sup\{\mathcal{N}(H)/\|H\| : \operatorname{rank} H \le r\}$.

THEOREM 5.57 (Bounded rank). *$\mathcal{N}(H) \le \sqrt{r}\,\|H\|$ whenever $\operatorname{rank} H \le r$, independently of the number of variables. Moreover*

$$r^{\log_2(2/\sqrt{\pi})} \lesssim C(r) \le \sqrt{r}, \qquad \log_2(2/\sqrt{\pi}) = 0.1742\ldots$$

PROOF. $\|H\|_{S_2} \le \sqrt{r}\|H\|$, and Theorem 5.55 with $S = \mathbb{N}$, $C = 1$. For the lower bound, let $P_d = z_1 + \cdots + z_d$ and $M_d = \|P_d\|_{L^1(\mathbb{T}^d)}$. The Helson matrix of the degree-one coefficients of P_d has rank 2 and norm $\sqrt{d}$, and every symbol φ satisfies $d = |\int \varphi \overline{P_d}| \le \|\varphi\|_\infty M_d$; the symbol $(d/M_d)P_d/|P_d|$ is admissible, by rotation and permutation symmetry, so $\mathcal{N} = d/M_d$ exactly. Tensoring k copies on disjoint variables, as in [69], gives rank 2^k, norm $d^{k/2}$ and $\mathcal{N} = (d/M_d)^k$; by conditional expectation, extra variables do not help. The central limit theorem gives $M_d/\sqrt{d} \to \sqrt{\pi}/2$. □

These tensor examples give the exponent $\log_2(\|Z\|_2/\|Z\|_1)$ for a complex Gaussian Z. Whether other constructions improve it remains open.

REMARK 5.58 (Maximal forms). Call $H_\varphi(F,G) = \int FG\overline{\varphi}\,\mathrm{d}m$, with $\varphi \in H^2 \cap L^\infty$, maximal if $\|H_\varphi\| = \|\varphi\|_\infty$ and minimal if $\|H_\varphi\| = \|\varphi\|_2$. The one-variable argument of [18] transfers to $\mathbb{T}^\infty$: if a maximal form attains its norm at F, G, equality in $|\int FG\overline{\varphi}| \le \|\varphi\|_\infty \|FG\|_1$ forces $|\varphi| = \|\varphi\|_\infty$ on $\{FG \neq 0\}$, which has full measure because a nonzero $H^1(\mathbb{T}^\infty)$ function vanishes only on a null set [1, Corollary 2]. So compact maximal Helson matrices are minimal.

Open

(1) Characterize the measures with $s_n(M_{\nu,R}) \asymp a_n^*$ without an index change; doubling profiles (Corollary 5.39) and separated atoms (Theorem 5.36) are two sufficient conditions.
(2) Determine the spectral type under only the $o(1)$ hypothesis of Proposition 5.7. The rate used in Theorem 5.18 is established for R_P, not for this whole class.
(3) The true exponent of $C(r)$, between 0.174 and 1/2.
(4) The spectral scale for R and its complement both infinite.
(5) Characterize when the union of limit spectra in Theorem 5.26 has no gaps. The gap for R_8 is certified in remark 5.8; general uniformly continuous profiles are covered by the hull formula.

Notes

Pushnitski–Sobolev [74, Theorem 1.2 and Sections 5–6] prove stronger band conclusions for unweighted periodic Hankel operators. For $\theta \neq 1$ their multiplicity and nonflat-band results do not apply directly to this weighted model, and none of them is used here.

The exclusion of flat bands in Lemma 5.13 is the argument of Thomas [90] for periodic Schrödinger operators, transferred to this model: it uses only a positive Laplace measure and $\theta < 1$, through the decay $r_Y \to 0$. At $\theta = 1$ that factor tends to 1, consistent with the need for the elliptic-function analysis of [74] in the unweighted case.

CHAPTER 6

Bergman and Fock spaces of Dirichlet series

Question. What changes when Haar measure on $\mathbb{T}^\infty$ is replaced by a Bergman or Gaussian product measure?

Less than one might expect. Separation along curves is the Hellinger–Bessel inequality, which holds in all three geometries (chapter 2). Symmetry under sign changes needs a substitute for inner-outer factorization, and a contractive embedding into a space with heavier weight supplies it: Carleman's inequality for Hardy, Kulikov's theorem [55] for Bergman, and its $\beta \to \infty$ limit, proved below, for Gauss. Each embedding gives a Helson-type inequality, a two-point lemma with constant $1 + O(|x|^4)$, and the approximation step of a Hartman theorem. With these, the Nehari theorem for partially averaged curve measures and its compactness counterpart hold in every geometry, and the Bergman multiplicative Hilbert form has a bounded symbol.

We derive the Gaussian contraction as a limit of Kulikov's Bergman inequality; the contraction itself also follows from Janson's theorem, as in Lemma 1.10. New here, to our knowledge, are the two-point constants outside the Hardy case, the compactness criterion for every $\beta \geq 1$, the sign-averaged endpoint in the Bergman and Gauss geometries, the bounded symbol for the Bergman multiplicative Hilbert form, and the absence of cancellation in the Bergman Dirichlet kernels, unconditional for $\beta > 2$ and conditional on a preprint of Gorodetsky and Wong for $1 < \beta \leq 2$.

6.1. Spaces of Dirichlet series and their symbols

We keep the three geometries of section 1.1 and identify a Dirichlet series $f(s) = \sum a_n n^{-s}$ with its Bohr lift $F(z) = \sum a_n z^{\kappa(n)}$, so that $f(s) = F(\gamma(s))$ with $\gamma(s) = (p^{-s})_p$. Write $\mathscr{H}^p$, $\mathscr{A}^p_\beta$ and $\mathscr{F}^p$ for the closures of the Dirichlet polynomials in L^p of Haar measure, of m_β^∞ and of γ^∞. The monomials are orthogonal in each L^2 space, with

$$\|n^{-s}\|^2_{\mathscr{H}^2} = 1, \qquad \|n^{-s}\|^2_{\mathscr{A}^2_\beta} = \frac{1}{d_\beta(n)}, \qquad \|n^{-s}\|^2_{\mathscr{F}^2} = [n]! := \prod_p \kappa_p(n)!,$$

where $\zeta(s)^\beta = \sum d_\beta(n) n^{-s}$. The Bergman spaces are those of Bayart, Brevig, Haimi, Ortega-Cerdà and Perfekt [12]. For $\beta = 2$ they are the spaces of Bailleul and Lefèvre [2]. The Gauss spaces are the Fock-type spaces of Dirichlet series; along the curve γ, point evaluation on $\mathscr{F}^2$ has norm $\exp\bigl(\sum_p p^{-2\Re s}/2\bigr)$.

Let X be one of these families and α a sequence. The *weighted Helson form* with moments α is

$$H_\alpha(f,g) = \sum_k (fg)_k\,\alpha(k) = \langle fg, \varphi\rangle_{X^2}, \qquad \varphi = \sum_n \frac{\overline{\alpha(n)}}{\|n^{-s}\|^2_{X^2}}\, n^{-s}.$$

For the Bergman spaces these are the forms of [12] and [58]. In our analytic-symbol convention, $\varrho(n) = d_\beta(n)\overline{\alpha(n)}$; for the real moments considered below the conjugation is immaterial. A *bounded symbol* is a function $g \in L^\infty(\lambda)$ with $\int z^{\kappa(n)} g\,\mathrm{d}\lambda = \alpha(n)$ for every n. The orthogonal projection of $\overline{g}$ onto X^2 is then φ.

LEMMA 6.1. *H_α has a bounded symbol if and only if $h \mapsto \sum_k h_k\alpha(k)$ is bounded on X^1, and then $\|H_\alpha\| \le \inf\|g\|_\infty$, which equals the norm of that functional.*

PROOF. As Lemma 4.1, using $\|FG\|_{L^1(\lambda)} \le \|F\|_{L^2(\lambda)}\|G\|_{L^2(\lambda)}$. □

6.2. Contractive embeddings

THEOREM 6.2 (Kulikov). *For every $\beta \ge 1$ and every analytic polynomial g in one variable, $\|g\|_{L^2(m_{2\beta})} \le \|g\|_{L^1(m_\beta)}$, where m_1 is arc length on $\mathbb{T}$.*

This is [55, Corollary 1.3] in the case $p = 1$, $q = 2$; our normalization of m_β agrees with Kulikov's. For $\beta = 1$ it is Carleman's inequality $\|g\|_{A^2} \le \|g\|_{H^1}$. For integer β it also follows from repeated applications of [12, Theorem 1].

Let $\gamma_a = (a/\pi)e^{-a|z|^2}\,\mathrm{d}x\,\mathrm{d}y$ on $\mathbb{C}$, so $\gamma = \gamma_1$.

PROPOSITION 6.3 (Gauss as the limit of Bergman). *For every analytic polynomial g in one variable, $\|g\|_{L^2(\gamma_2)} \le \|g\|_{L^1(\gamma_1)}$.*

PROOF. Apply Theorem 6.2 to $g_\beta(z) = g(\sqrt{\beta}z)$ and substitute $z = w/\sqrt{\beta}$. With $\mathrm{d}A$ normalized area,

$$\|g_\beta\|_{L^1(m_\beta)} = \frac{\beta-1}{\beta}\int_{|w|<\sqrt{\beta}} |g(w)|\Big(1-\frac{|w|^2}{\beta}\Big)^{\beta-2}\frac{\mathrm{d}x\,\mathrm{d}y}{\pi} \longrightarrow \|g\|_{L^1(\gamma_1)},$$

and similarly $\|g_\beta\|^2_{L^2(m_{2\beta})} \to \|g\|^2_{L^2(\gamma_2)}$, since $(2\beta-1)\beta^{-1}(1-|w|^2/\beta)^{2\beta-2} \to 2e^{-2|w|^2}$. Dominated convergence applies because $(1-x/\beta)^{\beta-2} \le e^{-x/2}$ for $\beta \ge 4$. □

The rescaling $z = w/\sqrt{\beta}$ is the general principle behind this: the Bergman geometry with weight β tends to the Gauss geometry as $\beta \to \infty$, coordinates measured in units of $\beta^{-1/2}$. Several statements in this chapter could be obtained for the Gauss geometry by passing to that limit, but the direct proofs are just as short.

Each embedding tensorizes. Applying it in one coordinate at a time and using Minkowski's integral inequality, as in [12], gives contractive embeddings on infinitely many coordinates. Reading off the L^2 norm in coefficients:

COROLLARY 6.4 (Helson inequalities). *For every $f = \sum a_n n^{-s}$:*

$$\sum_n \frac{|a_n|^2}{d(n)} \le \|f\|^2_{\mathscr{H}^1}, \qquad \sum_n \frac{|a_n|^2}{d_{2\beta}(n)} \le \|f\|^2_{\mathscr{A}^1_\beta}, \qquad \sum_n |a_n|^2\frac{[n]!}{2^{\Omega(n)}} \le \|F\|^2_{L^1(\gamma^\infty)}.$$

The first is Helson's inequality [44], and the second was proved for $\beta = 2$ by Bayart, Brevig, Haimi, Ortega-Cerdà and Perfekt [12]. The third uses that the monomials of $L^2(\gamma_2)$ have $\|z^k\|^2 = k!/2^k$.

Kulikov also records the full Hardy scale [55, Theorem 5.1]: for $0 < p \le 2$,

$$\sum_n \frac{|a_n|^2}{d_{2/p}(n)} \le \|f\|^2_{\mathscr{H}^p}.$$

His proof tensorizes the one-variable contraction $H^p \hookrightarrow A^2_{2/p}$. This is valid also for $p < 1$, because the Minkowski exponent in the induction is $2/p \ge 1$.

6.3. Sign averages

PROPOSITION 6.5 (Two-point lemmas). *Let $x \in \mathbb{C}$, with $|x| < 1$ in the first two cases, and let g be an analytic polynomial. Then $(|g(x)|+|g(-x)|)/2 \le \widetilde{c}(x)\|g\|_{L^1(\lambda)}$, with*

$$\widetilde{c}(x) = \frac{1}{1-|x|^4}, \qquad \widetilde{c}(x) = \Lambda_\beta(|x|^2)^{1/2}, \qquad \widetilde{c}(x) = \cosh(2|x|^2)^{1/2}$$

in the Hardy, Bergman and Gauss cases, where $\Lambda_\beta(t) = \big((1-t)^{-2\beta}+(1+t)^{-2\beta}\big)/2 = 1 + \beta(2\beta+1)t^2 + O(t^4)$. The Hardy constant is sharp.

PROOF. The Hardy case is Lemma 2.17 with $k = 2$. In the other two, by Cauchy–Schwarz and the contractive embedding into a space Y with kernel K^Y,

$$\frac{1}{2}\big(|g(x)| + |g(-x)|\big) \le \Big(\frac{1}{2}\big(|\langle g, K^Y_x\rangle|^2 + |\langle g, K^Y_{-x}\rangle|^2\big)\Big)^{1/2} \le \big(\frac{1}{2}\|G\|\big)^{1/2}\|g\|_Y,$$

where G is the Gram matrix of $K^Y_{\pm x}$. Its entries are real and positive: $K^Y(x,x)$ on the diagonal and $K^Y(x,-x)$ off it, so $\|G\| = K^Y(x,x) + K^Y(x,-x)$. For $Y = A^2_{2\beta}$ the kernel is $(1-\overline{w}z)^{-2\beta}$, giving $2\Lambda_\beta(|x|^2)$. For $Y = L^2(\gamma_2)$ it is $e^{2z\overline{w}}$, giving $e^{2|x|^2} + e^{-2|x|^2}$. □

All three constants are $1 + O(|x|^4)$. That is what lets them tensorize over the primes, where $\sum_p |x_p|^4 \le \sum_p p^{-2} < \infty$. A single point evaluation, by contrast, costs $\|K_x\|_2^2 = 1 + O(|x|^2)$. Comparing the Hardy cases is instructive: the route through Carleman's inequality gives $\Lambda_1(t)^{1/2} = 1 + 3t^2/2 + O(t^4)$ instead of the sharp $1 + t^2 + O(t^4)$. Passing through L^2 loses only in the coefficient of $|x|^4$. In the Bergman case the whole constant can be recovered from the L^1 norm near $x = 0$, and in both cases the obvious lower bound fails once $K_x + K_{-x}$ has a zero.

The same route works in every L^p. Kulikov's theorem holds in the full range [55, Corollary 1.3]: for $0 < p < q$ and $\beta > 1$, $\|g\|_{L^q(m_{q\beta/p})} \le \|g\|_{L^p(m_\beta)}$, and the rescaling in the proof of Proposition 6.3 turns the case $q = 2$ into $\|g\|_{L^2(\gamma_{2/p})} \le \|g\|_{L^p(\gamma_1)}$.

PROPOSITION 6.6 (Two-point lemmas in every L^p). *In the Bergman case with $\beta > 1$, let $x \in \mathbb{D}$; in the Gauss case, let $x \in \mathbb{C}$. Let $0 < p < \infty$, write $p = kq$ with $k = \lceil p/2 \rceil$ and $0 < q \le 2$, and let $t = |x|^2$. For every analytic polynomial g,*

$$\frac{1}{2}\big(|g(x)|^p + |g(-x)|^p\big) \le c_p(x)\|g\|^p_{L^p(\lambda)}, \qquad c_p(x) = \Lambda_{\beta/q}(t)^{q/2} \quad \text{or} \quad \cosh(2t/q)^{q/2},$$

and $c_p(x) = 1 + O(|x|^4)$.

PROOF. Apply the case q to g^k, since $|g^k|^q = |g|^p$. For $q \le 2$, the power-mean inequality gives

$$\frac{|g(x)|^q + |g(-x)|^q}{2} \le \left(\frac{|g(x)|^2 + |g(-x)|^2}{2}\right)^{q/2}.$$

As in the proof of Proposition 6.5, the inner average is at most half the Gram norm of the kernels at $\pm x$ in $A^2_{2\beta/q}$, respectively $L^2(\gamma_{2/q})$, times the square of the norm there, which is at most $\|g\|^2_{L^q}$. The Gram norms are $2\Lambda_{\beta/q}(t)$ and $2\cosh(2t/q)$. □

PROPOSITION 6.7 (Sharp two-point constants). *In the Bergman and Gauss cases, let $c(x)$ be the least constant in Proposition 6.5, let K be the kernel of X^2 (that is, of A^2_β, or of $L^2(\gamma)$ in the Gauss case), and let $\mu = \int |z|^4 \,\mathrm{d}\lambda$, which is $2/(\beta(\beta+1))$ and 2 respectively.*

1. *$c(x) \ge L(x) \coloneqq (K(x,x) + K(x,-x))/2$, that is, $\big((1-|x|^2)^{-\beta} + (1+|x|^2)^{-\beta}\big)/2$ in the Bergman case and $\cosh(|x|^2)$ in the Gauss case.*
2. *$c(x) = 1 + |x|^4/\mu + o(|x|^4)$ as $x \to 0$; that is, $1 + \beta(\beta+1)|x|^4/2 + o(|x|^4)$ and $1 + |x|^4/2 + o(|x|^4)$.*
3. *In the Bergman case, $c(x) = L(x)$ whenever $\beta \arctan|x| \le \pi/2$: for every x when $\beta \le 2$, and for $|x| \le \tan(\pi/(2\beta))$ when $\beta > 2$.*
4. *$c(x) > L(x)$ in the Bergman case when $|x| > \tan(\pi/(2\beta))$, which can happen only for $\beta > 2$, and in the Gauss case for every $x \ne 0$.*

PROOF. By rotation invariance we may take $x = r > 0$. Put $H = K_r + K_{-r}$, $A = K(r,r)$ and $B = K(r,-r)$, so that $H(\pm r) = A + B = 2L$ and $\|H\|_2^2 = 4L$.

(1) The function $g = H^2$ lies in the closure of the polynomials in L^1, and $\|g\|_1 = \|H\|_2^2 = 4L$, $g(\pm r) = 4L^2$. So its two-point quotient is L.

(2) The lower bound is (1). For the upper bound, let $g = \sum a_n z^n$ be a polynomial with $\|g\|_1 = 1$ and two-point quotient $N \ge 1$, rotated so that $a_0 \ge 0$; note $a_0 \le 1$. Let $w_n = \|z^n\|_Y^2$ in the space Y of the proof of Proposition 6.5, and $q = 1 - a_0^2$. The contractive embedding gives $\sum_{n\ge1} w_n |a_n|^2 \le q$, so by Cauchy–Schwarz

$$\sum_{n\ge k} |a_n|\rho^n \le \sqrt{q}\,\Phi_k(\rho), \qquad \Phi_k(\rho)^2 = \sum_{n\ge k} \rho^{2n}/w_n \le K^Y(\rho,\rho),$$

and $\Phi_k(r) = O(r^k)$ as $r \to 0$. Let $e = (g(r) + g(-r))/2$ and $o = (g(r) - g(-r))/2$. Then $e = a_0 + a_2 r^2 + O(\sqrt{q}\,r^4)$, $o = O(\sqrt{q}\,r)$, and

$$N \le \big(|e|^2 + |o|^2\big)^{1/2}.$$

Since $a_0 \le 1 - q/2$ and $|a_2| \le \sqrt{q/w_2}$, the condition $N \ge 1$ forces $\sqrt{q} = O(r^2)$. Hence $o = O(r^3)$ and $N \le a_0 + |a_2| r^2 + O(r^6)$.

It remains to bound $1 - a_0$ below. Let $h = g - a_0$. For complex h and $a_0 > 0$, $|a_0 + h| \ge a_0 + \Re h$, and $|a_0 + h| \ge a_0 + \Re h + (\Im h)^2/(2(a_0 + |h|))$. Since $\int \Re h \,\mathrm{d}\lambda = 0$, for any disc $D_\rho = \{|z| \le \rho\}$,

$$1 - a_0 = \int \big(|a_0 + h| - a_0 - \Re h\big)\,\mathrm{d}\lambda \ge \frac{1}{2(1+\eta)} \int_{D_\rho} (\Im h)^2 \,\mathrm{d}\lambda,$$

where $\eta = \sup_{D_\rho}|h| \le \sqrt{q}\,\Phi_1(\rho)$. On the circle of radius t the mean of $(\Im h)^2 = (|h|^2 - \Re h^2)/2$ is $(1/2)\sum_{n\ge1}|a_n|^2t^{2n} \ge |a_2|^2t^4/2$. Hence $1-a_0 \ge |a_2|^2\mu(\rho)/(4(1+\eta))$ with $\mu(\rho) = \int_{D_\rho}|z|^4\,\mathrm{d}\lambda$. Let ρ tend to the boundary slowly enough that $r^2\Phi_1(\rho)\to 0$. Then $\eta\to0$ and $\mu(\rho)\to\mu$, so

$$N \le 1 - \frac{\mu}{4}(1-o(1))|a_2|^2 + |a_2|r^2 + O(r^6) \le 1 + (1+o(1))\frac{r^4}{\mu}.$$

(3) By continuity in r for a fixed polynomial, we may assume $\beta\arctan r < \pi/2$. Here $H(z) = 2(1-r^2z^2)^{-\beta/2}\cosh(\beta\,\mathrm{artanh}(rz))$ and $|\Im\,\mathrm{artanh}(rz)| \le \arctan r$ on the closed disc, so H has no zeros near the closed disc. To prove that H^2 is extremal, we need $\int u|H|^2\,\mathrm{d}\lambda \ge 2L(u(r)+u(-r))$ for subharmonic u. We prove this by showing that the difference measure has nonnegative logarithmic potential. By reproduction, $\int h|H|^2\,\mathrm{d}\lambda = 2L(h(r)+h(-r))$ for h analytic near the closed disc. So the real measure $\sigma = |H|^2\lambda - 2L(\delta_r+\delta_{-r})$ has zero mass and zero analytic moments, and its logarithmic potential $U(w) = \int\log|w-z|\,\mathrm{d}\sigma(z)$ vanishes for $|w|\ge1$; it is continuous off $\pm r$, since the density of λ lies in L^p for some $p>1$, and tends to $+\infty$ at $\pm r$.

We show $U\ge0$. Angular integration gives, with $\rho_\beta(t) = (\beta-1)(1-t)^{\beta-2}$,

$$\int\frac{\overline{H(z)}}{w-z}\,\mathrm{d}\lambda(z) = \frac{1}{w}\int_0^{|w|^2} H(t/w)\rho_\beta(t)\,\mathrm{d}t = \sum_{\varepsilon=\pm1}\frac{1-(1-|w|^2)^{\beta-1}(1-\varepsilon r\overline{w})^{1-\beta}}{w-\varepsilon r},$$

and reproduction applied to $(H(z)-H(w))/(w-z)$ then gives

$$2\partial_w U(w) = -(1-|w|^2)^{\beta-1}H(w)\Big(\frac{(1-r\overline{w})^{1-\beta}}{w-r} + \frac{(1+r\overline{w})^{1-\beta}}{w+r}\Big).$$

At a critical point the bracket vanishes, so $|w+r|/|w-r| = (|1-r\overline{w}|/|1+r\overline{w}|)^{\beta-1}$; if $\Re w\ne0$ one side exceeds 1 and the other does not, so critical points lie on the imaginary axis. For $w=iy$, $0<y<1$, put $a=\arctan(ry)$ and $\gamma=\beta-1$. Then $H(iy) = 2(1+r^2y^2)^{-\beta/2}\cos(\beta a) > 0$ and

$$\partial_y U(iy) = \frac{2(1-y^2)^\gamma H(iy)}{(r^2+y^2)(1+r^2y^2)^{\gamma/2}}\big(r\sin(\gamma a) - y\cos(\gamma a)\big).$$

The ratio $\arctan(y/r)/\arctan(ry)$ decreases in y, because the ratio $(1+r^2y^2)/(r^2+y^2)$ of the derivatives does, so it exceeds its value $(\pi/2-\arctan r)/\arctan r > \gamma$ at $y=1$. Hence $\gamma a < \arctan(y/r)$, the bracket is negative, and $U(iy) > U(i) = 0$; by symmetry the same holds for $y<0$. If U were negative somewhere, it would have a negative minimum in the disc away from $\pm r$, which is a critical point; there are none off the imaginary axis, and $U>0$ on it. So $U\ge0$.

If u is smooth and subharmonic near the closed disc, then

$$\int u\,\mathrm{d}\sigma = \frac{1}{2\pi}\int U\Delta u\,\mathrm{d}x\,\mathrm{d}y \ge 0.$$

For a polynomial g put $f = g/H^2$ and apply this to $u = (|f|^2+\varepsilon^2)^{1/2}$. Letting $\varepsilon\to0$,

$$\|g\|_1 = \int|f||H|^2\,\mathrm{d}\lambda \ge 2L\big(|f(r)|+|f(-r)|\big) = \frac{|g(r)|+|g(-r)|}{2L}.$$

With (1), $c(r) = L$.

(4) Let $a = ic$ be a zero of H with $c > 0$ inside the domain: in the Bergman case $z = i\tau/r$, $\tau = \tan(\pi/(2\beta))$, is a zero, inside when $r > \tau$; in the Gauss case $H = 2\cosh(rz)$ vanishes at $i\pi/(2r)$. Let $F_t = H^2 + itH/(z-a)$ for real t, which lies in the closure of the polynomials. Then $F_t(\pm r) = 4L^2 + 2itL/(\pm r - a)$, and both moduli have derivative $-2Lc/(r^2+c^2)$ at $t = 0$. Since $|F_t| - |F_0| = O(t|H/(z-a)|)$, the norm has derivative $\Re(iJ)$, $J = \int \overline{H}/(z-a)\,\mathrm{d}\lambda$. Write $\mathrm{d}\lambda = w(u)\,\mathrm{d}u\;\mathrm{d}\theta/(2\pi)$ with $u = |z|^2$. On a circle $|z| < c$ expand $1/(z-a)$ in powers of z/a, and on $|z| > c$ in powers of a/z; only the inside contributes, and since H is even with real coefficients,

$$\Re(iJ) = -\frac{1}{c}\int_0^{c^2} H(iu/c)\,w(u)\,\mathrm{d}u.$$

Over the whole range of u this integral is known. With $\omega = r/c$,

$$\frac{1}{2}\int H(iu/c)w(u)\,\mathrm{d}u = \Re\int (1-i\omega u)^{-\beta}w(u)\,\mathrm{d}u = \Re\frac{1}{1-i\omega} = \frac{1}{1+\omega^2},$$

by Euler's integral for ${}_2F_1(\beta,1;\beta;i\omega)$, and in the Gauss case by $\int_0^\infty e^{i\omega u - u}\,\mathrm{d}u = (1-i\omega)^{-1}$. The quotient $Q_t = N_t/\|F_t\|_1$ equals L at $t = 0$, so

$$Q_0' = -\frac{c}{2(r^2+c^2)} + \frac{1}{4c}\int_0^{c^2} H(iu/c)w(u)\,\mathrm{d}u = -\frac{1}{4c}\int_{u>c^2} H(iu/c)w(u)\,\mathrm{d}u.$$

In the Gauss case $H(iu/c) = 2\cos(\omega u)$ and $rc = \pi/2$, so $Q_0' = re^{-c^2}/(2(r^2+c^2)) > 0$. In the Bergman case $H(iu/c)/2 = \Re(1+i\omega u)^{-\beta}$. Substitute $u = \tan\varphi/\omega$ and put $\tan\varphi_1 = \omega$; then $1 - u = \sin(\varphi_1 - \varphi)/(\sin\varphi_1\cos\varphi)$, the lower limit becomes $\varphi = \pi/(2\beta)$ because $\omega c^2 = \tau$, and

$$Q_0' = \frac{\beta-1}{2c\,\omega\sin^{\beta-2}\varphi_1}\int_0^{\alpha}\sin(\beta(\alpha-\psi))\sin^{\beta-2}\psi\,\mathrm{d}\psi = \frac{\sin^\beta\alpha}{2c\,\omega\sin^{\beta-2}\varphi_1},$$

where $\alpha = \varphi_1 - \pi/(2\beta)$. The integral equals $\sin^\beta\alpha/(\beta-1)$: both sides vanish to first order at $\alpha = 0$ and solve $y'' + \beta^2 y = \beta\sin^{\beta-2}\alpha$. Since $\tau < \omega < 1/\tau$, $0 < \alpha < \pi/2$, so $Q_0' > 0$. In both cases the quotient of F_t exceeds L for small $t > 0$. □

For $\beta = 3$, $Q_0' = c(3|x|^2-1)^3/(16|x|^2(3|x|^4+1))$ with $c^2 = 1/(3|x|^2)$.

REMARK 6.8 (A dual certificate). Near $x = 0$ there is a shorter proof of (3), which also covers the Hardy space. Here $K(z,w) = (1-z\overline{w})^{-\beta}$. Put $O = K_r - K_{-r}$ and $\varphi = O/H = \tanh(\beta\,\mathrm{artanh}(rz))$. The Taylor coefficients of artanh are nonnegative, and those of tanh have the moduli of those of tan, which are nonnegative; so on the disc

$$|\varphi| \le \tan(\beta\,\mathrm{artanh}\,r) \le 1/2.$$

In particular H has no zeros on the closed disc. For a polynomial g let $G = g/H$, which is analytic on a neighbourhood of the closed disc, and let e, o be as in the proof of (2). Since $\langle G, H\rangle = G(r) + G(-r)$ and $H(\pm r) = 2L$,

$$e = L\int g\,\frac{\overline{H}}{H}\,\mathrm{d}\lambda, \qquad o = L\big(G(r) - G(-r)\big).$$

Let $\rho = (2L/A)\Re\varphi$, a real function with $|\rho| \le L/A \le 1$. Then

$$\int G\overline{H}\rho\,\mathrm{d}\lambda = \frac{L}{A}\Big(\langle G\varphi, H\rangle + \langle G, \varphi H\rangle\Big) = \frac{L}{A}\big(1+\varphi(r)\big)\big(G(r)-G(-r)\big) = G(r)-G(-r),$$

because φ is odd, $\varphi H = O$ and $\varphi(r) = (A-B)/(A+B)$. So $o = L\int g(\overline{H}/H)\rho\,\mathrm{d}\lambda$. Now $(|g(r)|+|g(-r)|)/2 = \max_\theta |e\cos\theta - io\sin\theta|$, and

$$|e\cos\theta - io\sin\theta| = L\left|\int g\,\frac{\overline{H}}{H}(\cos\theta - i\rho\sin\theta)\,\mathrm{d}\lambda\right| \le L\|g\|_1,$$

since $|\cos\theta - i\rho\sin\theta| \le 1$. This needs $\tan(\beta\,\mathrm{artanh}|x|) \le 1/2$. With the Hardy measure and $\beta = 1$ it gives $|\rho| \le 2|x|/(1+|x|^2) \le 1$ for every x, a second proof of the Hardy constant.

In (3) and (4) together, the square bound is sharp exactly up to the first zero of $K_x + K_{-x}$: beyond it the squares H^2 stop being extremal because a zero of H can be moved in one of the two factors; in the Gauss case the gain $e^{-\pi^2/(4|x|^2)}$ is invisible to every order at $x = 0$, consistent with (2).

Remark 6.9 (Neumann ovals). For $\beta = 2$, where λ is normalized area measure, (3) holds on the whole disc and has a geometric meaning. With $\varphi(z) = z/(1-x^2z^2)$, so that $2\varphi' = H$, the inequality $U \ge 0$ says that the Neumann oval $\varphi(\mathbb{D})$ is a quadrature domain for subharmonic functions, with two equal nodes at $\pm x/(1-x^4)$: $\pi^{-1}\int_{\varphi(\mathbb{D})} u\,\mathrm{d}A \ge (L/2)(u(t)+u(-t))$. This classical fact (see Nishry–Wennman [66]) can also be proved by shrinking the oval to two tangent discs.

Proposition 6.10. *Let F be an analytic polynomial, let Q be a set of coordinates with points x_p, $p \in Q$, and fix the other coordinates at points w_p. With independent signs ε_p,*

$$\mathbb{E}_\varepsilon\big|F\big((\varepsilon_p x_p)_{p\in Q}, w\big)\big| \le \prod_{p\in Q} c(x_p) \int \big|F(\zeta, w)\big|\,\mathrm{d}\lambda^Q(\zeta).$$

For $x = \gamma_Q(s)$ with $\Re s \ge 1/2$ the constant is at most

$$C_{\mathrm{sign}} = \zeta(2), \qquad C_\beta = \prod_p \Lambda_\beta(1/p)^{1/2}, \qquad \prod_p \cosh(2/p)^{1/2} \le e^{\sum_p p^{-2}} < 1.58.$$

Proof. As Proposition 4.6, by induction on the coordinates of Q with Fubini. For the Gauss bound use $\cosh y \le e^{y^2/2}$. □

For comparison, the Carleman route at $\beta = 1$ gives $C_1 = \zeta(2)^{3/2}\zeta(4)^{-1/2} \approx 2.03$ instead of $\zeta(2)$, and $C_2 \approx 7.26$.

The constant in Proposition 6.10 is $\prod_{p\in Q} c(x_p)$ exactly, and on the Bergman side it can now be computed.

Corollary 6.11 (Sharp sign averages). *In the Bergman case with $1 < \beta \le \pi/(2\arctan(2^{-1/2}))$, the best constant in Proposition 6.10, uniformly over $\Re s \ge 1/2$,*

over Q and over F, is

$$C_\beta^{\mathrm{sharp}} = \prod_p L(p^{-1/2}) = \prod_p \frac{(1-p^{-1})^{-\beta} + (1+p^{-1})^{-\beta}}{2},$$

and it is attained in the closure of the polynomials. For $\beta = 2$, $C_2^{\mathrm{sharp}} = \zeta(2)^3/\zeta(4) = 5\pi^2/12 \approx 4.11$. In the Hardy case the constant $\zeta(2)$ is sharp in the same sense.

PROOF. For product polynomials $F = \prod_{p\in Q} g_p(z_p)$, with no other coordinates, both sides of Proposition 6.10 factor, so the best constant for a finite Q is $\prod_{p\in Q} c(x_p)$. Since $c \le \Lambda_\beta^{1/2}$, the factors are $1+O(p^{-2})$, and the product over all primes converges. The condition on β gives $\beta \arctan(p^{-1/2}) \le \pi/2$ for every prime, so $c(p^{-\sigma}) = L(p^{-\sigma})$ by Proposition 6.7, item (3), and L increases in $|x|$, so $\sigma = 1/2$ is the worst case. At $\beta = 2$, $L(p^{-1/2}) = (1-p^{-4})/(1-p^{-2})^3$. For attainment, let $h_p = (K_{r_p} + K_{-r_p})/(2L(r_p)^{1/2})$ with $r_p = p^{-1/2}$, so that $\|h_p\|_2 = 1$, $\int h_p \,\mathrm{d}\lambda = L(r_p)^{-1/2}$ and $h_p(\pm r_p)^2 = L(r_p)$. The products $H_N = \prod_{j\le N} h_{p_j}$ satisfy $\|H_N - H_M\|_2^2 = 2 - 2\prod_{M<j\le N} L(r_{p_j})^{-1/2} \to 0$, so H_N^2 converges in L^1 to some F with $\|F\|_1 = 1$, and the sign average of H_N^2 is the constant $\prod_{j\le N} L(r_{p_j})$. By Proposition 6.10, restriction to the sign points is bounded from cylinder polynomials in X^1 into L^1 of the sign probability space. Its continuous extension therefore sends F to the constant $\prod_p L(p^{-1/2})$, proving attainment without point evaluation outside ℓ^2. In the Hardy case $c(x) = (1-|x|^4)^{-1}$ for every x, and the same argument gives $\prod_p (1-p^{-2})^{-1} = \zeta(2)$. □

For larger β the finitely many primes $p < \cot^2(\pi/(2\beta))$ fall outside Proposition 6.7, item (3), and their factors exceed $L(p^{-1/2})$ by item (4).

6.4. Dirichlet kernels

The contractive embedding also measures cancellation. Let $D_N = \sum_{n\le N} n^{-s}$ and $d_\beta(n)$ the coefficients of ζ^β, so that $\|D_N\|_{\mathscr{A}^2_\beta}^2 = \sum_{n\le N} 1/d_\beta(n)$.

PROPOSITION 6.12. *Fix $\beta \ge 1$ and put*

$$G_\beta(1) = \prod_p (1-1/p)^{1/\beta} \sum_{k\ge 0} \frac{k!\, p^{-k}}{(\beta)_k}.$$

Then

$$\|D_N\|_{\mathscr{A}^2_\beta}^2 \sim \frac{G_\beta(1)}{\Gamma(1/\beta)}\, N(\log N)^{1/\beta - 1},$$

and

$$(c_\beta + o(1))(\log N)^{-1/(4\beta)} \le \frac{\|D_N\|_{\mathscr{A}^1_\beta}}{\|D_N\|_{\mathscr{A}^2_\beta}} \le 1, \qquad c_\beta^2 = \frac{G_{2\beta}(1)\Gamma(1/\beta)}{G_\beta(1)\Gamma(1/(2\beta))}.$$

PROOF. The Dirichlet series of $1/d_\beta$ is $\zeta(s)^{1/\beta} G_\beta(s)$ with G_β analytic and nonzero near 1, and Selberg–Delange gives the asymptotic. The lower bound is $\|D_N\|_{\mathscr{A}^1_\beta} \ge \|D_N\|_{\mathscr{A}^2_{2\beta}}$, by tensorized Kulikov, with the asymptotic at 2β. □

For $\beta = 2$ the Kulikov step is [12, Theorem 10]. At $\beta = 1$ Harper's theorem [40] gives the true ratio $(\log\log N)^{-1/4}$, so the lower bound is far from sharp there. For $\beta > 1$ it is not sharp either: the ratio does not tend to 0, as the subcritical chaos exponent $2/\sqrt{\beta}$ suggests. This is unconditional for $\beta > 2$ and rests on a recent preprint for $1 < \beta \le 2$.

The random model. Let (R_p) and (Z_p) be independent families of independent random variables, with R_p^2 distributed as $\mathrm{Beta}(1, \beta - 1)$ and Z_p uniform on $\mathbb{T}$, so that $R_p Z_p$ has law m_β. Extend R and Z completely multiplicatively and put $S_N = \sum_{n \le N} R(n)Z(n)$. Then $\|D_N\|_{\mathscr{A}_\beta^q}^q = \mathbb{E}|S_N|^q$, and $g(n) = \mathbb{E}R(n)^2 = 1/d_\beta(n)$. Write A_N^2 for $\|D_N\|_{\mathscr{A}_\beta^2}^2$, $\theta = 1/\beta$ in this section, and $\mathbb{E}_Z$ for the expectation over Z with R fixed.

THEOREM 6.13. *If $\beta > 2$, then $\mathbb{E}|S_N|^4 \ll_\beta A_N^4$, and hence $\|D_N\|_{\mathscr{A}_\beta^1} \gg_\beta \|D_N\|_{\mathscr{A}_\beta^2}$. If $\beta = 2$, then $\|D_N\|_{\mathscr{A}_2^1} \gg \|D_N\|_{\mathscr{A}_2^2}(\log\log N)^{-1/2}$.*

PROOF. By Hölder's inequality, $A_N^2 = \mathbb{E}|S_N|^2$ is at most $(\mathbb{E}|S_N|)^{2/3}(\mathbb{E}|S_N|^4)^{1/3}$, so it suffices to show $\mathbb{E}|S_N|^4 \ll A_N^4$, with an extra factor $\log\log N$ when $\beta = 2$. By orthogonality of Z, $\mathbb{E}|S_N|^4$ is the sum of $g(n_1 n_2)$ over $n_i \le N$ with $n_1 n_2 = n_3 n_4$. Put $a = (n_1, n_3)$, $n_1 = ab$, $n_3 = ac$; then $(b, c) = 1$, $n_2 = cd$ and $n_4 = bd$. This is a bijection onto quadruples with $(b, c) = 1$ and $b, c \le N/M$, $M = \max(a, d) \le N$.

Let $y^{(x)}$ be the largest divisor of y coprime to x. Since $g(p^k)$ decreases in k, the ratio of consecutive terms being $(k+1)/(\beta+k)$, we have $g(xy) \le g(x)g(y^{(x)})$, and so $g(abcd) \le g(a)g(d^{(a)})g(b^{(ad)})g(c^{(ad)})$. If h is multiplicative with $0 \le h(p^k) \le K$, the elementary argument behind the Halberstam–Richert inequality, bounding $\sum_{n \le x} h(n)\log n$ by Chebyshev's estimate and $\sum_{n \le x} h(n)\log(x/n)$ by $x\sum_{n \le x} h(n)/n$, gives

$$\sum_{n \le x} h(n) \ll_K \frac{x}{\log x} \prod_{p \le x} \Big(1 + \frac{h(p)}{p} + \frac{2K}{p^2}\Big).$$

Since $g(p) = \theta$ and $0 \le g \le 1$, this and Mertens' theorem give, uniformly in q and a, $\sum_{n \le x} g(n^{(q)}) \ll x(\log 2x)^{\theta-1} q/\varphi(q)$, $\sum_{d \le y} g(d^{(a)})(d/\varphi(d))^2 \ll y(\log 2y)^{\theta-1}\rho(a)$ with $\rho(a) = \prod_{p|a}(1 + C/p)$ for a suitable constant C, and $\sum_{a \le y} g(a)(a/\varphi(a))^2\rho(a) \ll y(\log 2y)^{\theta-1}$. Summing over b and c first, and using $ad/\varphi(ad) \le (a/\varphi(a))(d/\varphi(d))$,

$$\mathbb{E}|S_N|^4 \ll \sum_{a,d \le N} g(a)g(d^{(a)}) \Big(\frac{ad}{\varphi(a)\varphi(d)}\Big)^2 \Big(\frac{N}{M}\Big)^2 \Big(\log\frac{2N}{M}\Big)^{2\theta-2}.$$

Split into $2^j \le M < 2^{j+1}$, $0 \le j \le J = \lfloor \log_2 N \rfloor$. The other two bounds give $\ll 4^j(1+j)^{2\theta-2}$ for the weights with $a, d < 2^{j+1}$, so

$$\mathbb{E}|S_N|^4 \ll N^2 \sum_{j=0}^{J} (1+j)^{2\theta-2}(1+J-j)^{2\theta-2}.$$

If $\beta > 2$, then $2\theta - 2 < -1$ and the sum is $\ll J^{2\theta-2}$; if $\beta = 2$, it is $\ll J^{-1}\log J$. By Proposition 6.12, $A_N^4 \asymp N^2(\log N)^{2\theta-2}$. □

The argument does not give boundedness for $\beta \le 2$. Gorodetsky and Wong's multiplicative chaos for Steinhaus sums weighted by a multiplicative function covers the remaining range; their preprint has not yet been refereed.

THEOREM 6.14 (Conditional on [38, Theorem 1.3]). *Let $\beta > 1$. Almost surely $C(R) = \prod_p (1-1/p)^\theta (1 - R_p^2/p)^{-1}$ converges to a number in $(0,\infty)$ and $L(R) = \lim_N \mathbb{E}_Z|S_N|/(\mathbb{E}_Z|S_N|^2)^{1/2}$ exists in $(0, \sqrt{\pi}/2]$, and*

$$\lim_{N\to\infty} \frac{\|D_N\|_{\mathscr{A}^1_\beta}}{\|D_N\|_{\mathscr{A}^2_\beta}} = \kappa_\beta := \mathbb{E}\Big[L(R)\Big(\frac{C(R)}{G_\beta(1)}\Big)^{1/2}\Big] \in (0, \sqrt{\pi}/2].$$

PROOF. Let $W_N = \sum_{n\le N} R(n)^2 = \mathbb{E}_Z|S_N|^2$, so that $\mathbb{E}W_N = A_N^2$ and, with $\rho_N = \mathbb{E}_Z|S_N|/W_N^{1/2} \in [0,1]$,

$$\frac{\|D_N\|_{\mathscr{A}^1_\beta}}{A_N} = \mathbb{E}\Big[\rho_N \Big(\frac{W_N}{A_N^2}\Big)^{1/2}\Big].$$

Hypotheses. For fixed R, $f = R$ is completely multiplicative with $0 \le f(p^k) \le 1$, and Z is a Steinhaus random multiplicative function, so condition (b) of [38, Theorem 1.3] holds. Condition (a) asks that $\sum_{p\le t} R_p^2 = \theta t/\log t + E(t)$ with $E(t) = o(t/\log t)$ and $\int_2^\infty t^{-2}|E(t)|\,\mathrm{d}t < \infty$. The R_p^2 are independent, take values in $[0,1]$ and have mean θ. The strong law of large numbers gives the first condition almost surely; since $\mathbb{E}|\sum_{p\le t}(R_p^2-\theta)| \le \pi(t)^{1/2}$, Tonelli's theorem gives the second, the prime number theorem error being $O(t/\log^2 t)$.

Conditional limit. By [38, Theorem 1.3], for almost every R, $S_N/W_N^{1/2} \to V_\infty^{1/2} G$ in distribution under $\mathbb{E}_Z$, with G standard complex Gaussian independent of V_∞ and $0 < V_\infty < \infty$ almost surely. These variables have $\mathbb{E}_Z$-second moment 1, hence are uniformly integrable, and $\rho_N \to L(R) = \Gamma(3/2)\mathbb{E}_Z V_\infty^{1/2} > 0$. By Fatou, $\mathbb{E}_Z V_\infty \le 1$, so $L(R) \le \Gamma(3/2) = \sqrt{\pi}/2$.

Random normalization. Wirsing's theorem [97, Satz 1.1] applies to $h = R^2$: $0 \le h \le 1$, $\sum_{p\le x} h(p)\log p/p \sim \theta \log x$ almost surely by partial summation from the first condition, $\sum_p \sum_{k\ge 2} h(p^k)/p^k < \infty$, and the prime powers $p^k \le x$ with $k \ge 2$ contribute $\ll \sqrt{x}$. It gives

$$W_N \sim \frac{e^{-\gamma\theta}}{\Gamma(\theta)} \frac{N}{\log N} \prod_{p\le N} \Big(1 - \frac{R_p^2}{p}\Big)^{-1} \sim C_N(R) \frac{N(\log N)^{\theta-1}}{\Gamma(\theta)}$$

by Mertens' theorem, with C_N the partial products of C. The logarithms of the factors of C are $(R_p^2 - \theta)/p + O(p^{-2})$, independent and centred up to the error, with summable variances, so Kolmogorov's one-series theorem gives $C_N \to C(R) \in (0,\infty)$ almost surely. With Proposition 6.12, $W_N/A_N^2 \to C(R)/G_\beta(1)$ almost surely.

Limit. The integrand converges almost surely to $L(R)(C(R)/G_\beta(1))^{1/2} > 0$ and is bounded by $(W_N/A_N^2)^{1/2}$, whose second moment is 1. Vitali's theorem gives the limit $\kappa_\beta > 0$, and Fatou with Cauchy–Schwarz gives $\kappa_\beta \le \sqrt{\pi}/2$. □

For $\beta > 2$ the proof of [38, Theorem 1.3] rests on an earlier paper of the same authors, but there Theorem 6.13 needs neither. Whether $\kappa_\beta \to 0$ as $\beta \to 1^+$,

matching Harper's cancellation, is open. The ratio cannot tend to 0 uniformly in β in any case: it tends to 1 when $\beta/N \to \infty$.

6.5. Positive measures in three geometries

With R, Q, ν and the measure $\mu_{\nu,R}$ as in section 4.2, let $\alpha_{\nu,R}(n) = \mathbf{1}[n_Q \in \square] \int n^{-s}\,\mathrm{d}\nu(s)$ and $[\nu]_\kappa = \sup_{0<t\le 1} \nu((1/2, 1/2+t])t^{-\kappa}$.

THEOREM 6.15. *Let R have density $\theta \in [0,1]$ with constant B, let κ be the kernel exponent of the geometry, and let ν be a finite positive Borel measure on $(1/2,\infty)$. The following are equivalent:*

(1) $H_{\alpha_{\nu,R}}$ is bounded on $X^2 \times X^2$;
(2) $[\nu]_\kappa < \infty$;
(3) $\int |F|\,\mathrm{d}\mu_{\nu,R} \lesssim \|F\|_{X^1}$;
(4) $H_{\alpha_{\nu,R}}$ has a bounded symbol.

The ratio between the smallest symbol norm and $\|H_{\alpha_{\nu,R}}\|$ is bounded by a constant depending only on θ, B and the geometry. If $\theta = 0$, all four hold for every finite ν. Moreover $H_{\alpha_{\nu,R}}$ is compact if and only if $\nu((1/2,1/2+t]) = o(t^\kappa)$, which is automatic when $\theta = 0$, and then its bounded symbols lie in the $L^\infty(\lambda)$-closure of $C + N_\lambda$, where N_λ is the annihilator of the analytic polynomials and C is $C(\overline{\mathbb{D}}^\infty)$ in the Bergman geometry and the bounded continuous functions in the Gauss geometry.

PROOF. The structure is that of Theorems 4.2 and 4.10. We record the three geometry-dependent inputs.

Kernel test. Let K be the reproducing kernel of X^2 at $\gamma_R(1/2+\delta)$, restricted to the coordinates of R. Its coefficients are positive and supported on integers with $n_Q = 1$, so Tonelli's theorem gives $H_{\alpha_{\nu,R}}(K,K) = \int K(\gamma(s))^2\,\mathrm{d}\nu(s)$. Also $K(\gamma(s)) \ge \|K\|_2^2$ for $s \le 1/2+\delta$. Hence

$$\nu((1/2,1/2+\delta]) \le \|H_{\alpha_{\nu,R}}\|\,\|K\|_2^{-2} \le e^{\kappa' B}(2\delta)^\kappa\,\|H_{\alpha_{\nu,R}}\|,$$

with $\kappa' = \beta$ in the Bergman case and $\kappa' = 1$ otherwise, by the lower bound $\|K\|_2^2 \ge e^{-\kappa' B}(2\delta)^{-\kappa}$ that follows from (2.1). If $\theta > 0$, these normalized kernels converge weakly to zero: their norms diverge, while every fixed monomial coefficient stays bounded before normalization. Compactness makes their quadratic forms tend to zero, so the same lower bound gives $\nu((1/2,1/2+t]) = o(t^\kappa)$. When $\theta = 0$, this condition is automatic because ν is finite and the intervals decrease to the empty set.

Sufficiency. Average the Q-coordinates with Proposition 6.10. Then treat the remaining R-coordinates with Theorem 2.11 (2), with $p = 1$, if $\theta > 0$, or with the uniform bound $\|K_{\gamma_R(s)}\|_2^2 \le e^{\kappa'(1+B)}$ if $\theta = 0$. Treat $\Re s > 3/2$ with the point estimate. This gives $\int |F|\,\mathrm{d}\mu_{\nu,R} \le C_{\mathrm{sign}}\big(C[\nu]_\kappa + C'\nu((1/2,\infty))\big)\|F\|_{X^1}$. Symbols then come from Lemma 6.1.

Compactness. For the part of ν above $1/2+\varepsilon$, the moments satisfy $\alpha(n) \le \nu((1/2,\infty))n^{-1/2-\varepsilon}$, and Corollary 6.4 bounds the tail of the functional beyond

$n > K$ by the square root of

$$\sum_{n>K} d(n)\, n^{-1-2\varepsilon}, \qquad \sum_{n>K} d_{2\beta}(n)\, n^{-1-2\varepsilon}, \qquad \sum_{n>K} \frac{2^{\Omega(n)}}{[n]!}\, n^{-1-2\varepsilon}.$$

These are tails of convergent series, the last being $\prod_p \exp(2p^{-1-2\varepsilon})$. The part below $1/2+\varepsilon$ is small by the sufficiency estimate. The rest of the argument is that of Theorem 4.10. □

In the Bergman case with $R = \emptyset$, every $H_{\alpha_{\nu,\emptyset}}$ is bounded, compact, and has a bounded symbol of norm at most $C_\beta\, \nu((1/2,\infty))$. Without the sign average, Ma, Lu and Zu show that boundedness requires $\nu \lesssim t^\beta$ and compactness requires $\nu = o(t^\beta)$ [58].

COROLLARY 6.16. *In the Bergman geometry with $\beta > 1$ and in the Gauss geometry, the sign-averaged measure $\mu_{\nu,R}$ of Theorem 6.15 is a Carleson measure for X^p for some $p > 0$ if and only if it is for every $p > 0$, if and only if $[\nu]_\kappa < \infty$.*

PROOF. Sufficiency: Proposition 6.6 in each averaged coordinate, with Fubini, costs $\prod_q c_p(\gamma_q(s))$, which is bounded since $\sum_q |\gamma_q(s)|^4 \le \sum_q q^{-2}$; then part (2) of Theorem 2.11 on each slice when $\theta > 0$, as in Corollary 4.15. When $\theta = 0$, use instead the uniform bound $\|K_{\gamma_R(s)}\|_2^2 \le e^{\kappa'(1+B)}$, where $\kappa' = \beta$ in Bergman geometry and $\kappa' = 1$ in Gauss geometry. Finiteness of ν gives the embedding, and $[\nu]_0 < \infty$ is automatic. Necessity: test with $K^{2/p}$ for the kernel K at $\gamma_R(1/2+\delta)$ in the coordinates of R, as in Corollary 4.15. □

6.6. The Bergman multiplicative Hilbert form

Bayart, Brevig, Haimi, Ortega-Cerdà and Perfekt [12] introduced the form on $\mathscr{A}^2 \times \mathscr{A}^2$, $\mathscr{A} = \mathscr{A}_2$, with analytic symbol

$$\varphi(s) = \sum_{n\ge 2} \frac{d(n)}{\sqrt{n}\log^2 n}\, n^{-s},$$

the Bergman analogue of the multiplicative Hilbert matrix. They proved that it is bounded and not compact, and that φ defines a bounded functional on $\mathscr{A}^p$ for $1 < p < \infty$. Ma, Lu and Zu [58, Theorem 1.1] have since determined its spectrum on the subspace $\mathscr{A}^2_0$ of series with $a_1 = 0$: the essential spectrum is $[0,1]$, absolutely continuous of multiplicity one, and above it there are finitely many eigenvalues, at least one, all simple. Its moments are $\alpha(n) = 1/(\sqrt{n}\log^2 n) = \int_0^\infty n^{-1/2-u} u\, \mathrm{d}u$, so it is the form of the measure $u\, \mathrm{d}u$ on the curve: the Bergman counterpart of Lebesgue measure.

The Bergman analogue of Problem 3 asks whether this form has a bounded symbol. It does, with an explicit constant. Write $r(\sigma) = (p^{-1/2-\sigma})_p$.

PROPOSITION 6.17. *Let $\beta \ge 1$ and $0 < p < \infty$. For every $f \in \mathscr{A}^p_\beta$,*

$$\int_0^1 |f(1/2+\sigma)|^p \sigma^{\beta-1}\, \mathrm{d}\sigma \le C_\beta \|f\|^p_{\mathscr{A}^p_\beta}, \qquad C_\beta = \Big(\frac{3}{\sqrt{2}}\Big)^\beta B\Big(\frac{\beta}{4}, \frac{\beta}{4}\Big),$$

and C_2 can be replaced by $(\pi + 5/2 + 1/\sqrt{3})/2 < 3.11$.

PROOF. In one coordinate, with $a = p^{-1/2-\sigma}$ and $b = p^{-1/2-\tau}$, we have $1 - \rho(a,b)^2 = (1-a^2)(1-b^2)/(1-ab)^2$, so by Proposition 1.4 (4) the Gram kernel $\mathcal{G} = \|K\|\|K\|H$ of Theorem 2.1 satisfies

$$\mathcal{G}\big(r(\sigma), r(\tau)\big) \le \big(\zeta(1+2\sigma)\zeta(1+2\tau)\big)^{\beta/4}\zeta(1+\sigma+\tau)^{\beta/2}.$$

Apply Theorem 2.1 to the image of $\sigma^{\beta-1}\,\mathrm{d}\sigma$ on $(0,1)$ under r. After the unitary $c \mapsto \sigma^{(\beta-1)/2}c$ onto $L^2(0,1)$, and with $\zeta(1+x) \le (1+x)/x$ and $(1+2\sigma)^{1/4}(1+2\tau)^{1/4} \le (1+\sigma+\tau)^{1/2}$, the kernel is at most

$$M(\sigma,\tau) = 2^{-\beta/2}(1+\sigma+\tau)^{\beta}(\sigma\tau)^{\beta/4-1/2}(\sigma+\tau)^{-\beta/2}.$$

On $(0,1)^2$ the first factor is at most 3^β, and the rest is homogeneous of degree -1. Schur's test with the weight $\tau^{-1/2}$ bounds its norm by $\int_0^\infty x^{\beta/4-1}(1+x)^{-\beta/2}\,\mathrm{d}x = B(\beta/4,\beta/4)$. For $\beta = 2$, $M = (1/(\sigma+\tau)+2+\sigma+\tau)/2$. On $L^2(0,1)$ the first kernel has norm at most π by Hilbert's inequality, the constant 2 has norm 2, and the rank-two kernel $\sigma+\tau$ has norm $1/2+1/\sqrt{3}$. □

THEOREM 6.18. *For every $f \in \mathscr{A}^1$, $|\sum_{n\ge2} a_n/(\sqrt{n}\log^2 n)| \le 25\|f\|_{\mathscr{A}^1}$, where the series converges in its natural order. Consequently, the Bergman multiplicative Hilbert form has a bounded symbol in $L^\infty(m_2^\infty)$.*

PROOF. For polynomials, the Gamma integral gives $\sum_{n\ge2} a_n/(\sqrt{n}\log^2 n) = \int_0^\infty (f(1/2+u)-a_1)\,u\,\mathrm{d}u$. On $(0,1)$ apply Proposition 6.17 with $\beta = 2$ and $p = 1$, and $|a_1| \le \|f\|_{\mathscr{A}^1}$. For $u \ge 1$ use $|a_n| \le d(n)\|f\|_{\mathscr{A}^1}$, read off from the reproducing property; the integral over $(1,\infty)$ is then at most $(1/\log 2 + 1/\log^2 2)(\zeta(3/2)^2 - 1)\|f\|_{\mathscr{A}^1} < 20.6\|f\|_{\mathscr{A}^1}$. Altogether the constant is less than $3.11 + 0.5 + 20.6 < 25$. The bounded symbol then comes from Lemma 6.1.

For natural-order convergence, compare the partial sum up to N with the Abel sum at $\varepsilon = 1/\log N$. The Bergman Helson inequality with d_4 (Corollary 6.4) and the dyadic estimate $\sum_{e^u \le n < e^{2u}} d_4(n)/(n\log^4 n) \le 16e^2$ bound the contribution of the terms with $n > M$ by a constant times $(\sum_{n>M}|a_n|^2/d_4(n))^{1/2}$, which is small for fixed f and large M, while the terms with $n \le M$ tend to 0. So the difference tends to 0. Translations $f \mapsto f(\cdot+\varepsilon)$ are contractions of $\mathscr{A}^1$ [12, Theorem 11], so the Abel sums converge to the value of the bounded functional. □

The same subtraction as in Corollary 4.18 also handles the finite measure $u\mathbf{1}_{(0,1]}(u)\,\mathrm{d}u$, with correction terms controlled by the d_4 Helson inequality. Combined with Theorem 6.15, it gives bounded symbols for the sign-averaged variants: for instance, the Bergman Hilbert form restricted to pairs with $(mn)_Q$ a square.

6.6.1. Unimodality in the Bergman parameter

Ma, Lu and Zu [58, §6.2, Question 1] ask whether the norm of the multiplicative Hilbert operator decreases to a unique minimum and then increases. We retain their normalization and parameter α.

For $\alpha > 1$ let d_α be given by $\zeta(s)^\alpha = \sum_{n\ge1} d_\alpha(n)n^{-s}$, let $A^2_{\alpha,0}$ be the space of Dirichlet series $\sum_{n\ge2} a_n n^{-s}$ with $\sum_{n\ge2}|a_n|^2/d_\alpha(n) < \infty$, and let T_α be the

operator on $A^2_{\alpha,0}$ whose matrix in the orthonormal basis $(\sqrt{d_\alpha(n)}\,n^{-s})_{n\ge2}$ is

$$\frac{\Gamma(\alpha)\sqrt{d_\alpha(m)d_\alpha(n)}}{\sqrt{mn}\,(\log mn)^\alpha}, \qquad m,n\ge2.$$

THEOREM 6.19. *Conditional on [58, Theorems 3.1 and 1.1, Lemma 6.2 and §5.3] (facts (F1)–(F3) below), $N(\alpha)=\|T_\alpha\|$ has a unique $\alpha_{\min}\in(8/5,3)$ such that N is strictly decreasing on $(1,\alpha_{\min}]$ and strictly increasing on $[\alpha_{\min},\infty)$. Moreover $\log N$ is strictly convex on $[8/5,3]$.*

The proof is computer-assisted: Lemmas 6.20 and 6.22 rest on interval arithmetic, and the script is `bergman-unimodality/unimodality.py` (chapter A).

The certified values and convexity of $\log(N/\alpha)$ control the two ends of the parameter range. In the middle we dilate the companion kernel without changing its norm and prove strict pointwise log-convexity. Hölder's inequality transfers convexity to the norm; the positive top eigenfunction of (F2) supplies strictness above the essential band.

Facts from Ma, Lu and Zu

Put $\rho=(\alpha-1)/2$ and $\Lambda_\alpha=\Gamma(\alpha/2)^2/\Gamma(\alpha)$. We use the following results of [58] (arXiv version 3).

(F1) [58, Theorem 3.1]: $N(\alpha)=\|K_\alpha\|$, where K_α is the operator on $L^2(0,\infty)$ with the positive kernel $k_\alpha(x,y)=(\zeta(1+x+y)^\alpha-1)(xy)^\rho$.

(F2) [58, Theorem 1.1]: $\sigma_{\mathrm{ess}}(T_\alpha)=[0,\Lambda_\alpha]$, so $N(\alpha)\ge\Lambda_\alpha$; there is $\alpha_*\in(1,2)$ such that for $\alpha>\alpha_*$ the number $N(\alpha)$ is an eigenvalue of K_α above Λ_α. By the end of [58, §5.3] the corresponding eigenfunction can be taken strictly positive. For $1<\alpha\le\alpha_*$, $N(\alpha)=\Lambda_\alpha$ by [58, Theorem 1.1(5)].

(F3) [58, Lemma 6.2]: $F(\alpha)=\log(N(\alpha)/\alpha)$ is convex on $(1,\infty)$.

Certified values

LEMMA 6.20. *$N(19/10)<1.2313$, $N(13/5)<1.3438$ and $N(3)>1.6031$; also $\Lambda_{8/5}>1.5169$.*

PROOF. Upper bounds come from Schur's test. If $w>0$ and

$$\int_0^\infty k_\alpha(x,y)w(y)\,\mathrm{d}y\le Uw(x)\qquad(x>0),$$

then $\|K_\alpha\|\le U$, since k_α is symmetric and nonnegative. We take $w(y)=y^{-\sigma}e^{-\tau y}$ with $(\sigma,\tau)=(0.2734,0.5587)$ for $\alpha=19/10$ and $(-0.2552,0.6175)$ for $\alpha=13/5$, and bound $S(x)=w(x)^{-1}\int_0^\infty k_\alpha(x,y)w(y)\,\mathrm{d}y$ as follows.

- For $10^{-8}\le x\le80$: put $J(x)=\int_0^\infty(\zeta(1+x+y)^\alpha-1)y^{\rho-\sigma}e^{-\tau y}\,\mathrm{d}y$, so $S(x)=x^{\sigma+\rho}e^{\tau x}J(x)$. The function $\zeta(1+q)^\alpha-1=\sum_{n\ge2}d_\alpha(n)n^{-1-q}$ is completely monotone, hence convex and decreasing, and so is J. On a cell $[x_1,x_1+h]$ of a geometric grid of ratio 1.01, the concave function $\log(x^{\sigma+\rho}e^{\tau x})$ lies below its tangent at x_1 and J below its chord, so $S(x)\le S_1e^{\max(d,0)h}$ with $S_1=x_1^{\sigma+\rho}e^{\tau x_1}J_1$ and $d=(\sigma+\rho)/x_1+\tau+(J_2/J_1-1)/h$, where J_1,J_2 are upper bounds for J at the cell ends. These come from

chords of $\zeta(1+x+y)^\alpha-1$ on the cells of a y-grid on $[10^{-13},40]$, integrated exactly against $y^{\rho-\sigma}e^{-\tau y}$ (incomplete gamma functions in ball arithmetic), with values bounded in turn by chords of a table of ball-arithmetic values of $\zeta(1+q)^\alpha-1$; the tails $y<10^{-13}$ and $y>40$ are estimated with monotonicity and $\zeta(1+q)-1\le 2^{-q}$ $(q\ge 2)$.

- For $x\le 10^{-8}$: $\zeta(1+q)\le(1+q)/q$ gives $S(x)\le e^{\tau x}x^{\sigma+\rho}\int_0^\infty(x+y)^{-\alpha}(1+x+y)^\alpha y^{\rho-\sigma}e^{-\tau y}\,\mathrm{d}y$; splitting at $y=10^{-4}$ bounds this by $e^{\tau x}(1+x+10^{-4})^\alpha B(1+\rho-\sigma,\rho+\sigma)+e^{\tau x}M(x/10^{-4})^{\rho+\sigma}/(\rho+\sigma)$ with $M=\sup_{y>0}(1+x+y)^\alpha e^{-\tau y}$.
- For $x\ge 80$: $\zeta(1+q)^\alpha-1\le\alpha\zeta(1+q)^{\alpha-1}(\zeta(1+q)-1)\le 2\alpha 2^{-q}$ gives $S(x)\le 2\alpha x^{\sigma+\rho}e^{-(\log 2-\tau)x}\Gamma(1+\rho-\sigma)(\tau+\log 2)^{\sigma-\rho-1}$, decreasing for $x\ge 80$.

The lower bound is a Rayleigh quotient. For $f(y)=y^{-s}e^{-ty}$ the substitution $x=qz$, $y=q(1-z)$ gives

$$\frac{\langle K_\alpha f,f\rangle}{\|f\|^2}=\frac{B(1+\rho-s,1+\rho-s)(2t)^{1-2s}}{\Gamma(1-2s)}\int_0^\infty(\zeta(1+q)^\alpha-1)q^{\alpha-2s}e^{-tq}\,\mathrm{d}q,$$

and we take $(s,t)=(-0.5302,0.6229)$ at $\alpha=3$ and bound the integral below by its part over $[10^{-6},60]$, enclosed by rigorous Gauss–Legendre integration in ball arithmetic. All ζ-values, cell weights, scalar prefactors, and tails have outward arb enclosures. The grid points are exact binary64 numbers; every rounded sum $x+y$ is moved one binary64 step downward before the decreasing table is used. For a positive binary64 sum of n products, the script bounds the exact sum by the computed sum divided by $1-(n+3)2^{-53}$, and checks that all intermediates are positive, normal, and finite. The threshold comparisons are rigorous comparisons of balls with rational decimals. □

LEMMA 6.21. *N is strictly decreasing on $(1,8/5]$ and strictly increasing on $[3,\infty)$.*

PROOF. F is convex (F3), so its one-sided derivatives exist and are nondecreasing. For $1<\alpha\le 8/5$, by (F2) and Lemma 6.20,

$$F'_\pm(\alpha)\le\frac{F(19/10)-F(8/5)}{3/10}\le\frac{10}{3}\Big(\log\frac{1.2313}{1.9}-\log\frac{1.5169}{1.6}\Big)<-1.26,$$

so $(\log N)'_\pm=1/\alpha+F'_\pm<0$. For $\alpha\ge 3$,

$$F'_\pm(\alpha)\ge\frac{F(3)-F(13/5)}{2/5}\ge\frac{5}{2}\Big(\log\frac{1.6031}{3}-\log\frac{1.3438}{2.6}\Big)>0,$$

so $(\log N)'_\pm\ge 1/\alpha>0$. □

Convexity in the middle

Let $c(\alpha)=(\alpha-1)^{2/3}$ and, for $q>0$,

$$\Phi_q(\alpha)=\alpha\log c(\alpha)+\log\big(\zeta(1+c(\alpha)q)^\alpha-1\big).$$

LEMMA 6.22. *$\Phi''_q(\alpha)>0$ for all $q>0$ and $\alpha\in[8/5,3]$.*

PROOF. On $[8/5,3]$ we have $c\in[0.71,1.59]$, $c'=(2/3)(\alpha-1)^{-1/3}\in[0.52,0.80]$ and $c''=-(2/9)(\alpha-1)^{-4/3}\in[-0.45,-0.08]$.

The range $10^{-40}\le q\le 8$ is verified by interval arithmetic. With $p=cq$, $\ell(p)=\log\zeta(1+p)$, $y=\alpha\ell(p)$, $c'/c=2/(3(\alpha-1))$ and $c''/c=-2/(9(\alpha-1)^2)$, one has $\Phi_q=\alpha\log c+y+\log(1-e^{-y})$, so Φ_q'' is an explicit expression $F(\alpha,p)$ in α, ℓ, $p\ell'$ and $p^2\ell''$. The rectangle $\alpha\in[8/5,3]$, $p\in[7\cdot10^{-41},12.8]$ contains all $(\alpha,c(\alpha)q)$ with $10^{-40}\le q\le 8$; it is cut into boxes, and on each box $F>0$ is checked by the mean value form, with the gradient in $(\alpha,\log p)$ enclosed by forward differentiation in ball arithmetic. For $p\le 0.9$ we write $\zeta(1+p)=1/p+D(p)$ and use $\Phi_q=-\alpha\log q+\alpha\log H(p)+\log(1-(p/H(p))^\alpha)$, with $H=1+pD$; the term $-\alpha\log q$ is linear in α, and F is divided by p. The entire function D is evaluated from its Taylor polynomial of degree 40 at 0, with coefficients $(-1)^n\gamma_n/n!$ computed in arb, and the tail is bounded by Berndt's estimate $|\gamma_n|\le 4(n-1)!/\pi^n$ [13]. For $p\ge 0.9$ we write $\zeta(1+p)-1=\zeta(1+p,2)$ (Hurwitz). About 2400 boxes suffice; boxes are bisected at exact binary64 points, and each box encloses its endpoints.

The range $q\le 10^{-40}$. Here $p\le 2\cdot10^{-40}$. With $h=\log H$, the linear term dropped, $\Phi_q''=I_1+I_2$, where

$$I_1=\frac{\mathrm{d}^2}{\mathrm{d}\alpha^2}\alpha h(cq)=q\big(h'(p)(2c'+\alpha c'')+\alpha h''(p)c'^2q\big),$$
$$I_2=\frac{\mathrm{d}^2}{\mathrm{d}\alpha^2}\log(1-X),\qquad X=(p/H)^\alpha.$$

The script also certifies $D>0.57$, $0<D'<1$, and $|D''|<1$ on $[0,2\cdot10^{-40}]$ from the Taylor coefficients and Berndt tail. Since $D(0)=\gamma$, this gives $D\in[\gamma,1]$ on that interval. Thus $|h'(p)-\gamma|\le 3p$ and $|h''(p)|\le 4$. The function $2c'+\alpha c''=(\alpha-1)^{-4/3}(10\alpha/9-4/3)$ has its minimum on $[8/5,3]$ at an endpoint and is at least 0.79. Hence $I_1\ge 0.45q$. Next, $\log X=\alpha(\log p-h(p))$ has $|(\log X)'|\le|\log p|+4$ and $|(\log X)''|\le 8$, and $X\le p^\alpha\le p^{8/5}$ because $H\ge 1$, so $|I_2|\le 3X((\log X)'^2+|(\log X)''|)\le 3p^{8/5}((|\log p|+4)^2+8)<10^{-18}q$.

The range $q\ge 8$. The Dirichlet coefficients satisfy $d_\alpha(n)=\prod_{p^k\|n}(\alpha)_k/k!$. Put

$$b_n(\alpha)=c(\alpha)^\alpha d_\alpha(n)n^{-1-c(\alpha)q},\qquad f_n(\alpha)=\log b_n(\alpha),\qquad \Omega(n)=\sum_{p^k\|n}k.$$

Then $e^{\Phi_q(\alpha)}=\sum_{n\ge2}b_n(\alpha)$. The series and its first two α-derivatives converge locally uniformly for $q\ge 8$ and $8/5\le\alpha\le 3$: here $c>0.71$ and $d_\alpha(n)\le d_3(n)\le n^2$, while the differentiated factors grow at most polynomially in $\log n$. Writing $A=(\alpha\log c)''=(2/3)(\alpha-2)/(\alpha-1)^2$ and $c''=-(2/9)(\alpha-1)^{-4/3}$, we have

$$\begin{aligned}f_n''&=A+(\log d_\alpha(n))''-qc''\log n\\&\ge A+\Omega(n)C,\qquad C=-\frac{1}{\alpha^2}-qc''\log 2,\end{aligned}$$

because $(\log d_\alpha(n))''=-\sum_{p^k\|n}\sum_{j=0}^{k-1}(\alpha+j)^{-2}\ge-\Omega(n)/\alpha^2$ and $\log n\ge\Omega(n)\log 2$. For $8/5\le\alpha\le 2$, using $q\ge 8$ and $\log 2>2/3$, we get

$$A\ge-\frac{20}{27},\qquad C>\frac{32}{27}-\frac{25}{64},\qquad A+C>\frac{31}{576}.$$

For $2 \le \alpha \le 3$, using $2^{-4/3} > 1/3$, we get

$$A \ge 0, \qquad C > \frac{32}{81} - \frac{1}{4} = \frac{47}{324}.$$

Thus $C > 0$ and every $f_n'' > 31/576$. Differentiating the log of the positive sum gives $\Phi_q'' = \sum_n \pi_n f_n'' + \mathrm{Var}_\pi(f_n') > 31/576$, where $\pi_n = b_n / \sum_{m \ge 2} b_m$. □

LEMMA 6.23. *$\log N$ is strictly convex on $[8/5, 3]$.*

PROOF. The dilation $(V_c g)(u) = c^{1/2} g(cu)$ is unitary on $L^2(0,\infty)$. For $c = c(\alpha)$, the kernel of $V_c K_\alpha V_c^{-1}$ is

$$\tilde{k}_\alpha(u,v) = c\, k_\alpha(cu, cv) = e^{\Phi_{u+v}(\alpha)} (uv)^\rho.$$

The factor $(uv)^\rho$ is log-linear in α, so by Lemma 6.22 $\alpha \mapsto \log \tilde{k}_\alpha(u,v)$ is strictly convex on $[8/5,3]$ for all $u, v > 0$. Let $\alpha < \beta$ in $[8/5,3]$, $\theta \in (0,1)$ and $\gamma = (1-\theta)\alpha + \theta\beta$. Then $\tilde{k}_\gamma < \tilde{k}_\alpha^{1-\theta} \tilde{k}_\beta^{\theta}$ pointwise, and for $g \ge 0$ with $\|g\| = 1$, Hölder's inequality gives, as in [58, Lemma 6.2],

$$\langle \tilde{K}_\gamma g, g\rangle \le \langle \tilde{K}_\alpha g, g\rangle^{1-\theta} \langle \tilde{K}_\beta g, g\rangle^{\theta} \le N(\alpha)^{1-\theta} N(\beta)^{\theta},$$

with strict first inequality when $g > 0$ almost everywhere. Since $\tilde{K}_\gamma$ has a nonnegative kernel, $N(\gamma)$ is the supremum of the left side over such g, so $\log N$ is convex. If $\gamma > \alpha_*$, then (F2) supplies a strictly positive normalized eigenfunction at $N(\gamma)$, and the first inequality is strict. Thus $\log N$ cannot be affine on a nondegenerate interval containing a point above α_*. On $(1, \alpha_*]$, (F2) gives $N(\alpha) = \Lambda_\alpha$, and the identity

$$\Lambda_\alpha = \int_0^1 (t(1-t))^{\alpha/2 - 1}\, \mathrm{d}t$$

shows strict log-convexity by strict Hölder. Hence $\log N$ is affine on no nondegenerate interval in $[8/5, 3]$, so it is strictly convex. □

PROOF OF THEOREM 6.19. By Lemma 6.23, N has a unique minimizer $\alpha_{\min}$ on $[8/5, 3]$, and is strictly decreasing on $[8/5, \alpha_{\min}]$ and strictly increasing on $[\alpha_{\min}, 3]$. With Lemma 6.21, N is strictly decreasing on $(1, \alpha_{\min}]$ and strictly increasing on $[\alpha_{\min}, \infty)$. The one-sided derivative bounds in Lemma 6.21 give $(\log N)'_+(8/5) < 0$ and $(\log N)'_-(3) > 0$, so neither endpoint can minimize N. Thus $\alpha_{\min} \in (8/5, 3)$. □

The Gaussian form

The Gauss geometry has its own multiplicative Hilbert form, with moments $\alpha(n) = 1/(\sqrt{n} \log n)$, $n \ge 2$, on $\mathscr{F}^2 \times \mathscr{F}^2$: the form of Lebesgue measure on the curve, as in the Hardy case. Let M_0 be the Meissel–Mertens constant and γ_{E} Euler's constant.

PROPOSITION 6.24. *The Gauss multiplicative Hilbert form has a bounded symbol. On series with $a_1 = 0$ its essential spectrum is $[0, \pi e^{M_0 - \gamma_{\mathrm{E}}}]$, and $\pi e^{M_0 - \gamma_{\mathrm{E}}} = 2.291\ldots$.*

PROOF. Split Lebesgue measure at $u = 1$, in the shifted variable $u = s - 1/2$. The part on $(0,1]$ has $[\nu]_1 = 1$, and Theorem 6.15 gives a bounded symbol. The rest has moments $n^{-3/2}/\log n$, and by the Gauss Helson inequality of Corollary 6.4 and Cauchy–Schwarz its functional is bounded on $\mathscr{F}^1$, since $\sum_n n^{-3} 2^{\Omega(n)}/[n]! \le$

$\prod_p e^{2p^{-3}} < \infty$. In the orthonormal basis $n^{-s}/\sqrt{[n]!}$ the form is $\mathcal{N}^*\mathcal{N}$ as in Lemma 5.1, with companion kernel

$$h(u+v) = \sum_{n\geq 2} \frac{n^{-1-u-v}}{[n]!} = \exp\Big(\sum_p p^{-1-u-v}\Big) - 1.$$

By Mertens' theorem $\sum_p p^{-1-x} = \log(1/x) + M_0 - \gamma_{\mathrm{E}} + o(1)$, so $xh(x) \to e^{M_0-\gamma_{\mathrm{E}}}$, and $h(x) = O(2^{-x})$ for $x \geq 1$. Theorem 5.4 with $\kappa = 1$ and $b = 1$ gives the essential spectrum $[0, e^{M_0-\gamma_{\mathrm{E}}}\Gamma(1/2)^2]$. □

The spectral dictionary of chapter 5 carries over. For a finite positive ν on $(0,\infty)$, in the shifted variable, let $M^X_{\nu,R}$ be the matrix of the form with moments $\alpha_\nu(n) = \int n^{-1/2-u}\,\mathrm{d}\nu(u)$ on the integers composed of primes in R, in the orthonormal basis $\sqrt{d_\beta(n)}\,n^{-s}$ (Bergman) or $n^{-s}/\sqrt{[n]!}$ (Gauss).

THEOREM 6.25. *In the Bergman geometry with $\beta > 1$ and in the Gauss geometry, let R have density $\theta \in (0,1]$ and let κ be the kernel exponent. For $0 < p < \infty$ and $0 < q \leq \infty$,*

$$\|M^X_{\nu,R}\|_{S_{p,q}} \asymp \|(e^{j\kappa}\nu(I_j))_j\|_{\ell^{p,q}} + \nu((1,\infty)),$$

with constants depending only on θ, B, p, q and the geometry.

PROOF. The proofs of Theorems 5.31 and 5.38 use five inputs, and each holds here with κ in place of θ. The matrix is $\mathcal{N}^*\mathcal{N}$ for the synthesis by $k(u) = (w_n n^{-1/2-u})_n$, with w_n the basis weights, and k is analytic in $\Re u > 0$ with $\|k(u)\|^2$ equal to $\zeta_R(1+2\Re u)^\beta$ or $\exp(\sum_{p\in R} p^{-1-2\Re u})$, which is $\asymp (\Re u)^{-\kappa}$ on $(0,1]$ by Lemma 2.10; this gives the trace formula and the block estimate (5.15). The operator norm bound is Theorem 2.11 with $p = 2$. The normalized kernels at $\delta_j = e^{-j}$ have inner products decaying like $e^{-\kappa|j-k|/2}$, the βth power of the Hardy ones in the Bergman case and $\exp(-\sum_{p\in R} p^{-1}(p^{-\delta_j} - p^{-\delta_k})^2/2)$ in the Gauss case. Finally $\|k(w) - e_1\|^2 \leq C4^{-\Re w}$ for $\Re w \geq 1$. □

PROPOSITION 6.26 (The Hardy pairing is unbounded). *The functional $L(f) = \sum_{n\geq 2} a_n d(n)/(\sqrt{n}\log^2 n)$, the pairing of f with φ in $\mathscr{H}^2$, is unbounded on $\mathscr{H}^1$. With $f_\delta = \zeta(s+1/2+\delta)^2/\zeta(1+2\delta)$, one has $\|f_\delta\|_{\mathscr{H}^1} = 1$ and $L(f_\delta) \sim 1/(3\zeta(2)\delta)$.*

PROOF. $\|f_\delta\|_{\mathscr{H}^1} = \|\zeta(s+1/2+\delta)\|^2_{\mathscr{H}^2}/\zeta(1+2\delta) = 1$. By $1/\log^2 n = \int_0^\infty t n^{-t}\,\mathrm{d}t$ and $\sum d(n)^2 n^{-s} = \zeta(s)^4/\zeta(2s) \sim \zeta(2)^{-1}(s-1)^{-4}$,

$$\sum_{n\geq 2} \frac{d(n)^2}{n^{1+\delta}\log^2 n} \sim \frac{1}{\zeta(2)} \int_0^\infty \frac{t\,\mathrm{d}t}{(\delta+t)^4} = \frac{1}{6\zeta(2)\delta^2},$$

and $\zeta(1+2\delta) \sim 1/(2\delta)$. Truncated Euler products give polynomial witnesses. □

So the Bergman symbol is bounded against the Bergman pairing, where the weight $d(n)$ cancels, and the natural Hardy analogue fails by a power of δ; the pointwise count u^{-3} against $u\,\mathrm{d}u$ was already the right size.

6.7. Comparison

	Hardy	Bergman (β)	Gauss
transition maps	Möbius	Möbius	translations
energy $E(r,s)$	$\sum \rho_p^2$	$\sum \rho_p^2$	$\sum \lvert r_p - s_p\rvert^2$
Hellinger affinity	$\le e^{-E/4}$	$\le e^{-\beta E/4}$	$= e^{-E/4}$
kernel exponent κ	θ	$\beta\theta$	θ
contractive embedding	Carleman	Kulikov	Proposition 6.3
Helson weight	$1/d(n)$	$1/d_{2\beta}(n)$	$[n]!/2^{\Omega(n)}$
two-point constant	$(1-\lvert x\rvert^4)^{-1}$ (sharp)	$\Lambda_\beta(\lvert x\rvert^2)^{1/2}$	$\cosh(2\lvert x\rvert^2)^{1/2}$
sign-average constant	$\zeta(2)$	C_β	< 1.58

The Gauss column is the cleanest: its Hellinger affinity is exact, and its sign-average constant is the smallest. The Hardy column is the only one with a sharp two-point constant for every x. In the Bergman column the sharp constant is the square bound exactly up to the first zero of $K_x + K_{-x}$, and not beyond it; in the Gauss column only the leading term is known (Proposition 6.7).

Open

QUESTION 4. What are the sharp two-point constants in the Gauss geometry, where the square bound always fails (Proposition 6.7, item (4)), and in the Bergman geometry beyond the first zero of $K_x + K_{-x}$?

The absence of cancellation in the Bergman Dirichlet kernels for $1 < \beta \le 2$ rests on an unrefereed preprint (Theorem 6.14); an independent proof is open, and so is the value of κ_β.

Part 3

Where Nehari's theorem fails: Schatten classes

CHAPTER 7

Schatten-class forms without bounded symbols

The question of this chapter was posed by Brevig and Perfekt.

PROBLEM 6 (Brevig and Perfekt [20, Question 3]). *Is there a multiplicative Hankel form in S_p, $2 < p < \infty$, without a bounded symbol, and if so, for which p? Brevig and Perfekt remark that it is tempting to conjecture that such forms exist for every $p > 2$.*

Let p_* be the infimum of such p. Helson's theorem gives $p_* \geq 2$ in the sense that S_2 forms always have bounded symbols, and Brevig and Perfekt [20] proved $p_* \leq (1 - \log \pi / \log 4)^{-1} = 5.7388\ldots$. This chapter proves $p_* \leq 4$, which settles Problem 6 for every $p > 4$, by a construction in the Gaussian model, and shows that the threshold 4 is built into the method: for the natural blocks it is the identity $\|\varphi\|_{L^4} = \|A(\varphi)\|_{S_4}$ of Janson, Peetre and Rochberg.

Bampouras [4] was the first to prove $p_* \leq 4$. His symbols are $1 - t^2 + tF$, with F a product of normalized linear forms on finite tori, tested against perturbed squares $(1+tF)^2 - \eta t^2 F^2$; integrating out the scale t against $t^{-\kappa-1}\,\mathrm{d}t$ reduces everything to the moments $\mathbb{E}|F|^\kappa$. We found the same construction independently, in its Gaussian limit $1 + cP_n$, with a longer proof that fixes the scale and optimizes it by large deviations, at rates that are Legendre transforms of his moments. Section 7.2 presents Bampouras's scale-integration argument, extended to higher-order tests, and its Gaussian form (remark 7.8). Section 7.3 then gives our fixed-scale proof, with explicit parameters. New here are higher-order tests extending his criterion from L^4 to L^p (Theorem 7.5), the identity of Proposition 7.6, which shows that neither version can reach $p \leq 4$ on any finite torus, and the barriers of chapter 8.

7.1. Blocks and the Gaussian model

For an analytic polynomial φ on $\mathbb{T}^d$ write $H_\varphi(f, g) = \langle fg, \varphi \rangle$. Its matrix is $(\overline{\hat{\varphi}(\alpha + \beta)})_{\alpha,\beta}$.

LEMMA 7.1 (Brevig–Perfekt [20]). *If $p \geq 1$ and polynomials f, φ in finitely many variables satisfy $|\langle f, \varphi \rangle| > \|H_\varphi\|_{S_p} \|f\|_1$, then some multiplicative Hankel form in S_p has no bounded symbol.*

PROOF. The symbol can be written down. Normalize $\|H_\varphi\|_{S_p} = 1$ and put $R = |\langle f, \varphi \rangle| / \|f\|_1 > 1$. Let φ_m and f_m be the mth tensor powers of φ and f in a block of md new variables, disjoint for different m, and put $\Phi = \sum_m m^{-2} \varphi_m$. Pairing, L^1 norm and Schatten norm are multiplicative over products in disjoint variables, so $\|H_\Phi\|_{S_p} \leq \sum_m m^{-2}$ and $|\langle f_m, \varphi_m \rangle| = R^m \|f_m\|_1$. For $k \neq m$ the blocks

are disjoint, so $\langle f_m, \varphi_k\rangle = f_m(0)\overline{\varphi_k(0)}$, with $|f_m(0)| \le \|f_m\|_1$ and $|\varphi_k(0)| \le 1$. Hence $|\langle f_m, \Phi\rangle| \ge (m^{-2}R^m - \pi^2/6)\|f_m\|_1$, which is unbounded in m, while a bounded symbol ψ would give $|\langle f_m, \Phi\rangle| \le \|\psi\|_\infty \|f_m\|_1$. □

Let μ_d be the law of d independent standard complex Gaussians on $\mathbb{C}^d$. For $\varphi = \sum c_\alpha z^\alpha$ the Gaussian Hankel form $(f,g) \mapsto \langle fg, \varphi\rangle_{L^2(\mu_d)}$ has, in the basis $z^\alpha/\sqrt{\alpha!}$, the coefficientwise conjugate of the matrix $A(\varphi)_{\alpha\beta} = c_{\alpha+\beta}(\alpha+\beta)!/\sqrt{\alpha!\beta!}$. Analytic $L^2(\mu_d)$ is the Fock space, and these are the small Hankel forms studied by Janson, Peetre and Rochberg [47]. Let $D(\varphi)$, the dual norm, be the norm of $f \mapsto \langle f, \varphi\rangle_{L^2(\mu_d)}$ on analytic polynomials with the $L^1(\mu_d)$ norm; by the Hahn–Banach theorem it is the least sup norm of a bounded representative of φ. If $D(\varphi) > \|A(\varphi)\|_{S_p}$ we call φ a *strict block* at p.

PROPOSITION 7.2 (Transfer). *Let $p \ge 2$. If $D(\varphi) > \|A(\varphi)\|_{S_p}$ for a polynomial φ on $\mathbb{C}^d$, then some multiplicative Hankel form in S_p has no bounded symbol.*

PROOF. Replace z_j by $L_{N,j} = N^{-1/2}\sum_{\ell \le N} z_{j,\ell}$ on disjoint blocks of Steinhaus variables. For fixed k, squarefree terms contribute $k!(N)_k/N^k$ to $\|L_N^k\|_2^2$, where $(N)_k = N(N-1)\cdots(N-k+1)$. The remaining indices have support of size at most $k-1$: there are $O_k(N^{k-1})$ of them, and each contributes $O_k(N^{-k})$. In the Hankel Hilbert–Schmidt norm, an index ν occurs $\prod_\ell(\nu_\ell+1)$ times, equal to 2^k for squarefree indices and at most 2^k otherwise. Hence

$$\|L_N^k\|_2^2 = k! + O_k(N^{-1}), \qquad \|H_{L_N^k}\|_{S_2}^2 = 2^k k! + O_k(N^{-1}).$$

The multinomial expansion also gives $\mathbb{E}|L_N|^{2k} \le k!$. These calculations apply independently in each block; distinct multidegrees are orthogonal. Thus $\|H_{\varphi_N}\|_{S_2} \to \|A(\varphi)\|_{S_2}$. Let K be the total degree of φ, and let P_N project onto the span of

$$e_{N,\alpha} = \frac{\prod_{j=1}^d L_{N,j}^{\alpha_j}}{\|\prod_{j=1}^d L_{N,j}^{\alpha_j}\|_2}, \qquad |\alpha| \le K.$$

These vectors are orthonormal by orthogonality of the multidegrees in the separate blocks. Let Q_N project onto their conjugates in the anti-analytic codomain. The matrix of $Q_N H_{\varphi_N} P_N$ converges entrywise to $\overline{A(\varphi)}$. Its Hilbert–Schmidt norm therefore has the same limit as that of the full operator. Since $T \mapsto Q_N T P_N$ is an orthogonal projection in the Hilbert space of Hilbert–Schmidt operators,

$$\|H_{\varphi_N} - Q_N H_{\varphi_N} P_N\|_{S_2}^2 = \|H_{\varphi_N}\|_{S_2}^2 - \|Q_N H_{\varphi_N} P_N\|_{S_2}^2 \longrightarrow 0.$$

For $p \ge 2$ the S_p norm of this remainder is at most its S_2 norm. Hence $\|H_{\varphi_N}\|_{S_p} \to \|A(\varphi)\|_{S_p}$. For each fixed polynomial test f, the central limit theorem and $\sup_N \mathbb{E}|f(L_N)|^2 < \infty$ give $\|f(L_N)\|_1 \to \|f\|_{L^1(\mu_d)}$, and the pairings converge by orthogonality. Apply Lemma 7.1. □

7.2. The moment criterion and why it stops at 4

Bampouras's proof [4] integrates over the scale instead of fixing it. We record it for a general block and with higher-order tests, which shows where it stops. Call a polynomial W on $\mathbb{T}^N$ *admissible* if $\|W\|_2 = 1$, $W(0) = 0$, $\int W^2 \overline{W}\,\mathrm{d}m = 0$, $\int g|W|^2\,\mathrm{d}m = \hat{g}(0)$ for analytic g, and the law of W is rotation invariant.

Every normalized analytic homogeneous polynomial of positive degree is admissible: common rotation of the variables gives a circular law and annihilates the nonconstant analytic moments of $|W|^2$. Products of admissible blocks in disjoint variables are also admissible.

LEMMA 7.3. *For admissible F and $\varphi_t = 1 - t^2 + tF$, the span of 1 and F reduces H_{φ_t}, with compression $\left(\begin{smallmatrix} 1-t^2 & t \\ t & 0 \end{smallmatrix}\right)$, and $\|H_{\varphi_t}\|_{S_p}^p = 1 + t^{2p} + t^p(\|H_F\|_{S_p}^p - 2)$.*

PROOF. $H_F(F, g) = \int g|F|^2 = \hat{g}(0)$ and $H_F(F, F) = \int F^2\overline{F} = 0$, so H_F swaps 1 and F and maps their orthogonal complement into itself. □

For an integer $k \geq 2$ and $\eta > 0$ let

$$D_{\eta,k}(u) = 1 + u^2 - \int_{\mathbb{T}} |(1 + u\zeta)^2 - (-1)^k \eta u^k \zeta^k| \, \mathrm{d}m(\zeta),$$

$$I_{\kappa,k}(\eta) = \int_0^\infty D_{\eta,k}(u) u^{-\kappa-1} \, \mathrm{d}u.$$

For $k = 2$ these are Bampouras's quantities.

LEMMA 7.4 (Mellin tests). *If $k < \kappa < 2k$, then $I_{\kappa,k}(\eta)$ converges absolutely and $I_{\kappa,k}(\eta)/\eta \to 2/(\kappa(\kappa - 2))$ as $\eta \to 0$.*

PROOF. Put $a = (1 + u\zeta)^2$ and $b = -(-1)^k \eta u^k \zeta^k$. Since $\int |a| \, \mathrm{d}m = 1 + u^2$ and $\overline{a}/|a| = (1 + u\overline{\zeta})/(1 + u\zeta)$, the first variation is

$$\partial_\eta D_{\eta,k}(u)\big|_{\eta=0} = (-1)^k u^k \operatorname{Re} \int_{\mathbb{T}} \zeta^k \frac{1 + u\overline{\zeta}}{1 + u\zeta} \, \mathrm{d}m(\zeta).$$

For $u < 1$ the integrand has only positive frequencies, since $k \geq 2$, so the integral vanishes. For $u > 1$ expand $(1 + u\zeta)^{-1} = \sum_{n \geq 1} (-1)^{n-1} u^{-n} \zeta^{-n}$; the constant term is $(-1)^k u^{-k}(u^2 - 1)$. Hence $D_{\eta,k}(u)/\eta \to (u^2 - 1)_+$ for $u \neq 1$. Since $||a+b| - |a|| \leq |b|$, we have $|D_{\eta,k}(u)| \leq \eta u^k$. For $u \leq 1/2$ and $\eta \leq 1/4$ we have $|a| \geq 1/4$ and $|b| \leq |a|/2$, so $|a + b| = |a| + \operatorname{Re}(b\overline{a})/|a| + O(|b|^2)$; the linear term integrates to 0, as for $u < 1$ above, and $|D_{\eta,k}(u)| \leq C\eta^2 u^{2k}$. These two bounds are integrable against $u^{-\kappa-1} \, \mathrm{d}u$ on $(1/2, \infty)$ and $(0, 1/2)$ exactly when $k < \kappa < 2k$, and dominated convergence gives $\int_1^\infty (u^2 - 1) u^{-\kappa-1} \, \mathrm{d}u = 2/(\kappa(\kappa - 2))$. □

THEOREM 7.5 (The moment criterion). *Let W be admissible and $p > 2$. If $\|W\|_{L^p} > \|H_W\|_{S_p}$, then some multiplicative Hankel form in S_p has no bounded symbol.*

With $\|W\|_{L^4}$ in place of $\|W\|_{L^p}$ this is Bampouras's criterion [4], whose proof uses the test with $k = 2$ and so only moments of order $\kappa < 4$. Allowing all integers $k \geq 2$ reaches every $2 < \kappa < p$.

PROOF. By continuity of $\kappa \mapsto \|W\|_\kappa$ choose $\kappa \in (2, p)$ with $\|W\|_\kappa > \|H_W\|_{S_p}$, let k be the least integer larger than $\kappa/2$, so that $k \geq 2$ and $k < \kappa < 2k$, and by Lemma 7.4 choose η with $I_{\kappa,k}(\eta) > 0$. Suppose the conclusion fails. For $F = \prod_{j \leq m} W(z^{(j)})$, with $A = \|H_W\|_{S_p}^p$, the test $F_t = (1 + tF)^2 - (-1)^k \eta t^k F^k$, Lemma 7.1 and Lemma 7.3 give

$$\|F_t\|_1 \geq (1 + t^2)(1 + t^{2p} + (A^m - 2)t^p)^{-1/p},$$

since rotation invariance gives $\langle F_t, 1-t^2+tF\rangle = 1+t^2$. Rotation invariance also gives $\|F_t\|_1 = 1+t^2-\int D_{\eta,k}(t|F|)\,\mathrm{d}m$. The absolute convergence in Lemma 7.4 permits Fubini: substituting $u=t|F|$ gives $I_{\kappa,k}(\eta)(\mathbb{E}|W|^\kappa)^m$. Since $1-(1+y)^{-1/p}\le \min(y,1)$, the other side is bounded by

$$\int_0^\infty (1+t^2)\min\{t^{2p}+(A^m-2)t^p,1\}\,t^{-\kappa-1}\,\mathrm{d}t.$$

Here $A\ge 2$, since H_W has the two singular values supplied by Lemma 7.3. Put $t=A^{-m/p}x$. After extracting $A^{m\kappa/p}$, the integrand is at most $4x^{p-\kappa-1}$ for $x\le 1$ and $(1+x^2)x^{-\kappa-1}$ for $x\ge 1$. Thus

$$I_{\kappa,k}(\eta)\big(\mathbb{E}|W|^\kappa\big)^m \le \left(\frac{4}{p-\kappa}+\frac{1}{\kappa}+\frac{1}{\kappa-2}\right)A^{m\kappa/p}.$$

Taking mth roots gives $\|W\|_\kappa \le \|H_W\|_{S_p}$, a contradiction. □

PROPOSITION 7.6 (L^4 is dominated by S_4). *Let $\varphi=\sum_\alpha c_\alpha z^\alpha$ be a polynomial on $\mathbb{T}^d$, and for $\gamma\ge 0$ let $B_\gamma\varphi=\sum_\alpha c_{\alpha+\gamma}z^\alpha$ be its backward shift. Then, in the spirit of the one-variable integral identities of Janson, Upmeier and Wallstén [48],*

$$\|H_\varphi\|_{S_4}^4 = \sum_{\gamma\ge 0}\|B_\gamma\varphi\|_{L^4}^4.$$

In particular $\|\varphi\|_{L^4}\le\|H_\varphi\|_{S_4}$, with equality only for constants, and the defect satisfies

$$\|H_\varphi\|_{S_4}^4-\|\varphi\|_{L^4}^4 \ge \sum_{\alpha,\alpha'}\Big(\prod_{j\le d}(1+\min(\alpha_j,\alpha_j'))-1\Big)|c_\alpha|^2|c_{\alpha'}|^2 \ge \frac{1}{d}\|\varphi-\varphi(0)\|_2^4.$$

In the Gaussian model $\|\varphi\|_{L^4(\mu)}=\|A(\varphi)\|_{S_4}$ [47].

PROOF. $\|H_\varphi\|_{S_4}^4=\sum_{\beta,\beta'}|\sum_\alpha c_{\alpha+\beta}\overline{c_{\alpha+\beta'}}|^2$. Every pair (β,β') is uniquely $(\gamma+\delta^+,\gamma+\delta^-)$ with $\gamma=\min(\beta,\beta')$ coordinatewise and $\delta^\pm$ of disjoint supports, and for fixed γ the inner sums are the Fourier coefficients of $|B_\gamma\varphi|^2$ at $\delta^+-\delta^-$; each frequency occurs once. Parseval gives the identity, and $\gamma=0$ is $\|\varphi\|_4^4$. For the defect, use $\|B_\gamma\varphi\|_4\ge\|B_\gamma\varphi\|_2$ and count the $\gamma\ne 0$ with $\gamma\le\min(\alpha,\alpha')$. Keeping only $\gamma=e_j$ gives $\sum_j I_j^2$ with $I_j=\sum_{\alpha_j\ge 1}|c_\alpha|^2$, and $\sum_j I_j\ge\|\varphi-\varphi(0)\|_2^2$. □

For a Dirichlet polynomial $\sum a_n n^{-s}$ the product in the defect is $d(\gcd(m,n))$: approaching the Gaussian equality $L^4=S_4$ requires the coefficients to have negligible common divisors, which is why spreading a block over many independent primes helps.

COROLLARY 7.7 (The method's threshold). *(1) For $p\le 4$ the criterion fails for every admissible block on every finite torus, since $\|W\|_{L^p}\le\|W\|_{L^4}\le\|H_W\|_{S_4}\le\|H_W\|_{S_p}$.*

(2) For a normalized linear form $L=\sum_{i\le N}c_iz_i$ with at least two nonzero coefficients it succeeds exactly for $p>p_L$, where $p_L>4$ is the unique solution of $\mathbb{E}|L|^p=2$. With equal coefficients, numerically $p_L=5.157$, 4.542, 4.360, 4.120 for $N=2,3,4,10$, and $p_L\to 4$ as $N\to\infty$.

(3) *(Bampouras [4].) The condition of his L^4 version is $p > 4\log 2/\log(2 - \sum|c_i|^4)$; with equal coefficients this is 6.84, 5.43, 4.95, 4.32 for $N = 2, 3, 4, 10$, and $4 + 2/(N \log 2) + O(N^{-2})$ in general.*

(4) *If H_W has rank two, then W is linear: a monomial of top degree k and a nested chain of divisors give a triangular $(k+1) \times (k+1)$ submatrix. So the rank-two version of the criterion is exactly the linear case.*

PROOF. (2) and (3) $\|H_L\|_{S_p}^p = 2$ and $\mathbb{E}|L|^4 = 2 - \sum|c_i|^4 < 2$. Since $|L|$ is not constant, $\|L\|_p$ increases strictly to $\sum|c_i| > 1$, while $2^{1/p}$ decreases, so $\mathbb{E}|L|^p > 2$ exactly for $p > p_L$. For $N = 2$, $\mathbb{E}|L|^p = 2^{p/2}\Gamma((p+1)/2)/(\sqrt{\pi}\,\Gamma(1 + p/2))$; for instance $\mathbb{E}|L|^6 = 5/2$. For fixed $p > 4$ the central limit theorem with bounded moments, as in Proposition 7.2, gives $\mathbb{E}|L|^p \to \Gamma(1 + p/2) > 2$ as $N \to \infty$ with equal coefficients. □

So two variables already beat the Brevig–Perfekt exponent, while Bampouras's L^4 version needs three. For every nonconstant Gaussian polynomial W the crossing $\|W\|_{L^p} = \|A(W)\|_{S_p}$ happens exactly at $p = 4$, since $\|W\|_{L^p(\mu)}$ increases and $\|A(W)\|_{S_p}$ decreases in p, with equality at 4 by Janson–Peetre–Rochberg. This is the limit of the thresholds p_L in (2).

REMARK 7.8 (The criterion in the Gaussian model). For normalized homogeneous Gaussian polynomials of positive degree, Bampouras's argument runs verbatim, with Proposition 7.2 in place of Lemma 7.1. Common rotation of the variables gives the Gaussian analogues of admissibility, including $\int g|W|^2\,\mathrm{d}\mu = g(0)$ for every analytic polynomial g. Hence the reducing compression of Lemma 7.3 is unchanged. The criterion then reads: if $\|W\|_{L^\kappa(\mu)} > \|A(W)\|_{S_p}$ for some $\kappa < p$, there is a form in S_p without bounded symbol. For $W = Z_1$ one has $\|A(Z_1)\|_{S_p}^p = 2$ and $\mathbb{E}|Z_1|^\kappa = \Gamma(1 + \kappa/2)$, and $\kappa = 4$ gives every $p > 4$; Bampouras's tests with $k = 2$ suffice here, letting $\kappa \to 4$. This is Bampouras's argument with the limit $N \to \infty$ taken inside the criterion; it is the shortest proof of $p_* \le 4$ we know.

Going below 4 with Lemma 7.1 requires a block outside the admissible class, so that the $\{1, F\}$ splitting fails, or a test family not of the form $(1+tF)^2 - (-1)^k \eta t^k F^k$.

7.3. Every $p > 4$

Let $Z_1, Z_2, \ldots$ be independent standard complex Gaussians and $P_n = Z_1 \cdots Z_n$, so $\mathbb{E}|P_n|^2 = 1$ and $\mathbb{E}|P_n|^4 = 2^n$.

Brevig and Perfekt's block is the linear symbol z, for which $D(z) = 2/\sqrt{\pi}$ and $\|A(z)\|_{S_p} = 2^{1/p}$; this gives exactly $(1 - \log\pi/\log 4)^{-1}$. The preceding criterion proves $p_* \le 4$ without choosing a scale. We now give explicit constant-plus-product symbols and perturbed square tests. This longer argument identifies the scale and compares the gain in the dual quotient with the residual Schatten cost.

LEMMA 7.9 (Spectrum). *For $c > 0$ let $\lambda = (1 + \sqrt{1 + 4c^2})/2$. The nonzero singular values of $A(1 + cP_n)$ are λ, $\lambda - 1$, and c with multiplicity $2^n - 2$. Hence $\|A(1 + cP_n)\|_{S_p} - \lambda \le 2^n c^p/p$ for $p \ge 2$.*

PROOF. Only squarefree monomials contribute, each paired with its complement. The pair $\{1, P_n\}$ carries $\left(\begin{smallmatrix} 1 & c \\ c & 0 \end{smallmatrix}\right)$, and the other $2^{n-1}-1$ pairs carry $\left(\begin{smallmatrix} 0 & c \\ c & 0 \end{smallmatrix}\right)$. Since $\lambda - 1 \le c$ and $\lambda \ge 1$, concavity of $x^{1/p}$ gives the estimate. □

LEMMA 7.10 (First variation of L^1). *Let W be rotation invariant with $\mathbb{E}|W|^2 = 1$, $M = \mathbb{E}|W|^4$, and let $u = W/r$, $G = \mathbb{E}(|u|^2 - 1)_+$, $T = \mathbb{E}(|u|^2; |u| > 1/2)$. For $t \ge 0$,*

$$\mathbb{E}\big|(1+u)^2 - tu^2\big| \le 1 + r^{-2} - tG + 16Mt^2r^{-4} + 8t^{5/4}T.$$

PROOF. For $w \neq 0$,

$$|w - tv| = |w| - t\Re(v\overline{w})/|w| + R, \qquad 0 \le R \le \min(2t|v|, 4t^2|v|^2/|w|).$$

Take $w = (1+u)^2$, $v = u^2$. The main term has mean $\mathbb{E}|1+u|^2 = 1 + r^{-2}$. Averaging the linear term over the angle,

$$\int_{\mathbb{T}} \rho^2\zeta^2 \frac{1+\rho\overline{\zeta}}{1+\rho\zeta}\, \mathrm{d}m(\zeta) = (\rho^2 - 1)_+,$$

by expanding in powers of $\rho\zeta$ for $\rho < 1$ and of $1/(\rho\zeta)$ for $\rho > 1$; so it has mean $-tG$. On $|u| \le 1/2$, $R \le 16t^2|u|^4$. On $|u| > 1/2$, $R \le 2^{5/4}t^{5/4}|u|^{5/2}|1+u|^{-1/2}$, and for $u = \rho e^{i\theta}$ we have $|1+u| \ge \rho|\sin\theta|$. Thus

$$\int_{\mathbb{T}} \frac{\rho^{1/2}}{|1+\rho\zeta|^{1/2}}\, \mathrm{d}m(\zeta) \le \frac{1}{2\pi}\int_0^{2\pi} |\sin\theta|^{-1/2}\, \mathrm{d}\theta \le 2,$$

by symmetry and $\sin\theta \ge 2\theta/\pi$ on $[0, \pi/2]$. The outer remainder is at most $2^{9/4}t^{5/4}T \le 8t^{5/4}T$. □

LEMMA 7.11 (Large deviations). *Let $\Psi(s) = \log \mathbb{E}|Z_1|^s = \log\Gamma(1 + s/2)$. For $s > 2$ put $\beta = \Psi'(s)$, $I = s\beta - \Psi(s)$, $r_n = e^{\beta n}$, and let G_n, T_n be as in Lemma 7.10 for $W = P_n$, $r = r_n$. Then $\beta > 0$, $T_n \le 2^{s-2}e^{-nI}$ and $n^{-1}\log G_n \to -I$.*

PROOF. Ψ is strictly convex with $\Psi(0) = \Psi(2) = 0$, so $\beta > 0$. With $x = |P_n|/r_n$, $\mathbb{E}x^s = e^{-nI}$, and $(x^2-1)_+ \le x^s$, $x^2\mathbf{1}_{x>1/2} \le 2^{s-2}x^s$ give the upper bounds. For the lower bound put $S_n = \log|P_n|$ and, for $u > s$, introduce $\mathrm{d}\mathbb{P}_u/\,\mathrm{d}\mathbb{P} = e^{uS_n - n\Psi(u)}$. Under $\mathbb{P}_u$ the summands remain independent with mean $\beta_u = \Psi'(u) > \beta$ and finite variance. Fix $0 < h < \beta_u - \beta$. The event $E_n = \{|S_n/n - \beta_u| \le h\}$ has probability tending to one under $\mathbb{P}_u$, and restriction to this event gives

$$G_n \ge e^{n\Psi(u) - un(\beta_u + h)}\big(e^{2n(\beta_u - h - \beta)} - 1\big)\mathbb{P}_u(E_n).$$

Taking logarithms, then letting $h \downarrow 0$ and $u \downarrow s$, gives $\liminf_n n^{-1}\log G_n \ge \Psi(s) - s\beta = -I$. □

The budget. Take the symbol $\varphi = 1 + cP_n$ with $c = r/(r^2-1) \approx 1/r$, so that the top singular value of $A(\varphi)$ is $\lambda = r^2/(r^2-1)$ (Lemma 7.9), and test with the square $f = (1 + P_n/r)^2$. Then $\langle f, \varphi\rangle = \lambda(1 + r^{-2})$ and $\|f\|_1 = \mathbb{E}|1 + P_n/r|^2 = 1 + r^{-2}$, so the square alone ties with the operator norm. Subtracting $t(P_n/r)^2$ leaves the pairing unchanged, since $P_n^2 \perp \varphi$, and changes $\|f\|_1$ by $-tG$, with $G = \mathbb{E}(|P_n|^2/r^2 - 1)_+$, plus a second-order cost of order $t^2\,\mathbb{E}|P_n|^4/r^4 = t^2 2^n r^{-4}$ (Lemma 7.10). Optimizing t, the gain is of order $G^2r^4/2^n$. Against it stands the Schatten excess of $A(\varphi)$ over λ, which comes from the $2^n - 2$ residual singular values c and is of order $2^n r^{-p}$. At

$p = 4$, $(x-1)_+ \le x^2/4$ gives $G \le 2^n r^{-4}/4$, so the optimized gain is at most of the same order as the Schatten excess. This comparison supplies no strict improvement at that endpoint. For $p > 4$ the excess is smaller by the factor r^{4-p}, while G, the large-deviation cost of $|P_n| > r = e^{\beta n}$, is not. The rates I and $J = 4\beta - \log 2$ in the proof of Theorem 7.12 make this precise.

THEOREM 7.12. *For every $p > 4$ and all large n there are $c_n, r_n, t_n > 0$ such that $\varphi_n = 1 + c_n P_n$ and $f_n = (1 + P_n/r_n)^2 - t_n(P_n/r_n)^2$ satisfy $|\langle f_n, \varphi_n\rangle| > \|A(\varphi_n)\|_{S_p}\|f_n\|_1$. Consequently $p_* \le 4$.*

PROOF. Take $s > 4$ and the data of Lemma 7.11, and put $J = 4\beta - \log 2$, so $2^n/r_n^4 = e^{-nJ}$. Strict convexity and $\Psi(4) = \log 2$ give $I > J$, and $(p\beta - \log 2) - (2I - J) \to (p-4)\Psi'(4) > 0$ as $s \downarrow 4$; fix s and $\varepsilon > 0$ with $2I - J + \varepsilon < p\beta - \log 2$. Let $c_n = r_n/(r_n^2 - 1)$, so that $\lambda_n = r_n^2/(r_n^2 - 1)$, and $t_n = e^{-n(I-J+\varepsilon)}$.

By Lemma 7.9, the Schatten excess is $\|A(\varphi_n)\|_{S_p} - \lambda_n \le (2^p/p)e^{-n(p\beta - \log 2)}$. The pairing is $\langle f_n, \varphi_n\rangle = 1 + 2c_n/r_n = \lambda_n(1 + r_n^{-2})$, because $P_n^2 \perp \varphi_n$. By Lemmas 7.10 and 7.11, $(1+r_n^{-2}) - \|f_n\|_1 \ge t_n G_n(1 - o(1))$, the two error terms being $e^{-n(\varepsilon + o(1))}$ and $e^{-n((I-J+\varepsilon)/4 + o(1))}$ relative to $t_n G_n$. Hence $\langle f_n, \varphi_n\rangle/\|f_n\|_1 - \lambda_n \ge t_n G_n/4 = e^{-n(2I - J + \varepsilon + o(1))}$, which beats the excess. Apply Proposition 7.2. □

For $p \ge 1$ let $C_p(d)$ be the best constant in $|\langle f, \varphi\rangle| \le C\|H_\varphi\|_{S_p}\|f\|_{L^1(\mathbb{T}^d)}$, the minimal-symbol constant on $\mathbb{T}^d$. These constants detect the threshold.

PROPOSITION 7.13 (Dichotomy). *Let $p \ge 1$. Either $C_p(d) = 1$ for every d, or $C_p(d)$ grows exponentially in d; and $\sup_d C_p(d) < \infty$ if and only if every multiplicative Hankel form in S_p has a bounded symbol. Consequently the forms in S_p all have bounded symbols if and only if $p \le p_*$, the endpoint included, and $p_* = 4$ if and only if every multiplicative Hankel form in S_4 has a bounded symbol.*

PROOF. If $C_p(d_0) > 1$, disjoint copies give $C_p(kd_0) \ge C_p(d_0)^k$, and Lemma 7.1 gives a form in S_p without bounded symbol. If $\sup_d C_p(d) = C < \infty$ and $H_\varphi \in S_p$, then for a polynomial f in d variables, $\langle f, \varphi\rangle = \langle f, \varphi_d\rangle$ with φ_d the restriction to those variables; Fejér means of φ_d are averages of two-sided unitary rotations of a compression of H_φ, so $|\langle f, \varphi\rangle| \le C\|H_\varphi\|_{S_p}\|f\|_1$, and the Hahn–Banach theorem gives a bounded symbol. For the endpoint: if $p < p_*$ then $C_p(d) = 1$, so $|\langle f, \varphi\rangle| \le \|H_\varphi\|_{S_p}\|f\|_1$ for polynomial tests and symbols. For such a symbol, H_φ has finite rank, so $p \mapsto \|H_\varphi\|_{S_p}$ is continuous, and the inequality passes to $p = p_*$. The last claim uses $p_* \le 4$ from Theorem 7.12. □

COROLLARY 7.14 (Dimension growth). *For $p > 4$, $C_p(d) \ge A_p e^{c_p d}$ with $c_p > 0$.*

PROOF. A finite pair with ratio $R_p > 1$ in d_p variables exists by Theorem 7.12 and Proposition 7.2; k disjoint copies give $C_p(kd_p) \ge R_p^k$, and $C_p(d)$ is nondecreasing in d. □

The rate degenerates as $p \downarrow 4$. For $2 < p \le 4$, Proposition 7.13 shows that $C_p(d)$ either equals 1 for every d or grows exponentially; which of the two happens is open. In a fixed number of variables the constant is finite for every $p \le \infty$: Nehari's theorem holds on $\mathbb{D}^d$, by Ferguson and Lacey [34] for $d = 2$ and Lacey and Terwilleger [56] for $d > 2$, and $\|A\| \le \|A\|_{S_p}$ gives $1 \le C_p(d) \le C_\infty(d) < \infty$.

Ortega-Cerdà and Seip [69] show that $C_\infty(d)$ grows at least exponentially in d, and Brevig [18] improved their rate. For $d = 2$, Bampouras and Perfekt [9] give a direct proof in the product Hardy space of two half-planes that Hankel operators in S_p have symbols in product BMO. So the counterexamples of this chapter live in the limit $d \to \infty$, and for $p > 4$ the corollary gives the exponential rate.

7.4. Gaussian immanants

Normalized determinants have first absolute moment tending to zero. The same can hold for an irreducible character whose dimension tends to infinity. This extends the determinant moment phenomenon; it does not presently give a Schatten counterexample or improve the threshold.

Let Z have independent standard complex Gaussian entries, so $\mathbb{E}|Z_{ij}|^2 = 1$. For $n \geq 3$, put

$$\chi_n(\sigma) = \operatorname{sgn}(\sigma)(\operatorname{fix}(\sigma) - 1), \qquad \Phi_n(Z) = \frac{1}{\sqrt{n!}} \sum_{\sigma \in S_n} \chi_n(\sigma) \prod_i Z_{i,\sigma(i)}.$$

This is the sign twist of the standard irreducible representation, of dimension $n - 1$. Its irreducibility follows, for example, by restricting the permutation representation to vectors of coordinate sum zero: every commuting matrix on the permutation representation has constant diagonal and constant off-diagonal entries, so the restricted commutant consists of scalars. Distinct permutation monomials are orthogonal, and $\mathbb{E}(\operatorname{fix}(\sigma) - 1)^2 = 1$ for a uniform permutation. Thus $\mathbb{E}|\Phi_n|^2 = 1$. Define

$$D_n = \frac{\mathbb{E}|\det Z|}{\sqrt{n!}} = \prod_{j=1}^{n} \frac{\Gamma(j + 1/2)}{\Gamma(j)\sqrt{j}} \asymp n^{-1/8}.$$

THEOREM 7.15. *The high-dimensional character χ_n satisfies*

$$\frac{\sqrt{n-1}}{(n^2-1)^{3/2}} D_n \leq \mathbb{E}|\Phi_n| \leq \sqrt{\frac{2n+1}{n+1}} D_n.$$

In particular, $\mathbb{E}|\Phi_n| = O(n^{-1/8}) \to 0$.

LEMMA 7.16. *Let P be the orthogonal projection onto the diagonal traceless matrices inside $\mathfrak{sl}_n(\mathbb{C})$, and let D be a positive diagonal matrix. For independent Haar unitary matrices U, V, set*

$$a = \operatorname{Tr}(P \operatorname{Ad}_{UDV}), \quad L = \operatorname{Tr}(D^2) \operatorname{Tr}(D^{-2}), \quad d = n^2 - 1.$$

Then

$$\mathbb{E}_{U,V}|a|^2 = \frac{(n-1)(L-1)}{d^2}, \qquad |a| \leq \sqrt{n-1}\sqrt{L-1}.$$

Here $\operatorname{Ad}_A(X) = AXA^{-1}$ on traceless matrices.

PROOF. The conjugation representation of $U(n)$ on traceless complex matrices is irreducible. Indeed, an invariant complex subspace is stable under commutators with skew-Hermitian matrices by differentiation, hence with all complex matrices, and is an ideal of $\mathfrak{sl}_n(\mathbb{C})$. This algebra is simple: commutators with diagonal matrices isolate off-diagonal matrix units from any noncentral element, and their commutators generate all off-diagonal units and traceless diagonals.

For any irreducible unitary representation R of dimension d, Schur averaging gives

$$\int R(U)XR(U)^* \,\mathrm{d}U = \frac{\operatorname{Tr} X}{d} I.$$

The corresponding matrix-coefficient identity gives, by two integrations,

$$\mathbb{E}_{U,V}|\operatorname{Tr}(AR(U)BR(V))|^2 = \frac{\|A\|_{\mathrm{HS}}^2 \|B\|_{\mathrm{HS}}^2}{d^2}.$$

For completeness, the first integration is $\mathbb{E}_U|\operatorname{Tr}(AR(U)M)|^2 = \|MA\|_{\mathrm{HS}}^2/d$; the second follows by applying the displayed averaging identity to AA^*. Here $\|P\|_{\mathrm{HS}}^2 = n-1$. On all matrix units, the squared Hilbert–Schmidt norm of Ad_D is L; the scalar subspace contributes one, so its squared norm on traceless matrices is $L-1$. This proves the first equality. The second is the Hilbert–Schmidt Cauchy–Schwarz inequality. □

We write $\mathbb{E}_*$ for expectation under the probability measure

$$\mathrm{d}\mathbb{P}_* = \frac{|\det Z|}{\mathbb{E}|\det Z|}\,\mathrm{d}\mathbb{P}.$$

The next calculation evaluates the singular-value quantity in Lemma 7.16.

LEMMA 7.17. *Under the Gaussian law tilted by* $|\det Z|$*, with* $W = Z^*Z$*,*

$$\mathbb{E}_* \operatorname{Tr}(W)\operatorname{Tr}(W^{-1}) = 2n^3 + n^2 - 2n.$$

PROOF. Condition on all columns except column j, whose orthogonal complement is almost surely one-dimensional. Write that column as its parallel component plus ζ times a unit perpendicular vector. Before tilt, ζ is a standard complex Gaussian independent of the parallel component. The absolute determinant is the volume of the other columns times $|\zeta|$, whereas the Schur-complement identity gives $(W^{-1})_{jj} = |\zeta|^{-2}$. Under determinant tilt, $|\zeta|^2$ therefore has Gamma shape $3/2$, and its inverse mean is $\Gamma(1/2)/\Gamma(3/2) = 2$. Summing over j gives $\mathbb{E}_* \operatorname{Tr}(W^{-1}) = 2n$, including integrability.

For radial scaling, write $Z = \sqrt{T}\,Y$, where $T = \|Z\|_{\mathrm{HS}}^2 = \operatorname{Tr} W$ and $\|Y\|_{\mathrm{HS}} = 1$. The real dimension is $2n^2$, and $|\det Z| = T^{n/2}|\det Y|$. Consequently, under tilt T and Y remain independent, and T has Gamma shape $K = n^2 + n/2$. Set $B = \operatorname{Tr}((Y^*Y)^{-1})$. Since $\operatorname{Tr}(W^{-1}) = B/T$, it follows that

$$\mathbb{E}_* \operatorname{Tr}(W)\operatorname{Tr}(W^{-1}) = \mathbb{E}_* B = (K-1)\mathbb{E}_* \operatorname{Tr}(W^{-1}) = 2n(K-1).$$

This is the stated value. No matrix-density Jacobian is needed. □

PROOF OF THEOREM 7.15. Expanding the determinant along its diagonal entries gives

$$\sqrt{n!}\,\Phi_n(Z) = \sum_i Z_{ii} \det Z_{i^c,i^c} - \det Z = \det Z\Big(\sum_i Z_{ii}(Z^{-1})_{ii} - 1\Big).$$

The expression in parentheses equals $a(Z) = \operatorname{Tr}(P \,\mathrm{Ad}_Z)$: the trace on the full diagonal space is the displayed sum, and the scalar matrix is fixed and contributes

one. The determinant-tilted Gaussian law is biunitarily invariant. Averaging on each singular-value orbit and applying Lemmas 7.16 and 7.17 therefore gives

$$\mathbb{E}_*|a(Z)|^2 = \frac{n-1}{(n^2-1)^2}\mathbb{E}_*\big(\operatorname{Tr}(W)\operatorname{Tr}(W^{-1})-1\big) = \frac{2n+1}{n+1}.$$

Cauchy–Schwarz under the tilted law now yields

$$\mathbb{E}|\Phi_n| = D_n\mathbb{E}_*|a(Z)| \le D_n\big(\mathbb{E}_*|a(Z)|^2\big)^{1/2} = D_n\sqrt{\frac{2n+1}{n+1}}.$$

For the separate lower estimate, $\mathbb{E}|a|^2 \le \sqrt{n-1}\sqrt{L-1}\mathbb{E}|a|$ on each orbit, so

$$\mathbb{E}_{U,V}|a| \ge \frac{\sqrt{n-1}\sqrt{L-1}}{(n^2-1)^2} \ge \frac{\sqrt{n-1}}{(n^2-1)^{3/2}}.$$

The last step uses $L \ge n^2$, by Cauchy–Schwarz on the positive singular values. Multiplying by $|\det Z|$ and integrating proves the lower bound. Finally, successive Gaussian orthogonal projections give independent squared determinant factors of Gamma shapes $1,\dots,n$, proving the product formula for D_n. Stirling's expansion gives

$$\log\frac{\Gamma(j+1/2)}{\Gamma(j)\sqrt{j}} = -\frac{1}{8j} + O(j^{-2}),$$

which yields $D_n \asymp n^{-1/8}$. □

CHAPTER 8

Barriers: why 4, and why not 2

Question. Can the Gaussian-block method of chapter 7 reach any $p \le 4$? And what keeps every construction away from $p = 2$?

Throughout, μ is Gaussian probability measure, $A(\varphi)$ the Gaussian Hankel matrix and $D(\varphi)$ the norm of $f \mapsto \langle f, \varphi\rangle$ on analytic polynomials in $L^1(\mu)$, as in section 7.1. A symbol gives an S_p counterexample through Proposition 7.2 when $D(\varphi) > \|A(\varphi)\|_{S_p}$; we call this a *strict block* at p. The Gaussian Carleman inequality $\sum_\alpha \alpha! 2^{-|\alpha|}|a_\alpha|^2 \le \|f\|^2_{L^1(\mu)}$ (Proposition 6.3 with Minkowski's inequality) and Cauchy–Schwarz give $D(\varphi) \le \|A(\varphi)\|_{S_2}$, the Gaussian form of Helson's inequality, and $\|A(\varphi)\|_{S_4} = \|\varphi\|_{L^4(\mu)}$ [47].

The answers: the constant-plus-product family that reaches every $p > 4$ has no strict block at $p \le 4$, for every n and every test, so 4 is exactly its threshold. Bampouras [4] singles out the endpoint $p = 4$ as especially interesting; for the Gaussian limit of his family the answer is negative. Affine symbols, even quadratic symbols of rank at most two, and all zero-free Hilbert–Schmidt symbols, are equally stuck; and every family of bounded degree in a bounded number of variables is excluded on an interval $(2, 2+\varepsilon)$. Whether *any* Gaussian symbol beats S_4 is the L^1–L^4 question of section 8.3. The Paley–Wiener threshold 4 of Bampouras has a different origin (section 8.5). The sharpness of the product family at 4, the barriers for affine, even quadratic and zero-free symbols, the interval $(2, 2+\varepsilon)$, the disc blocks and the polydisc growth are, to our knowledge, new.

8.1. The product family is sharp at 4

Let $P_n = Z_1 \cdots Z_n$. By Lemma 7.9, $\|A(c_0 + c_1 P_n)\|^4_{S_4} = |c_0|^4 + 4|c_0 c_1|^2 + 2^n |c_1|^4$.

THEOREM 8.1. *For every $n \ge 1$ and all $c_0, c_1 \in \mathbb{C}$, $D(c_0 + c_1 P_n) \le \|A(c_0 + c_1 P_n)\|_{S_4}$. Consequently the infimum of exponents with a strict block in this family is exactly* 4.

The cases $n = 1$ and $n \ge 2$ are proved differently: for $n = 1$ there are no residual singular values, $M = 2$, and the envelope argument below, which needs $M \ge 4$, does not apply; that case is handled by an explicit bounded representative.

The affine case

For $k > 0$ and $R = |Z|$ put $u_0 = \mathbb{E}(1 - k^2 R^2)_+$ and $u_1 = \mathbb{E}[R \min(kR, 1)]$.

LEMMA 8.2. *$Q(k) = u_0^4 + 4u_0^2 u_1^2 + 2u_1^4$ increases strictly on $(0, \infty)$ from the limit 1 at $k = 0$.*

PROOF. With $B = \mathbb{E}[R^2\mathbf{1}_{kR<1}]$, $u_0' = -2kB$ and $u_1' = B$, so $Q' = 8B\Phi$ with $\Phi = u_1^3 + u_0^2u_1 - ku_0^3 - 2ku_0u_1^2$. Put $t = 1/k$, $x = t^2$, $e = e^{-x}$. Then $u_0 = 1 - (1-e)/x$, and the Mills bound $\int_t^\infty e^{-s^2}\,\mathrm{d}s \geq te^{-t^2}/(2t^2+1)$ gives $u_1 \geq u_e = t^{-1}(1 - e(x+1)/(2x+1))$. For $x \leq 1$, $u_e > 2ku_0$ reduces to $2 + 3x - 2x^2 > e(x^2 + 5x + 2)$, which follows from the fourth-order Taylor bound for e^{-x} and the identity $(2 + 3x - 2x^2) - (x^2 + 5x + 2)(1 - x + x^2/2 - x^3/6 + x^4/24) = x^2(1-x)(x^3 + 2x^2 - 4x + 24)/24$; then $\Phi = u_1^2(u_1 - 2ku_0) + u_0^2(u_1 - ku_0) > 0$. For $x \geq 1$, write $u_1 = \rho u_0$. Then $\partial_{u_1}\Phi = u_0^2(3\rho^2 - 4k\rho + 1)$, which is nonnegative for $\rho \geq k$ (its larger root $(2k + \sqrt{4k^2 - 3})/3$ is at most k, and there is no real root once $x > 4/3$), and $\partial_{u_0}\Phi = u_0^2(2\rho - 3k - 2k\rho^2)$. For $x < 6$ the last quadratic has no real root, so $\partial_{u_0}\Phi < 0$. For $x \geq 6$, $u_1 \leq k\mathbb{E}R^2 = k$ and $u_0 \geq 1 - k^2 \geq 5/6$ give $\rho \leq 6k/5$, while the smaller root satisfies $(1 - \sqrt{1 - 6k^2})/(2k) \geq 3k/2$, so again $\partial_{u_0}\Phi \leq 0$. Replacing e by the larger $1/T(x)$, $T(x) = \sum_{j\leq 4} x^j/j!$, raises u_0 and lowers u_e, so ρ only decreases along the way; at the end $\rho \geq k$ is $T(x)(2x+1) \geq x^2 + 3x + 1$, which holds. So both signs persist, and the replacement only lowers Φ. The resulting expression is $x^{3/2}(x+2)^3P(x)/(13824(2x+1)^3T(x)^3)$ with $P(x) = x^3[8x^2(x - 9/4)^2 + 75x^2/2 - 163x + 1326] + 2268x^2 + 6408x + 6912 > 0$. □

PROOF OF THEOREM 8.1 FOR $n = 1$. For $b = \min(kR, 1)$ the unimodular function

$$\psi_k(R\zeta) = \frac{1 + b\zeta}{1 + b\overline{\zeta}}$$

has analytic coefficients $1-b^2, b, 0, \dots$ on each circle, so it represents the functional of u_0+u_1z, and $D(1+(u_1/u_0)z) \leq 1/u_0 < (1+4(u_1/u_0)^2+2(u_1/u_0)^4)^{1/4}$ by Lemma 8.2. The ratio u_1/u_0 increases continuously from 0 to ∞, so scaling and rotation cover every symbol with $c_0c_1 \neq 0$; for $c_0 = 0$, $D(c_1z) = 2|c_1|/\sqrt{\pi} < 2^{1/4}|c_1|$. □

Products of at least two factors

For $M \geq 4$ put $S_M(c) = (1 + 4c^2 + Mc^4)^{1/4}$ and $L_M(b) = \sup_{c\geq 0}(1 + bc)/S_M(c)$.

Here $S_M(c)$ is the S_4 norm of $A(1 + cP_n)$ when $M = \mathbb{E}|P_n|^4 = 2^n$, and $L_M(b)$ is the smallest L^1 norm, relative to $|a_0|$, that a test function with coefficient ratio $|a_1/a_0| = b$ must have if it is to lose against every symbol $c_0 + c_1P_n$. The proof is an induction on n. Multiplying by one more Gaussian Z does two things at once: it replaces the coefficient ratio b by the random ratio $b|Z|$, and it doubles M. The envelope inequality (8.1) below says that averaging L_M over the random ratio $b|Z|$ costs at least as much as doubling M: these two changes are compatible. To prove it, the single multiplication by $|Z|^2$ is split into a continuum of fractional steps, the log-Gamma semigroup of Lemma 8.5, while M moves continuously from M to $2M$. The discrete step then becomes a comparison between the generator of the semigroup and the speed at which M moves (Theorem 8.6), and that comparison is a one-variable inequality, checked below.

LEMMA 8.3 (Reduction). *The theorem holds for $n \geq 2$ if every polynomial $q(w) = a_0 + a_1w + \cdots$ with $a_0 \neq 0$ satisfies $\mathbb{E}|q(P_n)| \geq |a_0|L_{2^n}(|a_1/a_0|)$.*

PROOF. Averaging a test over the rotations with $\zeta_1 \cdots \zeta_n = 1$ keeps exactly the monomials with equal exponents, preserves the pairing and contracts L^1; so tests

may be taken of the form $q(P_n)$. Then $|a_0| + c|a_1| \le S_M(c)\mathbb{E}|q(P_n)|$ is the claim. If $a_0 = 0$, coefficient extraction gives $\mathbb{E}|q(P_n)| \ge |a_1|(\sqrt{\pi}/2)^n \ge |a_1|M^{-1/4}$. □

LEMMA 8.4 (Circle estimate). *If G is analytic on a neighbourhood of $\overline{\mathbb{D}}$ with $G(0) = 1$ and $|G'(0)| = s$, then $\int_{\mathbb{T}}|G|\,\mathrm{d}m \ge h(s)$, where $h(s) = 1 + s^2/4$ for $s \le 2$ and $h(s) = s$ for $s \ge 2$.*

PROOF. Factor $G = uv$ with $\|u\|_2\|v\|_2 = \|G\|_1$ and pair with $1 + \lambda z$: $1 + \lambda s \le \|G\|_1(1 + \sqrt{1 + 4\lambda^2})/2$. Take $\lambda = 2s/(4 - s^2)$ for $s < 2$; for $s \ge 2$ extract the coefficient. □

For $n = 2$: with $\beta = |a_1/a_0|$, the circle estimate and rotation invariance give $\mathbb{E}|q(Z_1Z_2)| \ge |a_0|\mathbb{E}h(\beta|Z_1Z_2|)$. Since $x \mapsto h(\beta\sqrt{x})$ is concave, Jensen gives $\mathbb{E}h(\beta|W|) \ge \mathbb{E}h(\beta\sqrt{2}|\Re W|)$ for $W = Z_1Z_2$, and $\sqrt{2}|\Re W|$ is distributed as $X/\sqrt{2}$ with X exponential of mean one. Put $H(b) = \mathbb{E}h(bX) = 1 + (b^2/2)(1 - e^{-2/b})$. For $b > 0$,

$$H(b)^2 - (1 + b^2) = b^2e^{-2/b}\big(b^2\sinh^2(1/b) - 1\big) \ge 0.$$

Hence $\mathbb{E}|q(Z_1Z_2)| \ge |a_0|H(\beta/\sqrt{2}) \ge |a_0|\sqrt{1 + \beta^2/2} = |a_0|L_4(\beta)$.

The induction from n to $n+1$ conditions on Z_{n+1}: the polynomial $w \mapsto q(wZ_{n+1})$ has coefficients a_0 and a_1Z_{n+1}, so it suffices to prove the envelope inequality

$$\mathbb{E}L_M(b|Z|) \ge L_{2M}(b) \qquad (M \ge 4,\ b \ge 0). \tag{8.1}$$

To interpolate multiplication by $|Z|^2$, we work in logarithmic coordinates: sums of independent increments become products of positive variables. Infinite divisibility of $\log|Z|^2$ supplies fractional steps, and the Gamma moments identify their normalization.

LEMMA 8.5 (The log-Gamma semigroup). *There are positive random variables X_a, $a \ge 0$, with $X_0 = 1$, $X_1 \overset{d}{=} |Z|^2$, $\mathbb{E}X_a^s = \Gamma(1+s)^a$ for $s \ge 0$, and $X_{a+a'} \overset{d}{=} X_aX'_{a'}$. The semigroup $P_ag(t) = \mathbb{E}g(tX_a)$ has generator*

$$\mathcal{A}g(t) = \int_0^1 \frac{g(tu) - g(t) + (1 - u)tg'(t)}{(-\log u)(1 - u)}\,\mathrm{d}u,$$

with positive kernel and $\mathcal{A}(t^s) = \log\Gamma(1 + s)\,t^s$.

PROOF. $\log|Z|^2$ is infinitely divisible [65]: the Lévy measure $e^y\,\mathrm{d}y/(|y|(1 - e^y))$ on $(-\infty, 0)$ has Laplace exponent $\int_0^1(u^s - 1 + s(1 - u))/((-\log u)(1 - u))\,\mathrm{d}u$, whose second derivative is $\sum_{k\ge1}(k + s)^{-2}$ and which vanishes at $s = 0, 1$, so it equals $\log\Gamma(1 + s)$. Put $X_a = e^{\xi_a}$. □

Write $f_M(t) = L_M(\sqrt{t})$; then (8.1) is $P_1f_M \ge f_{2M}$.

THEOREM 8.6. *For $M \ge 4$ and $t > 0$, $\mathcal{A}f_M(t) > (\log 2)M\partial_Mf_M(t)$.*

PROOF. Let c be the maximizer in $L_M(\sqrt{t})$, so $\sqrt{t} = (2c + Mc^3)/(1 + 2c^2)$, and put $y = Mc^2$. The envelope theorem gives

$$tf'_M = \frac{c^2(2 + y)}{2(1 + 2c^2)S_M(c)}, \qquad M\partial_Mf_M = -\frac{y}{2(2 + y)}\,tf'_M,$$

where we used $1+\sqrt{t}c = S_M(c)^4/(1+2c^2)$, so that $f_M = S_M(c)^3/(1+2c^2)$. Since $d\sqrt{t}/dc = (2-4c^2+3Mc^2+2Mc^4)/(1+2c^2)^2 > 0$ for $M \geq 4$, the maximizer c, and with it y, increases with t, and f_M is increasing. For the next derivatives put $x = c^2$ and

$$C = 1+2x, \quad J = 2+Mx, \quad Q = 1+4x+Mx^2, \quad B = 2+(3M-4)x+2Mx^2.$$

Direct differentiation gives

$$-f_M''(t) = \frac{(M-4)C^3}{2J^3BQ^{1/4}} + \frac{C^4}{4JBQ^{5/4}} > 0.$$

The first summand is nonincreasing in x, since $(C/J)' = (4-M)/J^2 \leq 0$ and B, Q increase. The logarithmic derivative of the second is

$$\frac{8}{C} - \frac{M}{J} - \frac{B'}{B} - \frac{5}{4}\frac{Q'}{Q} < 0.$$

Indeed $M/J \geq 2/C$, $Q'/Q \geq 4/C$, and $B'/B > 1/C$, the last following from $CB' - B = 3M - 6 + (7M-4)x + 6Mx^2 > 0$. Since t increases with x, we obtain $f_M^{(3)}(t) > 0$. We also need the curvature bound

$$\frac{y}{2(1+y)} + \frac{tf_M''}{f_M'} = \frac{c^2\big(2M^2c^6 + M^2c^4 + 8Mc^4 + Mc^2(M-4) + 24c^2 + 4\big)}{2(1+y)S_M(c)^4(2Mc^4+3Mc^2-4c^2+2)} > 0.$$

If $y \leq 4/3$: since $f_M^{(3)} > 0$, the divided difference $(f_M(tu) - f_M(t) + (1-u)tf_M'(t))/(1-u)^2$ increases in u, so its value $1 - f_M(t) + tf_M'(t)$ at $u = 0$ bounds it below. Since $\int_0^1 (1-u)/(-\log u)\,\mathrm{d}u = \log 2$, this gives $\mathcal{A}f_M \geq (\log 2)(1 - f_M + tf_M')$. With the formulas above, the inequality $1 - f_M + tf_M' > M\partial_M f_M$ becomes $(1+2c^2)S_M(c) > 1 + 3c^2 + Mc^4/4$. Put $c^2 = y/M$ and raise to the fourth power; M^5 times the difference is

$$\begin{aligned}(M+2y)^4(M+4y+y^2) - M(M+3y+y^2/4)^4 = {}& \frac{y^2(y+4)(4-3y)}{8}M^3 \\ &+ \frac{y^3(y+4)(80-32y-y^2)}{16}M^2 \\ &+ \frac{y^4(y+4)(4032-688y-44y^2-y^3)}{256}M + 16y^5(y+4),\end{aligned}$$

and every bracket is nonnegative for $0 < y \leq 4/3$, the second one strictly.

If $y \geq 4/3$: let $\alpha = (2+y)/(2+2y) \in (1/2, 5/7]$. The curvature bound integrates to the minorant $f_M(tu) \geq f_M(t) + tf_M'(t)(u^\alpha - 1)/\alpha$, which touches at $u = 1$, so positivity of the kernel gives $\mathcal{A}f_M(t) \geq tf_M'(t)\log\Gamma(1+\alpha)/\alpha$. Put $\delta = 1-\alpha \in [2/7, 1/2]$. Since the trigamma function ψ' decreases, Taylor's formula at 2 gives

$$\log\Gamma(2-\delta) \geq -(1-\gamma_{\mathrm{E}})\delta + \frac{\zeta(2)-1}{2}\delta^2.$$

Consequently

$$\log\Gamma(1+\alpha) + \frac{\log 2}{2}(1-\alpha) \geq \delta\left(\gamma_{\mathrm{E}} + \frac{\log 2}{2} + \frac{\zeta(2)-1}{7} - 1\right) > 0.$$

For the last sign, convexity of $1/s$ gives $\gamma_{\mathrm{E}} > 47/24 - 2\log 2$. The series for $2\operatorname{arctanh}(1/3)$ gives $\log 2 < 25/36$, while $\zeta(2) > \sum_{k=1}^{5} k^{-2} + \int_6^\infty s^{-2}\,\mathrm{d}s > 13/8$. Thus the bracket is greater than $23/24 - (3/2)(25/36) + (13/8 - 1)/7 = 1/168$. Since $(1-\alpha)/\alpha = y/(2+y)$, this is the claim. □

Differentiating $s \mapsto P_s f_{2^{1-s}M}$ gives

$$P_1 f_M - f_{2M} = \int_0^1 P_s\big[\mathcal{A} f_N - (\log 2) N \partial_N f_N\big]_{N=2^{1-s}M}\,\mathrm{d}s.$$

The derivative formulas in the proof of Theorem 8.6 give $|f_M(t)| + |t f_M'(t)| + |t^2 f_M''(t)| + |M\partial_M f_M(t)| \le C(1+\sqrt{t})$ on compact parameter ranges. Taylor's formula bounds the generator numerator by $C(1+\sqrt{t})(1-u)^2$ near $u=1$ and by $C(1+\sqrt{t})$ near $u=0$. These bounds are integrable against its kernel. Truncation and dominated convergence therefore justify the identity; stopping times may be removed using Doob's inequality for the positive, square-integrable martingale $\sqrt{X_s}/\Gamma(3/2)^s$. Theorem 8.6 makes the integrand positive. Thus $P_1 f_M \ge f_{2M}$, which proves the envelope inequality and completes the induction in Theorem 8.1.

Clipped moments of products

The monotonicity in Lemma 8.2 does not extend to arbitrary products. Put $R_n = |P_n|$ and, for $k > 0$, define

$$A_n(k) = \mathbb{E}(1 - k^2 R_n^2)_+, \qquad K_n(k) = \mathbb{E}[R_n \min(kR_n, 1)],$$

$$Q_n = A_n^4 + 4A_n^2 K_n^2 + 2^n K_n^4.$$

Proposition 8.7. *For every $n \ge 194$ there is $k > 0$ such that $Q_n'(k) < 0$. For $194 \le n \le 3998$ one may take $k = e^{0.4276n}$.*

The following derivative estimate reduces the assertion to a lower bound for A_n.

Throughout, $X = R_n^2$ is a product of n independent exponential variables with mean one, so $\mathbb{E}X^{-s} = \Gamma(1-s)^n$ for $\operatorname{Re} s < 1$. We write A, K for $A_n(k), K_n(k)$, put $\varepsilon_n = \mathbb{E}R_n = \Gamma(3/2)^n$, and let

$$\Phi = A^2 K + 2^{n-1} K^3 - kA^3 - 2kAK^2.$$

Lemma 8.8. *For $n \ge 2$ and $k > 0$, $Q_n'(k) = 8B\Phi$ with $B = \mathbb{E}[R_n^2 \mathbf{1}_{kR_n<1}] > 0$, and*

$$\varepsilon_n - \frac{1}{4k} \le K \le \varepsilon_n, \qquad \Phi \le 2^n \varepsilon_n^3 - kA^3.$$

Proof. The law of R_n has no atoms, so differentiation under the expectation gives $A' = -2kB$ and $K' = B$, and then $Q_n' = 4A^3A' + 8AA'K^2 + 8A^2KK' + 2^{n+2}K^3K' = 8B\Phi$. Next $K = \varepsilon_n - \mathbb{E}[R_n(1-kR_n)_+]$, and $0 \le r(1-kr)_+ \le 1/(4k)$. Finally $-2kAK^2 \le 0$, $A \le 1$ and $K \le \varepsilon_n$ give $\Phi \le \varepsilon_n + 2^{n-1}\varepsilon_n^3 - kA^3$, and $\varepsilon_n \le 2^{n-1}\varepsilon_n^3$ because $2^{n-1}\varepsilon_n^2 = (\pi/2)^n/2 \ge 1$ for $n \ge 2$. □

Proposition 8.9. *Let $n \ge 3999$, $s = 3/20$, $x = 1 - s$, $m = n\psi(x)$ and $w = \sqrt{2n\psi'(x)}$, where ψ is the digamma function. For $\log k = -(m + w + 1)/2$ one has $\Phi < 0$, hence $Q_n'(k) < 0$.*

PROOF. Let $Y = \log X$ and let $\mathbb{Q}$ be the probability measure with $\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P} = e^{-sY}/\Gamma(x)^n$. Under $\mathbb{Q}$ the factors of X are independent Gamma(x) variables, so Y has mean m and variance $n\psi'(x) = w^2/2$, and Chebyshev's inequality gives $\mathbb{Q}(|Y-m| \le w) \ge 1/2$. On this event $k^2 e^Y \le k^2 e^{m+w} = e^{-1}$ and $e^{sY} \ge e^{s(m-w)}$. Hence

$$A = \Gamma(x)^n \, \mathbb{E}_{\mathbb{Q}}\big[e^{sY}(1-k^2e^Y)_+\big] \ge \frac{1-e^{-1}}{2}\,\Gamma(x)^n e^{s(m-w)}.$$

By Lemma 8.8 it suffices that $\log k + 3\log A > n\log 2 + 3\log \varepsilon_n$. The bound on A shows this holds if

$$n\rho > c_1\sqrt{n} + c_0, \qquad \rho = -\frac{\psi(x)}{2} + 3\log\Gamma(x) + 3s\psi(x) - \log 2 - 3\log\Gamma(3/2),$$

with $c_1 = (1/2+3s)\sqrt{2\psi'(x)}$ and $c_0 = 1/2 - 3\log((1-e^{-1})/2)$. Interval evaluation gives

$$\rho \in [0.0317484,\ 0.0317485],$$
$$c_1 \in [1.944943,\ 1.944944], \qquad c_0 \in [3.955466,\ 3.955468].$$

At $n = 3999$ the difference $n\rho - c_1\sqrt{n} - c_0$ lies in $[0.0127,\ 0.0128]$. It is increasing in n once $\sqrt{n} > c_1/(2\rho)$, that is, for $n \ge 939$. So the inequality holds for all $n \ge 3999$. □

PROPOSITION 8.10. *For* $194 \le n \le 3998$ *and* $k = e^{0.4276n}$ *one has* $\Phi < 0$*, hence* $Q_n'(k) < 0$.

PROOF. For $0 < c < 1$, Mellin inversion of $(1-y)_+$, whose Mellin transform is $1/(s(s+1))$, and Fubini's theorem, justified by $\mathbb{E}X^{-c} < \infty$, give

$$A = \frac{1}{2\pi}\int_{-\infty}^{\infty} \mathrm{Re}\,\frac{k^{-2s}\Gamma(1-s)^n}{s(s+1)}\,\mathrm{d}t, \qquad s = c + it.$$

Take $c = 3/20$. The identity $|\Gamma(x+iy)/\Gamma(x)|^2 = \prod_{j\ge 0}(1 + y^2/(x+j)^2)^{-1}$ and $|s(s+1)| \ge t^2$ bound the part of the integral with $|t| > 2$ by $2k^{-2c}\Gamma(1-c)^n(1 + 4/(1-c)^2)^{-n/2}/2$. The part over $[-2,2]$ is enclosed by rigorous Gauss–Legendre integration in ball arithmetic (arb, through python-flint). The integrand is declared non-holomorphic on any ball where $\mathrm{Re}(1-s) \le 1/20$, which keeps $\log\Gamma(1-s)$ off its branch cut; on a ball where $s(s+1)$ may vanish, the division returns a non-finite ball, which arb also treats as non-holomorphic. The tail bound, not divided by 2π (a harmless overestimate), widens the enclosure on both sides. This gives an interval $[A_-, A_+]$ with $A_- > 0$. By Lemma 8.8, with $K_- = \varepsilon_n - 1/(4k) > 0$,

$$\Phi \le A_+^2\varepsilon_n + 2^{n-1}\varepsilon_n^3 - kA_-^3 - 2kA_-K_-^2.$$

The script `clipped-gaussian/clipped_finite.py` (chapter A) verifies that the right side divided by $2^{n-1}K_-^3$ is at most -0.01165 for each n in the range; after the factor $k^{-2c}\Gamma(1-c)^n$ is taken out of A, every quantity is continuous in n, and the script encloses it with n replaced by a ball over each of 11 blocks of consecutive integers (single integers near $n = 194$). The largest certified value is -0.011656, at $n = 194$. The same script recomputes the constants of Proposition 8.9. □

PROOF OF PROPOSITION 8.7. Combine Propositions 8.9 and 8.10. □

This failure of monotonicity does not affect Theorem 8.1, whose proof uses the envelope inequality. Monotonicity for $3 \le n \le 193$ remains undecided.

8.2. Other symbol families

PROPOSITION 8.11 (Even quadratic symbols). *Every analytic polynomial $f = \sum a_n z^n$ in one Gaussian variable has $\|f\|_1^2 \ge |a_0|^2 + |a_2|^2$. Hence $D(c_0 + c_2 z^2) \le (|c_0|^2 + 4|c_2|^2)^{1/2} < \|A(c_0 + c_2 z^2)\|_{S_4}$ for $c_2 \ne 0$.*

PROOF. If $a_0 = 0$, extracting the coefficient gives $\|f\|_1 \ge |a_2|$. Otherwise pass to the even part $g(z^2)$. With $\beta = |a_2/a_0|$, Lemma 8.4 on each circle and $x = r^2$ give $\|f\|_1/|a_0| \ge \int_0^\infty h(\beta x) e^{-x}\,\mathrm{d}x = 1 + (\beta^2/2)(1 - e^{-2/\beta}) \ge \sqrt{1+\beta^2}$, as in the case $n = 2$ above. Then $\langle f, \varphi\rangle = a_0\overline{c_0} + 2a_2\overline{c_2}$ and Cauchy–Schwarz. The Hankel block has $\|A\|_{S_4}^4 = |c_0|^4 + 8|c_0 c_2|^2 + 24|c_2|^4$, which exceeds $(|c_0|^2 + 4|c_2|^2)^2$ by $8|c_2|^4$. □

The even case extends to every number of variables. Let $\varphi = c + z^T C z$ with C complex symmetric, let $\lambda_j \ge 0$ be the Takagi singular values of C, and put $r^2 = \sum_j \lambda_j^2$.

PROPOSITION 8.12. *$D(\varphi) \le (|c|^2 + 4r^2)^{1/2}$ and $\|A(\varphi)\|_{S_4}^4 = |c|^4 + 8|c|^2 r^2 + 8r^4 + 16\sum_j \lambda_j^4$. Hence φ has no strict block at any $p \le 4$ whenever $(\sum_j \lambda_j^2)^2 \le 2\sum_j \lambda_j^4$, in particular whenever C has rank at most two.*

PROOF. A unitary change of variables preserves μ, analytic L^1 norms and the pairing, so we may take $\varphi = c + \sum_j \lambda_j z_j^2$. Applying Proposition 8.11 in one variable at a time and Minkowski's inequality in the others gives $\|f\|_1^2 \ge \sum_{\alpha \in \{0,2\}^d} |a_\alpha|^2$. Since $\langle f, \varphi\rangle = a_0\overline{c} + 2\sum_j a_{2e_j}\lambda_j$, Cauchy–Schwarz gives the bound for D. For the norm, $\|A(\varphi)\|_{S_4}^4 = \mathbb{E}|\varphi|^4$ by Proposition 7.6; with $X = \sum_j \lambda_j z_j^2$ one has $\mathbb{E}X = \mathbb{E}X^2 = 0$, $\mathbb{E}|X|^2 = 2r^2$ and $\mathbb{E}|X|^4 = 24\sum_j \lambda_j^4 + 8\sum_{j\ne k} \lambda_j^2\lambda_k^2$. The condition is $(|c|^2 + 4r^2)^2 \le \|A\|_{S_4}^4$, and $(a^2 + b^2)^2 \le 2(a^4 + b^4)$. □

So an even quadratic counterexample must have effective rank above two.

We first record the translation invariance needed for zero-free symbols. Unitary changes of variables preserve μ. The Weyl operator $W_a f(z) = f(z - a) e^{\overline{a}\cdot z - |a|^2/2}$ is unitary on $L^2(\mu)$, and $W_a f W_a g = S(fg)$ with $Sh(z) = h(z - a) e^{2\overline{a}\cdot z - |a|^2}$, which the substitution $z = w + a$ shows to be an isometry of analytic $L^1(\mu)$ onto itself. Hence $H_\varphi(W_a f, W_a g) = \langle fg, S^*\varphi\rangle$, where moving the factor $e^{\overline{a}\cdot w}$ across the pairing as a translation gives

$$S^*\varphi(w) = e^{-|a|^2} e^{-\overline{a}\cdot w} \varphi(w + 2a).$$

Thus the two Hankel forms have the same singular values, and $D(S^*\varphi) = D(\varphi)$.

PROPOSITION 8.13 (Zero-free symbols). *Suppose that φ is entire on $\mathbb{C}^d$, without zeros, and that $\|A(\varphi)\|_{S_2} < \infty$. Then $D(\varphi)$ equals the operator norm of $A(\varphi)$, so φ has no strict block at any p.*

PROOF. By Borel–Carathéodory and Cauchy's estimates, the growth bound $|\varphi(z)| \le \|A\|_{S_2} e^{|z|^2/4}$ implies that $\log \varphi(z) = z^T C z + b^T z + c$, with C complex

symmetric. The Hilbert–Schmidt condition forces every Takagi singular value of C to be less than $1/4$.

Under the symbol action $S_a^*\varphi(z) = e^{-|a|^2}e^{-\overline{a}\cdot z}\varphi(z+2a)$, the linear coefficient becomes $b + 4Ca - \overline{a}$. The real-linear map $a \mapsto \overline{a} - 4Ca$ is invertible because $4\|C\| < 1$, so a Weyl translation removes the linear term. This action preserves D and the singular values of the Hankel form. A unitary change of variables reduces the symbol, up to a constant factor, to $\prod_j e^{\alpha_j z_j^2}$ with $0 \le \alpha_j < 1/4$.

Write $\alpha_j = q_j/(2(1+q_j^2))$, $0 \le q_j < 1$. The bounded function

$$\prod_j \sqrt{1+q_j^2}\,\exp\biggl(i\sum_j q_j \operatorname{Im} z_j^2\biggr)$$

represents this symbol, giving $D \le \prod_j \sqrt{1+q_j^2}$. For the reverse inequality, evaluate the bounded Hankel form at

$$g(z) = \exp\biggl(\frac{1}{2}\sum_j q_j z_j^2\biggr), \qquad \frac{H_\varphi(g,g)}{\|g\|_2^2} = \prod_j \sqrt{1+q_j^2}.$$

Indeed, Taylor approximation in the Fock space and the coefficient moments give $H_\varphi(g,g) = \prod_j (1-4q_j\alpha_j)^{-1/2}$, while $\|g\|_2^2 = \prod_j (1-q_j^2)^{-1/2}$. Hence the operator norm has the same lower bound. Since $\|A(\varphi)\| \le D(\varphi)$, equality follows. □

For monomials the computation is explicit: $D(z^m) = m!/\Gamma(1+m/2)$ by circular averaging, and $A(z^m)$ is anti-diagonal. For even m this equals $\|A\|_{S_\infty}$; for $m = 2k+1$ a strict block needs $p > \log 2/\log(\sqrt{k!(k+1)!}/\Gamma(k+3/2)) \ge (1-\log\pi/\log 4)^{-1}$. So no one-variable homogeneous symbol improves on Brevig and Perfekt.

8.3. The L^1–L^4 question

Problem 7 (The L^1–L^4 question). *Is $|\int f\overline{\varphi}\,\mathrm{d}\mu_d| \le \|f\|_{L^1(\mu_d)}\|\varphi\|_{L^4(\mu_d)}$ for all analytic polynomials f, φ on $\mathbb{C}^d$ and all d? Equivalently, does every polynomial φ have a bounded representative ψ with $\|\psi\|_\infty \le \|\varphi\|_{L^4}$?*

Since $\|\varphi\|_{L^4} = \|A(\varphi)\|_{S_4}$, a counterexample gives a strict block at every p slightly below 4, hence $p_* < 4$. A positive answer shows that 4 is the limit of the Gaussian method: it holds exactly when no Gaussian block is strict at any $p \le 4$. It would not decide p_*. Nothing transfers back from the Gaussian model to the torus, and the torus analogue of the inequality is false already on $\mathbb{T}$: $\varphi = (1-z)^{-1/5}$ lies in $L^4(\mathbb{T})$ but not in BMOA, so its dilates have $D(\varphi_r)/\|\varphi_r\|_4 \to \infty$. Blocks on a fixed finite torus that are far from Gaussian, like the Ortega-Cerdà–Seip examples for bounded forms, are therefore not covered. The product, affine, zero-free and one-variable even quadratic symbols, and the even quadratic symbols of rank at most two, all satisfy the inequality. The natural remaining candidates are cumulative products $c_0 + \sum_j c_j Z_1 \cdots Z_j$ and quadratic symbols of large rank.

Near constants the inequality is sharp to second order: for $\varphi = 1 + \varepsilon b$ and $f = 1 + \varepsilon h$ with $b(0) = h(0) = 0$,

$$\log\frac{|\int f\overline{\varphi}\,\mathrm{d}\mu|}{\|f\|_{L^1}\|\varphi\|_{L^4}} = -\frac{\varepsilon^2}{4}\|h-2b\|_2^2 + O(\varepsilon^3),$$

optimal but not positive at $h = 2b$.

8.4. Near $p = 2$

THEOREM 8.14 (Bounded degree and dimension). *For all $d, K \geq 1$ there is $\varepsilon_{d,K} > 0$ such that $D(\varphi) \leq \|A(\varphi)\|_{S_p}$ for $2 \leq p \leq 2 + \varepsilon_{d,K}$ and every polynomial φ on $\mathbb{C}^d$ of total degree at most K, against tests of any degree.*

PROOF. *Near constants.* We first prove $D(\varphi) \leq \|A(\varphi)\|_{S_4}$ in a neighbourhood of the constant symbol. The idea is to approximate each relevant test by a square, with an exponentially small error, and then absorb that error using a polynomial lower bound for the second singular value. Normalize $c_0 = 1$ and put $\tau = \max_{1 \leq |\alpha| \leq K} |c_\alpha|$. All constants below may depend on d, K, but not on the degree of the test. The case $\tau = 0$ is equality, so suppose $\tau > 0$ is small.

Write $f = \sum_\alpha a_\alpha z^\alpha$. If $f(0) = 0$, Gaussian Carleman and Cauchy–Schwarz give $|\langle f, \varphi \rangle| \leq C\tau \|f\|_1 < \|f\|_1$. Such a test cannot improve on the constant test, whose quotient is 1. Otherwise rescale so that $f(0) = 1$. Tests with quotient at most 1 already satisfy the desired bound, since $\|A\|_{S_\infty} \geq |A_{00}| = 1$. We therefore consider only quotients exceeding 1. Put

$$x^2 = \sum_{1 \leq |\alpha| \leq K} \alpha! 2^{-|\alpha|} |a_\alpha|^2.$$

The same inequalities give

$$1 + x^2 \leq \|f\|_1^2 < |\langle f, \varphi \rangle|^2 \leq (1 + C\tau x)^2.$$

For small τ this implies $x = O(\tau)$, so all coefficients of f of degrees 1 through K are $O(\tau)$. The remaining coefficients and the degree of f are unrestricted.

Let g be the formal square root of f, truncated after total degree K. Then $g(0) = 1$, and g^2 agrees with f through degree K. The recursion

$$2g_\alpha = a_\alpha - \sum_{\substack{\beta + \gamma = \alpha \\ |\beta|, |\gamma| \geq 1}} g_\beta g_\gamma \qquad (1 \leq |\alpha| \leq K)$$

shows that the nonconstant coefficients of g are $O(\tau)$. Choose $R = c\tau^{-1/K}$ with $c > 0$ sufficiently small and restrict τ so that $R \geq 1$. On the polydisc $|z_j| \leq R$ we then have $|g(z) - 1| \leq C\tau R^K \leq 1/2$, so g has no zeros there.

Fix radii $r_j \leq R$ and put $u(\zeta) = g(r\zeta)/\overline{g(r\zeta)}$ on $\mathbb{T}^d$. This function has modulus one. Since $1/g$ is analytic near the polydisc, the Fourier indices of u have the form $\alpha - \beta$, where $|\alpha| \leq K$ and $\beta \geq 0$ coordinatewise. Every nonnegative index therefore has total degree at most K. Pairing with an analytic function uses only those coefficients, which agree for f and g^2. Hence

$$\int_{\mathbb{T}^d} |f(r\zeta)| \, \mathrm{d}m \geq \left| \int_{\mathbb{T}^d} f(r\zeta) \overline{u(\zeta)} \, \mathrm{d}m \right| = \int_{\mathbb{T}^d} |g(r\zeta)|^2 \, \mathrm{d}m.$$

Thus the higher-degree part of the test contributes no error on these tori.

Integrating over the radii inside the polydisc gives $\|f\|_1 \geq \|g\|_2^2 - E_R$, where $E_R = \int_{\max_j |z_j| > R} |g(z)|^2 \, \mathrm{d}\mu(z)$. Radial orthogonality and the incomplete-gamma

formula give

$$E_R \le e^{-R^2} \sum_{j=1}^{d} \sum_{|\alpha|\le K} |g_\alpha|^2 \alpha! \sum_{\ell=0}^{\alpha_j} \frac{R^{2\ell}}{\ell!} \le Ce^{-R^2}.$$

The constant coefficient contributes a bounded factor; for the others, $|g_\alpha|^2 = O(\tau^2)$ and $R^{2\ell} \le R^{2K} = O(\tau^{-2})$. Moreover,

$$|\langle f, \varphi\rangle| = |\langle g^2, \varphi\rangle| \le \|A(\varphi)\|_{S_\infty} \|g\|_2^2.$$

Since $\|f\|_1 \ge 1$ and $\|A\|_{S_\infty}$ is bounded near the constant symbol, we obtain, uniformly over all tests,

$$D(\varphi) \le \|A\|_{S_\infty} + Ce^{-b\tau^{-2/K}}.$$

To estimate the second singular value, put $m_\alpha = \alpha! c_\alpha$, $m_0 = 1$, and $m_\alpha = 0$ for $|\alpha| > K$. For small τ, $\delta = \max_{1\le|\alpha|\le K} |m_\alpha| \le 1/2$. Set $\mu_j = m_{e_j}$ and

$$\eta = \max_{\substack{1\le j\le d \\ 1\le|\alpha|\le K}} |m_{\alpha+e_j} - \mu_j m_\alpha|.$$

Iterating along coordinate increments and using $|\mu_j| \le 1/2$ gives $|m_\alpha - \mu^\alpha| \le 2\eta$ for $1 \le |\alpha| \le K+1$. In particular, $|\mu_j|^{K+1} \le 2\eta$ and $\delta \le \max_j |\mu_j| + 2\eta$. Either $\eta \ge \delta/4$ or $\max_j |\mu_j| \ge \delta/2$; in both cases,

$$\eta \ge \frac{\delta^{K+1}}{2^{K+2}} \ge \frac{\tau^{K+1}}{2^{K+2}}.$$

The minor of A in rows $0, e_j$ and columns $0, \alpha$ has absolute determinant $|m_{\alpha+e_j} - \mu_j m_\alpha|/\sqrt{\alpha!}$, which is at most $s_1(A)s_2(A)$. Since $s_1(A) < 2$ near the constant symbol and $\alpha! \le K!$, it follows that

$$s_2(A) \ge \frac{\tau^{K+1}}{2^{K+3}\sqrt{K!}}.$$

Writing $s_1 = \|A\|_{S_\infty}$ and shrinking the neighbourhood so that $\|A\|_{S_4} < 2$, we obtain

$$\|A\|_{S_4} - s_1 \ge \frac{s_2(A)^4}{32} \gtrsim \tau^{4(K+1)}.$$

This power exceeds $Ce^{-b\tau^{-2/K}}$ for sufficiently small τ. Thus $D(\varphi) \le \|A\|_{S_4}$ on one fixed neighbourhood of the constant symbol, and Schatten monotonicity gives the comparison for every $2 \le p \le 4$ there.

Away from constants. Gaussian Carleman and coefficient Cauchy–Schwarz always give $D(\varphi) \le H$, where $H = \|A(\varphi)\|_{S_2}$. Suppose equality holds for a nonconstant φ, and take polynomial tests f_n with $\|f_n\|_1 = 1$ and $\langle f_n, \varphi\rangle \to H$. Near equality in Cauchy–Schwarz forces $f_n \to h = \varphi(2z)/H$ in the Hilbert space with squared norm $\sum_\alpha \alpha! 2^{-|\alpha|} |a_\alpha|^2$. This convergence is locally uniform, so Fatou gives $\|h\|_1 \le 1$. The squared coefficient norm of h equals one, and hence equality holds in Gaussian Carleman. Its nonzero equality cases are exponentials [87], contradicting that h is a nonconstant polynomial. Thus $D(\varphi) < H$.

Normalize $H = 1$. The symbol sphere is finite-dimensional and compact, and $|D(\varphi) - D(\psi)| \le \|A(\varphi - \psi)\|_{S_2}$. Outside the constant neighbourhood there is therefore a uniform gap $D(\varphi) \le 1 - \eta_0$. Since $\operatorname{rank} A \le \binom{d+K}{K} = r$, we have

$\|A\|_{S_p} \geq r^{1/p-1/2}$ on this sphere. Choose $0 < \varepsilon_{d,K} \leq 2$ so that $r^{1/(2+\varepsilon_{d,K})-1/2} > 1 - \eta_0$. This proves the comparison outside the neighbourhood for $2 \leq p \leq 2 + \varepsilon_{d,K}$; the local S_4 bound proves it inside. Homogeneity completes the proof. □

PROPOSITION 8.15 (Quadratic compression). *For $\varphi = c + b^T Z + Z^T C Z$ on $\mathbb{C}^d$ with $C = C^T$, the nonzero singular values of $A(\varphi)$ are those of*

$$\begin{pmatrix} c & b^T & \sqrt{2}\|C\|_F \\ b & 2C & 0 \\ \sqrt{2}\|C\|_F & 0 & 0 \end{pmatrix}.$$

If $\operatorname{rank} C = r$, a unitary change of variables reduces φ to $r+1$ Gaussians, preserving D and all singular values. So quadratic symbols approaching $p = 2$ need $\operatorname{rank} C \to \infty$.

PROOF. The quadratic row of the full matrix is a vector q with $\|q\|^2 = 2\|C\|_F^2$; a unitary congruence in the quadratic block rotates it to one coordinate. Takagi factorization diagonalizes C, and Theorem 8.14 applies with $d = r + 1$, $K = 2$. □

Only the rank of $A(\varphi)$ matters in Theorem 8.14. The rank identity below is classical: Janson, Peetre and Rochberg [47, Theorem 10.2] show that the range of a Fock-space Hankel form is spanned by the translates of its symbol, and for a polynomial these span the same space as its derivatives; up to diagonal scalings, $A(\varphi)$ is the catalecticant matrix of φ.

THEOREM 8.16 (Bounded rank). *Let φ be a nonconstant polynomial on $\mathbb{C}^d$ of degree K and $r = \operatorname{rank} A(\varphi)$. Then $r = \dim \operatorname{span}\{\partial^\alpha \varphi\}$, $K \leq r - 1$, and $\varphi(Uw)$ depends on at most $r - 1$ of the coordinates w_j for some unitary U. Consequently $D(\varphi) \leq \|A(\varphi)\|_{S_p}$ for $2 \leq p \leq 2 + \varepsilon_{r,r}$, every d, and every φ with $\operatorname{rank} A(\varphi) \leq r$.*

PROOF. Since $\partial^\alpha \varphi = \sum_\beta c_{\alpha+\beta}(\alpha + \beta)!\, z^\beta/\beta!$, row α of $A(\varphi)$ is $(\alpha!)^{-1/2}$ times the coefficients of $\partial^\alpha \varphi$, each scaled by $\sqrt{\beta!}$; so $r = \dim \operatorname{span}\{\partial^\alpha \varphi\}$. If $c_\gamma \neq 0$ with $|\gamma| = K$, a chain $0 = \alpha_0 < \cdots < \alpha_K = \gamma$ gives derivatives of the exact degrees $K, K-1, \ldots, 0$, so $K + 1 \leq r$. The derivatives of order at least one span a space W of dimension at most $r - 1$, which does not contain φ. So $v \mapsto \sum_j v_j \partial_j \varphi$, from $\mathbb{C}^d$ to W, has a kernel of codimension $m \leq r - 1$; a unitary U whose last $d - m$ columns lie in the kernel makes $\varphi(Uw)$ independent of $w_{m+1}, \ldots, w_d$.

Both D and the singular values of A are invariant under unitary changes of variables. They are also unchanged by adding or removing dummy variables: the matrix only gains zero rows and columns, and restricting a test to $z'' = 0$ does not increase its L^1 norm, by the mean value property. So φ may be regarded as a polynomial on $\mathbb{C}^r$ of degree at most r, and Theorem 8.14 with $d = K = r$ applies. □

So blocks succeeding at exponents $p_k \downarrow 2$ must have unbounded rank; the bound $m \leq r - 1$ is attained by $\varphi = z_1$. For the odd monomials of the paragraph after Proposition 8.13 the threshold is exact: for K odd there is a unique $p_K > 2$ with $D(z^K) > \|A(z^K)\|_{S_p}$ exactly when $p > p_K$, and $p_K/K \to c_* \approx 2.788$, the positive

root of $\log(2\sum_{\ell\ge 0} e^{-c\ell(\ell+1)}) = c/4$. Monomials therefore move away from $p=2$ as their degree grows.

REMARK 8.17 (Entropy form of a strict block). For a polynomial block, put $q_j = s_j^2/\|A\|_{S_2}^2$ and $\Delta = \log(\|A\|_{S_2}/D) \ge 0$. Then $D > \|A\|_{S_p}$ exactly when $\Delta < (1/2 - 1/p)H_{p/2}(q)$, with H_α the Rényi entropy; in particular rank r forces $\Delta < (1/2-1/p)\log r$. Both Δ and H_α add under tensor products. By Theorem 8.16, a sequence of strict blocks at $p \downarrow 2$ must have unbounded rank. The exact condition above requires the entropy gain $(1/2 - 1/p)H_{p/2}(q)$ to exceed the Helson deficit Δ.

The translation invariance proved before Proposition 8.13 shows that Theorem 8.1, Proposition 8.12 and Theorem 8.14 hold for every symbol $ce^{-\overline{b}\cdot z}\psi(z+2b)$ with $\psi(z) = \varphi(Uz)$ obtained from theirs, U unitary, $b \in \mathbb{C}^d$ and $c \ne 0$; for instance $D \le \|A\|_{S_4}$ for $e^{-\overline{b}\cdot z}(c_0 + c_1\prod_j (z_j + 2b_j))$.

8.5. The Paley–Wiener threshold

Bampouras [6] proved that Nehari's theorem fails in S_p, $p > 4$, for Hankel operators on Paley–Wiener spaces $PW(\Omega)$ of convex $\Omega \subset \mathbb{R}^n$ with a curved boundary piece. The cap counting of Proposition 8.19 reorganizes his proof around a scale-uniform L^1 bound, which suggests that the coincidence with the $\mathbb{T}^\infty$ threshold comes from a different mechanism.

LEMMA 8.18 (Lossless cap sums). *Let $\hat{b}$ be supported in the unit ball, $\rho > 0$, and $|\xi_i - \xi_j| \ge 2\rho$ for $i \ne j$. Then for every integer $k > n/4$,*

$$\Big\| \sum_j c_j e^{2\pi i\langle \xi_j, x\rangle} \rho^n b(\rho x) \Big\|_{L^1(\mathbb{R}^n)} \le \|(1+|x|^2)^k b\|_2 \, \|(1+|x|^2)^{-k}\|_2 \, \|c\|_{\ell^2}.$$

PROOF. Let f be the sum. By Cauchy–Schwarz with the weight $w(x) = (1+|\rho x|^2)^k$, $\|f\|_1 \le \|wf\|_2 \|1/w\|_2$, and $\|1/w\|_2 = \rho^{-n/2}\|(1+|x|^2)^{-k}\|_2$, finite since $4k > n$. Multiplication by w is the operator $(1 - \rho^2\Delta/4\pi^2)^k$ on the Fourier side, which preserves supports, so the terms of wf have the disjoint Fourier supports $B(\xi_j, \rho)$ and are orthogonal. Each has L^2 norm $\rho^{n/2}\|(1+|x|^2)^k b\|_2$, by scaling, and Plancherel finishes. □

PROPOSITION 8.19. *Let N caps $\hat{\varphi}_i = \beta((\cdot - z_i)/\rho)$ in 2Ω satisfy the hypotheses of the lemma and give orthogonal Hankel blocks, and let a bound $\omega_\Omega = \chi_\Omega * \chi_\Omega$ on them. For $\psi = \sum \varphi_i$ and $p \ge 2$,*

$$\frac{\langle \psi, \psi\rangle}{\|\psi\|_1 \|H_\psi\|_{S_p}} \ge c\, N^{1/2 - 1/p} (\rho^n/a)^{1/p}.$$

PROOF. The caps have disjoint supports, so the pairing is $\sum_i \|\hat{\varphi}_i\|_2^2 = N\rho^n\|\beta\|_2^2$, up to the normalization of the Fourier transform. Lemma 8.18, with $c_j = 1$ and $\rho^n b(\rho\,\cdot) = \varphi_i$ translated in frequency, gives $\|\psi\|_1 \lesssim \sqrt{N}$, independently of ρ: the L^1 norm of a sum of separated caps grows only like the ℓ^2 norm of the coefficients. Bampouras's bound

$$\|H_{\varphi_i}\|_{S_p} \le \|\hat{\varphi}_i\, \omega^{1/p}\|_{L^{p'}},$$

with orthogonality, gives $\|H_\psi\|_{S_p} \lesssim N^{1/p}\rho^{n/p'} a^{1/p}$. □

With $\rho^n/a \approx N^{-\gamma}$ the ratio grows like $N^{1/2-1/p-\gamma/p}$, so the threshold is $p = 2(1+\gamma)$. The factor 2 is square-root cancellation in L^1, and γ is geometric. For caps of radius ε^2 along a fully curved boundary, $N \approx \varepsilon^{-(n-1)}$ caps fit, ω is of size ε^{n+1} on them, and $\gamma = 1$: this is Bampouras's threshold 4, in every dimension. Heuristically, partial curvature, with $k < n-1$ curved directions, gives $\gamma = (n-1)/k$ and larger thresholds, for instance 6 for $n = 3$, $k = 1$; this assumes the corresponding vanishing rate of ω, and the count of curved directions alone does not decide it. For the solid cone $\{(x', z) : 0 < z < 1,\ |x'| < 1 - z\}$ in $\mathbb{R}^3$, whose lateral boundary has one curved direction, $N \approx \delta^{-1/2}$ caps of radius $\rho \approx \delta$ near the rim have $a \approx \delta^{5/2}$, so $\gamma = 1$ and the same counting gives failure for every $p > 4$. The mechanism has no analogue on $\mathbb{T}^\infty$: there $\|H_\psi\| \geq \|H_\psi 1\| = \|\psi\|_2$ for every symbol, so a sum of symbols cannot split into orthogonal blocks, and decoupling must come from tensor products, where 4 is $\mathbb{E}|Z|^4 = 2$. Accordingly the $\mathbb{T}^\infty$ barriers do not transfer verbatim. The Paley–Wiener analogue of Problem 7 asks whether $|\langle h, \varphi\rangle| \leq C\|h\|_1 \|H_\varphi\|_{S_4}$; at $p = 4$ the counting bound above stops growing.

At $p = 2$ the Paley–Wiener question goes back to Brevig and Perfekt.

PROBLEM 8 (Brevig and Perfekt [21]). *Does every Hankel operator on $PW(D)$, D the unit disc, that belongs to S_2 have a bounded symbol?*

The answer is yes. Bampouras and Perfekt [8] proved it for balls in every dimension, and Bampouras [5] then for every convex set free of affine lines. Both go through a duality that turns the problem into a Paley–Wiener form of Helson's inequality. Put $\omega(\zeta) = |D \cap (\zeta - D)| = (\chi_D * \chi_D)(\zeta)$. Then $\|H_\varphi\|_{S_2}^2 = \int_{2D} |\hat{\varphi}|^2 \omega$, and Cauchy–Schwarz, with equality for $\hat{\varphi} = \hat{h}/\omega$, shows that the best constant C in $|\langle h, \varphi\rangle| \leq C\|h\|_1 \|H_\varphi\|_{S_2}$ is given by

$$C^2 = \sup_h \frac{1}{\|h\|_1^2} \int_{2D} \frac{|\hat{h}|^2}{\omega}, \tag{8.2}$$

the supremum taken over nonzero $h \in L^1(\mathbb{R}^2)$ with $\hat{h}$ supported in $2\overline{D}$. By the Hahn–Banach theorem, $C < \infty$ is exactly a positive answer to Problem 8. Near $\partial(2D)$ the weight ω vanishes like $\operatorname{dist}(\zeta, \partial(2D))^{3/2}$, so $1/\omega$ is not integrable and (8.2) is not trivial. Bampouras and Perfekt construct $\psi \in L^\infty(\mathbb{R}^2)$ whose Fourier transform equals $1/\omega$ on $2D$, as a distribution, from the boundary values of $(1 - |x|^2 \pm i0)^{-3/2}$; then, for $\hat{h} \in C_c^\infty(2D)$,

$$\int_{2D} \frac{|\hat{h}|^2}{\omega} = \int |\hat{h}|^2 \hat{\psi} = \langle h * \psi, h\rangle \leq \|\psi\|_\infty \|h\|_1^2,$$

so $C^2 \leq \|\psi\|_\infty$. Bampouras's theorem for general convex $\Omega \subset \mathbb{R}^n$ comes with the explicit constant $2^n (2n)!\, \Gamma(n/2+1)^2/(n!)^3 \leq 8^n$ in Helson's inequality, for the weight $|\Omega \cap (2x - \Omega)|$. The constant cannot be 1, already at $p = 2$; this is the first assertion below, and with the symbol ψ it pins down the growth of the constant on products of discs. On $\mathbb{T}^\infty$, by contrast, Helson's inequality gives the constant 1 in S_2 (Theorem 5.55).

PROPOSITION 8.20 (Disc blocks). *Let $D \subset \mathbb{R}^2$ be the unit disc. For every $p \geq 2$ there are Schwartz functions h, φ with Fourier supports in $2\overline{D}$ and*

$$|\langle h, \varphi \rangle| > \|h\|_1 \|H_\varphi\|_{S_p}.$$

Consequently, for every $p \geq 2$, the best constant in $|\langle h, \varphi \rangle| \leq C\|h\|_1\|H_\varphi\|_{S_p}$ on $PW(D^m)$ is at least exponential in m (for $p > 4$ it is already infinite, by Bampouras [6]).

PROOF. Let $v = |D|$ and $k = \check{\chi}_D$, and put $h_0 = k^2$. Then $\widehat{h_0} = \chi_D * \chi_D$ is supported in $2\overline{D}$, $h_0 \geq 0$, $h_0(0) = v^2$ and $\|h_0\|_1 = \|\chi_D\|_2^2 = v$; moreover $h_0 > 0$ almost everywhere, since k is real analytic and not identically zero. Let $\beta \geq 0$ be smooth, even and nonzero, with support in $2D$ away from the origin, and $h_1 = \check{\beta}$. Then h_1 is real, $\int h_1 = \beta(0) = 0$ and $h_1(0) = \int \beta > 0$. By dominated convergence, $\|h_0 + \varepsilon h_1\|_1 = v + \varepsilon \int h_1 + o(\varepsilon) = v + o(\varepsilon)$, while $(h_0 + \varepsilon h_1)(0) = v^2 + \varepsilon h_1(0)$. So $h = h_0 + \varepsilon h_1$ satisfies $h(0) > v\|h\|_1$ for small $\varepsilon > 0$. Choose $\eta \in \mathcal{S}(\mathbb{R}^2)$ with $\hat{\eta} \in C_c^\infty(D)$ and $\eta(0) = 1$. Replacing h by $h(rx)\eta(\delta x)$, with $r \uparrow 1$ and $0 < \delta < 2(1-r)$, makes it Schwartz, preserves the Fourier support condition, and converges to h in L^1 with the same value at zero. Thus the strict inequality survives.

For the symbol take $\hat{\varphi}_\ell = a_\ell \in C_c^\infty(2D)$ with $0 \leq a_\ell \leq 1$ and $a_\ell \to 1$ on $2D$. The kernels $a_\ell(\xi + \zeta)$ converge in $L^2(D \times D)$ to the constant 1, whose operator has rank one and norm v; hence $\|H_{\varphi_\ell}\|_{S_p} \to v$ for every $p \geq 2$, while $\langle h, \varphi_\ell \rangle = \int \hat{h}\, a_\ell \to h(0)$. For large ℓ the quotient exceeds 1. As in Ortega-Cerdà and Seip [69], on D^m the tensor powers $h^{\otimes m}$ and $\varphi^{\otimes m}$ have pairing, L^1 norm and S_p norm equal to the mth powers of those on D. □

The proof uses only that $\check{\chi}_D$ is real, so that $h_0 \geq 0$, and that $2D \setminus \{0\}$ contains the support of a nonnegative bump. It therefore applies verbatim to every convex body symmetric about a point, after a modulation that moves the center to 0; polygons with this symmetry included. For such bodies the S_p Nehari constant on $PW(\Omega^m)$ is at least exponential in m for every $p \geq 2$, although for a polygon it is finite when $m = 1$, by the theorem of Bampouras and Perfekt below.

COROLLARY 8.21 (Polydiscs). *Let ψ be the symbol of Bampouras and Perfekt [8], and let C_m be the best constant in $|\langle h, \varphi \rangle| \leq C_m\|h\|_1\|H_\varphi\|_{S_2}$ on $PW(D^m)$. Then*

$$C_1^m \leq C_m \leq \|\psi\|_\infty^{m/2}, \qquad C_1 > 1.$$

PROOF. The lower bound is the tensor argument of Proposition 8.20 applied to near extremals for C_1, and $C_1 > 1$ is that proposition at $p = 2$. Since $\chi_{D^m} * \chi_{D^m} = \omega^{\otimes m}$, (8.2) holds on D^m with this weight. The Fourier transform of $\psi^{\otimes m}$ is $\hat{\psi}^{\otimes m}$, which equals $(1/\omega)^{\otimes m}$ on $(2D)^m$, since tensor products of test functions are dense there, and $\|\psi^{\otimes m}\|_\infty = \|\psi\|_\infty^m$. The computation above, with h ranging over the dense set of $h \in L^1$ with $\hat{h} \in C_c^\infty((2D)^m)$ and Fatou's lemma for the rest, gives $C_m^2 \leq \|\psi\|_\infty^m$. Bampouras's constant, with $n = 2m$, gives an upper bound of the same exponential form. □

Thus there is no dimension-free Paley–Wiener analogue of Helson's inequality: the constants on $PW(D^m)$ grow exponentially in m. For balls the tensor argument

is not available, and the argument of Proposition 8.20 gives only a constant larger than 1; whether the constant for the ball B^n grows with n is open.

For polygons there is no threshold at all. Bampouras and Perfekt [9, Corollary 1.2] prove that every Hankel operator on $PW(P)$, P a polygon, that lies in some S_p, $p < \infty$, has a bounded symbol, by lifting a theorem on the product Hardy space of two half-planes. This is the limit $k = 0$ of the counting above, where the threshold $2(1 + (n-1)/k)$ is infinite, which suggests that the obstruction above S_2 is curvature.

For $2 < p \leq 4$ the disc remains open. Bampouras and Perfekt show that for the disc and $2 \leq p < 4$, where the Besov description of S_p is available [7, Theorem 1.2(2)], Nehari's theorem for S_p implies $\int_{2D} |\hat{h}|^{p'}/\omega \lesssim \|h\|_1^{p'}$ [8, Proposition 7.4]. (The range $p < 2(n+1)/(n-1)$, that is $p < 6$ for the disc, quoted in [8, Section 7] and in the abstract metadata of [7] is larger than the range $p < 2n/(n-1)$ of the theorem itself.) A function h violating this inequality for some $p' \in (4/3, 2)$ would therefore show that Nehari's theorem fails for $S_p(PW(D))$ at the dual exponent p. Their proof of the case $p' = 2$ uses L^2 through the identity $\int |\hat{h}|^2 \hat{\psi} = \langle h * \psi, h \rangle$, which has no analogue for $p' < 2$.

Open

(1) Is $p_* = 4$? By Proposition 7.13 this is the same as asking whether multiplicative Hankel forms in S_4 have bounded symbols.
(2) Problem 7, which decides whether Gaussian methods can go below 4.
(3) A dimension- and degree-free interval $(2, 2+\varepsilon)$ with no strict blocks, or a family of blocks with exponents tending to 2; by Theorem 8.16 such a family must have unbounded rank.
(4) Nehari's theorem for $S_p(PW(D))$, $2 < p \leq 4$; by Bampouras and Perfekt, for $p < 4$ it would give $\int_{2D} |\hat{h}|^{p'}/\omega \lesssim \|h\|_1^{p'}$.
(5) Does the S_2 constant for the ball B^n grow with n?

Part 4

The borderline: the multiplicative Hilbert matrix

CHAPTER 9

Factorization and π

For a Dirichlet series $f = \sum a_n n^{-s}$ write $c_n = 1/(\sqrt{n}\log n)$ and

$$\ell(f) = \sum_{n\geq 2} \frac{a_n}{\sqrt{n}\log n} = \int_0^\infty \big(f(1/2+\sigma) - a_1\big)\,\mathrm{d}\sigma.$$

Let C be the norm of ℓ on $\mathscr{H}_0^1$, the series with $a_1 = 0$. Write $r(\sigma) = (p^{-1/2-\sigma})_p$ for the prime curve, F for the Bohr lift of f, and $J(F) = \int_0^\infty |F(r(\sigma))|\,\mathrm{d}\sigma$. Let C_J be the least constant in $J(F) \leq C_J \|F\|_1$ on H_0^1. Since $|\ell(f)| \leq J(F)$, we have $C \leq C_J$.

Part IV is about four inequalities that share the constant π in one variable (Proposition 9.1).

	One variable	On $\mathbb{T}^\infty$
Hilbert form on $H^2 \times H^2$	π	π [22]; also with the row $m = 1$ adjoined (Theorem 9.2)
Signed functional ℓ on H_0^1	π	finite [70]; $\pi \leq C \leq 5.515$ (Proposition 9.14)
Fejér–Riesz functional J on H_0^1	π	finite [70]; $C \leq C_J \leq 5.515$
Hardy: $\sum \lvert a_n\rvert c_n$ on H^1	π	open (Problem 10); below π on primes up to 10^{216}

PROBLEM 9 (The sharp constant). *Is $C = \pi$, the norm of the multiplicative Hilbert matrix on $\mathscr{H}^2$?*

PROBLEM 10 (Perfekt [70]). *Is $|a_1| + \sum_{n\geq 2} |a_n|/(\sqrt{n}\log n) \lesssim \|f\|_{\mathscr{H}^1}$ for every Dirichlet polynomial f?*

Several different gaps involving π appear in Part IV, and they answer different questions.

- *Operator norm against symbol norm.* The Hilbert matrix has norm π; its smallest symbol has norm $C \geq \pi$. Problem 9 asks whether these agree.
- *Finitely many primes.* On d primes the constant C_d exceeds the H^2 benchmark π_d^* by about six per cent, and for two large primes by $\pi/(2\sqrt{2}) - 1 \approx 11$ per cent (Proposition 9.10). This is an effect of linear symbols at a scale where both constants are far below π. It says nothing about C: for functions of the primes below 10^{216} even the absolute sum is below π.
- *The Gram constant.* The Hellinger–Bessel method gives $\Gamma(1/4)^2/\sqrt{2\pi} \approx 5.244$ near the critical line. That is a limitation of the method, which discards phases, not an excess of C.

- *The Gaussian constant.* $C \geq C_{\mathrm{Gauss}} \geq \pi$ (chapter 11). Here a strict inequality $C_{\mathrm{Gauss}} > \pi$ would be an actual excess, and it would survive the removal of every continuous symbol.

In one variable the four constants coincide for one reason: every H^1 function is a product of two H^2 functions with the same norm, and Hilbert's inequality bounds the product. On $\mathbb{T}^\infty$ the first row survives with the same constant, while factorization fails, even in its weak form, because Nehari's theorem does [69]. Part IV measures what that failure costs, row by row. This chapter treats the H^2 side: the multiplicative Hilbert matrix and its near-extremal vectors (section 9.2), and the equivalent forms of $C = \pi$, each with the reason it is not automatic (section 9.3). It ends with explicit bounds, $\pi \leq C \leq 5.515$ and a constant below π for functions of the primes below 10^{216} (section 9.4), and with an explicit bounded symbol (section 9.5). Chapter 10 treats the relatives of ℓ: absolute values, other weights and exponents, and partial sums; it shows in passing that a counterexample must mix degrees. Chapter 11 then studies the regime where C is decided, a Gaussian scaling limit in which π is very probably exceeded.

Brevig, Perfekt, Seip, Siskakis and Vukotić introduced the multiplicative Hilbert matrix, showed that its norm on $\mathscr{H}^2$ is π, and observed that it has a bounded symbol exactly when $C < \infty$ [22]; the question became Problem 3.2 of Saksman and Seip [80]. Partial answers replaced $\mathscr{H}^1$ by $\mathscr{H}^p$, $p > 1$, by Bayart and Brevig [11], and $\log n$ by $(\log n)^\beta$, $\beta > 1$, by Bondarenko, Brevig, Saksman and Seip [15]. Perfekt [70, Theorem 1.1 and Corollary 1.2] proved $C_J < \infty$, with a constant of order 10^6; so $C < \infty$ and the multiplicative Hilbert matrix has a bounded symbol. He also noted that Problem 10 is open. Perfekt and Pushnitski proved that the multiplicative Hilbert matrix has purely absolutely continuous spectrum $[0, \pi]$ [72]. Results below that carry no other attribution are, to our knowledge, new.

The *sharp Fejér–Riesz conjecture* $C_J = \pi$ implies $C = \pi$. Both are very probably false (remark 11.10).

9.1. One variable

PROPOSITION 9.1. *Let* $f = \sum_{k\geq 0} a_k z^k \in H^1(\mathbb{T})$, *and let* $g = \sum_j b_j z^j$ *and* $h = \sum_k d_k z^k$ *belong to* $H^2(\mathbb{T})$.

(1) (Hilbert) $|\sum_{j,k} b_j d_k/(j+k+1)| \leq \pi \|g\|_2 \|h\|_2$.
(2) (Fejér–Riesz) $\int_0^1 |f(x)|\,\mathrm{d}x \leq \pi \|f\|_1$.
(3) (Hardy) $\sum_k |a_k|/(k+1) \leq \pi \|f\|_1$.
(4) $|\sum_k a_k/(k+1)| \leq \pi \|f\|_1$.

In each, π *is best possible.*

PROOF. (1) The entries are positive, so we may take $b, d \geq 0$. Put $A = j + 1/2$. The function $y \mapsto 1/((A+y)\sqrt{y})$ is convex on $(0,\infty)$, so its value at $k+1/2$ is at most its integral over $(k, k+1)$, and

$$\sum_{k\geq 0} \frac{(k+1/2)^{-1/2}}{j+k+1} \leq \int_0^\infty \frac{\mathrm{d}y}{(A+y)\sqrt{y}} = \pi (j+1/2)^{-1/2}.$$

Schur's test with the weight $(k+1/2)^{-1/2}$ on rows and columns gives the bound. The same test applies to the smaller entries $\rho^{j+k+1}/(j+k+1)$, $0<\rho<1$.

(2) Write $f=IO$ with I inner and O outer, and put $g=IO^{1/2}$, $h=O^{1/2}$. Then $f=gh$ and $\|g\|_2^2=\|h\|_2^2=\|f\|_1$. For $0<\rho<1$, $\int_0^\rho |g(x)|^2\,\mathrm{d}x = \sum_{j,k} b_j\overline{b_k}\rho^{j+k+1}/(j+k+1) \le \pi\|g\|_2^2$ by (1), and similarly for h. Cauchy–Schwarz and $\rho\to 1$ give (2).

(3) With $f=gh$ as in (2), $|a_k|\le\sum_{i+j=k}|b_i||d_j|$, so (1) applied to $(|b_i|)$ and $(|d_j|)$ gives (3).

(4) For polynomials $\sum_k a_k/(k+1)=\int_0^1 f(x)\,\mathrm{d}x$; both sides are continuous on H^1 by (2) and (3), so (4) follows from either.

Sharpness. Let $b_k=(k+1)^{-1/2-\varepsilon}$, $g=\sum b_k z^k$ and $f=g^2$. The coefficients of f are positive, $\|f\|_1=\|g\|_2^2$, and each left side in parts (1)–(4), with $h=g$, equals $Q=\sum_{j,k} b_jb_k/(j+k+1)$. Since $j+k+1\le(j+1)+(k+1)$ and $y\mapsto y^{-1/2-\varepsilon}/(J+y)$ decreases, with $J=j+1$,

$$\sum_k \frac{b_k}{j+k+1} \ge \int_1^\infty \frac{y^{-1/2-\varepsilon}\,\mathrm{d}y}{J+y} \ge J^{-1/2-\varepsilon}\Big(I_\varepsilon - \frac{J^{-1/2+\varepsilon}}{1/2-\varepsilon}\Big),$$

where $I_\varepsilon=\int_0^\infty u^{-1/2-\varepsilon}(1+u)^{-1}\,\mathrm{d}u$. Summing against $b_j=J^{-1/2-\varepsilon}$ gives $Q\ge I_\varepsilon\|g\|_2^2-O(1)$. As $\varepsilon\to0$, $\|g\|_2^2=\sum_J J^{-1-2\varepsilon}\to\infty$ and $I_\varepsilon=\pi/\cos(\pi\varepsilon)\to\pi$, so $Q/\|f\|_1\to\pi$. □

The dictionary to Dirichlet series is $z^k\leftrightarrow n^{-s}$, $\int_0^1 x^k\,\mathrm{d}x=1/(k+1)\leftrightarrow\int_0^\infty n^{-1/2-\sigma}\,\mathrm{d}\sigma=c_n$, and $1/(j+k+1)\leftrightarrow c_{mn}$. Only the first step of the proof, the factorization $f=gh$, has no counterpart on $\mathbb{T}^\infty$.

9.2. The multiplicative Hilbert matrix

Let $\mathscr{M}=(c_{mn})_{m,n\ge2}$, of norm π [22], and let $\mathscr{R}=(c_{mn})_{m\ge1,n\ge2}$ be $\mathscr{M}$ with the row $m=1$ adjoined. For $g\in H^2$, $h\in H^2_0$ one has $\ell(gh)=\langle\mathscr{R}\hat{h},\bar{\hat{g}}\rangle$, so $C\ge\|\mathscr{R}\|$.

THEOREM 9.2. *$\|\mathscr{R}\|=\pi$. If π_d^* is the norm of $\mathscr{R}$ restricted to the integers $\mathcal{S}_d$ generated by the first d primes, then $\pi_d^*\uparrow\pi$, and*

$$\frac{C}{\pi}=\lim_{d\to\infty}\frac{C_d}{\pi_d^*},$$

where C_d is the norm of ℓ on the same restricted $\mathscr{H}_0^1$.

PROOF. We apply Schur's test with weights $u_n=(n\log n)^{-1/2}$ on columns and $v_1=1$, $v_m=(m\log m)^{-1/2}$ on rows; it suffices that $\sum_n c_{mn}u_n\le\pi v_m$ and $\sum_m c_{mn}v_m\le\pi u_n$.

Rows $m\ge2$. With $x=\log m$, the summand $1/(t\sqrt{\log t}(x+\log t))$ decreases in t, so $v_m^{-1}\sum_n c_{mn}u_n\le\sqrt{x}\int_1^\infty \mathrm{d}t/(t\sqrt{\log t}(x+\log t))=2\sqrt{x}\int_0^\infty \mathrm{d}r/(x+r^2)=\pi$.

Row $m=1$. With $g(t)=1/(t\log^{3/2}t)$ decreasing, $\sum_n g(n)\le g(2)+g(3)+2/\sqrt{\log 3}<3.091<\pi$, using $\log2>0.69$ and $\log3>1.09$.

Columns. With $y = \log n$ and $f_y(t) = 1/(t\sqrt{\log t}(\log t + y))$,

$$u_n^{-1} \sum_m c_{mn} v_m = \frac{1}{\sqrt{y}} + \sqrt{y} \sum_{m \geq 2} f_y(m) \leq \frac{1}{\sqrt{y}} + \sqrt{y}\Big(\frac{\pi}{\sqrt{y}} - \int_1^2 f_y + f_y(2) \Big),$$

since $\int_1^\infty f_y = \pi/\sqrt{y}$. So it suffices that $\int_1^2 f_y - f_y(2) \geq 1/y$ for $y \geq \log 2$. With $a = \log 2$, $A = \sqrt{a}$ and $u = \sqrt{a/y} \in (0, 1]$, this reads

$$\Psi(u) = 2A \frac{\arctan u}{u} - \frac{1}{2A(1+u^2)} - 1 \geq 0.$$

Put $t = u^2$. Since $\arctan u/u = \int_0^1 (1 + tx^2)^{-1}\,\mathrm{d}x$, completing the square gives $\arctan u/u \geq 2\int_0^1 P - \int_0^1 (1+tx^2)P^2$ for every polynomial P, and the best P in the span of 1 and x^2 gives

$$\frac{\arctan u}{u} \geq \frac{105 + 55t}{105 + 90t + 9t^2}.$$

Substituting, $\Psi \geq 0$ follows from the positivity on $[0, 1]$ of

$$(420A^2 - 210A - 105) + (640A^2 - 390A - 90)t + (220A^2 - 198A - 9)t^2 - 18At^3,$$

which is concave there, since $220A^2 - 198A - 9 < 0$, and positive at $t = 0$ and $t = 1$; the bounds $0.8325 < A < 0.8326$ suffice for all three signs.

Limits. Monotonicity in d is clear, and every finitely supported test vector lives in some $\mathcal{S}_d$. The same density argument gives $C_d \uparrow C$. □

Equivalently $\mathscr{R}^*\mathscr{R} = \mathscr{M}^2 + c \otimes c \leq \pi^2$: the rank-one term is absorbed although it is not zero. Since $\pi_d^* \leq C_d < 1.91$ for $d \leq 7$ (Proposition 9.14), finite-prime numerics are far from the regime in which C/π is decided.

9.2.1. Near-extremal vectors

Let $N : \ell^2(\{2, 3, \dots\}) \to L^2(0, \infty)$, $Nb(\sigma) = \sum b_n n^{-1/2-\sigma}$, so $\mathscr{M} = N^*N$, and let S_m be the isometry $e_n \mapsto e_{mn}$. Then $NS_m = w_m N$ with $w_m(\sigma) = m^{-1/2-\sigma}$.

THEOREM 9.3 (Slow variation). *If b_k are unit vectors with $\langle \mathscr{M} b_k, b_k\rangle \to \pi$, then for all positive integers m, m'*

$$\sum_n b_k(mn)\overline{b_k(m'n)} \to (mm')^{-1/2}.$$

PROOF. Put $g_k = Nb_k$. From $\|(\pi - \mathscr{M})b\|^2 \leq \pi\langle(\pi - \mathscr{M})b, b\rangle$, $b_k = \pi^{-1}N^*g_k + o(1)$. Weak limits are eigenvectors, and $\mathscr{M}$ has none [72], so $b_k \rightharpoonup 0$. As N restricted to (δ, ∞) is Hilbert–Schmidt, $\int_\delta^\infty |g_k|^2 \to 0$, hence $\|w_m g_k - m^{-1/2} g_k\| \to 0$. Therefore $S_m^* b_k = \pi^{-1} N^* w_m g_k + o(1) = m^{-1/2}\pi^{-1}\mathscr{M} b_k + o(1) = m^{-1/2} b_k + o(1)$. Hence $\sum_n b_k(mn)\overline{b_k(m'n)} = \langle S_m^* b_k, S_{m'}^* b_k\rangle \to (mm')^{-1/2}$. □

Since $\mathscr{M}$ is real and positive, $|\ell(h^2)| = |\langle \mathscr{M} b, \bar{b}\rangle| \leq \langle \mathscr{M} b, b\rangle$ for $h = \sum b_n n^{-s}$, so the theorem applies to every sequence with $\|h_k\|_2 = 1$ and $|\ell(h_k^2)| \to \pi$.

COROLLARY 9.4 (Marginals and fixed factors). *Let S be a finite set of primes and $r_0 = (p^{-1/2})_{p \in S}$. For such h_k:*

(1) the marginal of $|h_k|^2\,\mathrm{d}m$ *on* $\mathbb{T}^S$ *tends weakly to the Poisson measure* $P_{r_0}\,\mathrm{d}m_S$*;*
(2) for every polynomial $u \neq 0$ *in the variables of* S*, overlapping those of* h_k *or not,*

$$\limsup_k \frac{|\ell(uh_k^2)|}{\|uh_k^2\|_1} \le \pi\,\frac{|u(r_0)|}{\int |u| P_{r_0}\,\mathrm{d}m_S} \le \pi.$$

PROOF. (1) For coprime S-integers m, m', the Fourier coefficient at $\kappa(m)-\kappa(m')$ is $\sum_{t\ge1} b_k(mt)\overline{b_k(m't)}$, with $b_k(1)=0$. The term $t=1$ tends to zero by weak convergence, and Theorem 9.3 gives the limit $(mm')^{-1/2}$ for the remaining sum. These are the Fourier coefficients of P_{r_0}, and every character of $\mathbb{T}^S$ arises from such a reduced pair. This proves weak convergence.

(2) $\ell(uh_k^2) = \int_0^\infty u(r(\sigma))g_k(\sigma)^2\,\mathrm{d}\sigma$ and $|g_k|^2$ concentrates at $\sigma=0$, so this is $u(r_0)\ell(h_k^2)+o(1)$; the denominator is $\int|u||h_k|^2\,\mathrm{d}m \to \int|u|P_{r_0}\,\mathrm{d}m_S$ by (1). □

For a single low prime no extremality is needed: if $g \in H^1$ depends on one prime and $h \in H_0^2$ on the others, then $|\ell(gh^2)| \le \pi\|gh^2\|_1$, by inner-outer factorization of g and Theorem 9.2.

9.3. Equivalent forms and obstructions

Let $G = \widehat{\mathbb{Q}_{>0}} \cong \mathbb{T}^\infty$ and $P = \{q \in \mathbb{Q}_{>0} : q \ge 1\}$, a total positive cone containing $\mathbb{N}$. For $b : P \to \mathbb{C}$ let $\mathcal{H}_b(u,v) = \sum_{q,r\in P} \hat u(q)\hat v(r) b(qr)$ on $H^2_P(G)$.

THEOREM 9.5 (Completion principle).

$$C = \min\bigl\{\|\mathcal{H}_b\| : b(n) = c_n \text{ for all integers } n \ge 2\bigr\},$$

with $b(1)$ *free. So* $C=\pi$ *if and only if the multiplicative Hilbert coefficients extend to the ordered rationals with Hankel norm* π*.*

PROOF. Extend ℓ from $\mathscr{H}^1_0$ to $L^1(G)$ by Hahn–Banach, without increasing its norm, and put $b(q) = \Lambda(\chi_q)$. For analytic polynomials u, v with spectrum in P,

$$|\mathcal{H}_b(u,v)| = |\Lambda(uv)| \le C\|uv\|_1 \le C\|u\|_2\|v\|_2.$$

Conversely, ordered-group Nehari [94] gives a symbol of norm $\|\mathcal{H}_b\|$ representing ℓ on $\mathscr{H}^1_0$. The value $b(1)$ is free because the original domain omits constants. □

REMARK 9.6 (What the reformulation does and does not give). By compactness, $C>\pi$ would be witnessed by one finite set $F \subset P$ and one trace-class matrix X on F with $\sum_{qr=t} X(q,r) = 0$ for $t \notin \mathbb{N}_{\ge2}$; finitely supported rank-one $X = u \otimes v$ never works: by Gauss's lemma in the Laurent ring $u = z^\gamma u_0$ and $v = z^\delta v_0$ with $\gamma+\delta \ge 0$, so moving $z^{\gamma+\delta}$ onto v gives integer spectra, and the form is a compression of $\mathscr{R}$. By Helson–Lowdenslager factorization [45], $C=\pi$ is equivalent to $|\sum_n c_n \sum_{qr=n}\hat u(q)\hat v(r)| \le \pi\|u\|_2\|v\|_2$ for $u, v \in H^2_P(G)$ whose product is an integer Dirichlet polynomial with zero constant term; the factors need not have spectrum in $\mathbb{N}$, already for z_1+z_2. The natural completions all fail: continuous interpolation $b(q) = F(\log q)$, the integrated flat representatives, and $b=0$ off $\mathbb{N}$ are unbounded (localizing the flat representatives does give a bounded completion, Theorem 9.18), and no completely positive Schur damping fixes the integer entries

non-trivially. A bounded completion must cancel the non-square-summable entries $\sum_m c_{Am}c_{Bm}$, $2 \le A < B$ coprime, through its non-integer layers.

9.3.1. Curve factorization

PROPOSITION 9.7 (Curve factorization). *For $F \in H^1_0$ let $\Phi(F)$ be the infimum of $\|G\|_2\|H\|_2/\|F\|_1$ over $G \in H^2_0$, $H \in H^2$ with $|F(r(\sigma))| \le |G(r(\sigma))H(r(\sigma))|$ for all $\sigma > 0$. Then $J(F) \le \pi\Phi(F)\|F\|_1$.*

PROOF. $J(F) \le \sum_{m\ge1,n\ge2} c_{mn}|\hat H(m)||\hat G(n)| \le \pi\|G\|_2\|H\|_2$ by Theorem 9.2, which applies to absolute values since the matrix is positive. □

In one variable inner-outer factorization gives $\Phi \le 1$. On $\mathbb{T}^\infty$ global weak factorization fails [69], but Φ asks only for a majorant on the curve, which is far weaker.

PROPOSITION 9.8 (Curve factorization fails). *For every $F \in H^1_0$ and $\sigma > 0$, with $\zeta_\sigma = \zeta(1+2\sigma)$,*

$$\Phi(F)\|F\|_1 \ge \frac{|F(r(\sigma))|}{\sqrt{\zeta_\sigma(\zeta_\sigma-1)}}.$$

Consequently $\Phi(K_{r(\sigma)}(K_{r(\sigma)}-1)) > 1$ for every $\sigma > 0$, and $\Phi(F) > 1.0151$ for $f = 2^{-s} + (2/5)3^{-s}$.

PROOF. For admissible $G \in H^2_0$, $H \in H^2$, $|G(r)H(r)| \le \|G\|_2\sqrt{\zeta_\sigma - 1}\cdot\|H\|_2\sqrt{\zeta_\sigma}$. For $F = K(K-1)$ with $K = K_{r(\sigma)}$, $F(r(\sigma)) = \zeta_\sigma(\zeta_\sigma-1)$ while $\|F\|_1 < \|K\|_2\|K-1\|_2 = \sqrt{\zeta_\sigma(\zeta_\sigma-1)}$, strictly, because $|1-1/K|$ is not constant. For the binomial, Parseval applied to $(1+2z/5)^{1/2}$ gives $\|F\|_1 \le 1 + 1/25 + (2/5)^4/(64(1-(2/5)^2)) < 1.04048$. At $\sigma = 3$, $|f(7/2)| > 0.09694$, while $\sqrt{\zeta(7)(\zeta(7)-1)} < 0.09178$ by $\zeta(7) \le \sum_{n=1}^6 n^{-7} + 1/(6\cdot6^6)$. Thus $\Phi(F) > 0.09694/(1.04048\cdot0.09178) > 1.0151$. □

The mechanism is again Ortega-Cerdà–Seip: for $2^{-s} + \varepsilon 3^{-s}$ the evaluation gains to first order in ε at a suitable point while the L^1 norm pays only to second order. Near extremizers the picture is different: for the two-prime near-extremizer of Proposition 9.14, the convex program for Φ and its dual, solved numerically, agree at $\Phi \approx 0.994$. So the reduction survives in the weaker form that matters, and in that form it is an equivalence: $C = \pi$ if and only if $\Phi(F_k) \to 1$ along some extremizing sequence. For the converse, take unit vectors $h_k \in \mathscr{H}^2_0$ with nonnegative coefficients along which the multiplicative Hilbert form tends to π [22], and put $F_k = h_k^2$. Then $\|F_k\|_1 = 1$, the functional at F_k is that quadratic form, and $G = H = h_k$ shows $\Phi(F_k) \le 1$; if $C = \pi$ the sequence is extremizing, and Proposition 9.7 gives $\Phi(F_k) \to 1$.

REMARK 9.9 (A warning from one variable). Building a symbol from the phases of extremal squares, $\psi = \pi F/|F|$, does not work directly. In one variable, with $F = (1-rz)^{-1}$ and $r \to 1$, the limit symbol has analytic coefficients $1/(k+1/2)$, so it represents $\sum a_k/(k+1/2)$ rather than $\sum a_k/(k+1)$. The multiplicative analogue is the family $\ell_c(f) = \sum_{n\ge2} a_n/(\sqrt{n}(\log n + c)) = \int_0^\infty e^{-c\sigma} f(1/2+\sigma)\,\mathrm{d}\sigma$ on $\mathscr{H}^1_0$, for $c \ge 0$. Their rectangular Hankel forms all have norm π. A phase-limit symbol would settle one member ℓ_{c_*}; passing to $c = 0$ is the completion problem of Theorem 9.5.

9.3.2. Small primes: a linear-symbol effect

Numerically $C_d/\pi_d^* \approx 1.06$ for $d = 2, 3, 4$. For primes $p < q$ let $C_{p,q}$ and $\pi_{p,q}^*$ be the two-prime analogues.

PROPOSITION 9.10. *As $p, q \to \infty$ with $q/p \to 1$, $C_{p,q}/\pi_{p,q}^* \to \pi/(2\sqrt{2}) = 1.1107\ldots$.*

PROOF. Put $a = c_p$, $b = c_q$. The functional is $a\hat{f}(p)+b\hat{f}(q)+R(f)$. The remaining coefficients satisfy

$$\sum_{j+k\geq 2} (j+1)(k+1)c_{p^j q^k}^2 \leq \frac{1}{4\log^2 p} \sum_{j+k\geq 2} (j+1)(k+1)p^{-j-k}$$
$$= O\big(p^{-2}(\log p)^{-2}\big) = o(a^2).$$

Helson's inequality therefore gives $\|R\| = o(a)$. The same sum bounds the squared Hilbert–Schmidt norm of the remainder of the rectangular matrix, whose first row has entries a, b. Hence $C_{p,q} = D(az_1+bz_2)+o(a)$ and $\pi_{p,q}^* = \sqrt{a^2+b^2}+o(a)$, where D denotes the H^1 dual norm. Now $\|z_1+z_2\|_1 = 4/\pi$, and $(\pi/2)(z_1+z_2)/|z_1+z_2|$ represents z_1+z_2 with norm $\pi/2$. Testing on z_1+z_2 gives the reverse inequality, so $D(z_1+z_2) = \pi/2$. Since $b/a \to 1$, the claimed ratio follows. □

(p,q)	$(2,3)$	$(5,7)$	$(11,13)$	$(101,103)$	$(1009,1013)$
$C_{p,q}/\pi_{p,q}^* \geq$	1.0617	1.0905	1.1038	1.1101	1.1107

The same ratio appears exactly as a factorization gap: for $q_d = \prod_{j\leq d}(z_{2j-1} + z_{2j})$,

$$\|q_d\|_1 = (4/\pi)^d, \qquad \|q_d\|_{H^2\odot H_0^2} = 2^{d/2}, \qquad \frac{\|q_d\|_{H^2\odot H_0^2}}{\|q_d\|_1} = \Big(\frac{\pi}{2\sqrt{2}}\Big)^d.$$

Since $|\ell(f)| \leq \pi\|f\|_{H^2\odot H_0^2}$ by Theorem 9.2, this shows that the rectangular norm alone can never prove $C = \pi$: the projective norm and the H^1 norm are not equivalent on H_0^1.

9.4. Perfekt's theorem and explicit bounds

Corollary 2.6 reproves Perfekt's theorem with the explicit bound $\pi \leq C < 9$; this section lowers 9 to 5.515, and below π on primes up to 10^{216}.

REMARK 9.11 (The limit of the unrestricted Gram estimate). For Lebesgue measure on $(0, \delta)$ along the curve, the Gram kernel of Theorem 2.1 is asymptotically $2^{-1/2}(\sigma\tau)^{-1/4}(\sigma+\tau)^{-1/2}$, homogeneous of degree -1, whose Mellin multiplier is $\Gamma(1/4)^2/\sqrt{2\pi} = 5.2441\ldots$. The H^2 analogue gives the Carleman kernel $1/(\sigma+\tau)$, of norm π. So, up to uniformity of the asymptotics, the unrestricted Gram estimate of Theorem 2.1 stops at 5.244. This does not rule out arguments that use the special structure of $u = |F|^{1/2}$. Since J is subadditive and $\|\sum F_i\|_1 \geq \sum\|F_i\|_1 - 2\sum_{i<j}(\|F_i\|_1\|F_j\|_1)^{1/2}H(\nu_i,\nu_j)$, profiles at well-separated scales cannot create an excess either; extremizing sequences for $\mathscr{M}$ spread over a long range of $\log(1/\sigma)$, so the sharp constant is a per-unit-scale quantity.

The local constant of Theorem 2.9 handles $\sigma \le 2$. Beyond that, a second refinement uses Helson's inequality in the odd primes together with an exact factorization in z_2.

LEMMA 9.12 (Vector-valued outer factorization). *If V is a nonzero $\mathbb{C}^N$-valued analytic polynomial on $\mathbb{T}$, there are an outer $g \in H^2(\mathbb{T})$ and $W \in H^2(\mathbb{T};\mathbb{C}^N)$ with $V = gW$, $\|g\|_2^2 = \int \|V\|\,\mathrm{d}m$ and $\|W\|_2^2 \le \int \|V\|\,\mathrm{d}m$.*

PROOF. Let g be outer with $|g|^2 = \|V\|$ on $\mathbb{T}$. Since $\log\|V\|$ is subharmonic, $\|V\| \le |g|^2$ in $\mathbb{D}$, so $W = V/g$ is holomorphic with $\|W\| \le |g|$. $\square$

For a positive measure ν on $(0,\infty)$ put $c_N(\nu) = \int N^{-1/2-\sigma}\,\mathrm{d}\nu$ and let A_ν be the nonnegative matrix with rows $k \ge 0$, columns (j,m) with $j \ge 0$, m odd and $(j,m) \ne (0,1)$, and entries $\sqrt{d(m)}\,c_{2^{k+j}m}(\nu)$.

THEOREM 9.13 (Tail bound). *For $F \in H^1_0(\mathbb{T}^\infty)$, $\int |F(r(\sigma))|\,\mathrm{d}\nu(\sigma) \le \|A_\nu\|\,\|F\|_1$, and $\|A_\nu\|^2 \le \sum_{N\ge 2} d(N) c_N(\nu)^2 - \sum_{k\ge 1} c_{2^k}(\nu)^2$. The same holds on $\mathcal{P}$-smooth functions with the sums restricted accordingly.*

PROOF. First let F be a polynomial and write $F = \sum_{m\ \mathrm{odd}} F_m(z_2) w^{\kappa(m)}$. For each z_2, Helson's inequality in w gives

$$\left(\sum_m \frac{|F_m(z_2)|^2}{d(m)}\right)^{1/2} \le \int |F(z_2,w)|\,\mathrm{d}m(w).$$

Apply Lemma 9.12 to $V = (F_m/\sqrt{d(m)})_m$: $F_m = \sqrt{d(m)} g W_m$ with $\|g\|_2\|W\|_2 \le \|F\|_1$ and $\hat W_1(0) = 0$. Write $g(z) = \sum_{k\ge 0} g_k z^k$ and $W_m(z) = \sum_{j\ge 0} w_{jm} z^j$. Then $a_{2^e m} = \sqrt{d(m)} \sum_{k+j=e} g_k w_{jm}$, so Tonelli's theorem gives

$$\begin{aligned}\int |F(r(\sigma))|\,\mathrm{d}\nu &\le \sum_{n\ge 2} |a_n| c_n(\nu) \\ &\le \sum_{k,j,m} \sqrt{d(m)}\, c_{2^{k+j}m}(\nu) |g_k||w_{jm}| \le \|A_\nu\|\|g\|_2\|W\|_2.\end{aligned}$$

Here m is odd and $(j,m) \ne (0,1)$, since $w_{0,1} = 0$. Density and Fatou's lemma extend the bound to H^1_0. For the Hilbert–Schmidt bound, the entries with $k+j = e$ for $N = 2^e m$ number $e+1$, and $(e+1)d(m) = d(N)$; when $m = 1$ the excluded column removes one entry. This gives the stated subtraction. $\square$

PROPOSITION 9.14 (Computer-assisted bounds). *The following bounds hold.*

1. *For every $F \in H^1_0$, $J(F) \le 5.5150\,\|F\|_1$. Hence $\pi \le C \le C_J \le 5.5150$.*
2. *If $F \in H^1_0$ uses only primes $p \le P$ with $\prod_{p\le P}(1-1/p)^{-1} \le 890$, in particular $P \le 10^{216}$, then $J(F) \le \sum_{n\ge 2} |a_n|/(\sqrt{n}\log n) \le 3.1407\|F\|_1 < \pi\|F\|_1$.*
3. *For the first d primes,*

d	1	2	3	4	5	6	7
$C_d \le$	1.19914	1.58582	1.72142	1.80494	1.84758	1.88301	1.90873

Moreover $C_1 = \pi_1^*$, $1.54792 \le C_2 \le 1.58582$ and $1.0617 \le C_2/\pi_2^* \le 1.0878$.

PROOF. (1) On $(0,2)$ use Theorem 2.9. On $(2,\infty)$ use Theorem 9.13 with Lebesgue measure: the finite part of A_ν with $N \le 2000$ has norm at most 0.27086 by Schur's test with weights verified in ball arithmetic, and the rest has Hilbert–Schmidt norm below 10^{-6}.

(2) The proof of Theorem 9.13 bounds $\sum|a_n|c_n(\nu)$, not only $\int|F(r(\sigma))|\,\mathrm{d}\nu$: it estimates each coefficient $a_{2^e m} = \sqrt{d(m)}\sum_{k+j=e}\hat g(k)\widehat{W_m}(j)$ in absolute value, so that $\sum|a_n|c_n(\nu) \le \langle A_\nu|\widehat{W}|,|\hat g|\rangle$. With Lebesgue measure it gives $J(F)^2/\|F\|_1^2 \le \int_0^\infty \sigma(\Pi_P(\sigma)^2-1)\,\mathrm{d}\sigma - \lambda_0$, where $\Pi_P(\sigma) = \prod_{p\le P}(1-p^{-1-\sigma})^{-1} \le \min\{\zeta(1+\sigma),\Pi_P(0)\}$ and $\lambda_0 = \mathrm{Li}_2(1/2)/\log^2 2$. With $\Pi_P(0) \le 890$ the integral is at most 11.07528 (ball arithmetic), so the bound is $\sqrt{9.86343} < 3.1407$. The bound $\prod_{p\le x}(1-1/p)^{-1} < e^{\gamma_{\mathrm{E}}}\log x\,(1+1/(2\log^2 x))$ of Rosser and Schoenfeld [79] gives 885.9 at $x = 10^{216}$.

(3) Upper bounds: Theorem 9.13 with Schur weights verified in ball arithmetic and a Hilbert–Schmidt tail. $d=1$: Nehari's theorem in one variable identifies C_1 with the Hankel matrix that defines π_1^*. Lower bound for C_2: an explicit polynomial in z_2, z_3 of degree at most 11 in each variable, found by second-order cone programming, with $\ell(q) = 1.5487079\ldots$ enclosed in ball arithmetic and $\|q\|_1 \le 1.0005030$ by a midpoint rule with explicit first- and second-order Taylor errors on an 8192^2 grid. The grid values are computed in binary64 as products of small matrices of exponentials enclosed in ball arithmetic, with an a priori bound for the rounding error that uses only the IEEE model for addition and multiplication. □

The tail bound is nearly tight for its part: $F = z_2$ has $\int_2^\infty|F(r)|\,\mathrm{d}\sigma = 2^{-5/2}/\log 2 \approx 0.255$. No refinement of this scheme reaches $\Gamma(1/4)^2/\sqrt{2\pi}$ itself, and making finitely many primes exact does not lower the boundary constant: the modified kernel is still asymptotic to the kernel L_0 of Theorem 2.9 near 0.

By (2), a counterexample to $C = \pi$ must involve a prime above 10^{216}. The finite-prime constants themselves are much smaller, and by (3) the two-prime excess of section 9.3.2 is real and capped, between six and nine per cent.

A certified three-prime gap

Let $\mathcal{S}_3 = \{2^a3^b5^c : a,b,c \ge 0\}$ and $c_N = (\sqrt{N}\log N)^{-1}$. The benchmark π_3^* is the norm of $\mathscr{R}_3 = (c_{mn})$, with rows $m \in \mathcal{S}_3$ and columns $n \in \mathcal{S}_3 \setminus \{1\}$, as in Theorem 9.2.

THEOREM 9.15. *One has* $C_3 > 1.599 > 1.5624 > \pi_3^*$. *In particular* $C_3/\pi_3^* > 1.0234$.

PROOF OF $C_3 > 1.599$. Let $q(z_2,z_3) = \sum_{j+k>0} a_{jk} z_2^j z_3^k$ be the dyadic polynomial specified by the 143 coefficients in the file `explicit-bounds/c2_witness.json` of chapter A. Each hexadecimal binary64 coefficient is interpreted as its exact dyadic rational, not as an interval of possible inputs. The existing two-prime certificate in part (3) of Proposition 9.14 gives $\|q\|_1 \le 1.0005030$. Rerunning its midpoint-grid certificate `c2_lower.py` at grid size 8192 gives the upper bound 1.0005029281700757, which is below the stated rounded bound.

Direct finite sums in Arb ball arithmetic at 100-bit precision give

$$A = \ell(q) > 1.5487079279756398,$$
$$B = \ell(5^{-s}q) > 0.2877147579857604.$$

Explicitly, the terms in these two sums are respectively $a_{jk}/\big(\sqrt{2^j3^k}\log(2^j3^k)\big)$ and $a_{jk}/\big(\sqrt{5\cdot 2^j3^k}\log(5\cdot 2^j3^k)\big)$. The script `three-prime/pi3_upper.py` checks the exact input file's SHA-256 digest and certifies both strict inequalities without converting their endpoints to binary64.

Put $Q(z_2,z_3,z_5) = q(z_2,z_3)(1+z_5/3)$. This polynomial has zero constant term and uses only the first three primes. Independence of the Haar variables gives

$$\|Q\|_1 = \|q\|_1 M, \qquad \ell(Q) = A + B/3, \qquad M = \int_{\mathbb{T}} |1+z/3|\,\mathrm{d}m(z).$$

Expanding $(1+z/3)^{1/2}$ and using Parseval gives

$$M = \sum_{k=0}^{\infty} \binom{1/2}{k}^2 3^{-2k} \le 1 + \frac{1}{36} + \frac{1}{4608} = \frac{1579}{1536}.$$

Indeed, $\binom{1/2}{2}^2 = 1/64$ and these squared coefficients decrease for $k \ge 2$, since their successive ratio is $\big((k-1/2)/(k+1)\big)^2 < 1$. The binomial series converges absolutely on $|z|=1$, so Parseval applies, and the terms with $k \ge 2$ sum to at most $(1/64)\sum_{k\ge 2} 3^{-2k} = 1/4608$. Consequently

$$C_3 \ge \frac{A + B/3}{1.0005030\,(1579/1536)} > 1.5990217 > 1.599.$$

The last comparisons follow either by rational arithmetic with the displayed decimal lower bounds, or by the accompanying ball-arithmetic certificate. □

Proof of $\pi_3^* < 1.5624$. Put $N_0 = 10^{12}$ and split $\mathscr{R}_3 = T + E$, where T keeps the entries with $mn \le N_0$ and E the others. The 3429 elements of $\mathcal{S}_3 \cap [1, N_0]$ index the rows of T, and the 3428 other than 1 its columns; all other rows and columns of T vanish, so $\|T\|$ is the norm of this finite matrix.

The finite part. Let u and v be the moduli of the leading singular vectors of T computed in binary64, with entries below 10^{-12} times the maximum raised to that value, so that $u, v > 0$, and read as exact dyadic rationals. Schur's test states that if $(Tv)_m \le Lu_m$ and $(T^{\mathsf{T}}u)_n \le Lv_n$ for all m, n, then $\|T\| \le L$. These 6857 inequalities were verified for $L = 1.56179569435114$ (rounded up in the last digit), with membership $mn \le N_0$ decided in exact integer arithmetic: the 3429 distinct entries c_N are enclosed in Arb ball arithmetic at 80-bit precision and rounded up to binary64, each sum is a correctly rounded binary64 sum of rounded products, and the exact sum is at most the computed one divided by $(1-2^{-53})^2$. The floating-point leading singular value of T is 1.5617956943511; the weights are stored with the script.

The tail. Let $d(N)$ be the number of divisors of $N \in \mathcal{S}_3$, so that at most $d(N)$ entries of E equal c_N. Since $N^{-1/2}/\log^2 N$ decreases for $N > 1$,

$$\|E\|^2 \le \|E\|_{\mathrm{HS}}^2 \le \sum_{\substack{N \in \mathcal{S}_3 \\ N > N_0}} \frac{d(N)}{N \log^2 N} \le \frac{N_0^{-1/2}}{\log^2 N_0} \sum_{N \in \mathcal{S}_3} \frac{d(N)}{\sqrt{N}}$$
$$= \frac{N_0^{-1/2}}{\log^2 N_0} \prod_{p \in \{2,3,5\}} (1 - p^{-1/2})^{-2},$$

which gives $\|E\| < 0.000529$. Hence $\pi_3^* \le \|T\| + \|E\| < 1.5624$, and $C_3/\pi_3^* > 1.5990217/1.5624 > 1.0234$. The script `three-prime/pi3_upper.py` performs both steps and gives $\pi_3^* < 1.56232458$ and $C_3/\pi_3^* > 1.02348$. □

9.5. An explicit symbol

The symbol of Corollary 4.18 and Perfekt's symbol both come from the Hahn–Banach theorem. Perfekt's proof of $C_J < \infty$ does use explicit localized representatives (Lemma 1.2), but on $\mathbb{T}^d$ with d chosen after F, so that a prime tail is small; the symbol is then taken from Hahn–Banach. On $\mathbb{T}^\infty$ the energy bound holds with the full prime sum, so all scales can be used at once, and the representatives assemble into one bounded function.

We localize the flat representative at each logarithmic scale. The sets are chosen so that representatives from scales at least L apart have disjoint supports. At most L terms can then contribute at any point, and the factor $e^{-j-\theta}$ cancels the size of evaluation at that scale. Integrating in θ recovers all scales and hence the desired coefficients.

Fix $L = 10^5$ and $\varepsilon = 1/32$. For $\theta \in (0,1)$ and integers $j, k \ge 0$ put $\sigma_j = e^{-j-\theta}$, $r^j = r(\sigma_j)$ and

$$a_p^{jk} = \frac{r_p^j - r_p^k}{1 - r_p^j r_p^k}, \qquad E_{jk} = \sum_p (a_p^{jk})^2 = E(r^j, r^k),$$

all depending on θ; E is the energy of section 1.3. Let $D_j = D_j^\theta$ be the set of $w \in \mathbb{T}^\infty$ such that, for every $k \ge 0$ with $|k - j| \ge L$, the series $\sum_p a_p^{jk} w_p$ and $\sum_p (a_p^{jk} w_p)^2$ converge and

$$\Big|\sum_p a_p^{jk} w_p\Big| \le \varepsilon E_{jk}, \qquad \Big|\sum_p (a_p^{jk} w_p)^2\Big| \le \frac{E_{jk}}{2}. \tag{9.1}$$

The first lemma says that Lemma 1.2 needs no approximation by finitely many coordinates.

LEMMA 9.16. *Let $r \in [0,1)^\infty \cap \ell^2$ and let D be a Borel set of positive measure, invariant under common rotations. On $\mathbb{T}^\infty$ the product defining $K_r/\overline{K_r}$ converges almost everywhere, and $\psi_{r,D}$ represents evaluation at r, with $|\psi_{r,D}| = \|K_r\|_2^2/m(D)$ on $T_r(D)$.*

PROOF. In each coordinate $K_{r_p}/\overline{K_{r_p}} = e^{-2iX_p}$ with $X_p = \Im \log(1 - r_p z_p)$. The X_p are independent and centred, with $\mathbb{E}X_p^2 \le r_p^2/(1 - r_p^2)$, so $\sum_p X_p$ converges

almost everywhere by Kolmogorov's one-series theorem; we set the product equal to 0 elsewhere. By Corollary 1.5, $(T_{-r})_* m$ is equivalent to m, and by Kakutani's theorem its density is the almost everywhere limit J of the products of the one-variable densities. So $\int g\,\mathrm{d}m = \int (g\circ T_r) J\,\mathrm{d}m$ for $g \in L^1$, which is Lemma 1.1 on $\mathbb{T}^\infty$. As in the proof of Lemma 1.2,

$$\int F\,\overline{\psi_{r,D}}\,\mathrm{d}m = \frac{1}{m(D)}\int_D G\,\mathrm{d}m, \qquad G = (F\circ T_r)\,h, \quad h(w) = \prod_p (1+r_p w_p)^{-2}.$$

Since $\int |1+r_p w|^{-4}\,\mathrm{d}m(w) = (1+r_p^2)/(1-r_p^2)^3$, the partial products of h form an L^2-bounded martingale, so $h \in H^2(\mathbb{T}^\infty)$ with $\widehat{h}(0) = 1$. For a polynomial F, $F\circ T_r$ is a uniform limit of polynomials in finitely many variables, so $G \in H^2(\mathbb{T}^\infty)$ and $\widehat{G}(0) = F(r)$. Averaging $G(e^{i\vartheta}w)$ over ϑ gives $\widehat{G}(0)$ for almost every w, and Fubini's theorem gives $\int_D G\,\mathrm{d}m = F(r)\,m(D)$. □

The next lemma collects Perfekt's three estimates [70, Lemmas 2.3, 2.5 and 2.6] in the present setting.

LEMMA 9.17. *For every $\theta \in (0,1)$ the following hold.*

(1) $E_{jk} \ge |j-k| - 5$ for all $j,k \ge 0$.
(2) D_j is invariant under common rotations, and $m(D_j) \ge 1/2$.
(3) If $|j-k| \ge L$, then $T_{r^j}(D_j) \cap T_{r^k}(D_k) = \emptyset$.

PROOF. (1) Since $(x-y)/(1-xy) \ge x-y$ for $0 \le y \le x < 1$, the energy is at least $\sum_p (r_p^j - r_p^k)^2$. Let $\mathcal{P}(s) = \sum_p p^{-s}$. From $0 \le \log\zeta(1+\sigma) - \mathcal{P}(1+\sigma) \le 1/2$ and $1/\sigma \le \zeta(1+\sigma) \le 1 + 1/\sigma$ we get $-\log\sigma - 1/2 \le \mathcal{P}(1+\sigma) \le -\log\sigma + \log 3$ for $0 < \sigma \le 2$. With $u = \sigma_j$ and $v = \sigma_k$,

$$\sum_p (r_p^j - r_p^k)^2 = \mathcal{P}(1+2u) + \mathcal{P}(1+2v) - 2\mathcal{P}(1+u+v) \ge \log\frac{(u+v)^2}{4uv} - 1 - 2\log 3,$$

and $(u+v)^2/(4uv) \ge e^{|j-k|}/4$, while $\log 4 + 1 + 2\log 3 < 5$.

(2) Rotation by $e^{i\vartheta}$ multiplies the two sums in (9.1) by $e^{i\vartheta}$ and $e^{2i\vartheta}$. For real $b \in \ell^2$, Hoeffding's inequality [46] applied to the real and imaginary parts of the partial sums, with Fatou's lemma, gives $m(|\sum_p b_p w_p| > x) \le 4\exp(-x^2/(4\|b\|_2^2))$, and the series converges almost everywhere. Apply this with $b = a^{jk}$, $x = \varepsilon E_{jk}$, and with $b_p = (a_p^{jk})^2$, $x = E_{jk}/2$; the w_p^2 are again independent and uniform, and $\sum_p (a_p^{jk})^4 \le E_{jk}$. By (1),

$$m(\mathbb{T}^\infty \setminus D_j) \le \sum_{|k-j|\ge L} 8e^{-E_{jk}/4096} \le 16 e^{5/4096} \sum_{h\ge L} e^{-h/4096} < 10^{-5}.$$

(3) Suppose that $T_{r^j} w = T_{r^k} w'$ with $w \in D_j$ and $w' \in D_k$, and write $a = a^{jk}$, $E = E_{jk}$. Since $T_s \circ T_t = T_{(s+t)/(1+st)}$ in each coordinate, $w' = T_a w$, and a computation gives

$$a_p \Re(w_p' - w_p) = \frac{a_p^2(1-\Re w_p^2)}{|1+a_p w_p|^2} \ge \frac{a_p^2(1-\Re w_p^2)}{4}.$$

Summing, and using the second condition in (9.1) for w, we get $\Re\sum_p a_p(w'_p - w_p) \geq E/8$. Since $a^{kj} = -a$ and $E_{kj} = E$, the first conditions for w and w' give $|\sum_p a_p(w'_p - w_p)| \leq 2\varepsilon E = E/16$, a contradiction. □

THEOREM 9.18 (An explicit symbol). *Let* $\psi_j^\theta = \psi_{r^j, D_j^\theta}$ *and*

$$B(z) = \int_0^1 \sum_{j\geq 0} e^{-j-\theta}\,\overline{\psi_j^\theta(z)}\,\mathrm{d}\theta + \sum_{n\geq 2} \frac{\overline{z^{\kappa(n)}}}{n^{3/2}\log n}.$$

Then $B \in L^\infty(\mathbb{T}^\infty)$, $\|B\|_\infty \leq 10^5 + 7$, $\int B\,\mathrm{d}m = 1$ *and* $\int z^{\kappa(n)} B\,\mathrm{d}m = 1/(\sqrt{n}\log n)$ *for every* $n \geq 2$.

So B is a bounded symbol of the multiplicative Hilbert matrix, in the normalization of Corollary 4.18, and its definition involves no choice and no extension theorem.

PROOF. Fix θ. Since $\|K_{r^j}\|_2^2 = \zeta(1+2\sigma_j) \leq 1 + 1/(2\sigma_j)$, Lemma 9.16 and part (2) of Lemma 9.17 give $\sigma_j|\psi_j| \leq 2\sigma_j + 1$. By part (3), the indices j with $z \in T_{r^j}(D_j)$ lie in an interval of length less than L, so there are at most L of them, and

$$\sum_{j\geq 0} \sigma_j|\psi_j(z)| \leq L + 2\sum_{j\geq 0} e^{-j} < L + 4$$

for every z. All objects are Borel in (θ, z): the series in (9.1) and the products are limits of continuous functions where they exist, T_{-r^j} is continuous in (θ, z), and $\theta \mapsto m(D_j^\theta)$ is Borel by Tonelli's theorem. For a Dirichlet polynomial f with Bohr lift F, Lemma 9.16, dominated convergence and Fubini's theorem therefore give, with $\sigma = e^{-t}$,

$$\int F\,B\,\mathrm{d}m = \int_0^\infty e^{-t} f(1/2 + e^{-t})\,\mathrm{d}t + \sum_{n\geq 2} \frac{a_n}{n^{3/2}\log n} = a_1 + \sum_{n\geq 2} \frac{a_n}{\sqrt{n}\log n},$$

by integrating $n^{-1/2-\sigma}$ over $0 < \sigma < 1$. Finally, the last series in B is bounded by $\sum_{n\geq 2} n^{-3/2}/\log n \leq (\zeta(3/2) - 1)/\log 2 < 3$. □

REMARK 9.19. The construction and the three estimates of Lemma 9.17 are Perfekt's; what is new is that they run on $\mathbb{T}^\infty$ without a cutoff in the primes, which removes the Hahn–Banach step. The constant $10^5 + 7$ is far from $C \leq 5.515$ (Proposition 9.14), which is not constructive, and the normalizers $m(D_j^\theta) \in [1/2, 1]$ are not known in closed form. The Fourier coefficients $b(q) = \int \chi_q B\,\mathrm{d}m$, $q \in P$, give an explicit completion in Theorem 9.5, of norm at most $\|B\|_\infty$: the integrated flat representatives are unbounded, but their localized versions are not. No linear dual of the Gram estimate of Theorem 2.1 was found; that estimate bounds $|F|^{1/2}$, not a linear expression in F.

Open

(1) Is there a bounded completion of (c_n) to the ordered rationals with norm π (Theorem 9.5)?

(2) Does $\Phi(F_k) \to 1$ along every extremizing sequence? Along *some* extremizing sequence this is equivalent to $C = \pi$; the certificate of remark 11.10 predicts not. ($\Phi \leq 1$ fails in general, Proposition 9.8.)
(3) Close $1.0617 \leq C_2/\pi_2^* \leq 1.0878$. The three-prime benchmark is strictly exceeded by Theorem 9.15.
(4) Improve $C \leq 5.515$. The unrestricted Gram estimate of Theorem 2.1 stops at $\Gamma(1/4)^2/\sqrt{2\pi} \approx 5.244$; can the analytic structure of F improve this bound?
(5) Find an explicit symbol with norm much smaller than the $10^5 + 7$ of Theorem 9.18, for instance by a linear dual of the Gram estimate.

CHAPTER 10

Coefficient inequalities

The functional ℓ has relatives that ask for more: absolute values on the coefficients (Problem 10), other logarithmic weights, other $\mathscr{H}^p$, and partial sums. One arithmetic test runs through the first part of the chapter: by Helson's inequality, the absolute inequality holds on a set S of integers as soon as $B(S) = \sum_{n\in S} d(n)/(n\log^2 n)$ is finite, and each support restriction below is a way of taming the same Euler-product divergence $\zeta(1+\sigma)^2 \asymp \sigma^{-2}$. The second part replaces absolute values by cancellation: on prime sets of density θ the signed sum is bounded down to the weight $(\log n)^{-\theta}$, and the fractional primitives $\sum a_n/(\sqrt{n}(\log n)^{\beta})$ are classified on every $\mathscr{H}^p$, exactly as on the disc. Partial sums and the mean-value abscissa close the chapter. This chapter does not use chapter 11; it needs Helson's inequality, chapter 2, and, for the signed results, Theorem 3.1.

10.1. The absolute inequality

In one variable the absolute inequality is part (3) of Proposition 9.1; the same argument works for Paley–Wiener spaces whenever Nehari's theorem, equivalently weak factorization, holds [8, Lemma 7.5]. If $f = \sum_j g_j h_j$ with $\sum_j \|g_j\|_{\mathscr{H}^2}\|h_j\|_{\mathscr{H}^2} \lesssim \|f\|_{\mathscr{H}^1}$, then $|a_n|$ is dominated by the Dirichlet convolutions of the coefficient moduli, and the multiplicative Hilbert matrix, of norm π, bounds the weighted sum. On $\mathbb{T}^\infty$ this weak factorization fails, since Nehari's theorem does [69], and the S_2 substitute of chapter 5 controls only $\sum_j (\|g_j\|_{\mathscr{H}^2}\|h_j\|_{\mathscr{H}^2})^2$.

Problem 10 is open, although it holds below π on primes up to 10^{216} (Proposition 9.14). This section proves the inequality on every piece of $\mathscr{H}^1$ cut out by a degree constraint that the Euler product can see: one homogeneous degree, one degree in a subset of the primes, integers composed of a thin set of primes, and the balanced four-class diagonals together with summable unions of them. It also proves it for products across two complementary prime sets, and for sums of r such products at the sharp cost $\sqrt{r}$. All of these rest on Helson's inequality. It ends with the homogeneous dual constants, and with the unweighted sum $\sum_{n\le x}|a_n|$, where random multiplicative functions give a lower bound beyond the Dirichlet kernel (Theorem 10.7).

10.1.1. An arithmetic criterion

LEMMA 10.1. *Let $S \subseteq \mathbb{N}$ and $B(S) = \sum_{n\in S,\, n\geq 2} d(n)/(n\log^2 n)$. Every $f \in \mathscr{H}^1$ supported on S satisfies*

$$|a_1| + \sum_{n\geq 2} \frac{|a_n|}{\sqrt{n}\log n} \leq \big(1 + B(S)\big)^{1/2} \|f\|_{\mathscr{H}^1}.$$

Moreover $B(S) = \int_0^\infty \sigma \sum_{n\in S,\, n\geq 2} d(n) n^{-1-\sigma}\, \mathrm{d}\sigma$.

PROOF. Put $w_1 = 1$ and $w_n = (\sqrt{n}\log n)^{-1}$ for $n \geq 2$. Cauchy–Schwarz and Helson's inequality [44] give

$$\sum_{n\in S} |a_n| w_n \leq \left(\sum_n \frac{|a_n|^2}{d(n)}\right)^{1/2} \big(\mathbf{1}_{\{1\in S\}} + B(S)\big)^{1/2} \leq (1 + B(S))^{1/2} \|f\|_{\mathscr{H}^1}.$$

The integral formula follows from Tonelli's theorem and $\int_0^\infty \sigma n^{-\sigma}\, \mathrm{d}\sigma = 1/\log^2 n$. □

So every result in the next subsection is a bound on a Dirichlet series of $d(n)$ restricted to S, near $\sigma = 0$. The weight $\sigma\, \mathrm{d}\sigma$ is exactly what makes $\sigma^{-2+\varepsilon}$ integrable and σ^{-2} not: the full set $S = \mathbb{N}$ gives $\zeta(1+\sigma)^2 \asymp \sigma^{-2}$ and $B = \infty$, which is why the method cannot settle the full inequality.

10.1.2. Pieces

For a set $\mathcal{P}$ of primes let $\Omega_{\mathcal{P}}(n) = \sum_{p\in\mathcal{P}} v_p(n)$. Enumerate the primes $p_1 < p_2 < \cdots$ and let $\mathcal{P}_j$, $1 \leq j \leq 4$, be the residue classes of indices modulo 4.

THEOREM 10.2. *The absolute inequality $|a_1| + \sum_{n\geq 2} |a_n|/(\sqrt{n}\log n) \leq C\|f\|_{\mathscr{H}^1}$ holds for f supported on each of the following sets, with C as stated.*

1. *$\{\Omega(n) = m\}$, with C absolute and independent of m. For $m \geq 1$, and also for $\{\Omega(n) \in \{1, 2\}\}$, the sum alone satisfies $\sum_{n\geq 2} |a_n|/(\sqrt{n}\log n) \leq (2389/256)^{1/2}\|f\|_{\mathscr{H}^1} < \pi\|f\|_{\mathscr{H}^1}$, uniformly in m and in the number of primes.*
2. *$\{\Omega_{\mathcal{P}}(n) = k\}$, where $\sum_{p\in\mathcal{P}} p^{-1-\sigma} \leq \delta\log(1/\sigma) + C_0$ and $\sum_{p\notin\mathcal{P}} p^{-1-\sigma} \leq (1-\delta)\log(1/\sigma) + C_0$ on $(0, 1]$, $0 < \delta \leq 1$; C depends on δ, C_0 only, not on k. The degrees in the primes outside $\mathcal{P}$ are unrestricted.*
3. *The integers composed of primes in $\mathcal{P}$, where $\sum_{p\in\mathcal{P}} p^{-1-\sigma} \leq \delta\log(1/\sigma) + C_0$ with $\delta < 1$; for instance either alternating half of the primes. More generally, $\sum_{n\geq 2} |a_n|/(\sqrt{n}(\log n)^\beta) \leq C\|f\|_{\mathscr{H}^1}$ on these integers for every $\beta > \delta$, with C depending on β, δ, C_0.*
4. *The union over $a \in \mathcal{A} \subseteq \mathbb{Z}^3$ of the diagonals $\{\Omega_{\mathcal{P}_j}(n) - \Omega_{\mathcal{P}_4}(n) = a_j,\ j \leq 3\}$, with $C \lesssim (\sum_{a\in\mathcal{A}} (1 + |a|_\infty)^{-1})^{1/2}$; for a single diagonal C is absolute.*

PROOF. In each case we bound $B(S)$ through Lemma 10.1, by marking degrees in the Euler product $\prod_p (1 - u_p p^{-1-\sigma})^{-2}$ and comparing coefficientwise. For $0 < \sigma \leq 1$, expanding the logarithm gives the majorant $\exp(2\sum_p u_p p^{-1-\sigma}) E(u)$,

where E collects the terms $u^\ell/(\ell p^\ell)$, $\ell \ge 2$, and has nonnegative coefficients and $E(1) < \infty$; for $\sigma \ge 1$, every coefficient is at most $\zeta(1+\sigma)^2 - 1 \le C 2^{-\sigma}$.

(1) With one marker u, $\sum_p p^{-1-\sigma} \le \log(2/\sigma)$ gives $[u^m] \le [u^m](2/\sigma)^{2u} E(u)$. Integrating σ^{1-2u} over $(0,1)$ gives $4^u/(2(1-u))$, and the coefficient of u^m in $4^u E(u)/(1-u)$ is a partial sum of nonnegative coefficients, hence at most $4E(1)$. For the explicit constant, $a_1 = 0$ when $m \ge 1$, and the proof of Lemma 10.1 bounds the sum by $B_m^{1/2}\|f\|_{\mathscr{H}^1}$, where $B_m = \sum_{\Omega(n)=m} d(n)/(n \log^2 n)$. The argument just given bounds the part $\sigma \le 1$ of B_m by $2E(1)$, and $E(1) \le e$ because $\sum_p \sum_{\ell \ge 2} 1/(\ell p^\ell) \le \sum_{n \ge 2} 1/(2n(n-1)) = 1/2$. For $\sigma \ge 1$ the coefficient is at most $\zeta(1+\sigma)^2 - 1$, and with $x = \zeta(1+\sigma) - 1 \le 2^{-\sigma}(1/2 + 1/\sigma)$ the integrand $\sigma(2x + x^2)$ is at most $2^{-\sigma}(\sigma + 2) + 4^{-\sigma}(\sigma/4 + 2)$, whose integral over $[1, \infty)$ is below $981/256$ since $\log 2 > 2/3$. Hence $B_m < 2e + 981/256 < 2389/256 < \pi^2$, using $e < 11/4$ and $\pi^2 > 72 - 36\sqrt{3} > 48/5$. For the two layers $\{1, 2\}$ the same estimates give $B_1 + B_2 < (P_1 + P_2)/2 + 981/256 < 141/16$, where $P_1 = 1 + \log 4$ and $P_2 = P_1 + (\log 4)^2/2 + \sum_p p^{-2}$ are partial sums of the coefficients of $4^z E(z)$.

(2) Mark only the primes of $\mathcal{P}$. The majorant is $C\sigma^{-2(1-\delta)}\sigma^{-2\delta u} e^{2C_0 u} E(u)$, and $\int_0^1 \sigma^{-1+2\delta(1-u)}\,\mathrm{d}\sigma = 1/(2\delta(1-u))$; conclude as in (1).

(3) $\prod_{p \in \mathcal{P}} (1 - p^{-1-\sigma})^{-2} \le C\sigma^{-2\delta}$ and $\int_0^1 \sigma^{1-2\delta}\,\mathrm{d}\sigma < \infty$. For the alternating halves, $0 \le \sum_j (-1)^{j-1} p_j^{-s} \le 2^{-s}$ gives $\delta = 1/2$. For the weight $(\log n)^{-\beta}$, run the proof of Lemma 10.1 with

$$(\log n)^{-2\beta} = \frac{1}{\Gamma(2\beta)} \int_0^\infty \sigma^{2\beta - 1} n^{-\sigma}\,\mathrm{d}\sigma;$$

the bound becomes $\int_0^1 \sigma^{2\beta - 1 - 2\delta}\,\mathrm{d}\sigma < \infty$.

(4) Each class $\mathcal{P}_j$ satisfies $\sum_{p \in \mathcal{P}_j} p^{-1-\sigma} \le \log(1/\sigma)/4 + C_0$, by the same alternating comparison. With $T = \log(1/\sigma)$ and $\mu = T/2$, the coefficients of $\exp(\mu \sum_j u_j)$ are $e^{4\mu}$ times the joint law of four independent Poisson variables of mean μ, and the factor $e^{4\mu} = \sigma^{-2}$ cancels against $\sigma\,\mathrm{d}\sigma = e^{-2T}\,\mathrm{d}T$. So the contribution of a diagonal is bounded by a constant times $\int_0^\infty \mathbb{P}(N_j - N_4 = a_j,\ j \le 3)\,\mathrm{d}\mu$, convolved with the coefficients of E. Include the factors $e^{2C_0 \sum_j u_j}$ from the classwise bounds in E; its coefficients remain nonnegative and $E(3/2, \dots, 3/2) < \infty$. We claim that the integral is at most $C/(1 + |a|_\infty)$. Put $A = |a|_\infty$ and choose j with $|a_j| = A$. Conditioning on N_j, N_4, the other two Poisson point probabilities cost at most $C(1+\mu)^{-1}$. For independent Poisson variables X, Y of mean $\mu > 0$, tilting $X - Y$ by t has normalizer $\exp(2\mu(\cosh t - 1))$. Fourier inversion bounds each atom of the tilted law by

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-2\mu \cosh t (1 - \cos v)}\,\mathrm{d}v \le C(1+\mu)^{-1/2}.$$

For $A > 0$ choose $t = \min(1, A/(4\mu))$, using symmetry for a negative difference. Since $\cosh t - 1 \le t^2$ for $0 \le t \le 1$,

$$\mathbb{P}(N_i - N_4 = a_i,\ i \le 3) \le C(1+\mu)^{-3/2} e^{-\min(A, A^2/\mu)/8}.$$

For $A \geq 1$, split the integral at $\mu = A$ and $\mu = A^2$. The first part is $O(e^{-A/8})$; in the middle use $x = A^2/\mu$ to bound it by $CA^{-1}\int_1^\infty x^{-1/2}e^{-x/8}\,\mathrm{d}x$; the last part is $O(A^{-1})$. For $A = 0$, the bound $C(1+\mu)^{-3/2}$ is integrable. This proves the claim. The shifts b from E are absorbed by $(1+|a-b|_\infty)^{-1} \leq (1+|b|_\infty)/(1+|a|_\infty)$ and its exponential moment. For $\sigma \geq 1$, every selected integer has $\Omega(n) \geq |a|_\infty$, which costs $(3/2)^{-|a|_\infty}$. Different diagonals are disjoint; sum over $\mathcal{A}$. □

Since $J(F) \leq \sum_n |a_n|c_n$, part (1) proves the sharp Fejér–Riesz conjecture on every homogeneous layer: any counterexample mixes degrees in an essential way. Mixing a bounded number of degrees costs little: for $S = \{\Omega(n) \leq D\}$, Lemma 10.1 and the uniform bound $B_m < 2389/256$ give

$$|a_1| + \sum_{n\geq 2} \frac{|a_n|}{\sqrt{n}\log n} \leq \sqrt{1+(2389/256)D}\,\|f\|_{\mathscr{H}^1}.$$

The layer constants improve with the degree.

PROPOSITION 10.3. *With $B_m = \sum_{\Omega(n)=m} d(n)/(n\log^2 n)$, $B_m = 1/2 + O(\rho^m)$ for every $\rho > 2/3$. Hence the absolute inequality on the layer $\{\Omega(n) = m\}$ holds with constant at most $2^{-1/2} + o(1)$ as $m \to \infty$.*

PROOF. Since $\sum_{\Omega(n)=m} d(n)n^{-1-\sigma}$ is the coefficient of u^m in $\prod_p (1-up^{-1-\sigma})^{-2}$, Lemma 10.1 gives $\Phi(u) = \sum_m B_m u^m = \int_0^\infty \sigma\big(\prod_p (1-up^{-1-\sigma})^{-2} - 1\big)\,\mathrm{d}\sigma$. On $(0,1]$ write the product as $\sigma^{-2u}H(u,\sigma)$, where

$$H(u,\sigma) = \big(\sigma\zeta(1+\sigma)\big)^{2u} \exp\!\left(2\sum_p \sum_{k\geq 2} \frac{u^k - u}{k p^{k(1+\sigma)}}\right).$$

This is analytic for $|u| < 2$ and real analytic in σ at zero, locally uniformly in u. In particular $H(1,0) = 1$. For $|u| < 1$,

$$\begin{aligned}\Phi(u) = \frac{H(u,0)}{2(1-u)} - \frac{1}{2} &+ \int_0^1 \sigma^{1-2u}\big(H(u,\sigma) - H(u,0)\big)\,\mathrm{d}\sigma \\ &+ \int_1^\infty \sigma\left(\prod_p (1-up^{-1-\sigma})^{-2} - 1\right)\mathrm{d}\sigma.\end{aligned}$$

The first integral is analytic for $\operatorname{Re} u < 3/2$ within $|u| < 2$, because its bracket is $O(\sigma)$ locally uniformly. The last integral is analytic for $|u| < 2$. Since $H(1,0) = 1$, $\Phi(u) - 1/(2(1-u))$ is analytic in $|u| < 3/2$. Cauchy's coefficient estimates give $B_m = 1/2 + O(\rho^m)$ for every $\rho > 2/3$. The layer constant is $B_m^{1/2}$ by the proof of Theorem 10.2. □

Part (4) needs four classes for a reason that is easiest to see probabilistically. Along σ, the degrees in the four classes behave like independent Poisson variables of mean $\mu = \log(1/\sigma)/2$, and the weight $\sigma\,\mathrm{d}\sigma$ turns the contribution of a diagonal into the expected time $\int \mathbb{P}(N_j - N_4 = a_j,\ j \leq 3)\,\mathrm{d}\mu$ that the three differences spend at one point. That is a return probability of order $\mu^{-3/2}$, which is integrable. With three classes there are two differences, the return probability is of order μ^{-1}, and

the integral diverges, as for recurrent walks in two dimensions. That is a limit of the method, not a counterexample.

PROPOSITION 10.4 (Products across complementary prime sets). *Let $\mathcal{P}_0, \mathcal{P}_1$ be the primes of odd and even index.*

(1) If $g, h \in \mathscr{H}^1$ are supported on integers composed of primes in $\mathcal{P}_0$ and $\mathcal{P}_1$ respectively, then $f = gh$ satisfies the absolute inequality with an absolute constant, and $\|f\|_{\mathscr{H}^1} = \|g\|_{\mathscr{H}^1}\|h\|_{\mathscr{H}^1}$.

(2) If $f = \sum_{j \le r} g_j h_j$ with such factors, then the absolute inequality holds with constant $C\sqrt{r}$.

(3) The cost $\sqrt{r}$ of separated representations is sharp. For

$$f_N = \sum_{j \le N} (p_{2j-1} p_{2j})^{-s},$$

every representation of f_N as a sum of products $u_k v_k$ with separated supports costs at least $N/2$, while $\|f_N\|_{\mathscr{H}^1} \le \sqrt{N}$.

PROOF. (1) The lifts use disjoint variables. Group m, n into blocks $2^j \le \log n < 2^{j+1}$. By the bound $\prod_{p \in \mathcal{P}_0}(1 - p^{-1-\sigma})^{-2} \le K\sigma^{-1}$ from the proof of part (3) of Theorem 10.2, with $\sigma = 2^{-j}$, each block has $\sum d(n)/(n \log n) \le C$, and $\sqrt{\log m \log n}/(\log m + \log n) \le \sqrt{2}\, 2^{-|j-k|/2}$ across blocks. Cauchy–Schwarz per pair of blocks, a sum over diagonals $j - k$, and Helson's inequality for g and h finish.

(2) The span E of the lifts of the g_j in L^1 has dimension at most r, so the Kadets–Snobar theorem [50] gives a projection onto E of norm at most $\sqrt{2r}$, represented by a kernel $K(x) \in E$ with $\operatorname{ess\,sup}\|K(x)\|_1 \le \sqrt{2r}$. Since every w-section of the lift of f lies in E, $f(z, w) = \int K(x)(z) f(x, w)\, \mathrm{d}m(x)$, which is a representation with projective cost at most $\sqrt{2r}\|f\|_{\mathscr{H}^1}$. Apply (1) to each term.

(3) Projecting each u_k, v_k to degree one and Helson's inequality with $d(p) = 2$ give $\sum_j |\hat{u}_k(p_{2j-1})|^2 \le 2\|u_k\|^2$ and similarly for v_k; summing the coefficients at $p_{2j-1}p_{2j}$ gives $N \le 2\sum_k \|u_k\|\|v_k\|$, while $\|f_N\|_{\mathscr{H}^1} \le \|f_N\|_{\mathscr{H}^2} = \sqrt{N}$. □

REMARK 10.5 (Product Hardy inequality). The tensorized Hardy inequality

$$\sum_{\alpha} \frac{|a_\alpha|}{\prod_j (\alpha_j + 1)} \le \pi^d \|f\|_{H^1(\mathbb{T}^d)},$$

sharp by tensoring near-extremal squares for the Hilbert matrix. On the integers generated by d primes it reads $\sum_n |a_n|/d(n) \le \pi^d \|f\|_{\mathscr{H}^1}$. So no dimension-free inequality holds for the weight $1/d(n)$; the question is only about smaller weights such as $1/(\sqrt{n} \log n)$.

10.1.3. Homogeneous dual constants

Let c_m be the norm of $\varphi \mapsto \Lambda_\varphi$ from homogeneous degree-m symbols in $H^2(\mathbb{T}^\infty)$ to $(H^1)^*$; equivalently, the best constant in $\|f\|_2 \le c_m \|f\|_1$ for homogeneous f of degree m.

PROPOSITION 10.6. *$\sqrt{m!}/\Gamma(1+m/2) \le c_m \le 2^{m/2}$, so $c_m^{1/m} \to \sqrt{2}$; moreover $c_{m+n} \ge c_m c_n$. Consequently any inequality $\|\Lambda_\varphi\| \le C(\sum_m w_m \|\varphi_m\|_2^2)^{1/2}$ over homogeneous components needs $w_m \ge C^{-2} m!/\Gamma(1+m/2)^2$, so $\liminf w_m^{1/m} \ge 2$.*

PROOF. Projection onto degree m is an H^1 contraction, which gives the equivalence. The upper bound is Helson's inequality with $\prod(\nu_j+1) \le 2^m$. For the lower bound take $((z_1+\cdots+z_d)/\sqrt{d})^m$ and let $d \to \infty$: the moments converge to those of a complex Gaussian, $\mathbb{E}|Z|^q = \Gamma(1+q/2)$. Supermultiplicativity comes from disjoint variables, and $c_1 = 2/\sqrt{\pi}$ is Brevig's value [17]. □

10.1.4. Unweighted sums

Without the weight the natural quantity is the Sidon constant

$$S_1(x) = \sup\Big\{ \sum_{n\le x} |a_n| : \Big\| \sum_{n\le x} a_n n^{-s} \Big\|_{\mathscr{H}^1} \le 1 \Big\}.$$

By duality $S_1(x)$ is the supremum, over unimodular (ε_n), of the least $\|g\|_\infty$ with $\widehat{g}(n) = \varepsilon_n$ for $n \le x$; so it is a Nehari constant for unimodular data. Helson's inequality and the Cauchy–Schwarz inequality give $S_1(x) \le (\sum_{n\le x} d(n))^{1/2} \sim \sqrt{x \log x}$, and the Dirichlet kernel with Harper's estimate [40] gives $S_1(x) \gg \sqrt{x}(\log\log x)^{1/4}$. Divisor-type weights do better.

THEOREM 10.7. *There is an absolute constant $c > 0$ such that, for all large x,*

$$S_1(x) \ge \sqrt{x} \exp\big(c(\log\log x)^{1/3}\big).$$

Let us briefly describe the approach. The test polynomial is $\sum_{n\le x} g(n) n^{-s}$, where $g \ge 0$ is multiplicative with $\sum g(n) n^{-s} = \prod_p (1-p^{-s})^{-\alpha_p}$, and α_p decreases from $3/2$ to 1 over the primes up to some y and equals 1 beyond. Its $\mathscr{H}^1$ norm is $\mathbb{E}|\sum_{n\le x} g(n)\chi(n)|$ for a Steinhaus random multiplicative function χ. Harper's conditioning argument reduces the norm estimate to a half-moment of the Euler product over $p \le y$ on a unit segment. The weights enlarge the coefficient sum by $\exp(\sum_{p\le y}(\alpha_p - 1)/p)$. We tilt the prime variables by Poisson densities and center the resulting walk. The decreasing profile α_j gives an exponential penalty for paths below a barrier; a separate crossing estimate recovers the unrestricted half-moment. The resulting saving is $\exp(ct^{1/3})$.

Let $T = \log\log x$, $t = \lfloor T - 2\log T \rfloor$ and $y = \exp(e^t)$, so that $\log x/(eT^2) \le \log y \le \log x/T^2$. Let I_1 be the set of primes $p \le e^e$ and, for $2 \le j \le t$, I_j the set of primes in $(\exp(e^{j-1}), \exp(e^j)]$. By Mertens' theorem [89, Section I.1.6], $\sum_{p\in I_j} 1/p \le 2$ for all j, and $\sum_{p\in I_j} 1/p \ge 1/2$ for $j \ge j_0$, with j_0 absolute. Put $\alpha_j = 3/2 - (j-1)/(2t-2)$, $\alpha_p = \alpha_j$ for $p \in I_j$, and $\alpha_p = 1$ for $p > y$. Then $g(p^k) = \binom{\alpha_p + k - 1}{k} \le k+1$, and $g(n) = 1$ when every prime factor of n exceeds y. Let χ be a Steinhaus random multiplicative function, so that the $\chi(p)$ are independent and uniform on $\mathbb{T}$, and put

$$F_y(s) = \prod_{p\le y} \big(1 - \chi(p)p^{-s}\big)^{-\alpha_p}, \qquad \mathcal{Z} = \int_0^1 |F_y(1/2 + i\tau)|^2 \, \mathrm{d}\tau.$$

Constants C, c are absolute and may change from line to line.

LEMMA 10.8. *For large x,*

(1) $\sum_{n\le x} g(n) \ge (x/4)\exp\bigl(\sum_{p\le y}(\alpha_p-1)/p\bigr)$;

(2) $\mathbb{E}\bigl|\sum_{n\le x} g(n)\chi(n)\bigr| \le C\sqrt{x}\bigl(e^{-T^2/16}(\log x)^8 + te^{-t/2}\,\mathbb{E}\mathcal{Z}^{1/2} + e^{-3t/2}\bigr)$.

PROOF. (1) Write $g = 1*h$, where $h\ge 0$ is multiplicative, supported on y-smooth numbers, with $\sum h(n)n^{-s} = \prod_{p\le y}(1-p^{-s})^{-(\alpha_p-1)}$. Since $\lfloor x/m\rfloor \ge x/(2m)$ for $m\le x$, $\sum_{n\le x} g(n) \ge (x/2)(P - \sum_{m>x} h(m)/m)$ with $P = \prod_{p\le y}(1-1/p)^{-(\alpha_p-1)} \ge \exp(\sum_{p\le y}(\alpha_p-1)/p)$. With $\delta = 1/(2\log y)$, Rankin's trick gives $\sum_{m>x} h(m)/m \le x^{-\delta}\prod_{p\le y}(1-p^{\delta-1})^{-(\alpha_p-1)} \le CPx^{-\delta} \le CPe^{-T^2/2}$.

(2) This is Harper's conditioning argument [40], which applies verbatim because $g=1$ on y-rough numbers. Write $n = n_1n_2$ with n_2 y-smooth and n_1 y-rough, and $S(u) = \sum_{n_2\le u} g(n_2)\chi(n_2)$. Given $(\chi(p))_{p\le y}$, the $\chi(n_1)$ are orthonormal, so by conditional Jensen the left side is at most $\mathbb{E}(\sum_{n_1\le x}|S(x/n_1)|^2)^{1/2}$, the sum over y-rough n_1. For $n_1\le x^{3/4}$, Rankin's trick gives $\mathbb{E}|S(u)|^2 = \sum_{n_2\le u} g(n_2)^2 \le Cue^{-T^2/8}(\log y)^{14}$ for $u\ge x^{1/4}$, which gives the first term. For $n_1 > x^{3/4}$ we have $S(x/n_1) = S(r)$ with $r = \lfloor x/n_1\rfloor < x^{1/4}$, and Selberg's sieve [62, Section 3.2] bounds the number of y-rough n_1 with this r by $Cx/(r^2\log y)$. By Plancherel, this part of the sum is at most

$$\frac{Cx}{\log y}\int_1^\infty \frac{|S(u)|^2}{u^2}\,\mathrm{d}u = \frac{Cx}{2\pi\log y}\int_{\mathbb{R}} \frac{|F_y(1/2+i\tau)|^2}{|1/2+i\tau|^2}\,\mathrm{d}\tau.$$

Since $(\chi(p)p^{-i\tau_0})_p$ has the law of $(\chi(p))_p$, each $\int_n^{n+1}|F_y(1/2+i\tau)|^2\,\mathrm{d}\tau$ has the law of $\mathcal{Z}$. Split $\mathbb{R}$ into unit intervals, use $\mathbb{E}(\sum X_n)^{1/2} \le \sum \mathbb{E}X_n^{1/2}$ for $|n|\le(\log y)^{10}$, and Jensen with $\mathbb{E}|F_y(1/2)|^2 \le C(\log y)^8$ for the rest. With $\log y = e^t$ this gives the last two terms. □

For $h\in\mathbb{R}$ and $1\le j\le t$ put $\ell_p(h) = -\log|1-\chi(p)p^{-1/2-ih}|$, $Y_j(h) = \alpha_j\sum_{p\in I_j}\ell_p(h)$ and $W_j(h) = \sum_{i\le j} Y_i(h)$, so that $\mathcal{Z} = \int_0^1 e^{2W_t(h)}\,\mathrm{d}h$. For each h the process $(W_j(h))_j$ has the law of $(W_j)_j = (W_j(0))_j$, and the $Y_j = Y_j(0)$ are independent. Parseval's identity gives

$$\mathbb{E}\,e^{2\ell_p} = \Bigl(1-\frac{1}{p}\Bigr)^{-1}.$$

Let $\theta_j = 2/\alpha_j$ and $\Lambda = -\sum_{p\le y}\log(1-1/p)$, and define a probability measure $\mathbb{Q}$ by

$$\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}} = \exp\Bigl(\sum_{j\le t}\theta_jY_j - \Lambda\Bigr).$$

Under $\mathbb{Q}$ the $\chi(p)$, $p\le y$, are independent with Poisson densities $(1-1/p)|1-\chi(p)p^{-1/2}|^{-2} \ge 1/6$. The harmonic function $z\mapsto -\log|1-p^{-1/2}z|$ therefore has mean $-\log(1-1/p)$. Put $\mu_j = \mathbb{E}_{\mathbb{Q}}Y_j$, $\varphi(j) = \sum_{i\le j}\mu_i$, $\xi_j = Y_j-\mu_j$ and $B_j = \sum_{i\le j}\xi_i$. Writing $\lambda_j = \sum_{p\in I_j} -\log(1-1/p)$ and $\Lambda_J = \sum_{j\le J}\lambda_j$, we have exactly $\mu_j = \alpha_j\lambda_j$ and $\sum_{j\le J}\theta_j\mu_j = 2\Lambda_J$. Mertens' theorem gives $\Lambda_J = J + O(1)$ and $\varphi(t) = \sum_{p\le y}\alpha_p/p + O(1)$. Finally $\kappa_j = 1/\alpha_{j+1} - 1/\alpha_j \ge \kappa_0/t$ with $\kappa_0 = 2/9$.

LEMMA 10.9. *The following hold.*

(1) *If* $1 \le J \le t$ *and* $\Psi \ge 0$ *is a function of* $W_1, \dots, W_J$, *then* $\mathbb{E}\Psi$ *lies between* $C^{-1}e^{-J}\mathcal{E}$ *and* $Ce^{-J}\mathcal{E}$, *where* $\mathcal{E} = \mathbb{E}_{\mathbb{Q}}[\Psi \exp(-\theta_J B_J + 2\sum_{j<J} \kappa_j B_j)]$.

(2) *Under* $\mathbb{Q}$ *the* ξ_j *are independent and centred,* $\mathbb{E}_{\mathbb{Q}} e^{s\xi_j} \le e^{C_1 s^2}$ *for real* s, *and for every* $M > 0$ *there are* $p_M > 0$ *and* L_M *with* $\mathbb{Q}(B_{k+L} - B_k \ge M\sqrt{L}) \ge p_M$ *whenever* $L \ge L_M$, $k \ge j_0$ *and* $k + L \le t$.

(3) *For every* $b_0 \le 0$, *the walk* $B'_j = b_0 + B_j$ *satisfies*

$$\mathbb{E}_{\mathbb{Q}}\Big[\exp\Big(2\sum_{j<t} \kappa_j B'_j\Big);\, B'_j \le 0 \ (j \le t)\Big] \le Ce^{-ct^{1/3}}.$$

Proof. (1) The factors of $\mathrm{d}\mathbb{P}/\,\mathrm{d}\mathbb{Q}$ with $j > J$ have $\mathbb{Q}$-mean one and are independent of Ψ. Summation by parts gives $\sum_{j\le J} \theta_j Y_j = \sum_{j\le J} \theta_j \mu_j + \theta_J B_J - 2\sum_{j<J} \kappa_j B_j$; the remaining deterministic factor is $\exp(\Lambda_J - \sum_{j\le J} \theta_j \mu_j) = e^{-\Lambda_J} \asymp e^{-J}$.

(2) Each $\alpha_j \ell_p$ lies in an interval of length $16p^{-1/2}$, so Hoeffding's lemma [93, Example 2.5.8 (c), Propositions 2.5.2 and 2.6.1] and $\sum_{p\in I_j} 1/p \le 2$ give the moment generating function bound. Since the $\mathbb{Q}$-densities are at least $1/6$, $\mathrm{Var}_{\mathbb{Q}}(\alpha_j \ell_p) \ge \mathrm{Var}_{\mathbb{P}}(\ell_p)/6 \ge 1/(12p)$, so the variance of $B_{k+L} - B_k$ lies between $L/24$ and CL. Apply the Berry–Esseen inequality [73, Chapter V] to the independent centered prime summands $\alpha_j(\ell_p - \mathbb{E}_{\mathbb{Q}}\ell_p)$ in this block. Each has absolute value at most $16p^{-1/2}$, so their third absolute moments sum to at most $C\sum_p p^{-3/2} < \infty$. The normalized sum therefore approaches a standard Gaussian uniformly as $L \to \infty$, giving the last claim for each fixed M.

(3) Let $L = \lceil t^{2/3} \rceil$, $s_m = j_0 + mL$ and $N = \lfloor (t - 1 - j_0)/L \rfloor \ge t^{1/3}/2$. On the event every factor $e^{2\kappa_j B'_j}$ is at most 1. Keep only the blocks $(s_m, s_m + L]$ and condition successively on $B'_{s_m} = b \le 0$; the expectation is at most ρ^N, where ρ is the supremum over m and $b \le 0$ of the conditional expectation of $\exp(2\sum_{s_m<j\le s_m+L} \kappa_j B'_j)$ on $\{B'_j \le 0,\ s_m < j \le s_m + L\}$. Fix M with $e^{-M^2/(16C_1)} + e^{-\kappa_0 M} \le 1/2$. If $b > -M\sqrt{L}$, the walk must not rise by $M\sqrt{L}$, which by (2) has probability at most $1 - p_M$. If $b \le -M\sqrt{L}$, then either the walk rises by $M\sqrt{L}/2$, with probability at most $e^{-M^2/(16C_1)}$ by the exponential supermartingale bound and Doob's maximal inequality, or it stays below $-M\sqrt{L}/2$ and the weight is at most $\exp(-L(\kappa_0/t)M\sqrt{L}) \le e^{-\kappa_0 M}$, since $L^{3/2} \ge t$. So $\rho \le \max(1 - p_M, 1/2)$. □

Proof of Theorem 10.7. For $A \ge 0$ let G_A be the set of $h \in [0,1]$ with $W_j(h) \le \varphi(j) + A$ for all $j \le t$, let $\mathcal{Z}(A) = \int_{G_A} e^{2W_t(h)}\,\mathrm{d}h$, and let $\mathcal{B}_A$ be the event $[0,1] \not\subseteq G_A$.

First moment. By Fubini and stationarity, $\mathbb{E}\mathcal{Z}(A) = \mathbb{E}[e^{2W_t}; W_j \le \varphi(j) + A\ (j \le t)]$. Apply part (1) of Lemma 10.9 with $J = t$ and write $B_j = A + B'_j$, $b_0 = -A$. Since $\alpha_t = 1$ and $B'_t \le 0$, the exponent is at most $2A(1 - 1/\alpha_1) + 2\sum_{j<t} \kappa_j B'_j$, and part (3) gives

$$\mathbb{E}\mathcal{Z}(A) \le C\exp\big(2\varphi(t) - t + 2A(1 - 1/\alpha_1) - ct^{1/3}\big).$$

Crossings. Let $D_j = e^{-j}\sup_{h\in[0,1]} |\partial_h W_j(h)|$. The process $e^{-j}\partial_h W_j(h)$ has sub-Gaussian increments for the metric $C\min(1, e^j|h - h'|)$, by Hoeffding's inequality

and $\sum_{p\le z}(\log p)^m/p \le C_m(\log z)^m$. Dudley's integral is at most $C\sqrt{j}+C$, so [93, Theorem 8.1.6] gives $\mathbb{P}(D_j > C_2(\sqrt{j}+u)) \le 4e^{-u^2}$. Let J be the first j at which some h_0 has $W_j(h_0) > \varphi(j)+A$. On $\{J=j,\ D_j \le k\}$, every h with $|h-h_0| \le e^{-j}/k$ has $W_j(h) > \varphi(j)+A-1$ and $W_i(h) \le \varphi(i)+A$ for $i<j$; these h form a set of measure at least e^{-j}/k. By Fubini, stationarity and part (1),

$$\mathbb{P}(J=j) \le \mathbb{P}(D_j>k) + Ck\,\mathbb{E}_{\mathbb{Q}}\Big[e^{-\theta_j B_j + 2\sum_{i<j}\kappa_i B_i};\ B_i \le A\ (i<j),\ B_j > A-1\Big],$$

and on this event the exponent is at most $2-2A/\alpha_1$. With $u^2 = 2A+2\log t$ and $k = C_2(\sqrt{t}+u)$, summing over $j \le t$ gives $\mathbb{P}(\mathcal{B}_A) \le Ct^{3/2}(1+\sqrt{A})e^{-2A/\alpha_1}$.

Conclusion. On $\mathcal{B}_k \setminus \mathcal{B}_{k+1}$ we have $\mathcal{Z} = \mathcal{Z}(k+1)$, and $\mathcal{Z} = \mathcal{Z}(0)$ off $\mathcal{B}_0$. By the Cauchy–Schwarz inequality and the two bounds,

$$\mathbb{E}\mathcal{Z}^{1/2} \le (\mathbb{E}\mathcal{Z}(0))^{1/2} + \sum_{k\ge 0}\mathbb{P}(\mathcal{B}_k)^{1/2}(\mathbb{E}\mathcal{Z}(k+1))^{1/2} \le Ct^{3/4}e^{\varphi(t)-t/2-ct^{1/3}},$$

since $\sum_k (1+k)^{1/4}e^{-k(2/\alpha_1-1)} < \infty$ for $\alpha_1 = 3/2$. Insert this into part (2) of Lemma 10.8 and compare with part (1), using $\varphi(t) = \sum_{p\le y}\alpha_p/p + O(1)$ and $\sum_{p\le y}1/p = t + O(1)$. Since $g \ge 0$, this gives

$$S_1(x) \ge \frac{\sum_{n\le x} g(n)}{\|\sum_{n\le x} g(n)n^{-s}\|_{\mathscr{H}^1}} \ge c\sqrt{x}\,t^{-7/4}e^{ct^{1/3}},$$

and $t \ge T - 2\log T - 1$. □

The profile used here is not optimal. Heuristically, $\alpha^{3/2}$ linear from 2 to 1 gives $c \approx 1.33$, the first-moment constant of Maillard and Zeitouni [60]; multiplicative weights of this kind are not expected to give more than $\sqrt{x}\,e^{O((\log\log x)^{1/3})}$. To our knowledge, Theorem 10.7 is new; the order of $S_1(x)$ remains open between it and $(1+o(1))\sqrt{x\log x}$. The theorem also appears in the preprint [84].

10.2. Signed functionals and fractional primitives

10.2.1. Signed functionals on prime sets

Corollary 10.10. *Let R have density $\theta \in (0,1]$. For $f \in \mathscr{H}^1$ supported on integers composed of primes in R,*

$$\Big|a_1 + \sum_{n\ge 2}\frac{a_n}{\sqrt{n}(\log n)^{\theta}}\Big| \le C\|f\|_{\mathscr{H}^1},$$

and the series converges in its natural order.

Proof. For polynomials, $\sum_{n\ge2} a_n/(\sqrt{n}(\log n)^{\theta}) = \Gamma(\theta)^{-1}\int_0^\infty (f(1/2+\sigma)-a_1)\sigma^{\theta-1}\,d\sigma$. On $(0,1)$ use Theorem 3.1 with $p=1$; on $(1,\infty)$ use $|f(1/2+\sigma)-a_1| \le C2^{-\sigma}\|f\|_{\mathscr{H}^1}$. For natural-order convergence we follow Perfekt's Tauberian argument for $\theta=1$ [70]. Put $\varepsilon = 1/\log N$ and compare the partial sum with the Abel

sum $\sum_{n\ge 2} a_n n^{-1/2-\varepsilon}(\log n)^{-\theta}$. The density bound gives

$$\sum_{\substack{e^u\le n<e^{2u}\\ n\in\mathcal{S}_R}} \frac{d(n)}{n(\log n)^{2\theta}} \le C \qquad (u\ge 1).$$

Splitting into geometric intervals in $\log n$ therefore gives, uniformly in N,

$$\sum_{\substack{n\ge 2\\ n\in\mathcal{S}_R}} \frac{d(n)}{n(\log n)^{2\theta}} |\mathbf{1}_{n\le N} - n^{-\varepsilon}|^2 \le C.$$

Here $\mathcal{S}_R$ denotes the integers composed of primes in R. By Helson's inequality, $\sum_n |a_n|^2/d(n) < \infty$. Cauchy–Schwarz bounds the contribution of $n > M$ to the difference by $C(\sum_{n>M}|a_n|^2/d(n))^{1/2}$, while each fixed coefficient contributes a quantity tending to zero. Thus the difference tends to zero. The Abel sums converge to the bounded functional by continuity of horizontal translation on $\mathscr{H}^1$. □

Compare part (3) of Theorem 10.2: on a prime set of density $\theta < 1$ the *absolute* sum with weight $(\log n)^{-\beta}$ is bounded for every $\beta > \theta$, and the *signed* sum is bounded at the weight $(\log n)^{-\theta}$, which is critical by Corollary 5.3. So cancellation can matter only at the endpoint $\beta = \theta$, and whether the absolute inequality holds there is open. For $\theta = 1$ this endpoint question is exactly Perfekt's.

10.2.2. Fractional primitives of the zeta function

Bondarenko, Brevig, Saksman and Seip [15] considered, for $\beta > 0$, the functionals

$$\varphi_\beta(f) = a_1 + \sum_{n\ge 2} \frac{a_n}{\sqrt{n}(\log n)^\beta} = a_1 + \frac{1}{\Gamma(\beta)} \int_0^\infty \big(f(1/2+\sigma) - a_1\big)\sigma^{\beta-1}\,\mathrm{d}\sigma.$$

In one variable the analogue is $f \mapsto \Gamma(\beta)^{-1}\int_0^1 f(x)(1-x)^{\beta-1}\,\mathrm{d}x$, with coefficients $\Gamma(j+1)/\Gamma(j+1+\beta)$. It is bounded on $H^p(\mathbb{D})$ for $p > 1$ exactly when $\beta > 1/p$, and for $p \le 1$ exactly when $\beta \ge 1/p$ [15, Theorem 3.3].

Problem 11 (Conjecture of Bondarenko, Brevig, Saksman and Seip [15, Section 3]). *Let $0 < p \le 2$. Is $\varphi_{1/p}$ bounded on $\mathscr{H}^p$ if and only if $p \le 1$?*

The case $p = 1$ is Perfekt's theorem, and $p = 2$ is trivial. The answer for every p and β, and on prime sets, needs one more lemma.

Lemma 10.11 (Synthesis). *Let $1 \le p < \infty$, and let $P_j \ge 0$ be integrable functions with $\int \sqrt{P_i P_j} \le A e^{-b|i-j|}$. Then for finitely many $c_j \ge 0$,*

$$\int \Big(\sum_j c_j P_j^{1/p}\Big)^p \le C(A,b,p) \sum_j c_j^p.$$

Proof. Fix a point, and let j be an index with $c_j^p P_j$ largest there. Choose $a > 0$ with $a(p-1) < b$. Hölder's inequality with the weights $e^{-a|i-j|}$ gives

$(\sum_i c_i P_i^{1/p})^p \le C_a \sum_i e^{a(p-1)|i-j|} c_i^p P_i$, and $c_i^p P_i \le (c_i^p P_i)^{1/2}(c_j^p P_j)^{1/2}$ by the choice of j. Bounding the term of this j by the sum over all j and integrating,

$$\int \Big(\sum_i c_i P_i^{1/p}\Big)^p \le C_a A \sum_{i,j} e^{(a(p-1)-b)|i-j|} c_i^{p/2} c_j^{p/2},$$

and Schur's test bounds the right side by a constant times $\sum_j c_j^p$. □

For a set R of primes let $\mathscr{H}^p_R$ be the closure in $\mathscr{H}^p$ of the Dirichlet polynomials supported on integers composed of primes in R.

THEOREM 10.12. *Let R have density $\theta \in (0,1]$ with constant B, let $0 < p < \infty$ and $\beta > 0$. Then φ_β is bounded on $\mathscr{H}^p_R$ if and only if $\beta > \theta/p$, or $\beta = \theta/p$ and $p \le 1$. For R all primes, φ_β is bounded on $\mathscr{H}^p$ if and only if $\beta > 1/p$, or $\beta = 1/p$ and $p \le 1$, exactly as on the disc.*

PROOF. By Theorem 1.7 and Cauchy–Schwarz, $|f(1/2+\sigma)|^p \le \zeta(1+2\sigma)\|f\|^p$, and $\sigma \to \infty$ gives $|a_1| \le \|f\|$. For $\sigma \ge 1$ let F be the Bohr lift of a Dirichlet polynomial f, put $\rho = 2^{\sigma-1} r(\sigma)$, so that $\rho_q \le q^{-3/2}$, and $G(\lambda) = F(\lambda\rho)$. For $|\lambda| = 1$, plurisubharmonicity of $|F|^p$ gives $|G(\lambda)|^p \le \|f\|^p \sup P_{\lambda\rho} < 6\|f\|^p$, and the Schwarz lemma applied to $G - a_1$ gives $|f(1/2+\sigma) - a_1| \le 2 \cdot 6^{1/p} 2^{1-\sigma}\|f\|$; no triangle inequality is used, so this holds for $p < 1$ too. So the part $\sigma \ge 1$ of the integral and a_1 are bounded. By (2.1), $\zeta_R(1+2\sigma) \le C\sigma^{-\theta}$ on $(0,1]$, and $\nu = \sigma^{\theta-1}\,\mathrm{d}\sigma$ has $[\nu]_\theta = 1/\theta$, so part (2) of Theorem 2.11 gives $\int_0^1 |f(1/2+\sigma)|^p \sigma^{\theta-1}\,\mathrm{d}\sigma \le C\|f\|^p$.

Boundedness. If $p \le 1$ and $\beta \ge \theta/p$, write $|f| = |f|^p |f|^{1-p}$ and use $|f(1/2+\sigma)|^p \le \zeta_R(1+2\sigma)\|f\|^p \le C\sigma^{-\theta}\|f\|^p$ in the second factor:

$$|f|\sigma^{\beta-1} \le C|f|^p \sigma^{-\theta(1-p)/p + \beta - 1} \|f\|^{1-p},$$

and the exponent is at least $\theta - 1$. If $p > 1$ and $\beta > \theta/p$, Hölder's inequality gives

$$\int_0^1 |f|\sigma^{\beta-1}\,\mathrm{d}\sigma \le \Big(\int_0^1 |f|^p \sigma^{\theta-1}\,\mathrm{d}\sigma\Big)^{1/p} \Big(\int_0^1 \sigma^{(\beta-\theta/p)p'-1}\,\mathrm{d}\sigma\Big)^{1/p'},$$

with $p' = p/(p-1)$, and the last integral is finite.

Unboundedness. Let K_δ be the Bohr lift of $\sum n^{-1/2-\delta} n^{-s}$ over integers composed of primes in R, and $F_\delta = K_\delta^{2/p} \zeta_R(1+2\delta)^{-1/p}$, so that $|F_\delta|^p$ is the Poisson kernel P_δ of $r_R(\delta)$, $\|F_\delta\|_p = 1$, and F_δ has positive coefficients with constant term at most 1. For $\sigma \le \delta$, $F_\delta(1/2+\sigma) \ge \zeta_R(1+2\delta)^{1/p} \ge c\delta^{-\theta/p}$, so

$$\varphi_\beta(F_\delta) \ge \frac{1}{\Gamma(\beta)} \int_0^\delta \big(F_\delta(1/2+\sigma) - 1\big)\sigma^{\beta-1}\,\mathrm{d}\sigma \ge c'\delta^{\beta-\theta/p}$$

for small δ. If $\beta < \theta/p$ this is unbounded. If $\beta = \theta/p$ and $p > 1$, put $\delta_j = 2^{-j}$ and $g = \sum_{j=j_0}^{j_0+M} F_{\delta_j}$, so that $\varphi_\beta(g) \ge c'M$. By part (4) of Proposition 1.4 in each coordinate and (2.1),

$$\int \sqrt{P_{\delta_i} P_{\delta_j}} \le \Big(\frac{\zeta_R(1+\delta_i+\delta_j)^2}{\zeta_R(1+2\delta_i)\zeta_R(1+2\delta_j)}\Big)^{1/4} \le Ce^{-b|i-j|}, \qquad b = \theta(\log 2)/4,$$

so Lemma 10.11 gives $\|g\|_p^p \le CM$ and $\varphi_\beta(g)/\|g\|_p \ge cM^{1-1/p} \to \infty$. The functions F_δ involve infinitely many primes. First truncate their Euler products to finitely

many primes, keeping the normalizing constants. These truncations converge in $\mathscr{H}^p$, while their positive functional values increase to the values above. For each finite prime set, Taylor sections converge uniformly on the closed polydisc. Their norms and functional values therefore converge, giving the same unbounded ratios for Dirichlet polynomials. □

COROLLARY 10.13. *Let $0 < p \le 2$. The functional $\varphi_{1/p}$ is bounded on $\mathscr{H}^p$ if and only if $p \le 1$. This answers Problem 11.*

PROOF. The case R all primes and $\beta = 1/p$ of Theorem 10.12. □

For R all primes and $0 < p \le 2$, the cases $\beta \ne 1/p$ are [15, Theorem 3.1(b)]; what is new is the critical exponent, the range $p > 2$, and prime sets.

At the endpoint $\beta = 1/p$, $p > 1$, the growth on short polynomials is slow.

PROPOSITION 10.14. *Let $p > 1$. For Dirichlet polynomials f of length $N \ge 3$, $|\varphi_{1/p}(f)| \le C_p (\log\log N)^{1-1/p} \|f\|_{\mathscr{H}^p}$.*

PROOF. Put $\delta = 1/\log N$. The parts $\sigma \ge 1$ and a_1 are bounded as in the proof of Theorem 10.12. On $(\delta, 1)$, Hölder's inequality and Theorem 2.5 give $\int_\delta^1 |f| \sigma^{1/p-1}\, d\sigma \le C(\int_\delta^1 \sigma^{-1}\, d\sigma)^{1-1/p} \|f\|$. On $(0, \delta)$, vertical translations act isometrically on $\mathscr{H}^p$ and f has frequencies in $[0, \log N]$, so Bernstein's inequality for vector-valued functions of exponential type gives $\|f'\|_{\mathscr{H}^p} \le \log N \|f\|_{\mathscr{H}^p}$. Hence, by Theorem 2.5 for f' and Hölder, $|f(1/2+\sigma)| \le |f(1/2+\delta)| + C\delta^{1-1/p} \log N \|f\|$, and $|f(1/2+\delta)| \le \zeta(1+2\delta)^{1/p} \|f\| \le C\delta^{-1/p} \|f\|$. Integrating against $\sigma^{1/p-1}$ over $(0, \delta)$ gives $O(\|f\|)$. □

The order $(\log\log N)^{1-1/p}$ is attained: the test functions in the proof of Theorem 10.12, cut off at scale $1/\log N$, become polynomials of length N under the partial sum operators S_N, which keep the terms $n \le N$ and are uniformly bounded on $\mathscr{H}^p$ for $1 < p < \infty$ by Helson's theorem.

PROPOSITION 10.15. *Let $p > 1$. There are $c_p > 0$ and N_0 such that, for every $N \ge N_0$, some Dirichlet polynomial $f \ne 0$ of length N satisfies*

$$|\varphi_{1/p}(f)| \ge c_p (\log\log N)^{1-1/p} \|f\|_{\mathscr{H}^p}.$$

So the bound of Proposition 10.14 has the right order.

PROOF. Let J be the number of primes up to $X \ge N$, and order $\mathbb{Z}^J$ by $\alpha \mapsto \sum_j \alpha_j \log q_j$, which is injective since the $\log q_j$ are rationally independent. Helson's theorem [43] bounds the projection Q onto the characters z^α with $\prod_j q_j^{\alpha_j} > 1$ on $L^p(\mathbb{T}^J)$ by $C(p)$, independently of J. Multiplication by N^s is an isometry, and $I - S_N = N^{-s} Q N^s$, so $\|S_N\| \le 1 + C(p)$ uniformly in N and J [15, Section 5].

Let $\kappa = 2/p$, let $d_\kappa(n) > 0$ be the coefficients of ζ^κ, and $C_* = 2\log(2 \cdot 3^\kappa)$. Let g be the sum of the functions F_δ of the proof of Theorem 10.12, for R the primes up to X, over the scales $\delta = 2^{-j}$ with $2^{-p} \ge \delta \ge C_*/\log N$; there are $M \ge (\log_2 \log N)/2$ of them for large N. Choosing X after N and large enough, the affinity bound in that proof holds for these finitely many pairs with a constant independent of N, X and M, so Lemma 10.11 gives $\|S_N g\|_p \le C M^{1/p}$.

Fix one δ and let $x = e^{C_*/\delta} \le N \le X$. The coefficient of $n \le x$ in F_δ is at least $\zeta(1+2\delta)^{-1/p} d_\kappa(n) n^{-1/2-\delta}$, and since $1/t \le \zeta(1+t) \le 1 + 1/t$,

$$\sum_{n>x} \frac{d_\kappa(n)}{n^{1+\delta}} \le x^{-\delta/2} \zeta(1+\delta/2)^\kappa \le \frac{\delta^{-\kappa}}{2} \le \frac{\zeta(1+\delta)^\kappa}{2}.$$

So $\sum_{2\le n\le x} d_\kappa(n) n^{-1-\delta} \ge \delta^{-\kappa}/4$, as $\delta^{-\kappa} \ge 4$. With $(\log n)^{-1/p} \ge (\delta/C_*)^{1/p}$ for $n \le x$ and $\zeta(1+2\delta) \le 3/(2\delta)$, this gives $\varphi_{1/p}(S_N F_\delta) \ge (2/(3C_*))^{1/p}/4$. All coefficients are nonnegative, so $\varphi_{1/p}(S_N g) \ge cM$, and $f = S_N g$ has $\varphi_{1/p}(f)/\|f\|_p \ge cM^{1-1/p}/C$. □

The mechanism is the one-variable one: normalized kernels at dyadic scales are seen with weight of order one by $\varphi_{1/p}$, while their pth powers are almost orthogonal when $p > 1$. On $\mathbb{T}^\infty$ the almost orthogonality is the decay of the Hellinger affinity across scales.

10.3. Partial sums

Partial sums lie outside the story of π: the operators S_N are not coefficient functionals, and they share with the rest of Part IV only the tools, Helson's inequality and Harper's low moments. We include them because the same Euler products decide the $\mathscr{H}^1$ problem: a product of one-variable densities over the first d primes has entropy linear in d, and that entropy is what the partial sum operator cannot keep small.

Problem 12 (Saksman and Seip [80, Problem 4.5]). *What is the precise growth of $\|S_N\|_{\mathscr{H}^1\to\mathscr{H}^1}$, where $S_N \sum a_n n^{-s} = \sum_{n\le N} a_n n^{-s}$?*

For Problem 12, the order of growth was known for every p except $p = 1$. For $1 < p < \infty$ the operators S_N are uniformly bounded, by Helson's theorem on half-spaces of ordered groups [43] applied to $\mathbb{Q}_+$ ordered by size, as in the proof of Proposition 10.15. On $\mathscr{H}^\infty$, Balasubramanian, Calado and Queffélec [3] proved $\|S_N\| \lesssim \log N$, and Chen and Fu [29] have shown that $\|S_N\| \ge (1/(2\pi) + o(1)) \log N$, so the order is $\log N$. For $0 < p < 1$, Bondarenko, Brevig, Saksman and Seip [15] showed that $\log\|S_N\|_{\mathscr{H}^p\to\mathscr{H}^p}$ is of order $\log N/\log\log N$. On $\mathscr{H}^1$ the lower bound $\log\log N \lesssim \|S_N\|$ was known earlier, and they proved the upper bound $\|S_N\| \lesssim \log N/\log\log N$. The lower bound is a one-variable fact: on series in the powers 2^{-ks}, S_N is the partial sum of degree $\log_2 N$ on $H^1(\mathbb{D})$. Any larger lower bound must therefore use many primes. The next theorem shows that the upper bound gives the correct order.

10.3.1. The order on $\mathscr{H}^1$

On $\mathbb{T}^d$, let $\lambda \in \mathbb{R}^d$ satisfy $\lambda \cdot \gamma \ne 0$ for every $\gamma \in \mathbb{Z}^d \setminus \{0\}$, and let P_λ keep the Fourier coefficients with $\lambda\cdot\gamma \le 0$. With $\lambda_j = \log p_j$, the operator P_λ is a partial sum operator in disguise; this connection between partial sums and half-spaces of ordered groups is used by Konyagin, Queffélec, Saksman and Seip [54, Section 5]. The next lemma is the ordered-group form of the classical argument behind Zygmund's $L\log L$ converse for nonnegative functions [98, Chapter VII, Section 2].

LEMMA 10.16. *Suppose that $u \geq 0$ is a real trigonometric polynomial on $\mathbb{T}^d$ with $\int u \, \mathrm{d}m = 1$. Then $\|P_\lambda u\|_1 \geq \pi^{-1} \int u \log u \, \mathrm{d}m$.*

PROOF. Put $G = 2P_\lambda u - 1 = u + iv$; since $\widehat{u}(-\gamma) = \overline{\widehat{u}(\gamma)}$, v is real, and $\|P_\lambda u\|_1 \geq \|v\|_1/2$. All frequencies of $G - 1$ lie in the half-space $\lambda \cdot \gamma < 0$, which is closed under addition, so $\int A(G+\varepsilon) \, \mathrm{d}m = A(1+\varepsilon)$ for every polynomial A and $\varepsilon > 0$. Since $\Re(G + \varepsilon) \geq \varepsilon$, for R large we have $\|1 - (G+\varepsilon)/R\|_\infty < 1$, so $\log(G+\varepsilon)$ is a uniform limit of polynomials in $G + \varepsilon$ and

$$\int (G+\varepsilon) \log(G+\varepsilon) \, \mathrm{d}m = (1+\varepsilon)\log(1+\varepsilon).$$

Take real parts. Since $|G + \varepsilon| \geq u + \varepsilon$ and $|\arg(G+\varepsilon)| \leq \pi/2$, this gives $\int (u + \varepsilon)\log(u+\varepsilon) \, \mathrm{d}m \leq (1+\varepsilon)\log(1+\varepsilon) + (\pi/2)\|v\|_1$. Let $\varepsilon \to 0$. □

THEOREM 10.17. *Let $p_1, \ldots, p_d$ be the first d primes, Q_d their product, and $f_d = 2^{-d} \prod_{j \leq d} (1 + p_j^{-s})^2$. Then $\|f_d\|_{\mathscr{H}^1} = 1$ and $\|S_{Q_d} f_d\|_{\mathscr{H}^1} \geq d(1 - \log 2)/\pi$. Consequently there are absolute constants $c, C > 0$ such that, for all sufficiently large N,*

$$c \, \frac{\log N}{\log\log N} \leq \|S_N\|_{\mathscr{H}^1 \to \mathscr{H}^1} \leq C \, \frac{\log N}{\log\log N}.$$

PROOF. The Bohr lift of f_d is $z_1 \cdots z_d \, u_d$ with $u_d = \prod_j (1 + \cos\theta_j)$, $z_j = e^{i\theta_j}$. So $\|f_d\|_1 = \int u_d \, \mathrm{d}m = 1$. The coefficient at the frequency $\gamma \in \{-1, 0, 1\}^d$ of u_d belongs to the integer $Q_d \prod_j p_j^{\gamma_j}$, which is at most Q_d exactly when $\lambda \cdot \gamma \leq 0$, $\lambda_j = \log p_j$. Hence $S_{Q_d} f_d$ lifts to $z_1 \cdots z_d \, P_\lambda u_d$, and unique factorization gives $\lambda \cdot \gamma \neq 0$ for $\gamma \neq 0$. Entropy is additive over the coordinates, and with $\theta = 2t$,

$$\int_{\mathbb{T}} (1+\cos\theta)\log(1+\cos\theta) \, \mathrm{d}m(\theta) = \log 2 + \frac{8}{\pi} \int_0^{\pi/2} \cos^2 t \log\cos t \, \mathrm{d}t = 1 - \log 2,$$

since

$$\int_0^{\pi/2} \log\cos t \, \mathrm{d}t = -\frac{\pi}{2}\log 2, \qquad \int_0^{\pi/2} \cos 2t \log\cos t \, \mathrm{d}t = \frac{\pi}{4},$$

the second by an integration by parts. Now Lemma 10.16 gives the first claim.

For the lower bound in the second, each prime $p \leq 2n$ contributes at most $\log 2n$ to $\log \binom{2n}{n} \geq 2n \log 2 - \log(2n+1)$, so $\pi(x) \gtrsim x/\log x$ and $\log Q_d \leq C_0 d \log(d+1)$. Take $d = \lfloor c_0 \log N / \log\log N \rfloor$ with c_0 so small that $Q_d^2 \leq N/4$, and let $b = \lfloor N/Q_d \rfloor$. Then $b > Q_d$, so $\lfloor N/b \rfloor = Q_d$, and $S_N(b^{-s} f_d) = b^{-s} S_{Q_d} f_d$. Multiplication by b^{-s} preserves the $\mathscr{H}^1$ norm. The upper bound is [15, Theorem 5.2]. □

So the order in Problem 12 is $\log N/\log\log N$ for $p = 1$ as well; the leading constant is not determined. The output $S_{Q_d} f_d$ is supported on integers with $\Omega(n) \leq 2d$, so part (3) of Corollary 10.19 below is sharp up to the constant. The test function f_d is the case $p = 1$ of the product in the proof of [15, Theorem 5.3], and the passage from Q_d to general N through $\lfloor N/b \rfloor$ is also their argument. What is new, to our knowledge, is the lower bound for $\|S_{Q_d} f_d\|_{\mathscr{H}^1}$ by entropy (Lemma 10.16); the argument also appears in the preprint [86].

Optimizing ε in the inequality from the proof of [15, Theorem 5.2] gives an output-sensitive bound.

PROPOSITION 10.18. *There is an absolute C such that, for $f \in \mathscr{H}^1$ and $g = S_N f = \sum_{n\le N} a_n n^{-s} \neq 0$,*

$$\frac{\|g\|_{\mathscr{H}^1}}{\|f\|_{\mathscr{H}^1}} \le C\Big(1 + \log \frac{\|g\|_{\mathscr{H}^2}}{\|g\|_{\mathscr{H}^1}}\Big).$$

PROOF. Bondarenko, Brevig, Saksman and Seip [15] prove that

$$\|S_N f\|_{\mathscr{H}^{1-\varepsilon}} \le C_0 \varepsilon^{-1} \|f\|_{\mathscr{H}^1}.$$

Hölder's inequality gives

$$\|g\|_1 \le \|g\|_{1-\varepsilon}^{(1-\varepsilon)/(1+\varepsilon)} \|g\|_2^{2\varepsilon/(1+\varepsilon)}, \qquad \text{hence} \qquad \frac{\|g\|_1}{\|f\|_1} \le \frac{C_0}{\varepsilon}\Big(\frac{\|g\|_2}{\|g\|_1}\Big)^{2\varepsilon/(1-\varepsilon)}.$$

Take $\varepsilon = 1/(2(1 + \log(\|g\|_2/\|g\|_1)))$. □

COROLLARY 10.19. *Let $f \in \mathscr{H}^1$. The implied constants below are absolute; in particular they do not depend on the tail $f - S_N f$.*

(1) if $S_N f = \sum_{n\le N} n^{-s}$, then $\|S_N f\|_{\mathscr{H}^1} \lesssim (1 + \log\log\log N)\|f\|_{\mathscr{H}^1}$;
(2) if $S_N f = \sum_{n\le N} n^{-1/2-s}$, then $\|S_N f\|_{\mathscr{H}^1} \lesssim (1 + \log\log N)\|f\|_{\mathscr{H}^1}$;
(3) if every n in the support of $S_N f$ has $\Omega(n) \le m$, then $\|S_N f\|_{\mathscr{H}^1} \lesssim (m + 1)\|f\|_{\mathscr{H}^1}$.

PROOF. (1) Harper's theorem [40] gives $\|\sum_{n\le N} n^{-s}\|_1 \asymp \sqrt{N}(\log\log N)^{-1/4}$. (2) Gerspach's pseudomoment bound [35] gives $\|\sum_{n\le N} n^{-1/2-s}\|_1 \asymp (\log N)^{1/4}$ against an $\mathscr{H}^2$ norm $\asymp (\log N)^{1/2}$. (3) $d(n) \le 2^m$ on the support, so Helson gives $\|g\|_2 \le 2^{m/2}\|g\|_1$. □

By Proposition 10.18, an amplification of order d needs an output whose ratio of $\mathscr{H}^2$ to $\mathscr{H}^1$ norms is exponentially large in d. The outputs $S_{Q_d} f_d$ of Theorem 10.17 are of this kind: their $\mathscr{H}^2$ norm is $((1 + (3/2)^d)/2)^{1/2}$. The Dirichlet kernel is not, and neither is the partial sum of $\zeta(1/2 + s)$, which can at most give $\log\log N$. Vertical blow-up does not help either (section 3.2).

10.4. Order and mean-value abscissae

Helson's coefficient inequality also controls how far left a vertical mean recovers the Hardy norm. An arbitrarily small positive shift sends $\mathscr{H}^1$ into $\mathscr{H}^2$, where an order-to-mean theorem is available. Polynomial approximation then transfers that theorem to $1 \le p \le 2$. Sharpness needs one further fact: a bounded first mean on a line forces convergence of the Dirichlet series to its right.

Write $\mathbb{C}_\kappa = \{s = \sigma + it : \sigma > \kappa\}$. The Lindelöf order $\mu_f(\sigma)$ is the infimum of the exponents a such that $|f(\sigma + it)| = O((1 + |t|)^a)$. The numbers $\sigma_{\mathrm{c}}(f)$, $\sigma_{\mathrm{a}}(f)$ are the abscissae of convergence and absolute convergence. For $1 \le p < \infty$, $\mathscr{H}^p$ is the Hardy space of Dirichlet series $f(s) = \sum_{n\ge1} a_n n^{-s}$, normed through the Bohr lift (equivalently, for Dirichlet polynomials, by $\|P\|_{\mathscr{H}^p}^p = \lim_{T\to\infty}(2T)^{-1}\int_{-T}^{T}|P(it)|^p\,\mathrm{d}t$, see Bayart [10]). We write $f(\sigma + \cdot)$ for the Dirichlet series $\sum_{n\ge1} a_n n^{-\sigma} n^{-s}$. Saying that f has an analytic continuation *of finite order* to $\mathbb{C}_0$ means, as in [19], that for every $\varepsilon > 0$ there are $A, B > 0$ with $|f(s)| \le B(1 + |t|)^A$ for $\operatorname{Re} s \ge \varepsilon$.

For such f in $\mathscr{H}^p$, let $\sigma_{\mathrm{m}}^{(p)}(f)$ be the infimum of those $\kappa \ge 0$ such that, for every $\sigma > \kappa$, the limit

$$M_p(f;\sigma) = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} |f(\sigma+it)|^p \, \mathrm{d}t$$

exists and equals $\|f(\sigma+\cdot)\|_{\mathscr{H}^p}^p$. For $p = 2$ and f of finite order in $\mathbb{C}_0$ this is $\max(\sigma_{\mathrm{m}}(f), 0)$ in the notation of [19].

The external input is the following assertion of [19, Theorem 1]: if $h \in \mathscr{H}^2$ has an analytic continuation of finite order to $\mathbb{C}_0$ and $\mu_h(\sigma) \le \beta$ for every $\sigma > 0$, then $\sigma_{\mathrm{m}}(h) \le \beta/(1+2\beta)$. We assume this assertion throughout the next theorem.

THEOREM 10.20 (Conditional mean-value abscissa). *Assume [19, Theorem 1]. Fix $\beta > 0$ and $1 \le p \le 2$.*

(1) If f in $\mathscr{H}^p$ has an analytic continuation of finite order to $\mathbb{C}_0$ with $\mu_f(\sigma) \le \beta$ for every $\sigma > 0$, then $\sigma_{\mathrm{m}}^{(p)}(f) \le \beta/(1+2\beta)$.

(2) The function f_β of [19, Theorem 4] satisfies the hypotheses of (1) and $\sigma_{\mathrm{m}}^{(p)}(f_\beta) = \beta/(1+2\beta)$.

Hence the optimal bound is $\beta/(1+2\beta)$ for every $1 \le p \le 2$, and it is attained.

Part (2) rests on the following p-mean version of Titchmarsh's inequality $\sigma_{\mathrm{c}}(f) \le \sigma_{\mathrm{m}}(f)$ [91, §9.55]. Only L^1-means enter, so it holds for every $p \ge 1$.

LEMMA 10.21. *Let $f(s) = \sum_{n\ge1} a_n n^{-s}$ with $\sup_n |a_n| = K_0 < \infty$ have an analytic continuation of finite order to $\mathbb{C}_0$. Suppose that $\sigma_1 > 0$ and*

$$G(T) = \int_{-T}^{T} |f(\sigma_1+it)| \, \mathrm{d}t \le K_1(1+T) \qquad (T \ge 0). \tag{10.1}$$

Then the Dirichlet series of f converges to $f(\sigma_2)$ at every $\sigma_2 > \sigma_1$. In particular $\sigma_{\mathrm{c}}(f) \le \sigma_1$.

We first need a uniform form of (10.1) on a strip.

LEMMA 10.22. *Let f be as in Lemma 10.21, let $b > \max(1, \sigma_1)$, and set $L = b - \sigma_1$. There is K such that for all $U \ge \max(1, L)$ and all $\sigma_1 \le u \le b$,*

$$\int_{U}^{2U} |f(u+iV)| \, \mathrm{d}V \le KU \qquad \text{and} \qquad \int_{U}^{2U} |f(u-iV)| \, \mathrm{d}V \le KU.$$

PROOF. Let $S = \{\sigma_1 \le \operatorname{Re} s \le b\}$. By hypothesis $|f(s)| \le B(1+|t|)^A$ on S. On the right edge $|f(b+it)| \le K_0\zeta(b)$, so (10.1) also holds on $\operatorname{Re} s = b$, with some constant K_1'. For $s = u+it$ in the interior of S and $\theta = (u-\sigma_1)/L$, let

$$P_0(u,t) = \frac{1}{2L} \frac{\sin\pi\theta}{\cosh(\pi t/L) - \cos\pi\theta}, \qquad P_1(u,t) = \frac{1}{2L} \frac{\sin\pi\theta}{\cosh(\pi t/L) + \cos\pi\theta}$$

be the Poisson kernels of S for its left and right edges [95], so that $\int_{\mathbb{R}} (P_0 + P_1)(u,t) \, \mathrm{d}t = 1$, and set

$$\Phi(u+it) = \int_{\mathbb{R}} P_0(u, t-y) |f(\sigma_1+iy)| \, \mathrm{d}y + \int_{\mathbb{R}} P_1(u, t-y) |f(b+iy)| \, \mathrm{d}y.$$

Since the boundary data are continuous with polynomial growth and the kernels decay exponentially, $\Phi \ge 0$ is harmonic in the interior of S and continuous on S with

boundary values $|f|$. The function $h = |f| - \Phi$ is subharmonic in the interior of S, continuous on S, vanishes on ∂S, and $h \le |f| \le B(1+|t|)^A$. With $m = (\sigma_1 + b)/2$ and $\gamma = \pi/(2L)$, the function $\varphi(u+it) = \cos(\gamma(u-m))\cosh(\gamma t)$ is harmonic and $\varphi \ge 2^{-1/2}\cosh(\gamma t)$ on S. For $\eta > 0$, $h - \eta\varphi$ is subharmonic, nonpositive on ∂S and on $|t| = R$ for all large R, so $h \le \eta\varphi$ by the maximum principle; letting $\eta \to 0$ gives $|f| \le \Phi$ on S.

For $|t| \ge L$ we have $\cosh(\pi t/L) \pm \cos\pi\theta \ge \cosh(\pi t/L) - 1 \ge e^{\pi|t|/L}/4$, whence $P_j(u,t) \le (2/L)e^{-\pi|t|/L}$. Put $k_j(y) = \int_U^{2U} P_j(u, V-y)\,\mathrm{d}V$; then $k_j \le 1$, and if $|y| \ge 4U$ then $|V-y| \ge |y|/2 \ge L$ for $U \le V \le 2U$, so $k_j(y) \le (2U/L)e^{-\pi|y|/(2L)}$. Writing $g_0(y) = |f(\sigma_1 + iy)|$ and $g_1(y) = |f(b+iy)|$, Fubini gives

$$\int_U^{2U} |f(u+iV)|\,\mathrm{d}V \le \sum_{j=0}^{1} \int_{-4U}^{4U} g_j(y)\,\mathrm{d}y + \frac{4U}{L}\int_{|y|>4U} B(1+|y|)^A e^{-\pi|y|/(2L)}\,\mathrm{d}y,$$

which is at most KU by (10.1) on both edges. The case $u - iV$ is identical. □

Proof of Lemma 10.21. Fix $\sigma_2 > \sigma_1$ and put $\eta = \sigma_2 - \sigma_1$, $c = 2$, $b = \sigma_2 + c$, $L = b - \sigma_1$. Let $x = N + 1/2$ with $N \ge 1$ an integer, let $S(x) = \sum_{n<x} a_n n^{-\sigma_2}$ and put $U = x^6$, assuming x is so large that $U \ge L$. The series $\sum_n a_n n^{-\sigma_2} n^{-w} = f(\sigma_2 + w)$ converges absolutely on $\operatorname{Re} w = c$. By the truncated Perron formula [31, Chapter 17, Lemma], for $y > 0$, $y \ne 1$,

$$\left| \frac{1}{2\pi i}\int_{c-iV}^{c+iV} \frac{y^w}{w}\,\mathrm{d}w - \mathbf{1}_{y>1} \right| \le y^c \min\Bigl(1, \frac{1}{V|\log y|}\Bigr).$$

Apply this with $y = x/n$. For $n \le 2x$ we have $|\log(x/n)| \ge |x-n|/(2x) \ge 1/(4x)$, and for $n > 2x$ we have $|\log(x/n)| \ge \log 2$. Since $|a_n| n^{-\sigma_2} \le K_0$,

$$S(x) = \frac{1}{2\pi i}\int_{c-iV}^{c+iV} f(\sigma_2 + w)\frac{x^w}{w}\,\mathrm{d}w + R(V), \tag{10.2}$$

where

$$|R(V)| \le K_0\Bigl(\frac{8x^4}{V} + \frac{2x}{V}\Bigr) \le \frac{10K_0x^4}{V}.$$

The integrand is analytic on $\{-\eta \le \operatorname{Re} w \le c\}$ apart from a simple pole at $w = 0$ with residue $f(\sigma_2)$. The residue theorem on the rectangle with vertices $c \pm iV$ and $-\eta \pm iV$ gives

$$\frac{1}{2\pi i}\int_{c-iV}^{c+iV} f(\sigma_2 + w)\frac{x^w}{w}\,\mathrm{d}w = f(\sigma_2) + J(V) + H(V),$$

where $J(V)$ is the integral over $\operatorname{Re} w = -\eta$, $|\operatorname{Im} w| \le V$, and $H(V)$ collects the two horizontal segments. On those segments $|x^w/w| \le x^c/V$, so

$$|H(V)| \le \frac{x^c}{2\pi V}\sum_{\pm}\int_{\sigma_1}^{b} |f(u \pm iV)|\,\mathrm{d}u.$$

On $\operatorname{Re} w = -\eta$ we have $|x^w| = x^{-\eta}$ and $|w| \ge \max(\eta, |\operatorname{Im} w|)$. Splitting $|v| \le 1$ and the dyadic ranges $2^j < |v| \le 2^{j+1}$, $0 \le j \le \log_2 V$, and using (10.1),

$$|J(V)| \le \frac{x^{-\eta}}{2\pi} \int_{-V}^{V} \frac{|f(\sigma_1 + iv)|}{\max(\eta, |v|)}\,\mathrm{d}v \le x^{-\eta} K_1 \Big(\frac{2}{\eta} + 3(\log_2 V + 1)\Big).$$

Now average (10.2) over $U \le V \le 2U$. By Lemma 10.22,

$$\frac{1}{U}\int_U^{2U} |H(V)|\,\mathrm{d}V \le \frac{x^c}{2\pi U^2} \sum_{\pm} \int_{\sigma_1}^{b} \int_U^{2U} |f(u \pm iV)|\,\mathrm{d}V\,\mathrm{d}u \le \frac{LKx^2}{\pi U}.$$

Since $S(x)$ does not depend on V, we obtain, with $U = x^6$,

$$|S(x) - f(\sigma_2)| \le \frac{10K_0}{x^2} + x^{-\eta} K_1 \Big(\frac{2}{\eta} + 3(6\log_2 x + 2)\Big) + \frac{LK}{\pi x^4},$$

which tends to 0 as $x \to \infty$. Thus $\sum_{n \le N} a_n n^{-\sigma_2} \to f(\sigma_2)$. □

COROLLARY 10.23. *Let $1 \le p < \infty$ and let f in $\mathscr{H}^p$ have an analytic continuation of finite order to $\mathbb{C}_0$. Then $\sigma_{\mathrm{c}}(f) \le \sigma_{\mathrm{m}}^{(p)}(f)$.*

PROOF. The coefficients satisfy $|a_n| \le \|f\|_{\mathscr{H}^p}$, since a_n is a Fourier coefficient of the Bohr lift and $\|\cdot\|_{L^1} \le \|\cdot\|_{L^p}$ on the infinite polytorus. Let $\sigma_1 > \sigma_{\mathrm{m}}^{(p)}(f)$; then $\sigma_1 > 0$. Since $M_p(f;\sigma_1)$ exists, $(2T)^{-1}\int_{-T}^{T} |f(\sigma_1 + it)|^p\,\mathrm{d}t$ is bounded for $T \ge 1$, so Hölder's inequality and continuity of f on the line give (10.1). Lemma 10.21 yields $\sigma_{\mathrm{c}}(f) \le \sigma_1$. □

PROOF OF THEOREM 10.20. (1) Let $\sigma > \beta/(1+2\beta)$ and fix $0 < \varepsilon < \sigma - \beta/(1+2\beta)$. Put $g(s) = f(s+\varepsilon)$. By Helson's inequality [44], $\sum_n |a_n|^2/d(n) \le \|f\|_{\mathscr{H}^1}^2 \le \|f\|_{\mathscr{H}^p}^2$, where d is the divisor function. Since $d(n)n^{-2\varepsilon}$ is bounded, g belongs to $\mathscr{H}^2$. Let P_j be Dirichlet polynomials with $\|g - P_j\|_{\mathscr{H}^2} \to 0$. Each P_j is bounded on $\overline{\mathbb{C}_0}$, so $h_j = g - P_j$ is an element of $\mathscr{H}^2$ with an analytic continuation of finite order to $\mathbb{C}_{-\varepsilon} \supset \mathbb{C}_0$ and $\mu_{h_j}(\sigma') \le \max(\mu_f(\sigma'+\varepsilon), 0) \le \beta$ for every $\sigma' > 0$. By [19, Theorem 1], $\sigma_{\mathrm{m}}(h_j) \le \beta/(1+2\beta) < \sigma - \varepsilon$, so with $\sigma' = \sigma - \varepsilon$,

$$\lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} |h_j(\sigma' + it)|^2\,\mathrm{d}t = \|h_j(\sigma' + \cdot)\|_{\mathscr{H}^2}^2 \le \|g - P_j\|_{\mathscr{H}^2}^2.$$

Write $A_T(u)$ for the normalized L^p norm of $u(\sigma' + it)$ on $[-T, T]$. By Minkowski's inequality and $p \le 2$,

$$|A_T(g) - A_T(P_j)| \le A_T(h_j) \le \left(\frac{1}{2T}\int_{-T}^{T} |h_j(\sigma' + it)|^2\,\mathrm{d}t\right)^{1/2}.$$

Moreover $A_T(P_j) \to \|P_j(\sigma' + \cdot)\|_{\mathscr{H}^p}$ as $T \to \infty$ [10], and $|\|P_j(\sigma'+\cdot)\|_{\mathscr{H}^p} - \|g(\sigma' + \cdot)\|_{\mathscr{H}^p}| \le \|g - P_j\|_{\mathscr{H}^2}$. Hence

$$\limsup_{T\to\infty} |A_T(g) - \|g(\sigma' + \cdot)\|_{\mathscr{H}^p}| \le 2\|g - P_j\|_{\mathscr{H}^2} \to 0.$$

Since $g(\sigma' + it) = f(\sigma + it)$ and $g(\sigma' + \cdot) = f(\sigma + \cdot)$, $M_p(f;\sigma)$ exists and equals $\|f(\sigma + \cdot)\|_{\mathscr{H}^p}^p$.

(2) By [19, Theorem 4], f_β is an entire function in $\mathscr{H}^2 \subset \mathscr{H}^p$ with $\sigma_{\mathrm{c}}(f_\beta) = \sigma_{\mathrm{a}}(f_\beta) = \beta/(1+2\beta)$ and $\mu_{f_\beta}(\sigma) = \beta - (1+2\beta)\sigma \le \beta$ for $0 < \sigma \le \beta/(1+2\beta)$, while

$\mu_{f_\beta}(\sigma) = 0$ for $\sigma > \sigma_{\mathrm{a}}(f_\beta)$. It has finite order in $\mathbb{C}_0$: for an integer $r > \beta$, the tail estimate [19, (17)] in the construction of f_β bounds the tail by a constant times $\prod_{k=0}^{r-1}|s+k|$ uniformly on $\overline{\mathbb{C}_0}$; the remaining finite sum is a Dirichlet polynomial. By part (1), $\sigma_{\mathrm{m}}^{(p)}(f_\beta) \le \beta/(1+2\beta)$, and by Corollary 10.23, $\sigma_{\mathrm{m}}^{(p)}(f_\beta) \ge \sigma_{\mathrm{c}}(f_\beta) = \beta/(1+2\beta)$. □

REMARK 10.24. For $\beta = 0$, applying the upper bound with every $\beta' > 0$ gives $\sigma_{\mathrm{m}}^{(p)}(f) = 0$ for $1 \le p \le 2$, under the same external assumption. The optimal bound for $p > 2$ remains open.

Open

(1) The absolute inequality $\sum|a_n|/(\sqrt{n}\log n) \lesssim \|f\|_{\mathscr{H}^1}$ (Problem 10). By Lemma 10.1, any proof must use more than Helson's inequality on the full set of integers.
(2) What is the order of the Sidon constant $S_1(x)$ of section 10.1.4? It lies between $\sqrt{x}\exp(c(\log\log x)^{1/3})$ and $(1+o(1))\sqrt{x\log x}$.
(3) Determine the optimal order-to-mean abscissa for $p > 2$. The conditional sharp bound in Theorem 10.20 covers $1 \le p \le 2$.
(4) Does $\|S_N\|_{\mathscr{H}^1\to\mathscr{H}^1}\log\log N/\log N$ have a limit? Theorem 10.17 gives only the order.

Notes

The $\mathscr{H}^2$ upper bound and the sharp example f_β are due to Brevig–Kouroupis [19, Theorems 1 and 4]. The extension to $1 \le p \le 2$ uses Helson's inequality and the first-mean convergence argument proved here; it remains conditional on their Theorem 1.

CHAPTER 11

The Gaussian boundary model

Chapters 9 and 10 constrain a counterexample to $C = \pi$: it must involve primes beyond 10^{216} (Proposition 9.14) and mix degrees (Theorem 10.2). Moreover, multiplying near-extremal squares by a fixed polynomial in finitely many primes cannot produce an excess (Corollary 9.4). We therefore study kernels involving increasingly many primes as their evaluation points approach the critical line. This chapter finds the natural model for that regime. For quadratic combinations of Euler kernels at many scales near the critical line, the H^1 norm has an exact Gaussian limit (section 11.1), so C is at least a constant C_{Gauss} of a log-correlated Gaussian field, and a probabilistic certificate provides numerical evidence that this constant exceeds π. The same witnesses bound the essential size of the symbol and every boundary-weighted functional. Section 11.2 shows that extremizing sequences either concentrate at the critical line or produce an extremizer, and that above the essential size of the symbol the norm is attained.

11.1. The Gaussian scaling limit

We first compute the limiting H^1 norm of quadratic combinations of normalized Euler kernels at several scales.

Fix scales $a_1, \dots, a_m > 0$ and write $a_i = e^{t_i}$. For $\varepsilon > 0$ let

$$k_i^\varepsilon(z) = \zeta(1 + 2\varepsilon a_i)^{-1/2} \prod_p \bigl(1 - p^{-1/2-\varepsilon a_i} z_p\bigr)^{-1},$$

the normalized reproducing kernel at the point $r(\varepsilon a_i)$. For a real symmetric $m \times m$ matrix A put

$$F_\varepsilon = \sum_{i,j} A_{ij} k_i^\varepsilon k_j^\varepsilon, \qquad F_\varepsilon^0 = F_\varepsilon - F_\varepsilon(0) \in H_0^1.$$

PROPOSITION 11.1. *As $\varepsilon \to 0$, $\langle k_i^\varepsilon, k_j^\varepsilon \rangle \to G_{ij}$ and $\ell(F_\varepsilon^0) \to \sum_{i,j} A_{ij} H_{ij}$, where*

$$G_{ij} = \operatorname{sech} \frac{t_i - t_j}{2}, \qquad H_{ij} = \frac{t_i - t_j}{\sinh((t_i - t_j)/2)}, \quad H_{ii} = 2.$$

Moreover $H \preceq \pi G$ for every choice of scales.

PROOF. The Gram entry is $\zeta(1 + \varepsilon(a_i + a_j))/\sqrt{\zeta(1 + 2\varepsilon a_i)\zeta(1 + 2\varepsilon a_j)}$, and $\zeta(1 + t) = t^{-1} + O(1)$. The functional is $\ell(F) = \int_0^\infty (F(r(\sigma)) - F(0))\, \mathrm{d}\sigma$, and the entry for (i, j) equals $\int_0^\infty (\zeta(1+\varepsilon a_i+\sigma)\zeta(1+\varepsilon a_j+\sigma) - 1)\, \mathrm{d}\sigma$ divided by the same normalization. Replacing both zeta factors by their poles costs $O(\log(1/\varepsilon))$, which is $o(1)$ after

normalization. The substitution $\sigma = \varepsilon x$ then gives $2\sqrt{a_i a_j}\int_0^\infty \mathrm{d}x/((a_i+x)(a_j+x)) = H_{ij}$.

For the last claim, $\pi\operatorname{sech}(x/2) - x/\sinh(x/2)$ is a positive definite function on $\mathbb{R}$: the Fourier transforms of $\operatorname{sech}(x/2)$ and $x/\sinh(x/2)$ are $2\pi\operatorname{sech}(\pi\xi)$ and $2\pi^2\operatorname{sech}^2(\pi\xi)$, and $2\pi^2\operatorname{sech}(\pi\xi)(1-\operatorname{sech}(\pi\xi)) \geq 0$. □

The last statement explains why squares never beat π here. If $A = uu^T$, then $F_\varepsilon = (\sum u_i k_i^\varepsilon)^2$, its norm tends to $u^T G u$, and $u^T H u \leq \pi u^T G u$. Any excess must come from quadratics that are not squares. These include positive semidefinite A of higher rank: for $A = \sum_\ell u_\ell u_\ell^T$ the squares $(\sum_i u_{\ell i} k_i^\varepsilon)^2$ are complex and can cancel, so the norm can fall below $\operatorname{tr}(AG)$. Their norms are not determined by G, and the next theorem computes them.

The question is under which measure the normalized Euler kernels have a useful limit. Under Haar measure they have none: each k_i^ε tends to zero weakly while its modulus concentrates. The proof below therefore writes $\|F_\varepsilon\|_1$ as an average against the mixture $m^{-1}\sum_j |k_j^\varepsilon|^2$ of the harmonic measures of the m points. Under each $|k_j^\varepsilon|^2\,\mathrm{d}m$ the coordinates z_p are independent with explicit Poisson laws, and after dividing by the mixture the integrand depends only on the ratios $k_i^\varepsilon/k_0^\varepsilon$. The ratios remove what all the kernels share, and their logarithms are sums over primes of small independent terms, to which the central limit theorem applies. The integrand is bounded, so convergence in law gives convergence of the norm.

Fix a reference scale a_0 among the a_i. Let $D = (D_i)$ be a centered circular complex Gaussian vector with $\mathbb{E}D_i\overline{D_k} = \Sigma_{ik}$, where

$$\Sigma_{ik} = \log\frac{(a_i+a_0)(a_k+a_0)}{2a_0(a_i+a_k)} = \int_0^\infty (e^{-a_i x} - e^{-a_0 x})(e^{-a_k x} - e^{-a_0 x})\,\frac{\mathrm{d}x}{x},$$

and put $R_i^{(j)} = \sqrt{a_i/a_0}\,(a_0+a_j)(a_i+a_j)^{-1}e^{D_i}$.

Lemma 11.2 (The kernel ratios). *Fix j. Under the probability measure $|k_j^\varepsilon|^2\,\mathrm{d}m$, the vector $(k_i^\varepsilon/k_0^\varepsilon)_i$ converges in law to $R^{(j)}$ as $\varepsilon \to 0$.*

Proof. Under $|k_j^\varepsilon|^2\,\mathrm{d}m$ the coordinates z_p are independent and Poisson-kernel distributed, with $\mathbb{E}z_p^n = p^{-n(1/2+\varepsilon a_j)}$. The logarithm of $r_i = k_i^\varepsilon/k_0^\varepsilon$ is $\log(a_i/a_0)/2 + o(1)$ plus the linear sum

$$\sum_p p^{-1/2}\bigl(p^{-\varepsilon a_i} - p^{-\varepsilon a_0}\bigr)z_p$$

plus a remainder of degree at least two. The centered remainder tends to zero in L^2: its independent prime contributions have squared norms bounded by Cp^{-2}, and each fixed-prime contribution vanishes. Its expectation also tends to zero, by the Poisson moments and the same summable bound Cp^{-2}. The linear sum has mean

$$\sum_p p^{-1}\bigl(p^{-\varepsilon(a_i+a_j)} - p^{-\varepsilon(a_0+a_j)}\bigr) \to \log\frac{a_0+a_j}{a_i+a_j},$$

by $\sum_p p^{-1-t} = \log(1/t) + c + o(1)$. In the same way its covariances tend to Σ_{ik}, and its pseudo-covariances vanish because $\mathbb{E}(z_p - \mathbb{E}z_p)^2 = 0$. The linear coefficients $b_{i,p} = p^{-1/2}(p^{-\varepsilon a_i} - p^{-\varepsilon a_0})$ satisfy $\sup_p |b_{i,p}| \to 0$: each fixed-prime coefficient

vanishes, while the tail is bounded by $2p^{-1/2}$. Thus the centered summands are uniformly small, which verifies the Lindeberg condition and proves the claim. □

THEOREM 11.3.

$$\lim_{\varepsilon\to 0}\|F_\varepsilon^0\|_1 = \mathcal{N}(A) \coloneqq \frac{1}{m}\sum_{j=1}^{m}\mathbb{E}\,\frac{m\,|R^{(j)T}AR^{(j)}|}{|R^{(j)}|^2}.$$

Consequently $C \ge |\sum_{i,j} A_{ij}H_{ij}|/\mathcal{N}(A)$ *whenever* $\mathcal{N}(A) > 0$.

PROOF. With $q_\varepsilon = m^{-1}\sum_j |k_j^\varepsilon|^2$, a probability density, one has exactly

$$\|F_\varepsilon\|_1 = \frac{1}{m}\sum_j \int \frac{m\,|\sum A_{ik}r_ir_k|}{\sum_i |r_i|^2}\,|k_j^\varepsilon|^2\,\mathrm{d}m, \qquad r_i = k_i^\varepsilon/k_0^\varepsilon.$$

The integrand is a continuous function of r, bounded by $m\|A\|$, so Lemma 11.2 gives the convergence of each integral. The mixture is what makes the integrand bounded; a single reference measure would leave an unbounded integrand and a uniform integrability problem. Finally $F_\varepsilon(0) = O(\varepsilon)$. □

The mixture over j collapses to a single expectation. Put $g_i = G_{i0} = 2\sqrt{a_ia_0}/(a_i+a_0)$ and $R_i = g_ie^{D_i}$.

PROPOSITION 11.4. $\mathcal{N}(A) = \mathbb{E}\,|R^TAR|$. *Moreover* $\mathbb{E}[R^u\overline{R}^v] = g^{u+v}\exp(u^T\Sigma v)$ *for multi-indices* $u, v \ge 0$*; in particular* $\mathbb{E}[R_i\overline{R_k}] = G_{ik}$.

PROOF. Since D is circular, $\mathbb{E}[e^{u\cdot D+v\cdot\overline{D}}] = \exp(u^T\Sigma v)$, which gives the moments; with $e^{\Sigma_{ik}} = (a_i+a_0)(a_k+a_0)/(2a_0(a_i+a_k))$, the case $u = e_i$, $v = e_k$ is G_{ik}, and $\mathbb{E}|R_j|^2 = 1$. Under the probability measure $|R_j|^2\,\mathrm{d}\mathbb{P}$, the vector D is shifted by $\mathbb{E}[D\,2\Re D_j] = (\Sigma_{ij})_i$, so R_i becomes $g_ie^{\Sigma_{ij}}e^{D_i}$, and $g_ie^{\Sigma_{ij}} = \sqrt{a_i/a_0}\,(a_0 + a_j)(a_i+a_j)^{-1}$: this is the law of $R^{(j)}$. Hence $\mathcal{N}(A) = \sum_j \mathbb{E}[|R_j|^2|R^TAR|/|R|^2] = \mathbb{E}|R^TAR|$. □

The single expectation decides whether a square can be improved. Repeated scales may be merged, so we assume they are distinct. Let u be real with $u^TGu = 1$, so that $\mathcal{N}(uu^T) = 1$, put $\lambda = u^THu$, and

$$\Delta_u = \Re\,\mathbb{E}\Big[RR^T\,\overline{\frac{(u^TR)^2}{|u^TR|^2}}\Big], \qquad K_u = H - \lambda\Delta_u.$$

PROPOSITION 11.5. *For real symmetric* B*, the derivative at* $t = 0$ *of* $\operatorname{tr}(H(uu^T + tB))/\mathcal{N}(uu^T + tB)$ *is* $\operatorname{tr}(K_uB)$*, and* $\Delta_u u = Gu$*. If* $K_u = 0$*, the square* uu^T *maximizes the ratio over all real symmetric* A *with these scales; if* $K_u \preceq 0$*, it maximizes it over positive semidefinite* A.

PROOF. The covariance integral shows that Σ is positive definite on the nonreference coordinates: the functions $e^{-a_ix} - e^{-a_0x}$ are linearly independent. Hence D has a density there. The exponential polynomial u^TR is not identically zero, since $u^TGu = 1$, so its zero set has probability zero. Moreover, $|R^TBR| \le \sum|B_{ik}||R_iR_k|$ is integrable. Thus $\mathcal{N}$ is differentiable at uu^T with derivative $B \mapsto \operatorname{tr}(\Delta_uB)$, and $\Delta_u u = \Re\,\mathbb{E}[R\overline{u^TR}] = Gu$ by Proposition 11.4. Since $\mathcal{N}$ is convex and positively

homogeneous, $\mathcal{N}(A) \geq \operatorname{tr}(\Delta_u A)$ for every A. If $K_u = 0$ then $\operatorname{tr}(HA) = \lambda \operatorname{tr}(\Delta_u A) \leq \lambda \mathcal{N}(A)$; if $K_u \preceq 0$ and $A \succeq 0$ then $\operatorname{tr}(HA) \leq \lambda \operatorname{tr}(\Delta_u A)$ as well. □

So proving $C_{\text{Gauss}} > \pi$, hence $C > \pi$, is reduced to bounding one finite-dimensional Gaussian expectation from above. The anchored form of D makes this practical: with $L(t) = \log\cosh(t/2)$, the increments $D(t_j + s) - D(t_j)$ form a stationary field with covariance $L(s) + L(s') - L(s - s')$. Squares give exactly $u^T G u$: the square $|u^T R|^2$ has exact mean $u^T G u$, which provides an exact control variate.

Let C_{Gauss} be the supremum of $|\sum_{i,j} A_{ij} H_{ij}|/\mathcal{N}(A)$ over all m, all scales $a_1, \dots, a_m > 0$, and all real symmetric A with $\mathcal{N}(A) > 0$. It is a constant of the log-correlated Gaussian field D, and involves no arithmetic. Squares alone give $C_{\text{Gauss}} \geq \pi$: for $A = uu^T$ the ratio is $u^T H u / u^T G u$, and on fine grids of scales with long span this approaches the supremum π of the ratio $\pi \operatorname{sech}(\pi\xi)$ of the Fourier transforms in Proposition 11.1.

COROLLARY 11.6. $C \geq C_{\text{Gauss}}$.

PROOF. This is the last statement of Theorem 11.3. □

The same witnesses see past continuous symbols. Let $\mathscr{M}$ be the multiplicative Hilbert matrix and $\mathcal{E}_{\text{sym}}$ the essential size of its symbols, as in section 4.6.

COROLLARY 11.7. $\mathcal{E}_{\text{sym}}(\mathscr{M}) \geq C_{\text{Gauss}}$.

PROOF. Let g be a bounded symbol of $\mathscr{M}$, so that $\int F g \, \mathrm{d}m = \ell(F)$ for $F \in H^1_0$, and let $c \in C(\mathbb{T}^\infty)$ and $v \in N$. Since v annihilates analytic polynomials, $|\ell(F^0_\varepsilon)| \leq \|g - c - v\|_\infty \|F^0_\varepsilon\|_1 + |\int F^0_\varepsilon c \, \mathrm{d}m|$. The nth coefficient of $k^\varepsilon_i k^\varepsilon_j$ is at most $d(n) n^{-1/2} (\zeta(1 + 2\varepsilon a_i)\zeta(1 + 2\varepsilon a_j))^{-1/2}$, which tends to 0; so $\int F^0_\varepsilon P \, \mathrm{d}m \to 0$ for every trigonometric polynomial P, and, since $\|F^0_\varepsilon\|_1$ is bounded, also for every $c \in C(\mathbb{T}^\infty)$. Letting $\varepsilon \to 0$ in Proposition 11.1 and Theorem 11.3 gives $|\sum A_{ij} H_{ij}| \leq \|g - c - v\|_\infty \mathcal{N}(A)$. □

The witnesses also live at the critical line. For $\sigma \geq \delta$, $|k^\varepsilon_i k^\varepsilon_j(r(\sigma)) - k^\varepsilon_i k^\varepsilon_j(0)| \leq C\varepsilon(\zeta(1+\sigma)^2 - 1)$, so $\int_\delta^\infty |F^0_\varepsilon(r(\sigma))| \, \mathrm{d}\sigma = O(\varepsilon/\delta)$.

COROLLARY 11.8. *Let w be bounded on $[0, \infty)$, continuous at 0, with $w(0) = 1$, and $\ell_w(f) = \int_0^\infty w(\sigma)(f(1/2+\sigma) - a_1) \, \mathrm{d}\sigma$. Then $\|\ell_w\|_{(\mathscr{H}^1_0)^*} \geq C_{\text{Gauss}}$. In particular this holds for every ℓ_c of remark 9.9 and for the restriction of ℓ to any interval $(0, \delta)$.*

PROOF. $\ell_w(F^0_\varepsilon) - \ell(F^0_\varepsilon) = \int_0^\infty (w - 1) F^0_\varepsilon$. On $(0, \delta)$ this is at most $\sup_{(0,\delta)} |w - 1|$ times $\int_0^\delta |F^0_\varepsilon| \leq (5 + 2\sqrt{2}) \|F^0_\varepsilon\|_1$ (Theorem 2.5), and on (δ, ∞) it is $O(\varepsilon/\delta)$. Let $\varepsilon \to 0$, then $\delta \to 0$. □

Since $\|\mathscr{M}\|_{\text{e}} = \pi$ (Corollary 4.13), the inequality $C_{\text{Gauss}} > \pi$ suggested by remark 11.10 would give $\mathcal{E}_{\text{sym}}(\mathscr{M}) > \|\mathscr{M}\|_{\text{e}}$: the excess over π would survive removing every continuous symbol.

CONJECTURE 1. $C = C_{\text{Gauss}} > \pi$.

The evidence for the inequality is remark 11.10 below. The evidence for the equality is structural. Near-extremal vectors for the multiplicative Hilbert matrix tend weakly to zero, put asymptotically no mass at $\sigma > \delta$, and vary slowly under multiplication (Theorem 9.3). For ℓ on $\mathscr{H}_0^1$, section 11.2 proves that every extremizing sequence either concentrates at $\sigma = 0$ or produces an extremizer, and that the constant at $\sigma = 0$, suitably normalized, does not see any finite set of primes (Theorem 11.13 and Proposition 11.16). One route to the conjectured equality is to prove that ℓ does not attain its norm and that the concentrated problem is governed by the Gaussian limit of Theorem 11.3. Together these would identify C with C_{Gauss}. If the second fails for quadratics, the natural replacement allows kernel polynomials of higher degree, whose norms we have not computed.

REMARK 11.9 (The excess, numerically). Take the scales $t_i = i$, $|i| \le T/2$, and let A range over quadratics in the top few generalized eigenvectors of (H, G). Then minimize $\mathcal{N}(A)$ for fixed $\sum A_{ij}H_{ij}$ on one sample and evaluate on an independent one. The best ratios are then $\sum A_{ij}H_{ij}/\mathcal{N}(A) \approx 3.1386$, 3.1422, 3.1442, 3.1450 for $T = 120$, 160, 200, 240, against 3.1326, 3.1363, 3.1381, 3.1392 for squares. Indefinite quadratics thus gain about 0.006 over squares at every span. The squares' deficit decays with the span, and the two cross between $T = 120$ and $T = 160$. A refinement of the grid changes nothing.

REMARK 11.10 (A probabilistic certificate). For $T = 240$ and one explicit quadratic in six modes, normalized so that $\sum A_{ij}H_{ij} = 1$ (to 10^{-15}, in interval arithmetic), 10^8 samples drawn from a seed fixed in advance give $\mathcal{N}(A) \le 0.3179567$. The tool is a one-sided Bernstein inequality for the control-variate residual, with explicit Lipschitz, coupling and rounding budgets. The threshold is $1/\pi = 0.3183099$, so, if the certificate is right, $C \ge C_{\mathrm{Gauss}} \ge 3.1450 > \pi$. This is not a proof. It holds except on an event of probability at most $2 \cdot 10^{-30}$, assuming that the pseudo-random generator behaves like independent draws, that A was chosen without using the certification stream, and that floating-point arithmetic obeys its rounding model. A deterministic certificate needs a bound for one bounded Gaussian integral in 480 real dimensions, to about $4 \cdot 10^{-4}$. Grids are hopeless at that dimension, and quadratic majorants provably fail (their best bound is $\|G^{1/2}AG^{1/2}\|_{S_1}$).

The certificate is consistent with everything above. The kernels involve all primes, and the excess appears only as $\varepsilon \to 0$, so only primes far beyond 10^{216} matter. The quadratic is not a square, and it mixes all degrees. If it is right, then $C > \pi$, the sharp Fejér–Riesz conjecture fails, and $\Phi(F_k) \not\to 1$ along extremizing sequences (the two are equivalent; see the discussion after Proposition 9.7).

REMARK 11.11 (Fewer scales). Numerically, splitting the primes into K interleaved classes and summing the squares of the top eigenvectors on each class gives ratios 1.29, 0.61, 0.47 for $K = 1, 2, 3$: the Ortega-Cerdà–Seip gain from independent blocks is real, about $\sqrt{K}$, but sparse classes give trace-class matrices (Corollary 5.3) whose top eigenvalues fall like $1/K$. None of these constructions exceeds the square bound, and the Euler-product family $\prod_{p \le y}(1 + 2z_p/\sqrt{p}) - 1$ stays below $e^{\gamma_{\mathrm{E}}} < \pi$. All of them use a bounded number of scales; the excess of remark 11.9 appears numerically only from about one hundred and fifty scales.

11.2. Extremizing sequences

The obstruction to compactness is that mass can escape toward the critical line: an extremizing sequence may put almost all of $\int |f(1/2+\sigma)|\,\mathrm{d}\sigma$ near $\sigma = 0$, where no fixed coefficient sees it. The proof of Theorem 11.13 handles the other alternative with three tools, each doing one job. Komlós's theorem replaces a subsequence by Cesàro means that converge almost everywhere; Lemma 11.12 shows that such a limit of a bounded sequence in H^1 is again analytic, with the limiting coefficients; and dominated convergence splits the L^1 norm between the limit and the remainder, which lets the extremal ratio pass to the limit.

For the multiplicative Hilbert matrix, near-extremal vectors tend weakly to zero because $\mathscr{M}$ has no eigenvectors (Theorem 9.3). In H^1 weak-* limits do not split the norm, since L^1 norms are not additive under weak convergence. The substitute is an averaging argument and the following lemma, in which analyticity is what matters. Write E_d for the conditional expectation onto the first d coordinates, and P_x for the Poisson kernel of $x \in \mathbb{D}^d$, viewed as a function on $\mathbb{T}^\infty$.

LEMMA 11.12. *Let $F_N \in H^1(\mathbb{T}^\infty)$ with $\sup_N \|F_N\|_1 < \infty$ and $F_N \to u$ almost everywhere. Then $u \in H^1(\mathbb{T}^\infty)$, and the Fourier coefficients of F_N converge to those of u.*

PROOF. Let $K = \sup_N \|F_N\|_1$. By Fatou, $u \in L^1$. Let μ be a weak-* cluster point of $F_N\,\mathrm{d}m$; passing to a subsequence, $F_N\,\mathrm{d}m \to \mu$ weak-*. Since $|G|^{1/2}$ is plurisubharmonic for G analytic, and both sides are continuous on H^1,

$$|E_d G(x)|^{1/2} \le \int |G|^{1/2} P_x\,\mathrm{d}m \qquad (G \in H^1,\ x \in \mathbb{D}^d).$$

Fix M and put $g = |u - F_M|^{1/2}$. Apply this to $G = F_N - F_M$ and let $N \to \infty$. On the left, $E_d F_N(x) \to \int P_x\,\mathrm{d}\mu$. On the right, $|F_N - F_M|^{1/2} \to g$ almost everywhere with $\||F_N - F_M|^{1/2}\|_2^2 \le 2K$, hence weakly in L^2. So the Poisson extension h_d of the analytic measure $\nu = E_d\mu - E_d F_M$ on $\mathbb{T}^d$ satisfies $|h_d(x)| \le (P_x[E_d g])^2$. As $\rho \to 1$ the functions $(P_{\rho\zeta}[E_d g])^2$ converge in $L^1(\mathbb{T}^d)$, so $\{h_d(\rho\,\cdot)\}$ is uniformly integrable; therefore ν is absolutely continuous, and letting $\rho \to 1$ along a suitable sequence gives $|E_d\mu - E_d F_M| \le (E_d g)^2$ almost everywhere on $\mathbb{T}^d$.

By Doob's maximal inequality $\sup_d (E_d g)^2 \in L^1$, and $\{E_d F_M\}_d$ is uniformly integrable, so the martingale $(E_d\mu)_d$ is uniformly integrable. Hence $\mu = F\,\mathrm{d}m$ with $F \in L^1$, and martingale convergence gives $|F - F_M| \le |u - F_M|$ almost everywhere, for every M. Letting $M \to \infty$ gives $F = u$. So $\mu = u\,\mathrm{d}m$, which has analytic spectrum, and the whole sequence converges. □

THEOREM 11.13 (Concentration or an extremizer). *Let (f_k) be an extremizing sequence: $f_k \in \mathscr{H}^1_0$, $\|f_k\|_{\mathscr{H}^1} = 1$ and $\ell(f_k) \to C$. Then either*

$$\int_\delta^\infty |f_k(1/2+\sigma)|\,\mathrm{d}\sigma \to 0 \qquad \textit{for every } \delta > 0,$$

or ℓ attains its norm: there is $u \in \mathscr{H}^1_0$, $u \ne 0$, with $|\ell(u)| = C\|u\|_{\mathscr{H}^1}$.

PROOF. For $\|f\|_1 \le 1$ and $a_1 = 0$ we have $|f(1/2+\sigma)| \le \zeta(1+2\sigma)$ by Proposition 1.3 and $|f(1/2+\sigma)| \le \zeta(1/2+\sigma) - 1$ for $\sigma \ge 1$, an integrable majorant on $[\delta, \infty)$.

Helson's inequality $\sum|a_n|^2/d(n) \le \|f\|_1^2$ [44] makes the tails $\sum_{n>N}|a_n|n^{-1/2-\sigma}$ small uniformly on the unit ball. Hence, if the coefficients of f_k converge to those of u, then $f_k(1/2+\sigma) \to u(1/2+\sigma)$ for every $\sigma > 0$, and $\int_\delta^\infty |f_k| \to \int_\delta^\infty |u|$ by dominated convergence.

Suppose that $\int_\delta^\infty |f_k| \ge \eta > 0$ along a subsequence. Pass to a further subsequence whose coefficients converge, to those of a measure μ, and then, by Komlós's theorem [53] applied to the real and imaginary parts, to one whose Cesàro means F_N converge almost everywhere. Then $F_N \in \mathscr{H}_0^1$, $\|F_N\|_1 \le 1$ and $\ell(F_N) \to C$, so $\|F_N\|_1 \to 1$; the coefficients of F_N still converge to those of μ. By Lemma 11.12, $\mu = u\,dm$ with $u \in \mathscr{H}_0^1$ and $F_N \to u$ almost everywhere, and $\int_\delta^\infty |u| \ge \eta$ (apply the first paragraph to the f_k), so $u \ne 0$.

To split the norm, observe that

$$|F_N| - |F_N - u| \longrightarrow |u|, \qquad ||F_N| - |F_N - u|| \le |u|.$$

Since $u \in L^1$, dominated convergence gives $\|F_N\|_1 - \|F_N - u\|_1 \to \|u\|_1$. Thus $\|F_N - u\|_1 \to 1 - \|u\|_1$. Since $F_N - u \in \mathscr{H}_0^1$,

$$C = \lim|\ell(F_N)| \le |\ell(u)| + C\lim\|F_N - u\|_1 = |\ell(u)| + C(1 - \|u\|_1),$$

so $|\ell(u)| \ge C\|u\|_1$. □

Write $\mathcal{E} = \mathcal{E}_{\mathrm{sym}}(\mathscr{M})$. Since $(\mathscr{H}_0^1)^* = L^\infty/(N + \mathbb{C})$ and trigonometric polynomials are dense in $C(\mathbb{T}^\infty)$, $\mathcal{E}$ is the distance from ℓ to the closure of the functionals that depend on finitely many coefficients. Clearly $\mathcal{E} \le C$, and $\mathcal{E} \ge C_{\mathrm{Gauss}}$ by Corollary 11.7.

COROLLARY 11.14. *If $C > \mathcal{E}$, then ℓ attains its norm. Equivalently, if ℓ does not attain its norm, then $C = \mathcal{E}$, and moreover $C \le \Gamma(1/4)^2/\sqrt{2\pi} < 5.2442$.*

PROOF. Suppose ℓ does not attain its norm, and let (f_k) be extremizing; by Theorem 11.13 it concentrates. Then every coefficient of f_k tends to 0: otherwise a subsequence has coefficients converging to those of a nonzero Dirichlet series u, the first paragraph of the proof above gives $f_k(1/2+\sigma) \to u(1/2+\sigma)$ for $\sigma > 0$, and u, being nonzero, does not vanish identically on (δ, ∞), contradicting concentration. So $\lambda(f_k) \to 0$ for every functional λ in the closure, and $C = \lim|\ell(f_k)| \le \|\ell - \lambda\|$. Hence $C \le \mathcal{E}$. Finally $C = \lim|\int_0^2 f_k(1/2+\sigma)\,d\sigma|$ by concentration, and Theorem 2.9 bounds this by $\Gamma(1/4)^2/\sqrt{2\pi}$. □

An extremizer, if there is one, is rigid.

PROPOSITION 11.15 (Extremizers). *Let $F \in H_0^1$ with $\|F\|_1 = 1$ and $\ell(F) = C$.*

(1) $\ell(QF) = C\int Q|F|\,dm$ for every analytic polynomial Q.
(2) F depends on infinitely many variables.
(3) F is not of the form h^2 with $h \in H^2$.

PROOF. (1) For complex w, $|1+w| \le (1+|1+w|^2)/2 = 1 + \Re w + |w|^2/2$. So, for complex t,

$$C + \Re\big(t\,\ell(QF)\big) \le |\ell(F(1+tQ))| \le C\|F(1+tQ)\|_1 \le C + C\Re\Big(t\int Q|F|\,\mathrm{d}m\Big) + \frac{C|t|^2}{2}\|Q\|_\infty^2,$$

and letting $t \to 0$ in every direction gives (1).

(2) Suppose F depends only on the variables of a finite set S of primes. For $p \notin S$, $\int z_p|F|\,\mathrm{d}m = 0$, so by (1) $\int_0^\infty p^{-\sigma} g(\sigma)\,\mathrm{d}\sigma = 0$, where $g(\sigma) = F(r(\sigma))$. The function g is real analytic on $(-1/2,\infty)$, bounded on $[0,\infty)$, and not identically zero since $\ell(F) \ne 0$; if $g(\sigma) \sim c\sigma^m$ at 0, then Watson's lemma gives $\int_0^\infty e^{-x\sigma} g(\sigma)\,\mathrm{d}\sigma \sim c\,m!\,x^{-m-1} \ne 0$ as $x \to \infty$, which contradicts its vanishing at $x = \log p$ for every large prime p.

(3) If $F = h^2$ with $h = \sum b_n n^{-s}$, then $b_1 = 0$ and $C = |\langle \mathscr{M} b, \bar b\rangle| \le \langle \mathscr{M}|b|, |b|\rangle \le \pi\|b\|^2 = \pi$. Since $C \ge \pi$, $|b|$ would be an eigenvector of $\mathscr{M}$ with eigenvalue π, and there are none [72]. □

In the first alternative of Theorem 11.13 the constant is local. For $0 < \delta \le 1$ let $C^{(\delta)}$ be the norm of $f \mapsto \int_0^\delta f(1/2+\sigma)\,\mathrm{d}\sigma$ on $\mathscr{H}^1$, and $C^{(\delta)}_{>P}$ its norm on the series supported on integers with no prime factor $p \le P$. Put $c_P = \prod_{p\le P}(1-1/p)$.

PROPOSITION 11.16 (Small primes). *For every P there is K_P with $|C^{(\delta)} - C^{(\delta)}_{>P}/c_P| \le K_P\delta$ for $0 < \delta \le 1$. If ℓ does not attain its norm, then $C \le C^{(\delta)}$ for every δ, so*

$$C \le \liminf_{\delta\to 0} \frac{C^{(\delta)}_{>P}}{c_P} \qquad \text{for every } P.$$

PROOF. Let S be the primes up to P, write $z = (\zeta, w)$ and $r(\sigma) = (r_S(\sigma), r'(\sigma))$, and put $r_0 = r_S(0)$. For a polynomial F, $F(r(\sigma)) = \int_{\mathbb{T}^S} F(\zeta, r'(\sigma)) P_{r_S(\sigma)}(\zeta)\,\mathrm{d}m(\zeta)$, and $|P_{r_S(\sigma)} - P_{r_0}| \le K_P\sigma$ for $\sigma \le 1$ because r_0 lies inside the polydisc. The function $H(w) = F(r_0, w)$ has $\|H\|_1 \le \|F\|_1/c_P$: apply $|g(a)| \le \|g\|_{H^1(\mathbb{T})}/(1-|a|^2)$ in each variable of S and integrate. Splitting $P_{r_S(\sigma)} = P_{r_0} + (P_{r_S(\sigma)} - P_{r_0})$,

$$\Big|\int_0^\delta F(r(\sigma))\,\mathrm{d}\sigma\Big| \le \frac{C^{(\delta)}_{>P}}{c_P}\|F\|_1 + K_P\delta\int_{\mathbb{T}^S}\int_0^1 |F(\zeta, r'(\sigma))|\,\mathrm{d}\sigma\,\mathrm{d}m(\zeta),$$

and the last integral is at most $(5+2\sqrt{2})\|F\|_1$ by Theorem 2.5, applied for each ζ to $w \mapsto F(\zeta, w)$. Conversely, for g supported on integers free of primes $p \le P$, put $f = ug$ with $u = \prod_{p\le P}(1-1/p)(1-p^{-1/2}z_p)^{-2}$. The variables are disjoint, so $\|f\|_1 = \|u\|_1\|g\|_1 = \|g\|_1$, and $u(r_S(\sigma)) = c_P\prod_{p\le P}(1-p^{-1-\sigma})^{-2} = c_P^{-1} + O_P(\sigma)$. Hence $\int_0^\delta f(r)\,\mathrm{d}\sigma = c_P^{-1}\int_0^\delta g(r')\,\mathrm{d}\sigma + O_P(\delta)\|g\|_1$. For the last statement, apply Theorem 11.13: along an extremizing sequence $\int_0^\delta f_k(1/2+\sigma)\,\mathrm{d}\sigma \to C$. □

The factor $c_P = \|K_{r_0}\|_2^{-2}$ is the price of removing the small primes. To pursue the route to Conjecture 1 described above, one would need nonattainment and asymptotic optimality of kernels at many scales for the local constant

$\lim_P \liminf_{\delta\to 0} C^{(\delta)}_{>P}/c_P$. A possible approach is a central limit theorem for $\log|f|$ under $|f|\,\mathrm{d}m$ along concentrating sequences.

Open

(1) Prove $C > \pi$ deterministically, by bounding the Gaussian integral of Theorem 11.3 for one explicit quadratic.
(2) Prove Conjecture 1. Can one prove the equality by showing that ℓ does not attain its norm on $\mathscr{H}^1_0$ and that kernels at many scales determine the local constant at $\sigma = 0$ after removal of the small primes?
(3) Is the window scaling limit of J at $\sigma \asymp 1/\log P$ governed by critical multiplicative chaos, and does it give a per-unit-scale formula for C?

APPENDIX A

Computations

Several results rest on a computation. The scripts that check them are published at

`https://github.com/Oisov/math-preprints`, folder `numerics/book`,

and `python3 run_all.py` there reruns all of them and compares their output with the values printed here; it takes about two minutes. Interval computations use ball arithmetic (Arb, through `python-flint`); the others use exact rational or directed dyadic arithmetic in the Python standard library.

Result	What is computed	Arithmetic
Theorem 2.9	the analytic margin at $\sigma = 1/100$ and Schur rows on 268 cells of $[1/100, 2]$	Arb
Proposition 3.8	a 192-term Legendre witness for $C_0 > 2\pi + 2.8 \cdot 10^{-7}$, and the rank-two bound $C_0 < 2\pi + 10^{-3}$	exact dyadic
Proposition 4.23	a lower bound $17/100$ for the phase integral $\int \mathrm{Re}(\overline{U}/U)$ over $\mathbb{T}^3$, from 256^3 cells with Arb-enclosed kernels and outward-rounded binary64 sums	Arb and directed binary64
Theorem 5.23 and Corollary 5.24	the constants P, K, c_R and L_R from the primes below 10^6 with explicit tails, and the two scalar inequalities of the proof	Arb
Remark 5.8	entries of 128×128 Bloch matrices at 129 nodes, their inertia by exact LDL^* factorization with residual bounds, and the operator error budget	Arb, exact dyadic and rational
Theorem 6.19 (Lemmas 6.20 and 6.22)	Schur tests on 2291 cells with chord bounds from a table of ζ-values, a Rayleigh quotient by Gauss–Legendre integration, and positivity of Φ_q'' on about 2400 boxes by the mean value form	Arb and directed binary64
Proposition 8.10	the Mellin integral for A with n replaced by a ball over 11 blocks covering $194 \le n \le 3998$, and the constants of Proposition 8.9	Arb
Theorems 8.6 and 9.2	the displayed polynomial identities and the signs of their coefficients	exact (symbolic)
Proposition 9.14	Schur tests with verified weights, Hilbert–Schmidt tails, the integral behind 10^{216}, and the L^1 norm of the witness for C_2	Arb
Theorem 9.15	$\ell(q)$ and $\ell(5^{-s}q)$ for the two-prime witness, and a Schur test on the 3429×3428 truncation with stored weights and a Hilbert–Schmidt tail	Arb and directed binary64

The probabilistic certificate of remark 11.10 is published in the same folder, with its scripts, the fixed quadratic as exact binary64 data, and the log of the certified run. It is not a proof. Its failure bound $2 \cdot 10^{-30}$ assumes that the pseudo-random generator behaves like independent draws and that floating-point arithmetic obeys its rounding model, and an exact rerun needs the numpy version of the original run.

APPENDIX B

Problems and their status

The book states twelve numbered problems: nine from the literature and three that arise here. The table records what is known about each, as of October 2026; several of the answers rest on recent preprints. Of the twelve, four are answered in this book, one of them in the negative, one computer-assisted and one up to constants (Problems 2, 5, 11 and 12), three have been answered elsewhere, one of them also by a second proof here and with a new Bergman analogue (Problems 1, 3 and 8), two are answered in part, and three are open.

	Source	Status
Problem 1	Saksman–Seip [80, Problem 2.1]	Answered, modulo a preprint: yes exactly for $p \geq 2$. No for $0 < p < 2$ by Harper [40]; yes for $p > 2$ announced by Chen et al. [28]. Here: $N_p(\varepsilon) \leq \exp(C\sqrt{L \log L})$, $L = \log(1/\varepsilon)$, for every p (Theorem 3.4), and $N_p(\varepsilon) \gtrsim_p L^{kp/4}$, k the largest integer with $kp < 2$ (Proposition 3.5); the order for fixed $p < 2$ is open.
Problem 2	Perfekt–Pushnitski [71]	Answered: no (Theorem 4.21; a finite witness in three variables, computer-assisted, in Proposition 4.23). Yes for partially averaged curve measures in three geometries (Theorems 4.2 and 6.15) and for energy-separated families (Proposition 2.14).
Problem 3	Saksman–Seip [80, Problem 3.2]	Answered by Perfekt [70]; second proof in Corollary 4.18, explicit symbol in Theorem 9.18; Bergman analogue in Theorem 6.18.
Problem 4	Bayart et al. [12, Section 5]	Open. Equivalent to the sharp inequality $C_q \leq C_p$ for finite atomic measures (Theorem 4.19).
Problem 5	This book	Answered: yes under regular variation (Theorem 5.4; under C^2 regularity, with the finer spectral type, Ma–Lu–Zu [58]); comparability alone fails. The exact upper-limit criterion is Theorem 5.27; general translated profiles are covered by Theorem 5.26.
Problem 6	Brevig–Perfekt [20, Question 3]	Partial. Such forms exist for every $p > 4$ (Theorem 7.12; also Bampouras [4]); open for $2 < p \leq 4$, where the natural families are sharp at 4 (chapter 8). In two variables none exist, for any $p < \infty$ [9].
Problem 7	This book	Open. Every family treated in chapter 8 satisfies the inequality.
Problem 8	Brevig–Perfekt [21]	Answered, in preprints: for balls by Bampouras–Perfekt [8], for convex sets free of lines by Bampouras [5]. The constant exceeds 1 and grows exactly exponentially on polydiscs (Proposition 8.20 and Corollary 8.21). For S_p, $2 < p \leq 4$, open on the disc; no threshold for polygons.
Problem 9	This book	Open. $C < \infty$ by Perfekt [70]; $\pi \leq C \leq 5.515$, computer-assisted (Proposition 9.14); a probabilistic certificate, not a proof, indicates $C \geq 3.145$ (remark 11.10).
Problem 10	Perfekt [70]	Partial. True on every piece of Theorem 10.2, for products across prime sets (Proposition 10.4), and with constant $3.1407 < \pi$ on primes up to 10^{216} (Proposition 9.14).
Problem 11	Bondarenko–Brevig–Saksman–Seip [15]	Answered: yes, and for all p, β and prime densities (Corollary 10.13 and Theorem 10.12).
Problem 12	Saksman–Seip [80, Problem 4.5]	Answered up to constants. On $\mathscr{H}^1$ the order is $\log N / \log\log N$: upper bound by Bondarenko et al. [15], lower bound in Theorem 10.17; on $\mathscr{H}^\infty$ it is $\log N$ [29]. Leading constants open.

Further open questions

(1) *Two-point stability.* The leading constant is settled (Theorem 1.9), and the halved logarithmic bound fails for $p = 1$ (Theorem 1.12). For which other p does it fail, and what is the optimal defect function (chapter 1)?
(2) *What Hellinger–Bessel loses.* Is there a geometric condition, between ℓ^2-bounded affinities and separation, under which the Gram condition is necessary (Proposition 2.14)?
(3) *Vertical lines.* The order of $N_p(\varepsilon)$ for fixed $p < 2$, whether $\sup_p N_p(\varepsilon)$ is polylogarithmic in $1/\varepsilon$, and the exact value of C_0 in Proposition 3.7 (chapter 3).
(4) *Spectra.* Which measures have $s_n \asymp a_n^*$ without an index change? Certify further spectral gaps of Proposition 5.7; one is certified in remark 5.8. The exponent of the bounded-rank Nehari constant, between 0.174 and 1/2 (chapter 5).
(5) *Bergman spaces.* The sharp two-point constants: in the Bergman geometry beyond the first zero of $K_x + K_{-x}$ (up to it the square bound is sharp), and in the Gauss geometry, where the square bound always fails, the constant itself (Proposition 6.7); and an unconditional proof that Bergman Dirichlet kernels do not cancel for $1 < \beta \leq 2$ (Theorem 6.14; for $\beta > 2$ see Theorem 6.13).
(6) *Near $p = 2$.* Is there a dimension- and degree-free interval above 2 with no strict Gaussian blocks? There is one for each bounded rank (Theorem 8.16).
(7) *Fixed dimension.* The S_p Nehari constant on $\mathbb{T}^d$ is finite for every p, by Nehari's theorem on $\mathbb{D}^d$; does it grow with d for some $p \leq 4$ (chapter 7)? Does the S_2 constant for the ball B^n grow with n, as it does for polydiscs (chapter 8)?
(8) *Spectral type.* For prime sets of density θ, is the critical spectrum absolutely continuous, and when are there eigenvalues above it? For all primes in the Bergman geometry this is known, and for the explicit oscillating sets R_P the bands are absolutely continuous spectrum and there is no singular continuous spectrum (Theorem 5.18); the general case is open (chapter 5).
(9) *The sharp constant.* A deterministic proof of $C_{\mathrm{Gauss}} > \pi$; does ℓ attain its norm on $\mathscr{H}_0^1$ (Corollary 11.14); is $C = C_{\mathrm{Gauss}}$ (Conjecture 1)?

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